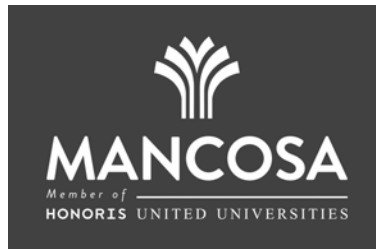


OPERATIONS MANAGEMENT

Module Guide

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This module guide,
Operations Management(NQF Level 8)
will be used across the following programmes:

- Bachelor of Commerce Honours in Supply Chain Management
- Postgraduate Diploma in Business Management

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Preface

A. Welcome

Dear Student

Welcome to the **Operations Management (OMN801)** module. You are encouraged to read this overview and module guide carefully, as it will aid you through your study journey. This module guide intends to develop your knowledge and proficiency.

The field of **Operations Management** is dynamic and innovative. The learning content and special features contained in this module guide will provide you with opportunities to explore current industry developments and theories.

As this is a distance-learning module, self-discipline and time management would need to be applied effectively. You will have the opportunity to engage with interactive digital tools via MANCOSACONnect to enhance your learning journey.

Through the inclusion of relevant content and industry aligned practices within the learning content, you will develop critical thinking and problem-solving skills, empowering you as change agents for a more sustainable world.

Please note that some special features may not have answers available, where answers are not available this can be further discussed with your lecturer during the webinar session.

We hope you enjoy the module.

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B. Module Overview

- The module is a **15 credit** module at **NQF level 8**
- The purpose of the module is to introduce you to some of the general and emerging market issues regarding each operations management topic

C. Learning Outcomes and Associated Assessment Criteria of the Module

Module Outcomes	Associated Assessment Criteria
• Appreciate the extent and qualities of operations management functions	• The scope and principles of operations management are detailed to afford the student an understanding of the nature and components of operations management • Operations management is holistically unbundled to enable the student to appreciate that operations management is core to management
• Apply the principles of the decision-making process to each part of the operations function	• Fundamental decision making process' and requirements are critiqued to effectively formulate strategy that will enhance sustainability • Principles of the decision-making process are illustrated to enhance the quality of business decisions
• Implement effective and sustainable managerial decisions in order to ensure the effective and sustainable utilisation of resources	• Effective managerial decisions and policies are applied through sustainable business practice to ensure the effective utilisation of resources • Implementation of management decisions and policies are in line with core strategic objectives to realise the vision and mission of the organisation

Analyse the manner in which decisions in operations management can guide the overall strategic objectives of an organisation	- Business unit decisions are assessed by the level of integration with the organisation's sustainability objectives to facilitate the going concern principle - Overall strategic objectives are analysed against the backdrop of sound business unit decision making policies to align with the overall business objectives
Examine and analyse the methodology behind developing effective operations strategies	- The methodology behind the development of effective and sustainable business strategies are examined in order to provide appropriate responses to current business challenges - Current and proposed strategies are analysed to ensure their ongoing relevance and practicality

D. How to Use this Module

This Module Guide was compiled to assist you with your study journey. The purpose of the Module Guide is to allow you the opportunity to integrate the theoretical concepts discussed within the content of the Module Guide. We suggest that you read through the entire module guide to get an overview of its contents. At the beginning of each Unit, you will find a list of Learning Outcomes This outlines the main points that you should understand when you have completed the Unit/s.

This Module Guide should be studied in conjunction with the prescribed, recommended textbook(s)/ reading(s) and other relevant study material. It is important that you make your own notes as you work through the prescribed, recommended textbook(s)/ reading(s), other relevant study material, and the module guide. You may obtain additional reading material by utilising publications referenced at the end module guide under the reference list and bibliography.

E. Study Material

The study material for this module includes, programme handbook, this Module Guide, and a list of prescribed and recommended textbooks/readings which may be supplemented by additional readings.

F. Prescribed Textbook

The prescribed and recommended reading(s)/textbook(s) presents a tremendous amount of material in a simple, easy-to-learn format. You should read ahead during your course. Make a point of it to re-read the learning content in your module textbook. This will increase your retention of important concepts and skills. You may wish to read more widely than just the Module Guide and the prescribed and recommended textbooks/readings, the Bibliography and Reference list provides you with additional reading.

The prescribed and recommended textbook(s)/reading(s) for this module are:

Prescribed Reading(s)/Textbook(s)

- Heizer, J., Render, B. and Munson, C. (2020). *Operations Management: Sustainability and Supply Chain Management*. Thirteenth Edition. United Kingdom: Pearson.

Recommended Reading(s)/Textbook(s)

- Venkataraman, R.R. and Pinto, J.K. (2020). *Operations Management: Managing Global Supply Chains*. Second Edition. Sage Publications.
- Parksoy, T. and Deveci, M. (2023). *Smart and Sustainable Operations and Supply Chain Management in Industry 4.0*. First Edition. Boca Raton: CRC Press.
- Slack, N., Jones, B. and Burgess, N. (2022). *Operations Management*. Tenth Edition. United Kingdom: Pearson.
- Kareska, PhD Katerina. 'Operational Management Strategies for Sustainable Productivity Growth' (August 5, 2023). Available at SSRN: <https://ssrn.com/abstract=4532740> or <http://dx.doi.org/10.2139/ssrn.4532740>
- *The Impact Of Ergonomics On Quality Of Life In The Workplace*. (2023). https://www.researchgate.net/publication/377267055_The_Impact_of_Ergonomics_on_Quality_of_Life_in_the_Workplace

G. Special Features

In the Module Guide, you will find the following icons together with a description. These are designed to help you study. It is imperative that you work through them as they also provide guidelines for examination purposes.

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Special Feature	Icon	Description
THINK POINT		A Think Point allows you to apply your analytical skills to reflect on the topic. You may be asked to apply a concept to your own experience or to think of an example.
ACTIVITY		An Activity tests your knowledge on content that is highlighted within specific content areas. The purpose of an activity is to give you an opportunity to apply what you have learned.
READINGS		The readings provided should be read in order to develop your knowledge of the content areas. If you are unable to acquire the suggested readings, then you may consult with any current sources that deal with the subject.
PRACTICAL APPLICATION OR EXAMPLES		A Practical Application or an Example provides insight and enhances your learning of the topic, allowing you to apply the theory learnt through real-life scenarios within this module
KNOWLEDGE CHECK QUESTIONS		Knowledge Check Questions (KCQ) appear in the form of True/False or Multiple Choice Questions throughout the Module Guide. KCQs will test your knowledge of the content areas covered.
REVISION QUESTIONS		Revision Questions will test your understanding of module and unit outcomes.
CASE STUDY		A Case Study provides different scenarios to illustrate how theory is practiced.
VIDEO ACTIVITY		A Video Activity with links are included in your module guide along with instructions to attempt after watching the video.

**Unit
1:**

**Operations and
Productivity**

Unit Learning Outcomes

CONTENT LIST	LEARNING OUTCOMES FOR THIS UNIT
1.1 Introduction	Introduces the topic areas covered for the unit
1.2 The History of Operations Management	Outline an historical view of the development of the studies of operations management
1.3 Ten Critical Decisions of Operations Management	Analyse the critical decisions that an operations manager must make
1.4 Operations and Productivity	Analyse the inter-relationship between different business functions
1.5 Exciting New Trends in Operations Management	Examine the Productivity of your organisation
1.6 Operations in the Service Sector	Outline the differences between the service and manufacturing industries
1.7 Productivity	Explain how to improve productivity within your organisation
1.8 Summary	Summarises the topic areas covered for the unit

Prescribed and Recommended Reading(s)/Textbook(s)

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- Heizer, J., Render, B. and Munson, C. (2020). *Operations Management: Sustainability and Supply Chain Management*. Thirteenth Edition. United Kingdom: Pearson.

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- 'The Impact Of Ergonomics On Quality Of Life In The Workplace'. (2023). https://www.researchgate.net/publication/377267055_The_Impact_of_Ergonomics_on_Quality_of_Life_in_the_Workplace



1.1 Introduction

Production is the creation of goods and services. Operations Management (OM) is the set of activities that creates goods and services through the transformation of inputs into outputs. Activities creating goods and services take place in all organisations. In manufacturing firms, the production activities that create goods are usually quite obvious. In them, we can see the creation of tangible products such as a TV or a motor vehicle. In organisations that do not create physical products, the production function may be less obvious. It may be hidden from the public and even from the customer. An example is the transformation that takes place at a bank, hospital, airline office, or college.

Often when services are performed, no tangible goods are produced; instead the product may take such forms as the transfer of funds from a savings account to a cheque account, the transplant of a liver, the filling of an empty seat on an airline, or the education of a student. Regardless of whether the end product is a good or a service, the production activities that take place in the organisation are referred to as operations or operations management.

We study OM for four reasons:

1. OM is one of the three major functions of any organisation, and it is integrally related to all the other business functions. All organisations market (sell), finance (account), and produce (operate), and it is important to know how the OM segment functions. Therefore, we study how people organise themselves for productive enterprise
2. We study OM because we want to know how goods and services are produced. The production function is the segment of our society that creates the products we use
3. We study OM to understand what operations managers do. By understanding what these managers do, you can develop the skills necessary to become such a manager. This will help you explore the numerous and lucrative career opportunities in OM
4. We study OM because it is such a costly part of an organisation. A large percentage of the revenue of most firms is spent in the OM function. Indeed, OM provides a major opportunity for an organisation to improve its profitability and enhance its service to society.



Think Point 1.1

Provide reasons for why you are studying Operations Management and how will it be of use in your chosen career?

1.2. The History of Operations Management

The field of OM is relatively young, but its history is rich and interesting. Our lives and the OM discipline have been enhanced by the innovations and contributions of numerous individuals. We introduce a few of these people in this section: and a summary of significant events in operations management is shown below.

Eli Whitney (1800) is credited for the early popularisation of interchangeable parts, which he achieved through standardisation and quality control in manufacturing. Through a contract he signed with the U.S. government for 10 000 muskets, he was able to command a premium price because of their interchangeable parts.

Frederick W Taylor (1881), known as the father of scientific management, contributed to personnel selection, planning and scheduling, motion study and the now popular field of ergonomics. One of his major contributions was his belief that management should be much more resourceful and aggressive in the improvement of work methods. Taylor and his colleagues, Henry L. Gantt and Frank and Lillian Gilbreth were amongst the first to seek the best way to produce using systematic methods. (For further reading, see “systems theory”)

Another of Taylor’s contributions was the belief that management should assume more responsibility for:

- Matching employees to the right job
- Providing the proper training
- Providing proper work methods and tools
- Establishing legitimate incentives for work to be accomplished

By 1913, Henry Ford and Charles Sorensen combined what they knew about standardised parts with the quasi-assembly lines of the meatpacking and mail-order industries and added the revolutionary concept of the assembly line where men stood still, and material moved.

Quality control is another historically significant contribution to the field of OM. Walter Shewhart (1924) combined his knowledge of statistics with the need for Quality Control and provided the foundation for statistical sampling in quality control. W. Edwards Deming (1950) believed, as did Frederick Taylor, that management must do more to improve the work environment and processes so that quality can be improved.

Significant events in the field of Operations Management (Heizer, J. et.al 2020)

- Division of labour (AdamSmith, 1776, and Charles Babbage, 1852)
- Standardised parts (Whitney, 1800)
- Scientific management (Taylor, 1881)
- Coordinated assembly line (Ford /Sorensen/Avery, 1913)
- Gantt charts (Gantt, 1916)
- Time and Motion study (Frank & Lillian Gilbreth, 1922)
- Quality control (Shewhart, 1924; Deming, 1950)
- Computer (Atanasoff, 1938)
- CPM/PERT (DuPont, 1957)
- Material requirements planning (Orlicky, 1960)
- Computer - aided design (CAD, 1970)
- Flexible mfg. system (FMS, 1975)
- Baldrige Quality Awards (1982)
- Computer integrated mfg. (CIM, 1990)
- Globalisation (1992)
- Internet (1995)
- Systems Theory

Operations management will continue to progress with contributions from other disciplines, including industrial engineering and management science. These disciplines, along with statistics, management, human resource management and economics, have contributed substantially to greater productivity. Innovations from physical science (biology, anatomy, physiology, chemistry, and physics) have also contributed to advances in OM. These advances include new adhesives, chemical processes for printed circuit boards, gamma rays to sanitise food products, and molten tin tables on which to float higher-quality molten glass as it cools. The designing of products and processes often depends on the biological and physical sciences.

An especially important contribution to OM has come from the information sciences, which we define as the systematic processing of data to yield information. The information sciences are contributing in a major way towards improved productivity while providing society with a greater diversity of goods and services.

Decisions in operations management require individuals who are well versed in management science, in information science, and often in one of the biological or physical sciences. In this chapter, we take a look at the diverse ways a student can prepare for careers in operations management.

The Functions and Duties of Operations Managers

All good managers perform the basic functions of the management process. The management process consists of planning, staffing, leading, organising and controlling. Operations managers apply this generic management process to the decisions they make in the OM function. Managers contribute to production and operations through the decisions shown in the Table 1. To address each of these decisions requires planning, organising, staffing, leading, and controlling.

1.3 Ten Critical Decisions of Operations Management

TABLE 1: TEN CRITICAL DECISIONS OF OPERATIONS MANAGEMENT

DECISION AREA	ISSUES
Quality Management	Who is responsible for quality? How do we define the quality we want in our service or product?
Service and Product design	What product or service should we offer? How should we design these products or services?
Process and capacity design	What equipment and technology is necessary for these processes?
Location	Where should we put our facility?
Layout design	How should we arrange our facility?
Human resources and job design	How do we provide a reasonable work environment?
Supply chain management	Should we make or buy a component? How many suppliers should we have?
Inventory, MRP, JIT	How much of inventory should we keep?
Intermediate, short-term and project scheduling	Is subcontracting production a good idea?
Maintenance	Who is responsible for maintenance?
Returns	How do we deal with returned goods?

Adapted: Heizer.et.al (2020)



Think Point 1.2

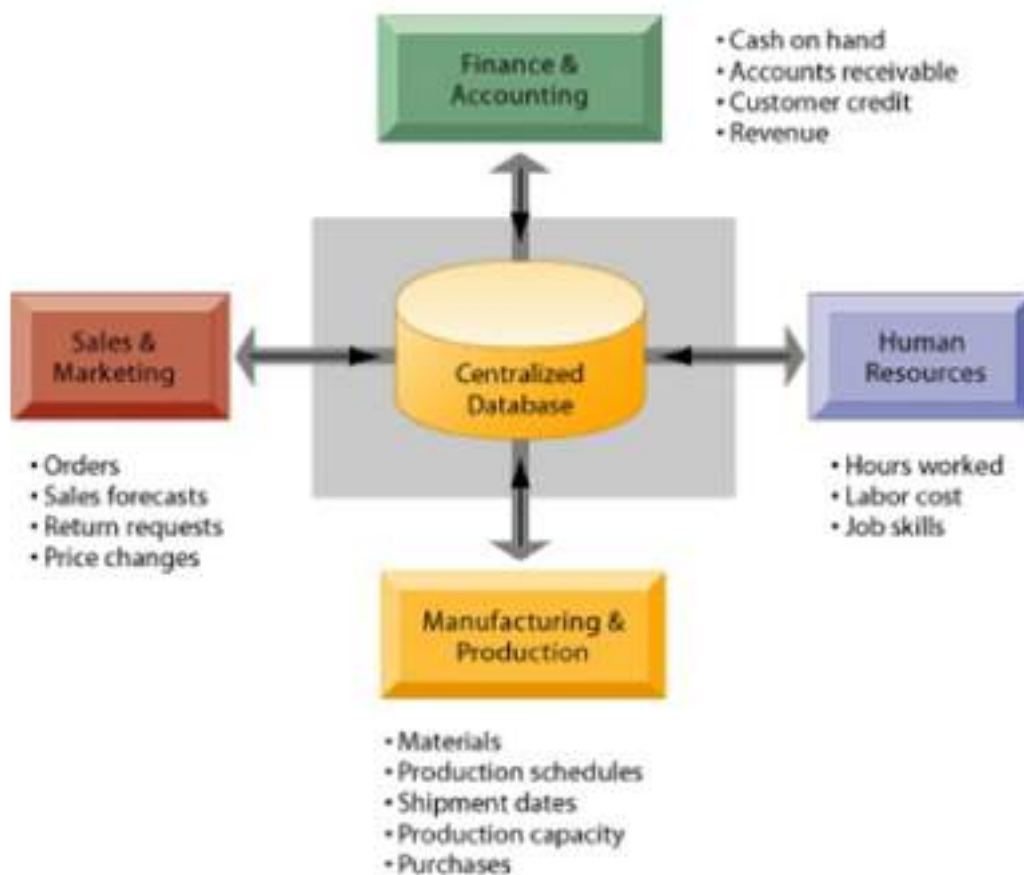
Select from the above list, the number of functions that you perform in your current job.

1.4 Operations and Productivity

There is a strong inter-relationship between the four major functions within an organisation:

- Marketing, which generates the demand, or at least takes the order for, a product or services (nothing happens until there is a sale)
- Production/operations, which creates the product
- Finance/accounting, which tracks how well the organisation is doing, pays the bills, and collects the money
- Human resource management, which attempts to motivate employees to do their jobs

The diagram below details the inter-relationship of the major functions.



Source: Online

Universities, churches, temples, mosques or synagogues, and businesses all perform these functions. Any institution, even a volunteer group such as the Boy Scouts of America is organised to perform these three basic functions.

Activity 1.1



1. A logistics company is looking to improve the efficiency and transparency of its supply chain operations. They are considering adopting blockchain technology. Discuss how blockchain technology can enhance supply chain transparency and efficiency for the logistics company?
2. A manufacturing firm wants to adopt Industry 4.0 technologies to stay competitive. They plan to implement IoT devices and data analytics. How can IoT devices and data analytics contribute to improved operational efficiency for the manufacturing firm?
3. A company aims to reduce its carbon footprint as part of its sustainable Operations Management strategy. They are considering investing in renewable energy sources for their production facilities. Determine the potential benefits and challenges of investing in renewable energy for the company's operations?
4. A clothing manufacturer is looking to adopt a circular economy approach to its operations. This includes recycling used garments into new products. Discuss how adopting a circular economy approach benefits the clothing manufacturer both environmentally and economically?

1.5 Exciting New Trends in Operations Management

Exciting New Trends in Operations Management

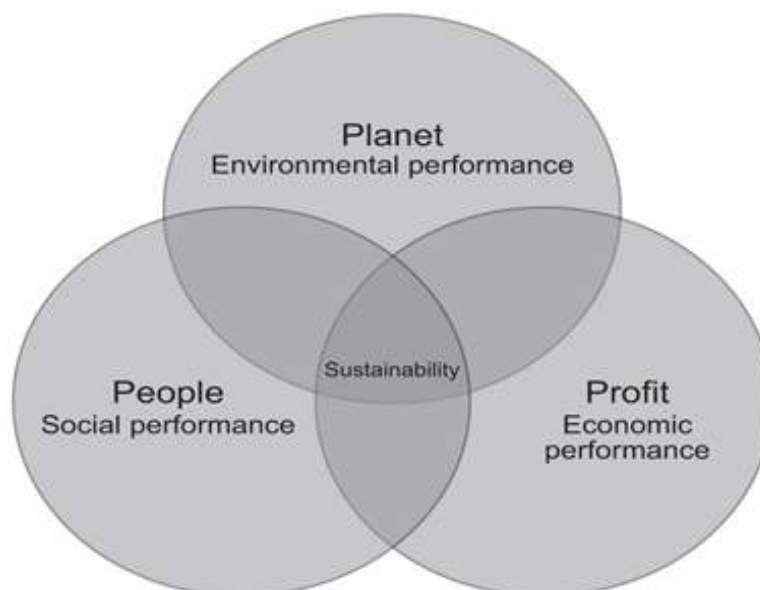
One of the reasons why OM is such an exciting discipline is that the operations manager is confronted with an ever-changing environment. Consequently, both the approach to and the results of the 10 OM decisions are subject to change. These dynamics are the result of a variety of forces, from globalisation of world trade to the transfer of ideas, products and money at electronic speeds. We now introduce some of the challenges:

- **Global Focus:** The rapid decline in communication and transportation costs has, of course, made global markets more accessible. However, at the same time, resources in the form of materials, talent, and labour have also become global. Contributing to this rapid globalisation are countries throughout the world that are going for economic growth and industrialisation
- Operations managers are responding with innovations that generate and move ideas, parts, and

finished goods rapidly, wherever and whenever needed.

- **Just-in-Time Performance:** Vast financial resources are committed to inventory. Inventory impedes the response of dynamic changes in the marketplace. Operations managers are slashing inventories at every level, from raw materials to finished goods.
- **Supply-chain Partnering:** Shorter product life cycles, as well as rapid changes in material and process technology, require more participation by suppliers. Suppliers usually supply up to 80% of the value of products. Consequently, operations managers are building long-term partnerships with critical players in the supply chain
- **Rapid Product Development:** Rapid international communication of news, entertainment, and lifestyles is dramatically chopping away at the life of products. Operations managers are responding with design technology that is faster and design management that is more effective
- **Mass Customisation:** Once we begin to consider the world as the marketplace, then the individual differences become quite obvious. Cultural differences, in a world where consumers are increasingly aware of options, places substantial pressure on firms to respond. Operations managers are responding with production processes that are flexible enough to cater to individual whims of consumers. The goal is to produce individual products, whenever and wherever needed
- **Empowered Employees:** More sophisticated employees and a more technical workplace have combined to require more competence at the workplace. Operations managers are responding by moving more decision-making to the individual worker
- **Triple bottom line reporting**

Triple bottom line can best be described by the diagram below:



Online Source: <https://www.aqa.org.uk/resources/business/as-and-a-level/business-7131-7132/teach/teaching-guide-elkingtons-triple-bottom-line>

This business concept emerged in the 1970s as a way to understand firms and industries beyond the underlying principle of profit making.

Companies are now required by the Code of Corporate Practices and Conduct (based on the King IV Report) to conduct business in an ethical manner. This means that the sole motive of profit must now be replaced with the sustainability concept where businesses are managing operations to ensure that staff, the community in which they operate, the environment and business profits are considered sustainably.

- **Social benefits** include legislated labour practices, worker remuneration, transparency, governance, training and uplifting the economic status of the community
- **Environmental benefits** include all the efforts that a firm can add value to the environment in which they operate and includes increasing resource efficiency, reducing its carbon footprint and undertaking actions that will correct any negative environmental impacts of their operations
- **Economic benefits** include growing shareholders' wealth, improving the company's share price, transparent profit sharing and other matters of financial sustainability
- Green Operations
- Green operation is closely linked to triple bottom line reporting. The main aim is to reduce the environmental impact of a business' operations

Several steps can be taken by firms to ensure that operations are "greened":

- Make green thinking a part of your company culture
- Use energy efficient light bulbs
- Minimise the use of plastic
- Store information in electronic format rather than on paper
- Partner with vendors and other businesses that are involved in greening their operations
- Encourage recycling



Think Point 1.3

Discuss the of Scope 3 emissions? How do they fit the "Green agenda?"



Readings

For further reading on the Triple Bottom line concept, here is a link to an online article: [The Triple Bottom Line: What It Is & Why It's Important \(hbs.edu\)](http://hbs.edu/The%20Triple%20Bottom%20Line%20What%20It%20Is%20&%20Why%20It%20s%20Important)

Case Study 1.1

Read the following article and answer the question below.

Google pledges carbon-neutral shipping, recycled plastic for all devices:

Paresh Dave

Online Source:

<https://www.reuters.com/article/us-alphabet-google-devices/google-pledges-carbon-neutral-shipping-recycled-plastic-for-all-devices-idUSKCN1UV1J2>

SAN FRANCISCO (Reuters) - Alphabet Inc's (GOOGL.O) Google on Monday announced that it would neutralise carbon emissions from delivering consumer hardware by next year and include recycled plastic in each of its products by 2022.

The new commitments step up the competition among tech companies aiming to show consumers and governments that they are curbing the environmental toll from their widening arrays of gadgets.

Anna Meegan, head of sustainability for Google's devices and services unit, said in an interview that the company's transport-related carbon emissions per unit fell 40% last year compared to 2017 by relying more on ships instead of planes to move phones, speakers, laptops and other gadgets from factories to customers across the world.

The company will offset remaining emissions by purchasing carbon credits, Meegan said. Three out of nine Google products for which the company has detailed disclosures online contain recycled plastic, ranging from 20% to 42% in the casings for its Google Home speakers and Chromecast streaming dongles.

In a blog post, Google committed to introducing some recycled plastic to 100% of products by 2022.

Meegan acknowledged that Google's 3-year-old hardware business trails far larger hardware rival Apple Inc. (AAPL.O) in some sustainability efforts. Apple said Thursday that it will open a "Material Recovery" lab to investigate new techniques using robotics and machine learning to rip apart its devices and recover valuable materials such as copper, aluminium and cobalt.

The 9,000-square-foot lab will be at the same Austin facility as "Daisy," an Apple-built robot that can now tear apart iPhones at the rate of 1.2 million per year. The lab is part of Apple's broader goal to make all of its products from



recycled or renewable materials.

Apple has not set a date for when it will reach that goal, though some products such as the MacBook Air already feature aluminium made from melted down iPhones traded in to Apple.

Apple, which in 2017 committed to “one day” only using recycled and renewable materials, has at least 50% recycled plastic in some parts of several products, recycled tin in at least 11 products and recycled aluminium in at least two.

But sustainability standards are now a part of Google’s hardware planning, Meegan said. Devices cannot clear the second checkpoint in the company’s design process unless they show that sustainable packaging and materials and ease of repair have been considered.

“We are fundamentally looking to build sustainability into everything we do,” she said. “It’s going to take us time to demonstrate progress.”

Question:

Outline the efforts Apple and Google have made towards greening their operations.



Activity 1.2

Explain if your organisation is involved in any new trends in the field of Operations Management.

1.6 Operations in the Service Sector

The service sector is defined differently by different people. Because definitions vary, much of the data and statistics generated about the service sector are inconsistent. However, we will define services as including repair and maintenance, government, food and lodging, transportation, insurance, trade, finance, Real Estate, education, legal, medical, entertainment, and other professional occupations.

Differences between Goods and Services

According to Heizer.et.al (2020), some of the differences between goods and services are as follows:

- Services are usually intangible (for example, your purchases of a ticket in an empty airline seat between two cities) as opposed to tangible goods
- Services are often produced and consumed simultaneously: there is no stored inventory. For instance, the beauty salon produces a haircut that is “consumed” simultaneously, or the doctor produces an operation that is “consumed” as it is produced. We have not yet figured out how to

inventory haircuts or appendectomies

- Services are often unique. Your mix of financial coverage, such as investments and insurance policies, may not be the same as anyone else's, just as the medical procedure or a haircut produced for you is not exactly like anyone else's
- Services have high customer interaction. Services are often difficult to standardise, automate, and make as efficient as we would like because customer interaction demands uniqueness. In fact, in many cases this uniqueness is what the customer is paying for: therefore, the operations manager must ensure that the product is designed so that it can be delivered in the required unique manner
- Services have inconsistent product definition. Product definition may be rigorous, as in the case of a vehicle insurance policy, but inconsistent because policyholders change cars and mature
- Services are often knowledge-based, as in the case of educational, medical, and legal services, and therefore hard to automate
- Services are frequently dispersed. Dispersion occurs because services are frequently brought to the client/customer via a local office, a retail outlet, or even a house call

Video Activity 1.1



Watch the video using the link below and answer the question that follows.

Operations Management in Goods and Services

Available from: <https://www.youtube.com/watch?v=d7PEFcbH-Bk> (Accessed: 04 May 2025)

Question:

Determine the difference between Goods vs Services.

There are additional differences between goods and services that impact on OM decisions. Although service products are different from goods, the operations function continues to transform resources into products. Indeed, the activities of the operations function are often very similar for both goods and services. For instance, both goods and services must have quality standards established, and both must be designed and processed on a schedule, in a facility where human resources are employed.

Table 2: Further Differences between Goods and Service

ATTRIBUTES OF GOODS	ATTRIBUTES OF SERVICES
(Tangible product)	(Intangible product)
Product can be resold.	Reselling a service is unusual.
Product can be inventoried.	Many services cannot be inventoried.
Site of facility is important for cost.	Site of facility is important for customer contact.
Often easy to automate.	Service is often difficult to automate.
Revenue is generated from the tangible product.	Revenue is generated from intangible services.
Some aspects of quality are measurable.	Many aspects of quality are difficult to measure.
Selling is distinct from production.	Selling is often a part of the service.
Product is transportable.	Provider, not product, is often transportable.

Source: Heizer.et.al (2020)

1.7 Productivity

The difference between production and productivity:

Production is the conversion of raw materials into finished goods that can be used by the customer, whereas

Productivity is a ratio of $\frac{\text{Output}}{\text{Input}}$

Productivity of Land

By improving methods of planting, fertilisation and harvesting of crops, we can increase the harvest tonnage from 1 to 1.5 tons per hectare. We can then say that the productivity of the land will have increased by 50%.

Productivity of Materials

A team of skilled carpenters with accurate equipment uses 4 metres of timber to construct a dining room suite. A second group of carpenters uses 4.5 metres of timber to produce the same dining room suite. Which group has the higher material productivity?

Productivity of Machines

A forming press has a process time of 7 minutes and a load and unload time of 3 minutes. Therefore, for every hour 6 units can be produced; $[60 \text{ minutes} / (7+3)]$. If we had to improve the method of loading and unloading, we may be able to reduce this time by 2 minutes in total. We can now produce 7.5 units per hour; $[60 \text{ minutes} / 8]$

Our machine productivity has now increased by 25% $(7.5 / 6)$. That is the new productivity divided by the old productivity.

Productivity of Labour

A bricklayer can lay 500 bricks per day. After improving the method, he can now lay 700 bricks per day. This is an increase of 40%.

Now let's identify some outputs and inputs:

COMPANY	OUTPUTS	INPUTS
Bakers Bread	Bread, cakes, biscuits	Man hours, machine hours
Toyota	Cars, bakkies	Man hours, machine hours
Accounting Dept.	No. of accounts processed	Hours spent processing



Practical Application 1.1

Bakers Bread produces 10 000 loaves of bread per day. The factory has 500 workers, each working 8 hours per day.

Remember our definition of productivity - $\frac{\text{Output}}{\text{Input}}$

Activity 1.3

Calculate the productivity using the information below.

	1996	1997	1998	1999
Fridges produced per annum	2 508 000	3 207 000	3 759 000	4 056 000
No.of workers	200	250	350	400
Hours per day	8	8	8	8
Working days per year	250	252	251	253



Firstly, let us plot our production figures on a graph.

- Analyse the graph and comment on our performance
- Now calculate the productivity
- Plot the productivity on a separate graph
- Now analyse the performance
- What can you conclude?



Activity 1.4

Calculate the productivity of your department.

Practical Application 1.2

Single Factor Productivity

You are the operations manager for the MTN Call Centre. You would like to compare the productivity of two employees for working one day, Jane and Michelle. Both Jane and Michelle have worked 8 hours each, whereas Jane has handled 150 customer calls and Michelle handled 300 customer calls.

1. Calculate Jane's productivity.
2. Calculate Michelle's productivity.
3. Select who is the most productive employee is.



Multi – Factor Productivity

$$\text{Multi – Factor Productivity} = \frac{\text{Output From The Operation}}{\text{All Inputs From The Operation}}$$

$$\text{Multi – Factor Productivity} = \frac{\text{Output From The Operation}}{\text{Input 1} + \text{Input 2} + \text{Input 3} \dots}$$

Multi-factor productivity is utilised to calculate productivity when it's hard to separate out the effects of various inputs. An output depends on multiple factors, such as labour, material, and machine costs.

In some instances, considering labour as the single input may be inappropriate, especially if labour costs could be driven down by driving some other cost up (such as machine cost). In situations like this, multi -factor productivity measures may be preferable.



Video Activity 1.2

Watch the video using the link below.

Ingredients for Total Factor Productivity

Available at: <https://youtu.be/Au2bs7NVT78?si=-nkZVkp0KQcSouAK>



Practical Application 1.3

A shoe manufacturing company produces a total of 100 000 shoes a month.

The total labour cost is R10 000, and the total materials cost is R25000 while the total machine cost is R6000.

Calculate the multifactor productivity of the shoe manufacture.

Practical Application 1.4

ElectroTech components produces a specific type of electronic device used to facilitate wireless communication and connectivity in various electronic devices such as smartphones, tablets, and smart home devices. During its two years of operation, the following measurements were gathered:



Measurement	Year 1	Year 2
Price per unit	R 50	R 50
Units produced	R 10 000	R 10 900
Material costs	R 8 000	R8 400
Total labour cost	R150 000	R 150 000
Overhead cost	R 12 000	R 12 200

Determine which of these two years was the manufacturing company efficient.

Factors that affect Productivity

Several factors can affect productivity and these include: Methods; Capital; Quality; Technology and Management. You have to type a lengthy report. If you are a typist of average speed, you can possibly complete three pages per hour. How then can you improve your productivity?

- Enrol on a short course to improve your typing skills
- Replace your typewriter with a computer
- Outsource the typing

A common mistake that people make is that they believe workers are the main determinant of productivity. According to that theory, productivity gains are achieved by getting employees to work harder. However, many of the productivity gains have come from technological improvements e.g.

- Paint rollers, power lawn mowers, copying machines, microwave ovens, washing machines, calculators, computers, email, and many other electric and electronic items/goods

Improving Productivity

There are a number of ways to improve productivity:

- Develop productivity measures for all operations; measurement is the first step in managing and controlling an operation
- Consider the system as a whole in deciding which operations to concentrate on. Find bottleneck

operations in the system and try to improve them

- Develop methods for achieving productivity improvements such as getting ideas from the workers, studying how other companies have improved productivity and re-examining the way work is done
- Establish reasonable goals for improvement
- Obtain support of top management
- Measure improvements and publicise them

Knowledge Check Questions 1.1

Select the correct answer:

1. Which of the following is a key trend in Operations Management that leverages data analysis for decision-making?

- Lean manufacturing
- Automation
- Big data analytics
- Outsourcing



2. The adoption of Industry 4.0 technologies in Operations

Management primarily focuses on:

- Reducing labour costs
- Enhancing digital connectivity and automation in manufacturing
- Outsourcing production to low-cost countries
- Increasing the size of production facilities

3. Which trend in Operations Management is characterised by integrating sustainability into business operations?

- Lean production
- Circular economy
- Globalisation
- Mass customisation

Knowledge Check Questions 1.2

State if the following is True or False by selecting the correct answer.



1. Sustainable Operations Management aims to balance economic, environmental, and social factors in decision-making.
True
False
2. Implementing energy-efficient technologies in manufacturing processes is a key component of sustainable Operations Management.
True
False
3. Sustainable Operations Management typically results in higher long-term operational costs.
True
False

1.8 Summary

Operations and Productivity focus on enhancing the efficiency of business processes to maximise output with minimal input. Operations management involves overseeing the production of goods and services, optimising supply chains, managing resources, and ensuring quality. By streamlining operations and adopting effective management strategies, organisations can boost productivity, reduce costs, and improve overall performance.

Revision Questions

1. Mance Fraily, the Production Manager at Ralts Mills, can currently expect his operation to produce 1000 square yards of fabric for each ton of raw cotton. Each ton of raw cotton requires 5 labour hours to process. He believes that he can buy a better quality raw cotton, which will enable him to produce 1200 square yards per ton of raw cotton with the same labour hours. Outline the impact on productivity (measured in square yards per labour-hour) if he purchases the higher quality raw cotton.

2. C. A. Ratchet, the local auto mechanic, finds that it usually takes him 2 hours to diagnose and fix a typical problem. Illustrate his daily productivity (assume an 8-hour day).

3. Mr. Ratchet believes he can purchase a small computer trouble-shooting device, which will allow him to find and fix a problem in the incredible (at least to his customers!) time of 1 hour. He will, however, have to spend an extra hour each morning adjusting the computerised diagnostic device. Explain the impact on his productivity if he purchases the device.

4. Joanna French is currently working a total of 12 hours per day to produce 240 dolls. She thinks that by changing the paint used for the facial features and fingernails that she can increase her rate to 360 dolls per day. Total material cost for each doll is approximately R3.50; she has to invest R20 in the necessary supplies (expendables) per day; energy costs are assumed to be only R4.00 per day; and she thinks she should be making R10 per hour for her time. Viewing this from a total (multi-factor) productivity perspective, outline her productivity at present and with the new paint.

5. Illustrate how total (multi-factor) productivity would change if using the new paint raised Ms. French's material costs by R0.50 per doll.

6. If she uses the new paint, outline by what amount Ms. French's material costs could increase without reducing total (multi-factor) productivity.



Answers

Activity 1.1

1. A logistics company is looking to improve the efficiency and transparency of its supply chain operations. They are considering adopting blockchain technology. Discuss how blockchain technology can enhance supply chain transparency and efficiency for the logistics company?

Answer: Blockchain technology can provide a secure and immutable ledger of all transactions and movements within the supply chain. This ensures transparency by allowing all stakeholders to access real-time data on the status and history of shipments, reducing the risk of fraud, errors, and inefficiencies.

2. A manufacturing firm wants to adopt Industry 4.0 technologies to stay competitive. They plan to implement IoT devices and data analytics. How can IoT devices and data analytics contribute to improved operational efficiency for the manufacturing firm?

Answer: IoT devices can collect real-time data from machinery and production processes, which can then be analysed to identify inefficiencies, predict maintenance needs, and optimise operations. Data analytics can provide actionable insights, enabling the firm to make data-driven decisions that enhance productivity and reduce downtime.

3. A company aims to reduce its carbon footprint as part of its sustainable Operations Management strategy. They are considering investing in renewable energy sources for their production facilities. Determine the potential benefits and challenges of investing in renewable energy for the company's operations?

Answer: Benefits include reduced greenhouse gas emissions, lower energy costs in the long term, and an enhanced corporate image. Challenges may involve high initial investment costs, the need for technological adaptation, and potential reliability issues depending on the energy source.

4. A clothing manufacturer is looking to adopt a circular economy approach to its operations. This includes recycling used garments into new products. Discuss how adopting a circular economy approach benefits the clothing manufacturer both environmentally and economically?

Answer: Environmentally, it reduces waste and conserves resources by recycling materials. Economically, it can lower material costs, create new revenue streams from recycled products, and enhance the brand's reputation among environmentally conscious consumers.

Activity 1.2

Explain if your organisation is involved in any new trends in the field of Operations Management.

Answer: To be discussed during the webinar(s) with lecturer

Activity 1.3

Calculate the productivity below:

Analyse the graph and comment on our performance

Now calculate the productivity

Plot the productivity on a separate graph

Now analyse the performance

What can you conclude?

Answer:

Output	2 508 000	3 207 000	3 759 000	4 056 000
Input	40 0000	504 000	702 800	809 600
Productivity	6.27	6.36	5.35	5.01

Activity 1.4

Calculate the productivity of your department.

Answer: The next step is to now calculate the productivity of your own departments, plot the results and then comment on the performance. Remember by using productivity ratios, you will be able to identify problem areas and thereby take any action if necessary.

Single factor productivity

Single-factor productivity measures output levels relative to a single input. That is, the relationship between a single input and output is calculated. The assumption is that there is one-to-one relationship between the output and input of interest that can be managed. Single-factor productivity is easy to analyse, as there is only one input in consideration. Therefore, a high productivity can be isolated to the effective use of the single input.

$$\text{Single Factor Productivity} = \frac{\text{Output From The Operation}}{\text{Single Input To The Operation}}$$

Three simple steps can be followed in order to calculate productivity:

1. Step One: Determine The Total Worker Output Either in Units or in Monetary Value.
2. Step Two: Determine The Total Number of Man Hours (input) Associated with The Production.
3. Step Three: Divide Output by The Number of Man Hours (input).

Case Study 1.1

Outline the efforts Apple and Google have made towards greening their operations.

Answer: To be discussed during the webinar(s) with lecturer.

Knowledge Check Questions 1.1

1. Which of the following is a key trend in Operations Management that leverages data analysis for decision-making?

Answer: c. Big data analytics

2. The adoption of Industry 4.0 technologies in Operations Management primarily focuses on:

Answer: b. Enhancing digital connectivity and automation in manufacturing

3. Which trend in Operations Management is characterised by integrating sustainability into business operations?

Answer: c. Circular economy

Knowledge Check Questions 1.2

1. Sustainable Operations Management aims to balance economic, environmental, and social factors in decision-making.

Answer: True

2. Implementing energy-efficient technologies in manufacturing processes is a key component of sustainable Operations Management.

Answer: True

3. Sustainable Operations Management typically results in higher long-term operational costs.

Answer: False

Practical Application 1.1

Bakers Bread produces 10 000 loaves of bread per day. The factory has 500 workers, each working 8 hours per day.

Answer: To be discussed during the webinar(s) with lecturer.

Practical Application 1.2

1. Calculate Jane's productivity.

Answer:

$$\text{Single Factor Productivity} = \frac{\text{Output From The Operation}}{\text{Single Input To The Operation}}$$

$$\text{Jane's Productivity} = \frac{\text{Number of customer calls handled}}{\text{Support staff hours}}$$

Step One: Determine The Total Worker Output Either in Units or in Monetary Value.

Output = 150 customer calls

Step Two: Determine The Total Number of Man Hours (input) Associated with The Production.

Input = 8 hours

Step Three: Divide Output by The Number of Man Hours (input).

$$\text{Jane's Productivity} = \frac{150}{8}$$

$$\text{Jane's Productivity} = 18.75$$

2. Calculate Michelle's productivity.

Answer:

$$\text{Single Factor Productivity} = \frac{\text{Output From The Operation}}{\text{Single Input To The Operation}}$$

$$\text{Michelle's Productivity} = \frac{\text{Number of customer calls handled}}{\text{Support staff hours}}$$

Step One: Determine The Total Worker Output Either in Units or in Monetary Value.

Output = 300 customer calls

Step Two: Determine The Total Number of Man Hours (input) Associated with The Production.

Input = 8 hours

Step Three: Divide Output by The Number of Man Hours (input).

$$\text{Michelle's Productivity} = \frac{300}{8}$$

Michelle's Productivity = 37.5

3. Select who is the most productive employee is.

Answer: Michelle has a higher productivity; therefore, she is the more productive employee. This means she accomplishes more work in the 8 hours than Jane did.

Practical Application 1.3

Calculate the multifactor productivity of the shoe manufacture.

Answer:

$$\text{Multifactor productivity} = \frac{\text{OUTPUT FROM OPERATION}}{\text{INPUT1} + \text{INPUT2} + \text{INPUT3}}$$

Multifactor productivity

$$= \frac{\text{Shoe output}}{\text{Labour cost} + \text{materials cost} + \text{machine cost}}$$

$$\text{Multifactor productivity} = \frac{1000000}{10000 + 25000 + 6000}$$

Multifactor Productivity = 24.4%

Practical Application 1.4

Determine which of these two years was the manufacturing company efficient.

Answer:

Year 1:

$$\begin{aligned} \text{Multifactor productivity} &= \frac{\text{Units} \times \text{Price per unit}}{\text{Labour costs} + \text{materials costs} + \text{overhead costs}} \\ \text{Multifactor productivity} &= \frac{10\,000 \times 50}{150\,000 + 8\,000 + 12\,000} \end{aligned}$$

Multifactor Productivity = 2.9%

Year 2:

$$\text{Multifactor productivity} = \frac{10\,900 \times 50}{150\,000 + 8\,400 + 12\,200}$$

Multifactor Productivity = 3.1%

Therefore year 2 was the most efficient year.

Think Point 1.1

Provide reasons for why you are studying Operations Management and how will it be of use in your chosen career?

Answer: To be discussed during the webinar(s) with lecturer.

Think Point 1.2

Select from the above list, the number of functions that you perform in your current job.

Answer: To be discussed during the webinar(s) with lecturer.

Think Point 1.3

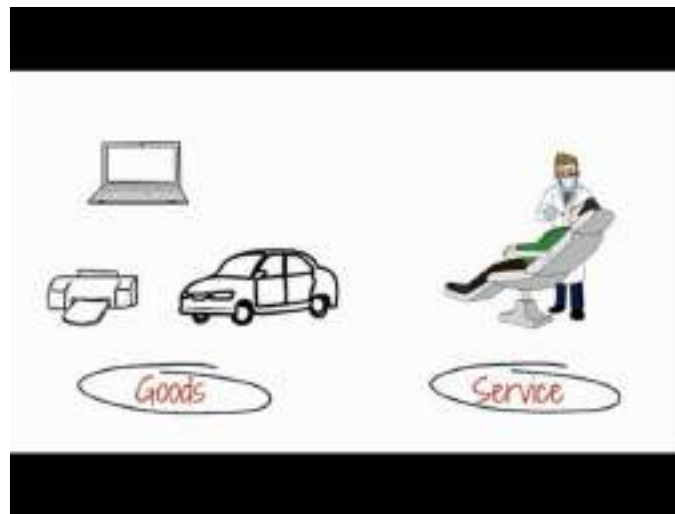
Discuss the of Scope 3 emissions? How do they fit the “Green agenda?”

Answer: Scope 3 emissions: These are indirect emissions from a company's value chain, such as those from suppliers or product use. They are crucial in the “Green agenda” as they often represent the largest share of a company’s carbon footprint and need to be reduced to meet sustainability goals.

Video Activity 1.1

Determine the difference between Goods vs Services.

Answer:



There are additional differences between goods and services that impact on OM decisions. Although service products are different from goods, the operations function continues to transform resources into products. Indeed, the activities of the operations function are often very similar for both goods and services. For instance, both goods and services must have quality standards established, and both must be designed and processed on a schedule, in a facility where human resources are employed.

Unit 2:

Operations Strategy in a Global Environment

Unit Learning Outcomes

CONTENT LIST	LEARNING OUTCOMES FOR THIS UNIT
2.1 Introduction	Introduces the topic areas covered for the unit
2.2 Mission	Identify the concept of a mission statement
2.3 Strategy	Analyse strategies that are available to global business
2.4 Operations in a Global Environment	Analyse the critical factors that affect global business operations
2.5 The Importance of Global Operations	Outline the importance of international business and the global economic village
2.6 Global Issues in Service Operations	Analyse challenges that face the global service industry
2.7 Summary	Summarises the topic areas covered for the unit

Prescribed and Recommended Reading(s)/Textbook(s)

Prescribed Reading(s)/Textbook(s)

- Heizer, J., Render, B. and Munson, C. (2020). *Operations Management: Sustainability and Supply Chain Management*. Thirteenth Edition. United Kingdom: Pearson.

Recommended Reading(s)/Textbook(s)

- Venkataraman, R.R. and Pinto, J.K. (2020). *Operations Management: Managing Global Supply Chains*. Second Edition. Sage Publications.
- Parksoy, T. and Deveci, M. (2023). *Smart and Sustainable Operations and Supply Chain Management in Industry 4.0*. First Edition. Boca Raton: CRC Press.
- Slack, N., Jones, B. and Burgess, N. (2022). *Operations Management*. Tenth Edition. United Kingdom: Pearson.
- Kareska, PhD Katerina. 'Operational Management Strategies for Sustainable Productivity Growth' (August 5, 2023). Available at SSRN: <https://ssrn.com/abstract=4532740> or <http://dx.doi.org/10.2139/ssrn.4532740>
- 'The Impact Of Ergonomics On Quality Of Life In The Workplace'. (2023). https://www.researchgate.net/publication/377267055_The_Impact_of_Ergonomics_on_Quality_of_Life_in_the_Workplace



2.1 Introduction

In the contemporary business landscape, globalisation has transformed the way organisations operate, compelling them to develop robust operations strategies that transcend national boundaries. This unit, Operations Strategy in a Global Environment, delves into the intricacies of formulating and executing operations strategies that align with an organisation's mission while effectively navigating the complexities of global markets. A clear and compelling mission serves as the foundation of any organisation's strategic endeavours. Aligning operations strategy with the organisational mission ensures coherence in decision-making and resource allocation. This alignment facilitates the translation of strategic objectives into operational capabilities, enabling organisations to respond adeptly to global challenges and opportunities.

Engaging in global operations offers organisations access to new markets, diversified talent pools, and cost efficiencies. However, it also introduces challenges such as cultural differences, regulatory compliance, and geopolitical risks. The recent global supply chain disruptions have underscored the necessity for resilient and adaptable operations strategies. For instance, the COVID-19 pandemic revealed vulnerabilities in global supply chains, prompting companies to reassess their operational frameworks to enhance resilience. Service operations in a global context face unique challenges, including varying customer expectations, service standardisation difficulties, and logistical complexities. The rise of global capability centres (GCCs) exemplifies how organisations are centralising services like finance, human resources, and customer support to streamline operations and maintain consistent service quality across regions. This approach not only enhances efficiency but also fosters innovation by leveraging diverse perspectives.



Think Point 2.1

Explain your organisation's mission statement

2.2 Mission

According to Heizer.et.al (2020), a mission can be defined as the purpose and the company's contribution to the broader society. Economic success and survival are the result of identifying missions to satisfy a customer's needs. Developing a good strategy would be easier if the organisation's mission is clearly defined.



Activity 2.1

1. Define mission statement.
2. Discuss the term "strategy".
3. Determine the three main strategies that companies employ.



Readings

Find an example of a company's mission statement on the following link:

Available at:

<https://investors.coca-colacompany.com/about/our-purpose#:~:text=The%20Coca%2DCola%20Company%20purpose,This%20purpose%20is%20uniquely%20us.>

2.3 Strategy

Once an organisation's mission has been established, it can begin to identify its strategy and implement it. Strategy is an organisation's action plan to achieve its mission. Heizer.et.al (2020) suggests that firms can achieve their missions in three conceptual ways:

- Differentiation
- Cost Leadership
- Quick Response

This means that operations managers are required to deliver goods and services that are better or at least different, cheaper and more responsive.

Achieving Competitive Advantage through Operations

The following strategies provide operations managers with an opportunity to develop competitive advantages:

- **Competing on Differentiation**

Differentiation is concerned with providing uniqueness. A company's opportunity for creating uniqueness is not to play the game better than the competition, but to develop and play a totally different 'game'.

- **Competing on Cost**

The low-cost strategy ensures that the facilities are optimally utilised. Identifying the optimum size will allow firms to spread their overhead costs over sufficient units to drive down costs thereby providing a cost advantage. Low-cost leadership entails achieving maximum value as defined by your customer. This does not imply low value or low quality.

- **Competing on Response**

Response is not only flexible response but also refers to reliable and quick response. Response includes the entire range of values related to timely product development and delivery, as well as reliable scheduling and flexible performance.

Case Study 2.1

Read the following article and watch the video using the link provided and answer the question below.

Taking Wal-Mart Global: Lessons From Retailing's Giant

The primary objective of any international strategy should be to effectively manage the significant differences that arise at the borders of markets. This case delves into the core aspects of the business model adopted by Wal-Mart, both within the U.S. and internationally. It focuses particularly on the company's approaches to managing suppliers, their strategies when dealing with Wal-Mart, and the company's performance in global markets.

Becoming a global player is not merely the result of a grand plan, but neither can it be achieved through small, unplanned, or random actions. The most prudent approach would be one of "directed opportunism" - maintaining flexibility while adhering to a systematic framework.

A prime example illustrating the effectiveness of a deliberate and systematic process in navigating the complexities of becoming a global entity is Wal-Mart Stores Inc., the world's largest retailer. Since opening its first international store in Mexico City in 1991, Wal-Mart now operates in all 50 states of the U.S., Puerto Rico, Canada, China, Mexico, Brazil, Germany, Britain, Argentina, and South Korea. By July 1999, it employed over 130,000 people in 729 facilities outside the U.S., out of a global workforce exceeding 950,000. Before expanding into supermarkets, Wal-Mart operated three main types of



stores: 1) Wal-Mart stores offering a variety of goods like clothing, household items, and sports equipment; 2) Sam's Clubs selling bulk items to members; and 3) Supercentres combining discount stores with full-scale supermarkets. Wal-Mart has aggressively pursued globalisation since its initial international venture in 1991. In 1993, only 1 percent of its stores were located outside the U.S., growing to 18 percent by 1998. Between 1995 and 1998, international operations accounted for 5 percent of the company's sales growth and 4 percent of its profit growth.

Video: Walmart's Global expansion Strategy | Globalization | MBA Case study examples with solutions

Available at: <https://youtu.be/rCxlZJ4tFrA?si=mcRPVrAQp6lJ6Nsp>

Question:

Having read the above article and watched the video why do you think Walmart's global expansion strategy is experiencing challenges? Determine what made Walmart successful in the US market? Bringing this case study closer to home, why do you think the Massmart Group a subsidiary of Walmart struggled to thrive in South Africa?



Think Point 2.2

Explain the advantages and disadvantages of globalisation.



Video Activity 2.1

Watch the video using the link below and answer the question that follows.

Burger King's Secret STRATEGY to compete with McDonald's: Burger Wars Ep 1

Available at: <https://youtu.be/6Yh1f2nat9o?si=9DtBBonc0RhBQ2it>

Question:

Discuss their turnaround strategy in improving their operations. Discuss what the CEO did in addressing some of the challenges faced by the business?

2.4 Operations in a Global Environment

The signing of the GATT (General Agreement on Trade and Tariffs) agreement in 1994 meant that South African companies were for the first time exposed to international competitors. Foreign products and services flooded South African markets and many organisations were not in a position to react to the challenges facing them (Naidoo, 2000).

Global competition is here to stay. There are new standards of global competitiveness that include variety, quality, customisation, speed and costs. Although Globalisation forces companies to become more efficient, it tends to complicate the operations manager's job.

International operations management refers to the process by which global firms transform inputs to outputs. According to (Heizer et al 2020), the questions facing global competitors are:

- How to design products and where to build factories and offices?
- Where and how do we obtain the resources needed to create goods or services?
- What modes of transportation to use and how to manage inventory?



Source: Heizer and Render (2020)

An international firm is any business that engages in international trade or investment. A multinational corporation is a firm with extensive international business involvement. A global company views the world as a single marketplace and a transnational company, sometimes called a world company is an internationalised firm whose country identity is not as important as its interdependent network of worldwide operations.

For further reading on GATT, find link to article: <https://www.saflii.org/za/journals/LDD/1999/6.pdf>



Think Point 2.3

Explain if your organisation is geared to compete with global competitors.

Knowledge Check Questions 2.1

Select the correct answer:

1. Which of the following is a key advantage of globalisation for operations management?
 - a. Increased regulatory complexity
 - b. Higher tariffs
 - c. Access to a wider range of resources and markets
 - d. Limited consumer base
2. Operations management strategies typically aim to achieve all of the following except:
 - a. Cost reduction
 - b. Improved customer satisfaction
 - c. Increased complexity in production processes
 - d. Enhanced product quality
3. Which global issue in service operations is primarily concerned with adapting to varying cultural norms and expectations?
 - a. Supply chain optimisation
 - b. Quality control
 - c. Cross-cultural service delivery
 - d. Regulatory compliance
4. The importance of global operations in a business primarily includes:
 - a. Reducing competition
 - b. Limiting service capacity
 - c. Enhancing flexibility in meeting diverse customer needs
 - d. Increasing transaction costs
5. A firm employing a low-cost strategy in its operations management is most likely to focus on:
 - a. Investing in premium raw materials
 - b. Increasing labour costs
 - c. Streamlining processes to minimise expenses
 - d. Expanding without considering costs



2.5 The Importance of Global Operations

The following are some of the reasons:

- **Reduction of Costs**

International operations seek to take advantage of the tangible opportunities in reducing their costs. Foreign countries have lower wage structures and lower direct and indirect labour cost. Less restrictive government regulations on a wide variety of operations practices (e.g. environmental control, health, safety regulations, etc.) can reduce the costs of operations in a foreign country. Tax and tariff incentives are additional reasons used to establish operations in a foreign country.

- **Reduce Risks**

Globalisation has become easier and less hazardous for international operations because of international trade operations. GATT, for example, seeks to reduce tariffs and promote conditions of fair competition and increased investment opportunities by lowering barriers to create free flow of goods across international borders.

- **Improving the Supply Chain**

The supply chain can be improved by locating facilities in countries where unique resources are available. These resources may include expertise, labour or raw materials.

Provision of Superior Goods and Services

Improved understanding of local conditions and differences in culture permits firms to customise products and services to meet unique needs. Closeness to foreign customers helps to improve response times to meet customers' changing product and service requirements. It also offers a better after-sales service.

- **Establishing New Markets**

As international operations require local interaction with foreign customers, suppliers and other competitors, they are in a position to explore new and unique opportunities for new products or services. Knowledge of these markets helps to increase sales and diversify their customer base. Global operations also add production flexibility so that products and services can be switched between economies that are booming and those that are not.

- **Improving Operations**

Learning does not take place in isolation. The world is full of ideas and firms can learn from their customers by allowing the free flow and exchange of ideas.

- **Attracting and Retaining Talented Employees**

A Global operation is in a position to identify, attract and retain talented employees from across the world. It needs people in all functional areas and areas of expertise worldwide. Global firms can recruit and retain talented employees because they provide greater growth opportunities and insulation against unemployment during recession times.

- **Product Design and Process Technology**

A basic product or service is designed, wherever possible, to fit global tastes. If local variation is needed, it is handled as an option rather than as a separate product. Process technology is also standardised globally (Schroeder et al., 2018).



Video Activity 2.2

Watch the video using the link below explaining what globalisation is:

(New) what is globalization? 4 drivers of globalization | International Business

Available at: https://youtu.be/Z9uj81Mng78?si=wZjOR3kMRFbH_HwB



Activity 2.2

1. Mention some of the reasons why globalisation is important.
2. Define Globalisation.
3. Explain how OM plays a pivotal role in ensuring that companies become dominant players on the global scale.

Knowledge Check Questions 2.1

State if the following is True or False and explain your answer.



1. Globalisation has led to increased interdependency among nations in operations management.
2. Global issues in service operations are less challenging than in manufacturing due to less complex supply chains.
3. A critical OM strategy is aligning production schedules with demand forecasts to reduce waste.
4. The importance of global operations lies solely in cost advantages from international labor markets.
5. Globalisation in OM enables firms to innovate more quickly by accessing a global talent pool.

2.6. Global Issues in Service Operations

Globalisation affects services just as it does manufacturing. An international service provider is a firm that creates business for global customers by transforming resources into services. Heizer.et.al (2020) argue that the procedure for establishing global service operations involves four steps:

- Determine if sufficient people or facilities exist to support the service, which includes technology such as fax, modems, internet, voice-link technology, and foreign-language personnel needed to support it
- Identify foreign markets that are open – those that are not protected by tariff barriers
- Determine what services are of most interest to foreign customers. Start with services that the local operation performs efficiently
- Determine how to reach global customers. This includes exploring the internet, buying clientele lists, using existing business suppliers and sourcing information from local government departments and embassies

Activity 2.3



Compare your organisation's performance with both local and international competitors in the following categories:

- Profit
- Market Share
- Number of Staff
- Technology

Video Activity 2.3



Watch the video using the link below and answer the questions that follows.

Did Russia win the war against US & EU? Economic Case Study

Available at: <https://youtu.be/Q073TTZx7aM?si=78semYmBywi5snQQ>

Question:

Discuss the implications of conflict/ war on globalisation. How is Russia surviving the 17000 sanctions imposed on it by both the US and EU? What are the pros and cons of globalisation through the lens of an OM?

2.7 Summary

Throughout this unit, we have explored the multifaceted aspects of operations strategy within a global environment, focusing on understanding the critical role of aligning operations strategy with the organisational mission to ensure cohesive and effective global operations, analysing the benefits and challenges associated with global operations, including market expansion, cultural considerations, and the imperative for supply chain resilience in light of recent global disruptions.

We have also explored examining the complexities inherent in global service delivery, such as standardisation versus localisation, managing diverse customer expectations, and the strategic utilisation of global capability centres to optimise service operations. By integrating contemporary case studies and recent developments, this unit has provided a comprehensive understanding of how organisations can strategically manage their operations to thrive in a dynamic global environment.

Revision Questions

1. Explain how changes in the external environment may affect the OM strategy for a company. For example, examine the impact that the following factors are likely to have on OM strategy.

- a. The occurrence of a major storm or hurricane.
- b. Terrorist attacks of 9/11/01.
- c. The much-discussed decrease in the quality of American primary and secondary school systems.
- d. Trade Legislation such as WTO and NAFTA and changes in tariffs and quotas.
- e. The rapid rate at which the cost of health insurance is increasing.
- f. The Internet.

2. Explain how the changes in the internal environment affect the OM strategy for a company. For example, examine the impact that the following factors are likely to have on OM strategy.

- a. The increased use of Local and Wide Area Networks (LANs and WANs).
- b. An increased emphasis on service.
- c. The increased role of women in the workplace.
- d. The seemingly increasing rate at which both internal and external environments change.

3. Operations managers are called upon to support the organisation's strategy. OM does this with some combination of one of three strategies.

Outline these three strategies.

4. Determine how globalisation impacts the standardisation of products in OM?

5. Discuss the challenges of managing global operations.

6. Analyse how trade policies and tariffs influence operations management under globalisation.

7. Explain how globalisation impacts the standardisation of products in OM.

8. Analyse some of the challenges of managing global operations.

9. How do trade policies and tariffs influence operations management under globalisation?

10. Explain the advantages and disadvantages of globalisation.



Answers

Activity 2.1

1. Define mission statement.

Answer: A mission statement is a concise declaration that outlines an organisation's core purpose, values, and primary objectives. It communicates the company's reason for existence and guides its strategic decision-making and actions.

2. Discuss the term "strategy".

Answer: A strategy is a comprehensive plan that outlines how an organisation will achieve its long-term goals and objectives. It involves making decisions on resource allocation, competitive positioning, and actions to gain a sustainable advantage in the market.

3. Determine the three main strategies that companies employ.

Answer:

- **Cost Leadership:** Focusing on becoming the lowest-cost producer in the industry, enabling the company to offer products or services at a lower price than competitors
- **Differentiation:** Offering unique products or services that stand out from competitors, often through innovation, quality, or branding, to attract customers who value distinct features
- **Quick Response:** Emphasising speed and flexibility in operations to rapidly respond to customer demands and market changes, thereby reducing lead times and enhancing customer satisfaction

Activity 2.2

1. Mention some of the reasons why globalisation is important.

Answer:

- **Economic Growth and Development:** Globalisation promotes economic growth by enabling countries to access larger markets, attract foreign investment, and facilitate the exchange of goods, services, and technology. This expansion fosters increased productivity, innovation, and efficiency
- **Access to Resources and Technology:** Globalisation allows countries to access resources and technologies that may not be available domestically. This access can lead to advancements in industries, improved production processes, and overall economic development
- **Cultural Exchange and Diversity:** Through globalisation, people from different cultures and backgrounds can share ideas, traditions, and lifestyles. This exchange fosters understanding, tolerance, and the appreciation of cultural diversity, which enriches societies

- **Improved Standards of Living:** By providing access to a broader range of goods and services, globalisation can lead to improved living standards. Consumers benefit from greater choices and lower prices, while businesses can access more efficient supply chains and reduce costs
- **Employment Opportunities:** Globalisation creates job opportunities by opening up new markets and encouraging multinational companies to invest in various regions. This can lead to skill development, higher wages, and improved working conditions
- **Technological Advancement and Innovation:** Global competition encourages innovation and technological advancement as companies strive to maintain a competitive edge. This results in the development of new products, services, and processes that can drive economic growth
- **Political and Social Stability:** Economic interdependence created by globalisation can foster political stability and reduce the likelihood of conflict. Countries with strong economic ties are more likely to cooperate and work towards resolving issues peacefully
- **Environmental Awareness and Global Cooperation:** Globalisation has raised awareness about environmental issues and the need for sustainable development. International cooperation and agreements can lead to collective efforts in addressing global challenges such as climate change and pollution

2. Define Globalisation.

Answer: Globalisation refers to the process by which economies, societies, and cultures become increasingly interconnected and interdependent through the exchange of goods, services, information, technology, and capital across international borders. It involves the integration of markets, the flow of investment and knowledge, and the movement of people, leading to greater economic, cultural, and political collaboration on a global scale.

3. Explain how OM plays a pivotal role in ensuring that companies become dominant players on the global scale.

Answer:

- **Supply Chain Optimisation:** Operations Management (OM) focuses on streamlining and optimising supply chains to ensure that companies can deliver products and services efficiently and cost-effectively. This is crucial for competing on a global scale, where supply chain efficiency can make a significant difference in terms of cost and delivery speed
- **Quality Management:** OM emphasises the importance of maintaining high-quality standards in products and services. By implementing Total Quality Management (TQM) practices and continuous improvement initiatives, companies can enhance their global reputation, reduce defects, and increase customer satisfaction
- **Process Improvement and Innovation:** OM encourages the adoption of innovative practices and

the continuous improvement of processes. This helps companies remain competitive by reducing waste, improving efficiency, and staying ahead of technological advancements

- **Risk Management:** Effective OM involves identifying and mitigating risks associated with global operations, such as political instability, supply chain disruptions, and compliance with international regulations. By managing these risks, companies can ensure smooth operations and maintain their competitive edge
- **Cost Efficiency:** OM focuses on cost reduction through various methods, such as lean manufacturing, just-in-time inventory, and efficient resource allocation. Cost efficiency enables companies to offer competitive pricing in the global market while maintaining profitability
- **Customer Satisfaction and Responsiveness:** OM ensures that companies can meet customer demands effectively and respond to market changes swiftly. By focusing on customer satisfaction and being agile, companies can build a loyal customer base and adapt to global market trends
- **Sustainability and Corporate Responsibility:** OM integrates sustainable practices into business operations, addressing environmental and social concerns. This not only helps in complying with international regulations but also enhances the company's reputation and appeal to socially conscious consumers
- **Technology Integration:** OM plays a pivotal role in integrating new technologies into business processes, such as automation, artificial intelligence, and data analytics. These technologies enhance operational efficiency, reduce costs, and provide valuable insights for decision-making on a global scale
- **Strategic Global Planning:** OM involves strategic planning to align operations with global market demands and trends. This includes identifying key markets, understanding regional requirements, and developing strategies to penetrate and grow in international markets
- **Scalability and Flexibility:** OM helps companies design scalable and flexible operations that can adapt to changes in demand and market conditions. This flexibility is crucial for expanding into new markets and scaling operations to meet global demand

Activity 2.3

Compare your organisation's performance with both local and international competitors in the following categories:

- **Profit**
- **Market Share**
- **Number of Staff**
- **Technology**

Answer: To be discussed during the webinar(s) with lecturer

Case Study 2.1

Having read the above article and watched the video why do you think Walmart's global expansion strategy is experiencing challenges? Determine what made Walmart successful in the US market? Bringing this case study closer to home, why do you think the Massmart Group a subsidiary of Walmart struggled to thrive in South Africa?

Answer: The correct way of answering the above question is by dissecting Walmart's dominance in the US and their growth strategy in other countries. The video has partially provided some answers to the above question except for the South African Analysis; mainly because of stiff competition within the food retailing sector, unions discouraging Walmart to digitalise local outlets etc.

Knowledge Check Questions 2.1

1. Which of the following is a key advantage of globalisation for operations management?

Answer: c. Access to a wider range of resources and markets

2. Operations management strategies typically aim to achieve all of the following except:

Answer: c. Increased complexity in production processes

3. Which global issue in service operations is primarily concerned with adapting to varying cultural norms and expectations?

Answer: c. Cross-cultural service delivery

4. The importance of global operations in a business primarily includes:

Answer: c. Enhancing flexibility in meeting diverse customer needs

5. A firm employing a low-cost strategy in its operations management is most likely to focus on:

Answer: c. Streamlining processes to minimise expenses

Knowledge Check Questions 2.2

1. Globalisation has led to increased interdependency amongst nations in operations management.

Answer: True - Globalisation fosters interdependency as firms rely on international supply chains, labour, and markets.

2. Global issues in service operations are less challenging than in manufacturing due to less complex supply chains.

Answer: False - Service operations face unique global challenges such as language barriers, cultural differences, and varied customer expectations.

3. A critical OM strategy is aligning production schedules with demand forecasts to reduce waste.

Answer: True - Aligning production with demand helps prevent overproduction and reduce costs.

4. The importance of global operations lies solely in cost advantages from international labour markets.

Answer: False - Global operations provide broader benefits, including market expansion, resource access, and innovation opportunities.

5. Globalisation in OM enables firms to innovate more quickly by accessing a global talent pool.

Answer: True - Access to diverse talent globally can drive faster innovation and enhance product development.

Think Point 2.1

Explain your organisation's mission statement

Answer : To be discussed during the webinar(s) with lecturer

Think Point 2.2

Explain the advantages and disadvantages of globalisation.

Answer: To be discussed during the webinar(s) with lecturer

Think Point 2.3

Explain if your organisation is geared to compete with global competitors.

Answer: To be discussed during the webinar(s) with lecturer

Video Activity 2.1

Discuss their turnaround strategy in improving their operations. Discuss what the CEO did in addressing some of the challenges faced by the business?

Answer: The CEO took it upon himself to understand the challenges the business was experiencing

by getting involved in the tasks the employees were doing and spoke to the customers. This he did to identify the problems that were causing operational deficiencies and developed a strategy that could directly resolve those problems. For instance, he reduced the size of the menu and offered customers three options: the “Whopper burger” being the star of their offering. He did this because of the mess that would occur in the kitchen with sauces being mixed up thus causing order delays to customers. Furthermore, the Whopper burger gave the company good profit margins that resulted in increased sales profits. So, in OM terms the CEO improved operations and reduced their menu size to offer 3 products with minor variations (Process design and Process technology). Additionally, he went on to employ other cost-effective business tactics such as “conflict brings attention” a marketing ploy that worked for the company in its rivalry with McDonalds etc.

Video Activity 2.3

Discuss the implications of conflict/ war on globalisation. How is Russia surviving the 17000 sanctions imposed on it by both the US and EU? What are the pros and cons of globalisation through the lens of an OM?

Answer: Conflict and war have profound effects on the dynamics of globalisation. The video discusses how conflicts like the Russia-Ukraine war challenge the interconnectedness of the global economy. Here are the key implications:

1. Disruption of Supply Chains:

- Wars often disrupt global supply chains by cutting off access to crucial resources and markets. The Russia-Ukraine conflict, for example, has led to significant disruptions in energy supplies, particularly in Europe, which relied heavily on Russian gas and oil
- These disruptions force countries to seek alternative suppliers, leading to increased costs and inefficiencies in the global supply chain

2. Shift in Trade Policies:

- Conflicts prompt countries to reassess their trade policies and dependencies. Sanctions and trade barriers are commonly imposed, which can lead to a shift towards protectionism and a reduction in international trade
- The US and EU sanctions on Russia have led to a reorientation of trade flows, with Russia turning to other markets, such as China and India, for trade and economic partnerships

3. Economic Sanctions and Isolation:

- Sanctions can isolate countries from the global financial system, making it difficult for them to

engage in international trade and investment

- For Russia, the imposition of approximately 17,000 sanctions has curtailed its access to Western markets and financial systems, impacting its economic growth and stability

4. Impact on Multinational Corporations:

- Multinational corporations often must navigate complex geopolitical landscapes, leading to increased costs and risks. They may need to divest from conflict regions or restructure their operations to comply with international sanctions
- Companies with operations in or around Russia and Ukraine have faced significant challenges in maintaining their business activities

5. Re-evaluation of Globalisation Benefits:

- Conflicts raise questions about the sustainability and benefits of globalisation, as countries reconsider the risks of deep economic interdependencies
- There is a growing trend towards restoring or nearshoring manufacturing and production to reduce reliance on politically unstable regions

How Russia is Surviving the Sanctions?

Despite the heavy sanctions imposed by the US and EU, Russia has adopted several strategies to mitigate their impact:

1. Diversification of Trade Partners:

- Russia has shifted its trade focus towards non-Western countries, such as China, India, and Turkey. This shift has helped maintain some level of economic activity and reduce the impact of lost trade with Western nations
- The trade in energy, agriculture, and minerals has found new markets, compensating for the loss of Western business

2. Strengthening Domestic Industries:

- Russia has invested in domestic industries to reduce dependence on imported goods and technologies. This has included efforts to localise production in sectors like agriculture, pharmaceuticals, and manufacturing
- The government has also supported import substitution initiatives to encourage the production of goods that were previously imported

3. Utilising Sovereign Wealth Funds:

- Russia has utilised its sovereign wealth funds to stabilise the economy and support key industries. These funds have been critical in maintaining government spending and mitigating the economic downturn caused by sanctions
- The Central Bank of Russia has implemented measures to control inflation and stabilise the currency

4. Bilateral Agreements and Barter Trade:

- Russia has entered into bilateral trade agreements and engaged in barter trade with friendly nations, allowing it to bypass some of the sanctions-related restrictions on financial transactions
- These agreements have enabled Russia to continue trading essential goods and services despite the sanctions

5. Alternative Financial Systems:

- Russia has developed and expanded alternative financial systems, such as the SPFS (System for Transfer of Financial Messages) and MIR payment system, to reduce reliance on Western financial infrastructure
- Partnerships with countries like China have helped in establishing financial arrangements that are less vulnerable to sanctions

Advantages and Disadvantages of Globalisation from an OM Perspective:

Advantages:

1. Cost Efficiency:

- Globalisation allows companies to source materials and labour from regions where they are cheapest, reducing production costs
- It facilitates economies of scale, enabling businesses to operate more efficiently and compete effectively in the global market

2. Access to New Markets:

- Companies can expand their customer base and increase revenues by entering international markets
- Globalisation provides opportunities for businesses to diversify their market risks and tap into emerging markets with high growth potential

3. Innovation and Knowledge Sharing:

- Exposure to global competition and collaboration fosters innovation and the exchange of knowledge and technologies
- Companies can leverage global talent and expertise to improve their products and processes

4. Improved Supply Chain Management:

- Global supply chains can offer greater flexibility and redundancy, allowing companies to source from multiple suppliers and mitigate risks associated with local disruptions
- Advanced logistics and transportation networks support efficient global operations

Disadvantages:**1. Supply Chain Vulnerability:**

- Global supply chains can be susceptible to disruptions from geopolitical conflicts, natural disasters, and pandemics, leading to delays and increased costs
- Dependence on distant suppliers can result in long lead times and reduced responsiveness to market changes

2. Ethical and Regulatory Challenges:

- Operating in multiple countries exposes companies to varying regulatory environments and ethical standards, complicating compliance and ethical management
- Issues such as labour exploitation, environmental degradation, and corruption may arise in different regions

3. Cultural and Communication Barriers:

- Managing operations across diverse cultural contexts can lead to misunderstandings and conflicts, affecting collaboration and efficiency
- Language barriers and differences in business practices can hinder effective communication and coordination

4. Increased Complexity and Risk:

- Global operations add layers of complexity to supply chain management, logistics, and legal compliance
- Companies must navigate political risks, currency fluctuations, and trade restrictions, which can

impact their operations and profitability

In conclusion, while globalisation offers significant opportunities for growth and efficiency, it also presents challenges that require careful management and strategic planning in the face of geopolitical uncertainties and economic disruptions. The discussion above highlights the multifaceted impact of conflicts on globalisation and provides insights into how Russia is navigating the sanctions landscape. It also underscores the need for a balanced approach to globalisation in operations management, considering both the benefits and risks involved.

**Unit
3:**

**Total Quality
Management**

Unit Learning Outcomes

CONTENT LIST	LEARNING OUTCOMES FOR THIS UNIT
3.1 Introduction	Introduces the topic areas covered for the unit
3.2 The Origins of TQM	Outline the origins of TQM
3.3 TQM Concepts	Analyse TQM concepts against the backdrop of current operations management trends
3.4 The Costs of Quality	Differentiate between the different costs of quality and the interrelationship thereof
3.5 TQM in Services	Analyse the application of TQM principles in the service industry
3.6 Summary	Summarises the topic areas covered for the unit

Prescribed and Recommended Reading(s)/Textbook(s)

Prescribed Reading(s)/Textbook(s)

- Heizer, J., Render, B. and Munson, C. (2020). *Operations Management: Sustainability and Supply Chain Management*. Thirteenth Edition. United Kingdom: Pearson.

Recommended Reading(s)/Textbook(s)

- Venkataraman, R.R. and Pinto, J.K. (2020). *Operations Management: Managing Global Supply Chains*. Second Edition. Sage Publications.
- Parksoy, T. and Deveci, M. (2023). *Smart and Sustainable Operations and Supply Chain Management in Industry 4.0*. First Edition. Boca Raton: CRC Press.
- Slack, N., Jones, B. and Burgess, N. (2022). *Operations Management*. Tenth Edition. United Kingdom: Pearson.
- Kareska, PhD Katerina. 'Operational Management Strategies for Sustainable Productivity Growth' (August 5, 2023). Available at SSRN: <https://ssrn.com/abstract=4532740> or <http://dx.doi.org/10.2139/ssrn.4532740>
- 'The Impact Of Ergonomics On Quality Of Life In The Workplace'. (2023). https://www.researchgate.net/publication/377267055_The_Impact_of_Ergonomics_on_Quality_of_Life_in_the_Workplace



3.1 Introduction

Total Quality Management (TQM) is a strategic approach that integrates quality principles into all aspects of an organisation to achieve long-term success through customer satisfaction. It emphasises a culture of continuous improvement, data-driven decision-making, and proactive problem-solving. By implementing TQM, organisations enhance efficiency, reduce waste, and improve overall performance.

This unit explores key TQM concepts such as Six Sigma, continuous improvement, benchmarking, Just-in-Time (JIT), and the Taguchi approach to quality design. Six Sigma focuses on reducing defects through statistical methods, while continuous improvement fosters incremental enhancements in processes. Benchmarking enables organisations to compare performance against industry leaders, JIT minimises inventory waste by synchronising production with demand, and Taguchi methods emphasise robust design to improve product and process quality. By the end of this unit, students will understand how these TQM principles contribute to organisational excellence and competitive advantage.

3.2 The Origins of TQM – The Quality Gurus

- Armand Feigenbaum was a doctoral student at the Massachusetts Institute of Technology in the 1950s. He defined TQM as “an effective system for integrating the quality development, quality maintenance and quality improvement efforts of the various groups in an organisation to enable production and service at the most economical levels which allow for full customer satisfaction”
- W.E. Deming was considered as the father of quality management in Japan. Deming introduced the 14 points for quality improvement whereby he emphasised the need for statistical quality control, participation, education, openness and purposeful improvement
- J.M. Juran was also a key educator in the Japanese quality management systems. He advocated that although a dangerous product could conform to specification, it would not be fit to use. He coined the phrase “fitness for use”. He advocated the motivation and involvement of the workforce in quality improvement activities
- K. Ishikawa based his work on that of Deming, Juran and Feigenbaum and has been credited with originating the concept of quality circles and the cause-and-effect diagram (fishbone). Ishikawa realised that worker participation was the key to the successful implementation of TQM
- G. Taguchi was the director of the Japanese Academy of Quality and was concerned with engineering quality into the product through the optimisation of product design. His concept of Quality Loss Function (QLF) included such factors as warranty costs, customer complaints and loss of customer goodwill
- P.B. Crosby is best known for his work in trying to quantify the costs of quality. He stated that

many organisations did not know how much they spend on quality

What Is TQM?

The overall qualities and attributes of a product or service that impact its capacity to meet identified or inferred requirements (Heizer, J. et al 2020). TQM is a philosophy, a way of thinking and working, that is concerned with meeting the needs and expectations of customers. It tries to move the focus of quality away from being a purely operations activity into a major concern for the entire organisation. Through TQM, quality becomes the responsibility of all departments and sections in the organisation. It also targets the costs of quality by trying to reduce particular failure costs. TQM also espouses the process of continuous improvement.

TQM places the customer in the forefront of decision-making. It also looks at the concept of internal customer and supplier. It advocates that everyone in the organisation is a customer and consumer of goods and services provided by other internal suppliers.



Think Point 3.1

Explain why TQM is important.



Activity 3.1

1. Discuss the term 'TQM'.
2. Name three quality gurus and the concepts they pioneered.

3.3 TQM Concepts

Heizer et al (2020) identified the six listed TQM concepts below:

Continuous Improvement

TQM requires a never-ending process of continuous improvement that embraces people, suppliers, material, equipment and procedures. The argument of this philosophy is that every aspect of an operation can be improved – the end objective is perfection, which is always sought but seldom achieved. The Japanese use the word kaizen to describe this on-going process of improvement. In continuous improvement it is not the size of each step which is important – rather it is the likelihood that improvements will be on-going. We can therefore say it is not the rate of improvement but the momentum of improvement.

Lean Continuous Improvement is a strategic approach that focuses on minimising waste and boosting efficiency in manufacturing processes. It falls under the wider Lean Manufacturing framework, which aims to deliver greater value to customers using fewer resources (Salman et al.). For further information on lean manufacturing and sustainable development goals, here is a link to the article

<https://www.emerald.com/insight/content/doi/10.1108/JRPC-10-2023-0009/full/html>

Six Sigma

According to Heizer et al. (2020) this programme is designed to reduce defects to help lower costs, save time, improve quality and improve customer satisfaction. Furthermore, in the context of Total Quality Management (TQM), Six Sigma, as popularised by Motorola, Honeywell, and General Electric (GE), refers to a statistical measure denoting a process, product, or service that exhibits exceptionally high capability.

Moreover, Six Sigma constitutes a holistic system encompassing strategy, discipline, and a toolkit aimed at achieving and maintaining business success.

Practical Application 3.1



Motorola developed Six Sigma in the 1980s to address customer complaints and intense competition. Initially targeting a 90% reduction in defects, the company achieved rapid success by benchmarking competitors, soliciting employee ideas, revising reward structures, enhancing training, and overhauling critical processes. This led to the formalisation of Six Sigma procedures. While rooted in manufacturing, General Electric (GE) later extended Six Sigma principles to encompass services such as human resources, sales, customer service, and financial/credit services. The core objective of eliminating defects proved equally applicable in both manufacturing and service contexts.

Question:

Examine how the Six Sigma principles, as initially developed by Motorola, can be applied by operations managers to enhance process efficiency and quality.

Employee Empowerment

This refers to the involvement of the workforce in every step of the production process. To design equipment and processes that consistently produce the desired quality would require the involvement of those who understand the shortcomings of the system. People working on the system

on a daily basis understand it better than anyone else does. Techniques for building employee empowerment include:

- Building communication networks that include employees
- Developing open, supportive supervisors
- Moving responsibility from both managers and staff to production employees
- Building high-morale organisations
- Creating formal organisation structures such as teams and quality circle

Benchmarking

Benchmarking is a crucial component of an organisation's TQM initiative. It entails identifying a proven standard in products, services, costs, or practices that exemplify top-tier performance in processes or activities closely resembling one's own. Heizer et al (2020) describe benchmarking as the process of setting a target to aim for and then establishing a standard or benchmark against which your own performance can be compared. The steps involved in benchmarking are:

1. Identifying what to benchmark.
2. Forming a benchmark team.
3. Selecting benchmarking partners.
4. Gathering and analysing benchmarking data.
5. Taking action to meet or surpass the benchmark.

Benchmarking in TQM typically involves using performance measures such as defect percentage, unit or order cost, processing time per unit, service response time, return on investment, customer satisfaction levels, and customer retention rates.

Types of Benchmarking

According to Pycraft et al. (2010), the following are some of the types of benchmarking one can use:

- Internal benchmarking is a comparison between operations or parts of operations which are within the same organisation
- External benchmarking is a comparison between an operation and other operations which are part of a different organisation
- Non-competitive benchmarking is benchmarking against external organisations which do not compete directly in the same markets
- Competitive benchmarking is a comparison directly between competitors in the same or similar markets

- Performance benchmarking is a comparison between the levels of achieved performance in different operations
- Practice benchmarking is a comparison between an organisation's operations practices, or ways of doing things, and those adopted by another operation



Practical Application 3.2

Xerox found that L.L. Bean could process orders three times quicker than they could. After adopting similar practices, Xerox quickly managed to cut warehouse expenses by 10%. Similarly, Mercedes-Benz noticed that L.L. Bean's warehouse staff used flowcharts to eliminate unnecessary movements. Mercedes-Benz implemented this approach, leading to a greater emphasis on worker-level problem solving. Benchmarks often represent "best practices" identified in other companies or departments.

Question:

Analyse how Xerox utilises benchmarking to improve their warehouse operations, and determine the outcomes.

Just In Time (JIT)

The concept behind just-in-time (JIT) is centred on continuous improvement and enhanced problem-solving. JIT systems are designed to manufacture or deliver goods precisely when they are required. JIT is closely linked to quality in several ways:

- JIT reduces the cost of quality: This happens because expenses like wastage, reworking, inventory holding, and damages are directly tied to the amount of inventory held. With less inventory under JIT, these costs are minimised. Additionally, holding inventory can mask poor quality, whereas JIT promptly exposes any quality issues
- JIT enhances quality: By reducing lead times, JIT ensures that errors are noticed quickly and limits the potential sources of mistakes. JIT acts like an early warning system for quality concerns, both internally and with suppliers
- Better quality leads to reduced inventory and facilitates a more effective JIT system: Often, companies maintain inventory to safeguard against production issues stemming from inconsistent quality. With consistent quality assured, JIT enables firms to decrease the costs associated with inventory across the board



Video Activity 3.1

Watch the video using the link below to see how Toyota implements this concept in its operations.

How Toyota Maintains 99.9999% Quality | Toyota Quality Management System | Invensis Learning

Available at: <https://youtu.be/K3gaqsQcbfY?si=zLOEmw5L8YDPfLkv>

Taguchi Concepts

Most quality problems stem from inadequate product and process design. Genichi Taguchi introduced three concepts aimed at enhancing both product and process quality: quality robustness, target-oriented quality, and the quality loss function. Quality robust products are those that can be manufactured consistently and uniformly even under challenging manufacturing and environmental conditions. Taguchi's approach advocates mitigating the impacts of adverse conditions rather than eliminating their causes. He argues that addressing these impacts is often more cost-effective and efficient in achieving a robust product. This strategy ensures that minor fluctuations in materials and processes do not compromise product quality.



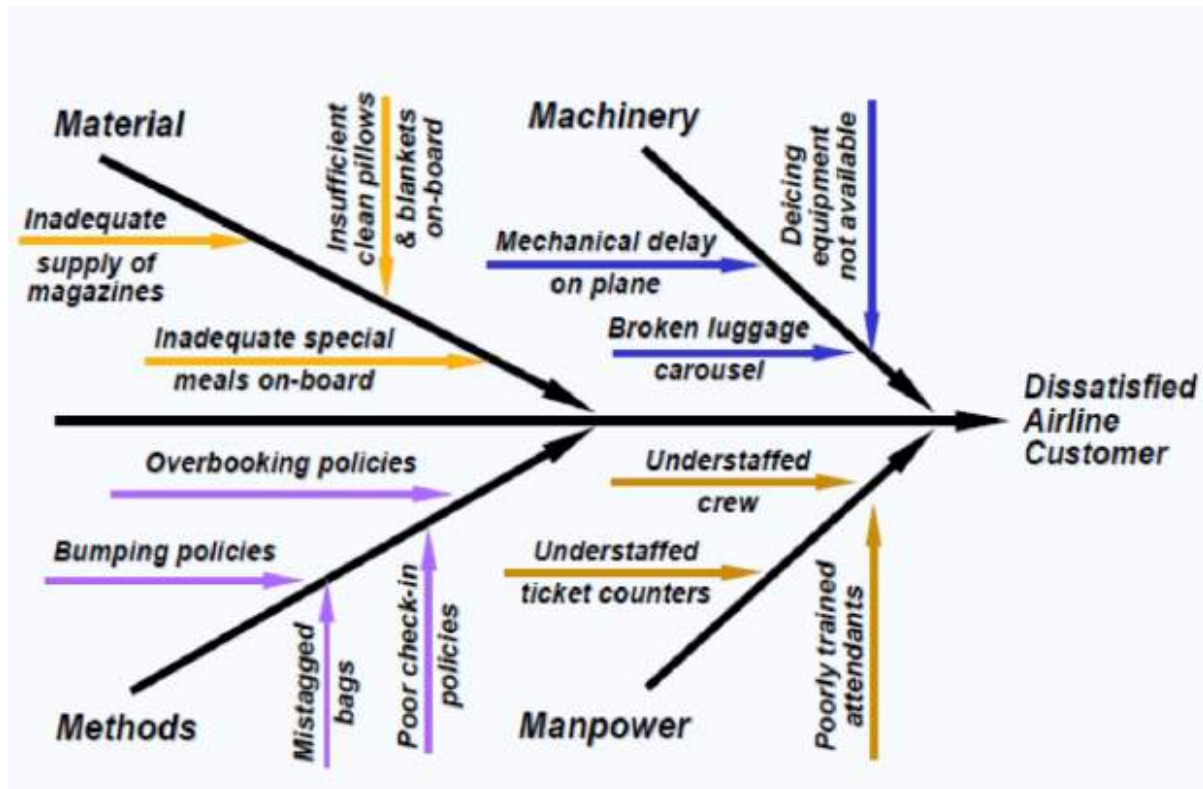
Activity 3.2

Explain why it would be difficult to apply JIT in Southern African countries.

2. Knowledge of TQM Tools

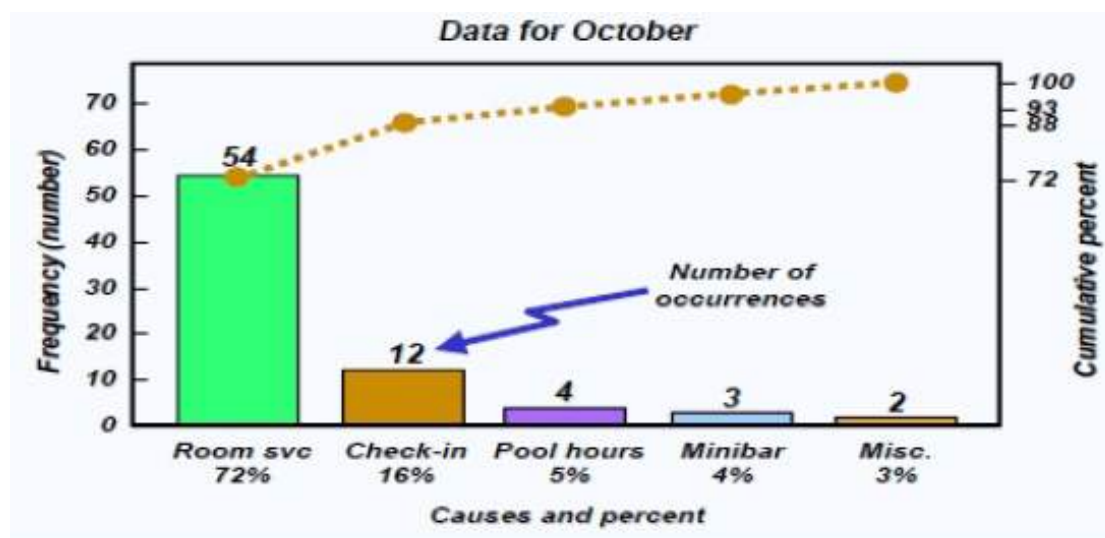
To empower employees and implement TQM everyone in the organisation must be trained in the techniques of TQM. Heizer.et.al (2020) suggest seven tools or techniques that can aid the TQM drive:

- Check Sheets – a check sheet is any kind of form that is designed for the recording of data. In many cases the recording is done so the patterns are easily seen while the data are being taken. Check sheets help analysts find the facts or patterns that may aid subsequent analysis. An example might be a drawing that shows a tally of the areas where defects are occurring or a check sheet showing the type of customer complaints
- Scatter Diagrams – show the relationship between two measurements. An example is the positive relationship between length of a service call and the number of trips a repair person makes back to the truck for parts
- Cause and Effect Diagrams - are another tool for identifying quality issues and inspection points. This type of diagram is also known as Ishikawa or fish-bone chart



Source: (Heizer and Render 2020)

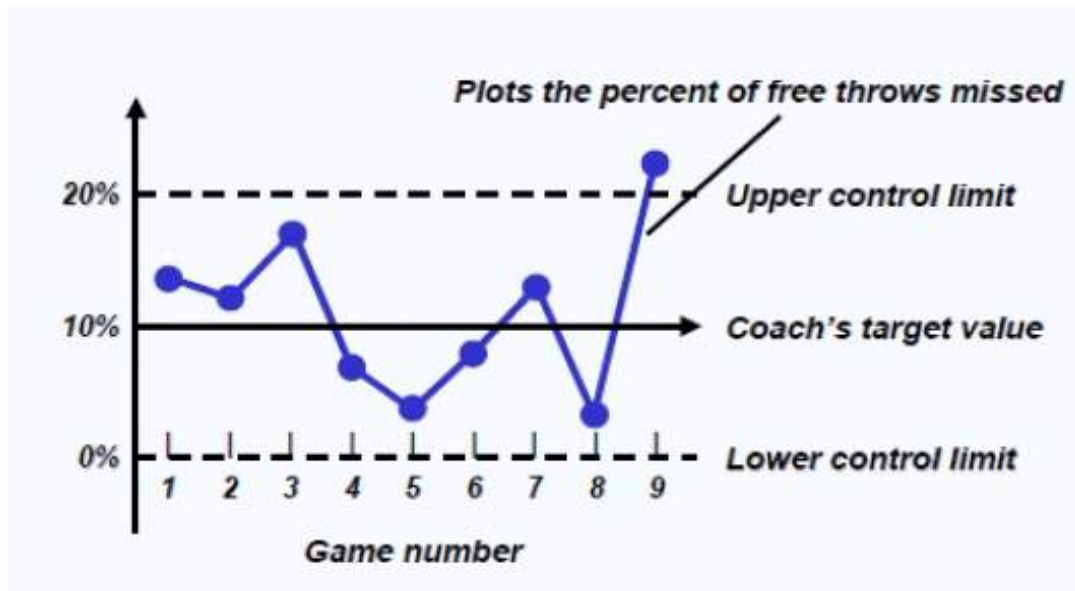
- **Pareto Charts** – a graphic way of identifying the few critical items as opposed to several less important ones



Source: Heizer.et.al (2020)

- **Flowcharts** – graphically represent a process or system using annotated boxes and interconnected lines. They are a simple, but great tool for trying to make sense of a process or explain a process

- **Histograms** – show the range of values of a measurement and the frequency with which each value occurs. They show the most frequently occurring readings as well as the variations in the measurements
- **Statistical Process Control** – is used to monitor standards, making measurements and taking corrective action as a product or service is being produced



Source: Heizer.et.al (2020)



Activity 3.3

Identify the TQM concepts that your organisation incorporates/utilises.

3.4 The Costs of Quality

The costs of quality can be categorised as prevention costs, appraisal costs, internal failure costs and external failure costs. Let's examine each of these:

1. Prevention Costs – these are costs incurred in trying to prevent problems, failures and errors from occurring in the first place and include:

- Identifying potential problems and rectifying them before poor quality occurs
- Improving the design of products, services and processes to reduce quality problems
- Training and developing personnel in the best way to perform their jobs

2. Appraisal Costs – are those costs associated with controlling quality to check if problems have occurred during or after the production process. They include:

- The setting up of statistical process control programmes and acceptance sampling plans
- The time and effort required to inspect outputs, inputs and processes
- Obtaining processing inspection and test data
- Investigating quality problems and providing quality reports
- Conducting customer surveys and quality audits

3. Internal Failure Costs – are costs associated with errors whilst the product is within the operation, and include:

- Costs of scrapped parts and material
- Reworked parts and material
- Lost production time as a result of errors
- Lack of concentration due to time spent troubleshooting rather than improvement

4. External Failure Costs – are those which are associated with errors that have reached the customer. These costs include:

- Loss of customer goodwill
- Aggrieved customers
- Litigation
- Guarantee and warranty costs

Case Study 3.1

Read the following article and answer the question below.

A Study of Quality Management in Small Organisations Providing Services

Directed at People Joseph N. Khamalah

Case study: adapted from – Clutejournals.com

Source: <https://clutejournals.com/index.php/JBER/article/view/2973>

Small business firms today employ more people than big firms. For instance, McDonald's franchises are collectively the largest employer in the U.S. Quality in small businesses attracts as much attention as is the case in large organisations. For instance, the International Standards organisation (ISO) has publications addressing the implementation of ISO 9001-based quality management systems for small organisations. This is even reflected in the Malcolm Baldrige National Quality Awards – out of the 49 organisations applying for the 2002 awards, only 8 were large manufacturers. The rest

were: 3 service, 10 educational, 17 health care, and 11 small business firms [2]. The intersection of service and small is indeed important as evidenced by frequent coverage in both the popular press and practitioner periodicals of instances of poor service [7, 13]. Despite this, there is a dearth of material in the literature regarding the status of total quality management in small firms. Complaints about poor quality and customer service seem to be universal. Paton refers to this as “aggressively bad customer service” [13]. Poor quality is very expensive for firms. This cost can be even more onerous for service firms given the proclivity of people to sue and/or recount their good or bad service experiences to friends, family and colleagues [6]. This penalises the “offending” service firm on multiple fronts – loss of repeat business, unfavourable word of mouth, and a costly legal battle on its hands. The stakes are especially high where the service act is directed at people’s bodies and minds for instance health care, food services, and beauty salons. Medical mal-practice suits, litigation against restaurants, and liability suits for negligence in settings such as beauty salons are commonplace and quite often do lead to bankruptcies. Hence, quality and its deployment in services directed at people is even more crucial than in other types of services. The study also sought to establish respondents perceptions regarding their firms practices in employee training, commitment, responsibility, culture, continuous improvement, and reward systems in connection with quality. In general, the majority of respondents indicated that their organisations were continually making improvements in their work processes, employees are committed to achieving excellence and are responsible for their own work, their organisational culture builds mutual trust, and had employees at all levels undergo extensive training on quality issues. Table 4 summarises these findings.



Personnel and Work Processes			
Statement		Percent of Respondents	
Agree/	Neutral	Disagree/	
Firm is always making improvements in work processes	93.7	2.9	3.4

Employees committed to achieving excellence in TQM	90.2	7.5	2.3
Employees responsible for their own work	87.9	5.7	6.4
Firm's organisational culture builds mutual trust	78.1	16.7	5.2
Firm's employees undergo extensive quality training	75.9	14.9	9.2
Employees who successfully apply TQM are rewarded/recognised	54.0	29.9	16.1

Responses to the first five statements in this section were not at all surprising. The overwhelming majority (more than three quarters of all respondents) agreed or strongly agreed with the statements as posed in the survey instrument. In marked contrast to these, however, a little over half of the respondents (54.0%) agreed with the statement that employees who successfully applied TQM were rewarded or recognised. This does not match the agree/strongly agree responses on top management commitment to TQM, continuous improvements in work processes, employee quality training, culture of mutual trust, employee commitment to TQM, and employees being responsible for their own work. It suggests that the bulk of agree/strongly agree responses to the first five survey statements in this category may simply have been "lip service" or the "politically correct" thing to say about the firm, or that the majority of firms in this sector are yet to follow through on their implementation of TQM.

The foregoing also signifies that most firms' commitment to TQM was higher than their commitment to rewarding employees for quality performance. In quality-conscious organisations, high quality is everybody's business [16] and it has to be buttressed by a reward system that recognises successful instances of quality accomplishments. The malaise that these results point to, however, is not the exclusive domain of people-oriented service firms. Other studies have shown that even in the manufacturing sector, most firms TQM programs are yet to evolve to that stage and time when the importance of tying reward systems to quality performance is recognised.

Question:

Through discussion, analyse the impact the ten general attributes or

determinants of service quality has on improving the public image of the service industry.

Knowledge Check Questions 3.1

State if the following is True or False by selecting the correct answer.

1. TQM focuses solely on improving the quality of products.
True
False
2. Continuous improvement is a key principle of TQM.
True
False
3. In TQM, the responsibility for quality lies only with the quality control department.
True
False
4. The primary focus of Six Sigma is to reduce process variation and defects.
True
False
5. The Taguchi method emphasises cost reduction through robust product design.
True
False
6. The JIT production is primarily concerned with increasing inventory levels to avoid stockouts.
True
False



Video Activity 3.1



Watch the video using the link below and answer the question that follows.

The Cost of Quality - CoQ (Prevention costs & Appraisal Costs) | Lean Six Sigma Complete Course.

Available at: <https://youtu.be/T416tLqIkHA?si=WKbCJKZ9GzWiZ1vN>

Question:

Determine 3 appraisal costs.



Think Point 3.2

Explain in financial terms the cost of quality within your company.



Activity 3.4

Determine the implications of high failure costs for an organisation practicing TQM?

3.5 TQM in Services

Before we continue to discuss TQM in Services think about the following:



Think Point 3.3

Do you think TQM processes in manufacturing and services are similar?

It is more difficult to measure quality in services than that of manufactured goods. Heizer.et.al (2020) identifies ten general attributes or determinants of service quality:

- **Reliability** – Performing the service right the first time and honouring your promises
- **Responsiveness** – Willingness or readiness of employees to provide a service
- **Competence** – Possessing the required skills and knowledge to perform the service

- **Access** – Involves ease of contact
- **Courtesy** – Politeness, respect and consideration
- **Communication** – Keeping customers informed in a language they understand
- **Credibility** – Trustworthiness and honesty
- **Security** – Freedom from danger, risk or doubt
- **Understanding** – Knowing the customer and
- **Tangibles** – Include the physical evidence of the service

Case Study 3.3

Read the following article and answer the question below.

Total Quality Management (TQM) at Ramaiah Institute of Management Studies (RIMS)

Ramaiah Institute of Management Studies (RIMS), located in Bangalore, India, is a leading business school known for its commitment to academic excellence and innovative management education. To maintain and enhance its quality standards, RIMS has integrated Total Quality Management (TQM) principles into its operational framework. This case study explores how RIMS has successfully implemented TQM to improve educational outcomes, administrative efficiency, and overall stakeholder satisfaction.

RIMS was established with a vision to provide world-class management education in India. The institute has consistently been at the forefront of academic innovation, offering a curriculum that aligns with global standards. Recognising the importance of continuous improvement and customer satisfaction, RIMS adopted TQM practices to ensure that it meets the expectations of students, faculty, and industry partners.

RIMS's leadership demonstrates a strong commitment to quality by integrating TQM principles into the institution's core values. This commitment is reflected in the establishment of a dedicated Quality Assurance Cell (QAC) that oversees the implementation and monitoring of quality initiatives across the institute. The QAC is responsible for setting quality benchmarks, conducting regular audits, and ensuring compliance with academic standards. RIMS places a significant emphasis on engaging with all stakeholders, including students, faculty, administrative staff, and industry partners. The institute regularly seeks feedback from these groups to identify areas for

improvement. For instance, students are encouraged to provide feedback on courses and teaching methodologies, which is then used to refine the curriculum and enhance learning outcomes.

The principle of continuous improvement is central to RIMS's TQM approach. The institute adopts a Plan-Do-Check-Act (PDCA) cycle to systematically improve its processes. For example, new teaching methods and technologies are piloted in classrooms, their effectiveness is evaluated, and successful practices are implemented across the board. This iterative process ensures that the institute continuously evolves to meet the changing needs of the education sector.

To ensure that its programs remain relevant and competitive, RIMS employs a robust curriculum development process that incorporates industry trends and technological advancements. The institute collaborates with industry experts to co-create course content, ensuring that students gain practical knowledge and skills that are in demand. This proactive approach helps RIMS maintain high academic standards and produce graduates who are ready to meet industry challenges.

Recognising that faculty are pivotal to delivering quality education, RIMS invests heavily in their professional development. The institute offers regular training programs, workshops, and opportunities for faculty to engage in research and attend international conferences. This not only enhances their teaching capabilities but also keeps them abreast of the latest developments in their fields.

RIMS leverages technology to streamline its administrative processes and enhance operational efficiency. The institute has implemented an Enterprise Resource Planning (ERP) system that integrates various functions such as admissions, finance, and academic management. This system provides real-time data and analytics, enabling informed decision-making and efficient resource allocation.

One of the key successes of RIMS's TQM implementation is the establishment of a pervasive quality culture. This culture is fostered through continuous communication, training, and the active involvement of all members of the institute in quality initiatives. The commitment to quality is evident in every aspect of RIMS's operations, from classroom instruction to campus maintenance.



The implementation of TQM at RIMS has yielded significant benefits. The institute has seen an improvement in student satisfaction, faculty performance, and academic outcomes. It has also strengthened its reputation among industry partners, leading to enhanced placement opportunities for graduates. The continuous focus on quality has enabled RIMS to maintain its position as a leading business school in India and has set a benchmark for others to follow.

Ramaiah Institute of Management Studies demonstrates how TQM principles can be effectively applied in an educational context to drive continuous improvement and achieve excellence. By fostering a culture of quality, engaging stakeholders, and leveraging technology, RIMS has created a dynamic and responsive educational environment that meets the needs of its students and prepares them for successful careers in management.

For more detailed information on RIMS's TQM practices, you can explore the original article on European CEO (<https://www.europeanceo.com>)

Questions:

1. How does RIMS ensure continuous improvement in its educational practices through TQM?
2. How does RIMS engage stakeholders in its TQM process?
3. What role does technology play in the TQM framework at RIMS?

Knowledge Check Questions 3.2

Select the correct answer:



1. Which of the following is a key aspect of Six Sigma?
 - a. Increasing inventory levels
 - b. Reducing process variation
 - c. Ignoring customer feedback

2. The primary focus of the Just-in-Time (JIT) production is to.
 - a. Maximising production regardless of demand
 - b. Minimising inventory and waste
 - c. Increasing lead times

3. Benchmarking helps organisations by:
 - a. Setting arbitrary performance targets
 - b. Comparing their processes to best practices and industry standards
 - c. Ignoring competitors' strategies

3.6 Summary

Total Quality Management (TQM) is perhaps the most significant of the new ideas which have swept across the operations management scene over the last few years. There are two reasons for this: first the ideas of TQM have great perceptive attraction for many people – most of us want to achieve 'high quality', and secondly, a TQM approach to improvement can sometimes result in dramatic increases in operational effectiveness.

This unit provided an in-depth exploration of TQM principles and their role in driving operational and strategic improvements. The key topics included TQM Concepts with a deep dive into Six Sigma, continuous improvement, benchmarking, Just-in-Time (JIT), and the Taguchi approach, with a focus on their application in improving process efficiency and product quality. The costs of quality was analysed, the financial impact of quality management, including the costs of prevention, appraisal, internal failure, and external failure, and how organisations can optimise spending to maximise quality. TQM in service was explored for understanding how quality principles apply in service industries, ensuring customer satisfaction through process standardisation, error reduction, and service delivery excellence. Through real-world examples and case studies, this unit demonstrated how organisations use TQM methodologies to enhance performance, minimise defects, and gain a competitive edge in both manufacturing and service sectors.

Revision Questions

1. The accounts receivable department has documented the following defects over a 30-day period:

Category	Frequency
Invoice amount does not agree with the check amount	108
Invoice not on record (not found)	24
No formal invoice issued	18
Check (payment) not received on time	30
Check not signed	8
Invoice number and invoice referenced do not agree	12

Explain the techniques that you would use, and the conclusions drawn about defects in the accounts receivable department.

2. Use a Pareto chart to organise the defects and conclude that the obvious problem (about half the defects) is the failure of the check to agree with the company's records as to the correct amount. Other problems are late payments and an apparent invoice-filing problem in the office. Notice that 27% of these common errors appear to be the result of procedural problems within accounts receivable (invoice not on record, no invoice issued, and invoice numbering problems). This value could be considerably higher depending on how much of the problem of disagreement between invoice and check amounts is the result of accounts receivable process problems.

3. Outline a flow chart for purchasing a Big Mac at the drive-through window at McDonald's.

4. Illustrate a fishbone chart detailing reasons why a part might not be correctly machined.

5. Examine how Mercedes-Benz benefits from benchmarking L.L. Bean's warehouse practices, and outline the changes that were implemented.

6. Discuss the primary goal of Just-in-Time (JIT) production.

7. Determine how Benchmarking help a organisations improve their performance.

8. Discuss the reasons why the Taguchi method is important in quality management.



9. Examine the implications of high failure costs for an organisation practising TQM.

Answers

Activity 3.1

1. Discuss the term 'TQM'.

Answer: It is the overall qualities and attributes of a product or service that impact its capacity to meet identified or inferred requirements.

2. Name three quality gurus and the concepts they pioneered.

Answer: It is the overall qualities and attributes of a product or service that impact its capacity to meet identified or inferred requirements.

1. Genichi Taguchi developed the Taguchi concept focusing on the engineering quality into the product through optimisation of product design.
2. W.E. Deming is often regarded as the pioneer of quality management in Japan. He introduced a framework of 14 principles aimed at enhancing quality, highlighting the importance of statistical quality control, active involvement, continuous learning, transparency, and deliberate improvement.
3. P.B. Crosby is renowned for his efforts to quantify the costs associated with quality. He observed that many organisations were often unaware of the actual amount they were spending on ensuring quality.

Activity 3.2

Explain why it would be difficult to apply JIT in Southern African countries.

Answer: To be discussed during the webinar(s) with lecturer

Activity 3.3

Identify the TQM concepts that your organisation incorporates/utilises

Answer: To be discussed during the webinar(s) with lecturer

Activity 3.4

Determine the implications of high failure costs for an organisation practicing TQM?

Answer: High failure costs, particularly external failure costs, can have several negative implications for an organisation practicing TQM:

- Financial Impact: High failure costs can result in significant financial losses due to increased expenses for warranty claims, product recalls, repairs, and customer refunds. This can erode

profit margins and strain the company's resources

- **Reputation Damage:** Frequent quality issues can damage an organisation's reputation, leading to a loss of customer trust and loyalty. Negative publicity and dissatisfied customers can hurt the brand image and make it challenging to attract and retain customers
- **Operational Disruptions:** Internal failure costs such as rework and scrap can disrupt production schedules, reduce efficiency, and increase operational complexity. This can lead to higher production costs and delays in delivering products or services to customers
- **Regulatory and Legal Risks:** Persistent quality problems can attract regulatory scrutiny and legal challenges, especially in industries with strict compliance requirements. This can result in fines, penalties, and increased oversight, further adding to the costs and complexities of doing business
- By effectively managing and reducing failure costs, organisations can enhance their overall quality, financial performance, and market competitiveness

Case Study 3.1

Through discussion, analyse the impact the ten general attributes or determinants of service quality has on improving the public image of the service industry.

Answer: To be discussed during the webinar(s) with lecturer

Case Study 3.2

Through discussion, analyse the impact the ten general attributes or determinants of service quality has on improving the public image of the service industry.

Answer: To be discussed during the webinar(s) with lecturer.

Case Study 3.3

1. How does RIMS ensure continuous improvement in its educational practices through TQM?

Answer: Ramaiah Institute of Management Studies (RIMS) employs the Plan-Do-Check-Act (PDCA) cycle to ensure continuous improvement in its educational practices. This approach involves planning new initiatives, implementing them on a small scale, checking the results through feedback and assessment, and then acting to integrate successful practices into the broader system. For instance, RIMS pilots new teaching methods and technologies in selected classrooms, evaluates their effectiveness, and then scales up the successful ones across the institute. This iterative process helps RIMS to continuously refine its educational practices and stay ahead of industry trends and academic standards.

2. How does RIMS engage stakeholders in its TQM process?

Answer: RIMS places a significant emphasis on stakeholder engagement, which includes students, faculty, administrative staff, and industry partners. The institute regularly seeks feedback from these

groups through surveys, focus groups, and regular meetings. This feedback is critical for identifying areas of improvement and ensuring that the needs and expectations of stakeholders are met. For example, student feedback on courses and teaching methods is used to refine the curriculum and improve teaching practices. Additionally, RIMS collaborates with industry experts to ensure that its programs are relevant and aligned with current industry needs, thereby enhancing the employability of its graduates.

3. What role does technology play in the TQM framework at RIMS?

Answer: Technology plays a crucial role in the TQM framework at RIMS by streamlining administrative processes and enhancing operational efficiency. The institute has implemented an Enterprise Resource Planning (ERP) system that integrates various functions such as admissions, finance, and academic management. This system provides real-time data and analytics, which support informed decision-making and efficient resource allocation. By leveraging technology, RIMS is able to maintain a high level of quality in its operations and ensure that it can quickly adapt to changing educational and administrative needs. The use of technology also helps in maintaining transparency and accountability within the institution.

Knowledge Check Questions 3.1

1. TQM focuses solely on improving the quality of products.

Answer: False

2. Continuous improvement is a key principle of TQM.

Answer: True

3. In TQM, the responsibility for quality lies only with the quality control department.

Answer: False

4. The primary focus of Six Sigma is to reduce process variation and defects.

Answer: True

5. The Taguchi method emphasises cost reduction through robust product design.

Answer: True

6. The JIT production is primarily concerned with increasing inventory levels to avoid stock-outs.

Answer: False

Knowledge Check Questions 3.2

1. Which of the following is a key aspect of Six Sigma?

Answer: b. Reducing process variation

2. The primary focus of the Just-in-Time (JIT) production is to.

Answer: b. Minimising inventory and waste

3. Benchmarking helps organisations by:

Answer: b. Comparing their processes to best practices and industry standards

Practical Application 3.1

Examine how the Six Sigma principles, as initially developed by Motorola, can be applied by operations managers to enhance process efficiency and quality.

Answer: Operations managers can utilise Six Sigma principles to significantly enhance process efficiency and quality in various business contexts. Here's how:

1. **Define and Measure:** Operations managers should start by clearly defining process goals and identifying key performance indicators (KPIs). For example, in a manufacturing setting, they could set a target for reducing production cycle time or lowering defect rates. In a service-oriented business, the focus might be on reducing customer wait times or improving service accuracy. These goals should be measured using data collection and statistical analysis tools, similar to how Motorola targeted a 90% reduction in defects through meticulous measurement and benchmarking against competitors.
2. **Analyse and Improve:** Managers should analyse data to identify root causes of inefficiencies or defects. For instance, in a supply chain operation, analysis might reveal bottlenecks or areas of waste that are hindering performance. Once identified, they can implement targeted improvements, such as streamlining workflow processes or enhancing training programs, much like Motorola revamped critical processes and solicited employee input to drive rapid success.
3. **Control:** To maintain improvements, operations managers should implement control measures, such as standard operating procedures (SOPs) and continuous monitoring systems. This helps ensure that the enhanced processes remain effective and efficient over time, mirroring the control mechanisms used in Six Sigma to sustain high quality and minimise variations.

By applying these Six Sigma principles, operations managers can drive substantial improvements in process efficiency and quality, leading to cost savings, increased customer satisfaction, and a stronger competitive edge in the market.

Practical Application 3.2

Analyse how Xerox utilises benchmarking to improve their warehouse operations, and determine the outcomes.

Answer: Xerox employed benchmarking by studying L.L. Bean's order fulfilment processes, discovering that L.L. Bean could "pick" orders three times faster. By adopting similar practices, Xerox streamlined its warehouse operations, leading to a significant 10% reduction in warehouse costs. This benchmarking process involved identifying best practices from L.L. Bean, such as efficient order picking methods, and implementing these practices within Xerox's operations. The successful outcome of this benchmarking effort demonstrates how adopting proven methods from other organisations can lead to immediate cost savings and operational improvements.

Xerox noticed that L.L. Bean was able to "pick" orders three times faster. After benchmarking, Xerox was immediately able to pare warehouse costs by 10%. Mercedes-Benz observed that L.L. Bean warehouse employees used flowcharts to spot wasted motions. The auto giant followed suit and now relies more on problem solving at the worker level. Benchmarks often take the form of "best practices" found in other firms or in other divisions.

Think Point 3.1

Explain why TQM is important.

Answer: Importance of TQM: Ensures quality, reduces waste, and enhances customer satisfaction

Think Point 3.2

Explain in financial terms the cost of quality within your company.

Answer : To be discussed during the webinar(s) with lecturer.

Think Point 3.3

Do you think TQM processes in manufacturing and services are similar?

Answer: Processes differ as manufacturing focuses on product quality, while services focus on customer interaction and experience.

Video Activity 3.2

Determine 3 appraisal costs.

Answer:

1. Inspection
2. Testing
3. Auditing

**Unit
4:**

Forecasting

Unit Learning Outcomes

CONTENT LIST	LEARNING OUTCOMES FOR THIS UNIT
4.1 Introduction	Introduces the topic areas covered for the unit
4.2 What is Forecasting?	Defining forecasting and its significance
4.3 Features Common to all Forecasts	Outline the common features of different forecasts
4.4 Forecasting Time Horizons	Explain time horizons and their influence on the forecasting methods and processes
4.5 Types of Forecasts	Analyse the different types of forecasting methods and their relevance to specific business process
4.6 The Strategic Importance of Forecasting	Analyse the importance of forecasting in achieving business objectives and strategy
4.7 Approaches to Forecasting	Outline the different approaches to forecasting and their applicability to specific business objectives
4.8 Summary	Summarises the topic areas covered for the unit

Prescribed and Recommended Reading(s)/Textbook(s)

Prescribed Reading(s)/Textbook(s)

- Heizer, J., Render, B. and Munson, C. (2020). *Operations Management: Sustainability and Supply Chain Management*. Thirteenth Edition. United Kingdom: Pearson.

Recommended Reading(s)/Textbook(s)

- Venkataraman, R.R. and Pinto, J.K. (2020). *Operations Management: Managing Global Supply Chains*. Second Edition. Sage Publications.
- Parksoy, T. and Deveci, M. (2023). *Smart and Sustainable Operations and Supply Chain Management in Industry 4.0*. First Edition. Boca Raton: CRC Press.
- Slack, N., Jones, B. and Burgess, N. (2022). *Operations Management*. Tenth Edition. United Kingdom: Pearson.
- Kareska, PhD Katerina. 'Operational Management Strategies for Sustainable Productivity Growth' (August 5, 2023). Available at SSRN: <https://ssrn.com/abstract=4532740> or <http://dx.doi.org/10.2139/ssrn.4532740>
- 'The Impact Of Ergonomics On Quality Of Life In The Workplace'. (2023). https://www.researchgate.net/publication/377267055_The_Impact_of_Ergonomics_on_Quality_of_Life_in_the_Workplace



4.1 Introduction

In most organisations, the responsibility for preparing the demand forecasts lies with the marketing or sales department, rather than operations. Since forecasts are a major input for operations managers, it is wise to harness their input.

Forecasting plays a crucial role in the sales and supply chain processes of any organisation. As companies expand and their operations become more complex, traditional template-based tools fall short in delivering accurate forecasts. This challenge is effectively addressed by leveraging AI-based tools, which offer superior forecast accuracy.



Readings

Find link to a video explaining what forecasting and demand planning are
Available from: https://youtu.be/bFK-9Kv_cZY?si=uxdD4d8dD424QWhp

4.2 What is Forecasting?

Heizer et al (2020) describe forecasting as the art of predicting future events. This usually involves taking historical data and projecting it into the future using a mathematical model sometimes adjusted by a manager's good judgement. It must be noted that what works best in one firm under one set of conditions may be a complete disaster in another organisation.

There are also limits as to what can be expected from forecasts. We must remember that forecasts are seldom, if ever, perfect and are costly and time-consuming to prepare and monitor.

However, we cannot afford to avoid the process of forecasting by adopting a wait-and-see attitude. Effective forecasting in both the short and long term will depend on the demand for your company's products or services. Business forecasts are used to predict sales, profits, costs, prices, interest rates, and other variables. In spite of the use of computers and sophisticated mathematical models in forecasting, it is not an exact science.

Experience, judgement and technical expertise all play a role in developing useful forecasts.



Think Point 4.1

Outline the reasons why accurate forecasting is critical?

4.3 Features common to all Forecasts

There are certain features common to all forecasts and Stevenson (2018) explains them as follows:

- Forecasting techniques generally assume that the same underlying casual system that existed in the past will continue to exist in the future
- Forecasts are rarely perfect; actual results may differ from predicted values – therefore allowances should be made for inaccuracies
- Forecasts for group items are more accurate than forecasts for individual items
- Forecast accuracy decreases as the time period covered by the forecast increases

Activity 4.1



Analyse the types of forecasts that your organisation uses and classify them into:

- Short-term
- Medium-term, and
- Long-term

4.4 Forecasting Time Horizons

Forecasts can fall into three categories:

- **Short-range Forecast** – Has a time span up to one year but is generally less than three months. It is used for planning purchasing, job-scheduling, workforce levels, job assignments and production levels
- **Medium-range Forecast** – Lasts from three months to three years. It is useful in sales planning, production planning and budgeting, cash budgeting and analysing various production plans
- **Long-range Forecast** – Is generally of a duration of three years or more. It is used in planning for new products, capital expenditure, facility location or expansion and research and development

4.5 Types of Forecasts

Heizer.et.al (2020) concludes that there are three major types of forecasts in planning future operations:

- **Economic Forecasts:** Address business cycles by predicting inflation rates, money supplies and economic indicators

- **Technological Forecasts:** Concerned with rates of technological progress which can result in the birth of exciting new products, requiring new plants and equipment
- **Demand Forecasts:** Projections of demand for a company's products or services. These forecasts are also known as sales forecasts, which determine a company's production capacity and scheduling systems and serve as inputs to financial, marketing and personnel planning

4.6 The Strategic Importance of Forecasting

We must remember that forecasts are only an estimate of demand until actual demand becomes known. Forecasts of demand, therefore, impact upon several areas of operations:

- **Human Resources**

Hiring, training and laying-off the workforce depends on anticipated demand. One needs to give ample warning to the Human Resources Department to hire, train or lay off workers. Inadequate training could result in poor quality products or services.

- **Capacity**

If capacity is inadequate, our ability to meet customer orders may be compromised which could result in loss of customers, loss of market share and loss of goodwill. If excess capacity is built on the other hand, we could end up in a situation whereby huge amounts of capital are tied up with little or no return.

- **Supply-Chain Management**

Good supplier relations and price discounts for materials and parts depend on accurate forecasting. The forecast must strive to get the right amount of material at the right time in the right quantity at the right price.



Think Point 4.2

Discuss how strategic forecasting helps in ensuring that decision making within an organisation becomes realisable.

Activity 4.2

Explain the strategic importance and impact of forecasts in the following areas in your organisation:

- Human Resources
- Capacity
- Supply Chain Management

Steps in The Forecasting Process

Forecasting follows the seven basic steps:

- Determine the use for the forecast - What are our objectives? This will provide an indication to the level of detail required in the forecast, the amount of resources we are willing to commit (manpower, computer time, Rands) and the level of accuracy necessary
- Select the items to be forecasted - Are we going to forecast for individual or group items?
- Determine the time horizon of the forecast - Is it short, medium or long term? Keep in mind that accuracy decreases as the time horizon increases
- Select the forecasting technique
- Gather data to make the forecast - Identify any assumptions that are made in conjunction with preparing and using the forecast
- Make the forecast
- Monitor the forecast to see if is performing in a satisfactory manner. If not, re-examine the method, assumptions, validity of data, and other significant issues, and perhaps prepare a revised forecast

4.7 Approaches to Forecasting

There are two general approaches to forecasting: qualitative and quantitative. Qualitative methods consist mainly of subjective inputs and incorporate such factors as the decision maker's intuition, emotions, and personal experiences amongst other things. Quantitative forecasts use a variety of mathematical models that are determined by historical data. In practice either or both approaches are used to determine a forecast.

**Think Point 4.3**

Explain the methods used to forecast together with their impact on sustainable business decisions.

Overview of Qualitative Methods

- **Forecasts based on Judgements or Opinions**

Under this method, the opinions of a group of high-level experts or managers, often in combination with statistical models are grouped to arrive at an estimate of demand. This approach is often used as part of long-range planning and new product development.

- **Sales Force Composite**

The sales staff is often a good source of information because of their direct contact with customers. Thus, they may be aware of customer's future plans. In this approach each salesperson estimates what projected sales will be in his or her area or region.

- **Customer Surveys**

This method solicits inputs from customers or potential customers regarding future purchasing plans. It can assist in not only preparing a forecast but also in improving product/service design and planning for new products. You should ensure you exercise a great deal of care in constructing a survey, administer it and correctly interpret the results in order to obtain valid information.

- **Outside Opinion**

Occasionally, outside opinions are needed to make a forecast. These may include advice on political or economic conditions within the region or in a global environment.

- **Delphi Method**

There are three different types of participants in the Delphi Method: decision makers, staff personnel and respondents. Decision-makers usually consist of a group of five to ten experts who will make the actual forecast. Staff personnel assist decision-makers by preparing, distributing, collecting and summarising a series of questionnaires and survey results. The respondents are a group of people,

located in different places, whose judgement is highly respected. This group provides input to decision-makers before the forecast is made.

Overview of Quantitative Methods

Five quantitative forecasting methods, all of which use historical data will be described. They fall into two categories:

Category A - Time-series Models: These models predict on the assumption that the future is a continuation of the past. In other words, they look at what has happened over a period of time and use a series of past data to make a forecast.

1. Naive Approach
2. Moving Averages
3. Exponential Smoothing

Category B - Casual Model: incorporate the variable or factors that might influence the quantity being forecast. If the cause-effect relationship between variables can be modelled, then predictions of the factors, which influence whatever we are trying to forecast, will enable a forecast to be made.

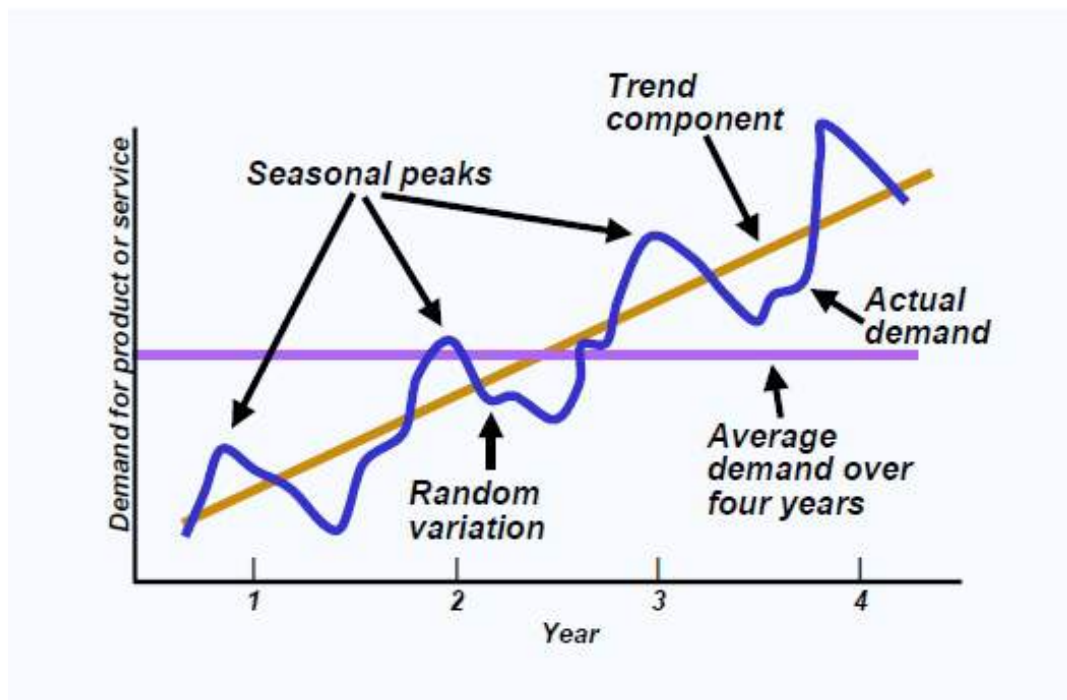
1. Trend Projection
2. Linear Regression

Forecasts based on Time Series Data

A time series is based on a sequence of evenly spaced data points (daily, weekly, monthly, quarterly, and annually, etc.). Forecasts using time series data means that future values are predicted only from past values and other variables, no matter how valuable they are, may be ignored. Analysis of time series data requires you to identify the underlying behaviour of the series. This can often be done by merely plotting the data and visually examining the plot or graph.

These behaviours can be described as follows:

- **'Trend'** refers to a gradual, long-term movement in the data over time. Changes in income, population, age distribution or cultural views may account for movement in trend
- **'Seasonality'** is a data pattern that repeats itself after a period of days, weeks, months or years. Restaurants, supermarkets and theatres experience weekly or even daily 'seasonal' variations
- **'Cycles'** are patterns in the data that occur every several years. These are often related to a variety of economic and political factors or even agricultural conditions
- **'Random' or 'irregular variations'** are due to unusual circumstances such as severe weather conditions, strikes, or a major change in a product or service



Source: Heizer.et.al (2020)



Practical Application 4.1

Illustrate the sales and production figures for the last three years on a graph and then analyse it using the Time Series Data Method.

Naive Approach

This is a forecasting technique that assumes demand in the next period is equal to demand in the most recent period. For example, if Vodacom sold 1550 cellular phones in January, then we assume that they will sell 1550 cellular phones in February.

Moving Averages

This is a forecasting method that uses an average of the n most recent periods of data to forecast the next period. If n represents 3 months, then we simply add up the actual sales in the last three months and divide by 3.

The table below shows how moving averages are calculated.

MONTH	ACTUAL SALES	3-MONTH MOVING AVERAGE
January	10	
February	12	
March	13	$(10+12+13) / 3 = 11.67$
April	16	$(12+13+16) / 3 = 13.67$
May	19	$(13+16+19) / 3 = 16$
June	23	$(16+19+23) / 3 = 19.33$
July	26	$(19+23+26) / 3 = 22.67$
August	30	$(23+26+30) / 3 = 26.33$
September	28	$(26+30+28) / 3 = 28$
October	18	$(30+28+18) / 3 = 25.33$
November	16	$(28+18+16) / 3 = 20.67$
December	14	

Exponential Smoothing

This is a weighted moving-average forecasting technique in which data points are weighted by an exponential function. It is useful when very little past data is available. The formula can be shown as follows:

New forecast = last period's forecast + a (last period's actual demand – last period's forecast), where a is a weight, or smoothing constant, chosen by the forecaster, that has a value between 0 and 1.

This equation can be written mathematically as:

$$F_t = F_{t-1} + a(A_{t-1} - F_{t-1}) \text{ where}$$

$$F_t = \text{new forecast}$$

$$F_{t-1} = \text{previous forecast}$$

$$a = \text{smoothing constant} \quad (0 \leq a \leq 1)$$

$$A_{t-1} = \text{previous period's actual demand.}$$



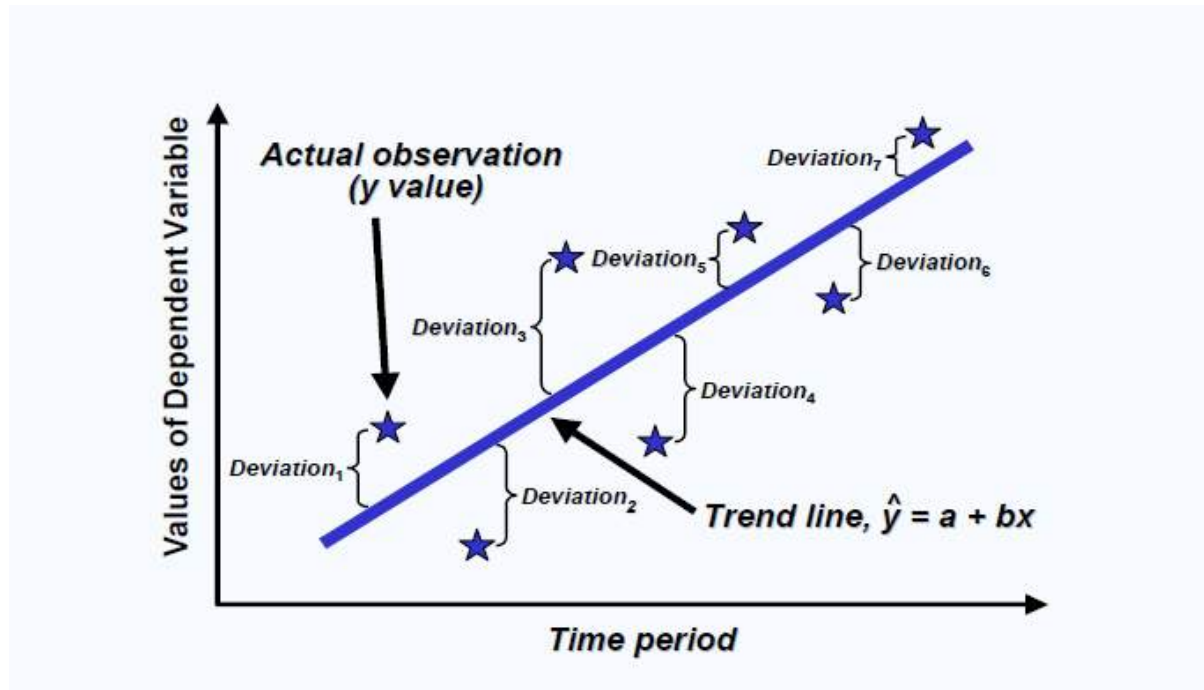
Practical Application 4.2

In January a car dealer predicted February demand for 142 new cars. Actual February sales were 153 new cars. Using a smoothing constant chosen by management of $a = .20$, we can forecast March demand using the exponential smoothing model.

Trend Projections

This is a time series forecasting method that fits a trend line to a series of historical data points and then projects the line into the future for forecasts.

The figure below depicts the Least Squares Method for finding the Best-Fitting Straight Line, Where the Asterisks are the Locations of the Seven Actual Observations or Data Points.



Source: Heizer.et.al (2020)

Regression Models

Linear Regression Analysis is a straight-line mathematical model used to describe the functional relationships between independent and dependant variables. We use this technique to verify whether our sales might be related to our advertising budget, our prices, competitor's prices or promotional strategies.

We use the equation:

$$y = a + bx$$

where y = value of the dependant variable (e.g. Sales)

a = y-axis intercept

b = slope of the regression line

x = independent variable

Knowledge Check Questions 4.1

State if the following is True or False by selecting the correct answer.



1. Forecasting methods, all of which use historical data fall into three categories.

True

False

2. 'Cycles' are patterns in the data that occur every several years.

These are often related to a variety of economic and political factors or even agricultural conditions.

True

False

3. Linear Regression Analysis uses the equation: $y = a + bx$

True

False

Case Study 4.1

Read the following article and answer the question below.

Forecasting in Operations Management at ABC Manufacturing

ABC Manufacturing, a medium-sized company producing electronic components, has been experiencing fluctuations in demand, leading to inefficiencies in inventory management and production scheduling. To address these challenges, the company decided to implement a forecasting system aimed at improving its operations management. This case study explores how ABC Manufacturing leveraged forecasting techniques to enhance its operational efficiency, reduce costs, and improve customer satisfaction.

ABC Manufacturing specialises in producing custom electronic components for a diverse range of industries, including automotive, consumer electronics, and telecommunications. The company has historically relied on reactive measures to manage inventory and production schedules, often leading to either stock-outs or excess inventory. Recognising the need for a more proactive approach, ABC Manufacturing decided to adopt a forecasting

system to predict future demand and align its operations accordingly.

- **Demand Variability:** The company faced significant variability in customer demand, making it difficult to maintain optimal inventory levels
- **Production Scheduling:** Without accurate forecasts, production schedules were often misaligned with actual demand, resulting in either idle production lines or unmet customer orders
- **Inventory Management:** The company struggled with either excess inventory, leading to increased holding costs, or insufficient inventory, resulting in stockouts and lost sales

To address these challenges, ABC Manufacturing implemented a comprehensive forecasting system, following these steps:

1. The company began by gathering historical sales data, market trends, and customer order patterns. Advanced data analytics tools were used to analyse this data, identifying key trends and seasonal patterns. This initial analysis provided a foundation for developing more accurate forecasts.
2. ABC Manufacturing evaluated various forecasting models, including time series analysis, regression models, and machine learning algorithms. After thorough testing, the company selected a combination of time series and machine learning models that best suited its data characteristics and demand patterns.
3. The selected forecasting models were integrated with the company's existing Enterprise Resource Planning (ERP) system. This integration allowed for seamless data flow and real-time updates, ensuring that forecasts were always based on the latest data.
4. Employees were trained on how to use the new forecasting tools and interpret the forecasts. The company implemented the system gradually, starting with pilot projects in select departments before rolling it out across the entire organisation.
5. The implementation of the forecasting system led to significant improvements in ABC Manufacturing's operations management:
6. The forecasting models provided more accurate predictions of customer demand, allowing the company to better align its production schedules with market needs. This reduced the occurrence of stockouts and excess inventory.



7. With accurate forecasts, ABC Manufacturing optimised its production schedules, leading to more efficient use of resources and a reduction in idle time on production lines. This resulted in lower operational costs and increased productivity.
8. Improved demand forecasting enabled the company to maintain optimal inventory levels, reducing holding costs and minimising the risk of obsolescence. This also led to better cash flow management.
9. By aligning production with actual demand, ABC Manufacturing was able to fulfil customer orders more reliably and quickly. This improved customer satisfaction and strengthened the company's reputation in the market.
10. The case of ABC Manufacturing demonstrates the significant benefits of implementing a robust forecasting system in operations management. By accurately predicting demand and aligning production schedules, the company was able to enhance operational efficiency, reduce costs, and improve customer satisfaction. This case highlights the importance of data-driven decision-making in achieving operational excellence and sustaining competitive advantage in a dynamic market environment.

Questions

1. Outline how ABC Manufacturing's adoption of forecasting models improve their inventory management, and what specific outcomes were achieved?
2. In what ways did the integration of forecasting tools with ABC Manufacturing's ERP system contribute to the company's operational efficiency.



Activity 4.4

Exponential Smoothing

A company uses exponential smoothing to forecast its monthly sales. The smoothing constant (α) is 0.3. The actual sales for June were 500 units, and the forecasted sales for June were 480 units. Calculate the forecast for July using exponential smoothing.

4.8 Summary

Forecasting is essential in operations management because it helps organisations anticipate future demand, plan resources and optimise production processes. While forecasting helps reduce uncertainty, it also faces challenges like data inaccuracies, changing market conditions, and unforeseen events. Accurate forecasting is crucial for effective operations and productivity, as it allows businesses to plan, allocate resources, and respond proactively to market shifts, ensuring they remain competitive and efficient. In summary, forecasting enables organisations to align their operations with anticipated demand, ensuring they remain agile, cost-effective, and capable of meeting customer expectations.

Revision Questions

1. List the qualitative methods of forecasting.
2. Auto Sales at Carmen's Chevrolet are shown below. Develop a 3-week moving average.

Week	Auto Sales
1	8
2	10
3	9
4	11
5	10
6	13
7	-

3. Carmen decides to forecast auto sales by weighting the three weeks as follows:

Weights Applied	Period
3	Last week
2	Two weeks ago
1	Three weeks ago
6	Total



4. A firm uses simple exponential smoothing with $\alpha = 0.1$ to forecast demand. The forecast for the week of January 1 was 500 units whereas the actual demand turned out to be 450 units. Calculate the demand forecast for the week of January 8.

5. Exponential smoothing is used to forecast auto mobile battery sales. Two values of $\alpha = 0.8$ and $\alpha = 0.5$.

Evaluate the accuracy of each smoothing constant. Which is preferable? (Assume the forecast for January was 22 batteries.) Actual sales are given below:

Month	Actual Battery Sales	Forecast
January	20	22
February	21	
March	15	
April	14	
May	13	
June	16	

6. To Minimise computations, transform the value of x (time) to simpler numbers. In this case, designate year 1996 as year 1, 1997 as year 2, etc.

Activity 4.3

Use the sales data given below to determine:

- the least squares trend line, and
- the predicted value for 2003 sales.



Year	Sales (Units)
1996	100
1997	110
1998	122
1999	130
2000	139
2001	152
2002	164

To Minimise computations, transform the value of x (time) to simpler numbers. In this case, designate year 1996 as year 1, 1997 as year 2, etc.

Answer

Activity 4.1

Analyse the types of forecasts that your organisation uses and classify them into:

- Short-term
- Medium-term, and
- Long-term

Answer: To be discussed during the webinar(s) with lecturer

Activity 4.2

Explain the strategic importance and impact of forecasts in the following areas in your organisation:

- Human Resources
- Capacity
- Supply Chain Management

Answer: To be discussed during the webinar(s) with lecturer

Activity 4.3

Use the sales data given below to determine:

- a. the least squares trend line, and
- b. the predicted value for 2003 sales

Year	Sales (Units)
1996	100
1997	110
1998	122
1999	130
2000	139
2001	152

2002	164
------	-----

To Minimise computations, transform the value of x (time) to simpler numbers. In this case, designate year 1996 as year 1, 1997 as year 2, etc

Answer:

Year	Time Period (X)	Sales (Units) (Y)	x^2	XY
1996	1	100	1	100
1997	2	110	4	220
1998	3	122	9	366
1999	4	130	16	520
2000	5	139	25	695
2001	6	152	36	912
2002	7	164	49	1148
	$\Sigma X = 28$	$\Sigma Y = 917$	$\Sigma x^2 = 140$	$\Sigma XY = 3961$

$$\bar{x} = \frac{\sum x}{n} = \frac{28}{7} = 4$$

$$\bar{y} = \frac{\sum y}{n} = \frac{917}{7} = 131$$

$$b = \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n\bar{x}^2} = \frac{3961 - (7)(4)(131)}{140 - (7)(4^2)} = \frac{293}{28} = 10.46$$

$$a = \bar{y} - b\bar{x} = 131 - (10.46 \times 4) = 89.16$$

Therefore, the least squares trend equation is:

$$\hat{y} = a + bx = 89.16 + 10.46x$$

To project demand in 2003, we denote the year 2003 as and:

$$\text{Sales in 2003} = 89.16 + 10.46 \times 8 = 172.84$$

Activity 4.4

Exponential Smoothing

A company uses exponential smoothing to forecast its monthly sales. The smoothing constant (α) is 0.3. The actual sales for June were 500 units, and the forecasted sales for June were 480 units. Calculate the forecast for July using exponential smoothing.

Answer: The formula for exponential smoothing is: $F_{t+1} = \alpha \times A_t + (1 - \alpha) \times F_t$ Where:

- F_{t+1} is the forecast for the next period (July).
- α is the smoothing constant.
- A_t is the actual sales for the current period (June).
- F_t is the forecast for the current period (June).

Plugging in the values: $F_{\text{July}} = 0.3 \times 500 + (1 - 0.3) \times 480$
 $F_{\text{July}} = 0.3 \times 500 + (1 - 0.3) \times 480 = 150 + 336 = 150 + 336 = 486 \text{ units}$

Case Study 4.1

1. Outline how ABC Manufacturing's adoption of forecasting models improve their inventory management, and what specific outcomes were achieved?

Answer: ABC Manufacturing's adoption of advanced forecasting models significantly improved their inventory management by providing more accurate predictions of customer demand. This led to several specific outcomes:

- **Optimal Inventory Levels:** By accurately predicting demand, ABC Manufacturing was able to maintain optimal inventory levels, avoiding both stockouts and excess inventory. This balance reduced the costs associated with holding excess stock and minimised the risk of obsolescence. For instance, the company experienced a 10% reduction in inventory costs, as they could better align inventory levels with actual demand patterns
- **Improved Cash Flow:** With a reduction in inventory holding costs, ABC Manufacturing experienced improved cash flow. The company was able to free up capital that would have otherwise been tied up in unsold inventory, allowing for better financial flexibility and investment in other areas of the business
- **Enhanced Customer Satisfaction:** By preventing stockouts, the company was able to meet customer orders more reliably, leading to higher customer satisfaction and a stronger reputation in the market. This reliability is crucial for maintaining customer loyalty and ensuring repeat business. Overall, the implementation of forecasting models enabled ABC Manufacturing to streamline their inventory management, leading to significant cost savings and operational

efficiency

2. In what ways did the integration of forecasting tools with ABC Manufacturing's ERP system contribute to the company's operational efficiency.

Answer: Integrating forecasting tools with ABC Manufacturing's ERP system contributed to the company's operational efficiency in several keyways:

- **Real-Time Data Access:** The integration allowed for real-time access to forecasting data, ensuring that the latest information was always available for decision-making. This real-time access enabled the company to respond swiftly to changes in demand, adjust production schedules, and manage inventory levels more effectively. The ERP system's seamless data flow helped to eliminate delays and errors associated with manual data entry and updates
- **Automated Processes:** By linking forecasting tools directly to the ERP system, many operational processes, such as order management, inventory replenishment, and production planning, could be automated. This automation reduced the need for manual intervention, thereby decreasing the likelihood of human errors and freeing up staff to focus on more strategic tasks
- **Enhanced Coordination and Collaboration:** The integration facilitated better coordination and collaboration across different departments, such as sales, production, and logistics. For instance, production schedules could be dynamically adjusted based on updated sales forecasts, ensuring that all departments worked in alignment with current demand forecasts. This improved coordination led to more efficient resource utilisation and reduced operational bottlenecks
- **Improved Reporting and Analytics:** The ERP system's robust reporting and analytics capabilities were enhanced by the integration of forecasting tools. This provided deeper insights into trends, performance metrics, and areas for improvement. Operations managers could use these insights to make informed decisions, optimise workflows, and enhance overall operational efficiency. Through these improvements, ABC Manufacturing achieved a more streamlined and responsive operation, leading to reduced costs, increased productivity, and better service delivery

Knowledge Check Questions 4.1

1. Forecasting methods, all of which use historical data fall into three categories.

Answer: False

2. 'Cycles' are patterns in the data that occur every several years. These are often related to a variety of economic and political factors or even agricultural conditions.

Answer: True

3. Linear Regression Analysis uses the equation: $y = a + bx$

Answer: True

Practical Application 4.1

Illustrate the sales and production figures for the last three years on a graph and then analyse it using the Time Series Data Method.

Answer: To be discussed during the webinar(s) with lecturer

Practical Application 4.2

In January a car dealer predicted February demand for 142 new cars. Actual February sales were 153 new cars. Using a smoothing constant chosen by management of $\alpha = .20$, we can forecast March demand using the exponential smoothing model.

Answer:

New forecast for March demand

$$= 142 + .2(153 - 142)$$

$$= 144.2$$

Therefore, our March demand for new cars is rounded off to 144.

Think Point 4.1

Outline the reasons why accurate forecasting is critical?

Answer: Strategic forecasting: Helps align operational plans with long-term goals.

Think Point 4.2

Discuss how strategic forecasting helps in ensuring that decision making within an organisation becomes realisable.

Answer: Strategic forecasting plays a crucial role in making decision-making within an organisation more actionable and realistic. Below are some of the ways in which it plays a role in:

- **Anticipating Future Trends:** Strategic forecasting involves predicting future trends, market conditions, and potential challenges. By understanding these future scenarios, decision-makers can develop strategies that are aligned with anticipated changes, ensuring that decisions are based on informed projections rather than assumptions
- **Setting Realistic Goals:** Forecasting helps organisations set achievable goals by providing a clearer picture of future conditions. This allows managers to set targets that are both ambitious and attainable, thus making strategic plans more practical and grounded in reality
- **Allocating Resources Efficiently:** By forecasting future needs and market conditions, organisations can allocate resources more effectively. This includes budgeting, staffing, and investing in technology. Forecasting helps ensure that resources are directed towards areas with the highest potential for return, thereby improving operational efficiency

- **Mitigating Risks:** Strategic forecasting helps identify potential risks and uncertainties. By anticipating these risks, organisations can develop contingency plans and make informed decisions that minimise the impact of adverse events, thus making the decision-making process more robust and resilient
- **Enhancing Strategic Alignment:** Forecasting ensures that strategic decisions are aligned with long-term objectives. By integrating forecasts into the strategic planning process, organisations can ensure that their decisions support their overall vision and goals, making the strategy more coherent and actionable

Think Point 4.3

Explain the methods used to forecast together with their impact on sustainable business decisions.

Answer:

Quantitative Methods:

- **Time Series Analysis:** Uses historical data to predict future trends. Impact: Helps in making data-driven, sustainable decisions by identifying long-term patterns
- **Regression Analysis:** Examines relationships between variables. Impact: Assists in understanding factors affecting sustainability and making informed adjustments

Qualitative Methods:

- **Expert Judgment:** Relies on insights from industry experts. Impact: Provides context-specific predictions, enhancing decision-making in uncertain environments
- **Delphi Method:** Gathers opinions from a panel of experts through multiple rounds. Impact: Builds consensus on sustainable practices by leveraging diverse expert views.

Scenario Planning:

- **Scenario Analysis:** Explores different future scenarios and their impacts. Impact: Prepares organisations for various outcomes, aiding in resilient and adaptable business strategies.

Simulation Models:

- **Monte Carlo Simulation:** Uses random sampling to model complex systems. Impact: Assesses risks and uncertainties, supporting more robust and sustainable decision-making.

**Unit
5:**

**Design of Goods and
Services**

Unit Learning Outcomes

CONTENT LIST	LEARNING OUTCOMES FOR THIS UNIT
5.1 Introduction	Introduces the topic areas covered for the unit
5.2 Objectives of Product and Service Design	Analyse business objectives that determine the process of design
5.3 Goods and Services Selection	Analyse the appropriate selection of goods and services based on customer satisfaction
5.4 The Design Process	Analyse the inter-relationship between different business functions
5.5 Product Life Cycles	Analyse the product life cycle to improve the strategic decision-making process
5.6 Key Terms	Outline key terms to improve your vocabulary in operations management
5.7 Summary	Summarises the topic areas covered for the unit

Prescribed and Recommended Reading(s)/Textbook(s)

Prescribed Reading(s)/Textbook(s)

- Heizer, J., Render, B. and Munson, C. (2020). *Operations Management: Sustainability and Supply Chain Management*. Thirteenth Edition. United Kingdom: Pearson.

Recommended Reading(s)/Textbook(s)

- Venkataraman, R.R. and Pinto, J.K. (2020). *Operations Management: Managing Global Supply Chains*. Second Edition. Sage Publications.
- Parksoy, T. and Deveci, M. (2023). *Smart and Sustainable Operations and Supply Chain Management in Industry 4.0*. First Edition. Boca Raton: CRC Press.
- Slack, N., Jones, B. and Burgess, N. (2022). *Operations Management*. Tenth Edition. United Kingdom: Pearson.
- Kareska, PhD Katerina. 'Operational Management Strategies for Sustainable Productivity Growth' (August 5, 2023). Available at SSRN: <https://ssrn.com/abstract=4532740> or <http://dx.doi.org/10.2139/ssrn.4532740>
- 'The Impact Of Ergonomics On Quality Of Life In The Workplace'. (2023). https://www.researchgate.net/publication/377267055_The_Impact_of_Ergonomics_on_Quality_of_Life_in_the_Workplace



5.1 Introduction

Global firms know that the basis for an organisation's existence is the goods or services it provides to society. Great products are the keys to great success. To maximise the potential for success, top companies tend to focus on a few key products and spend enormous amounts of time and resources in improving them.

Because most products have limited or even predictable life cycles, companies must be constantly on the lookout for new products to design, develop and introduce to the market. Good operations managers insist on strong communications between customers, products, processes and suppliers.

An effective product strategy links product decision with investment, market share and product life cycle. The objective of product design is to develop and implement a product strategy that meets the demands of the marketplace with a competitive advantage. Product strategy often focuses on developing a competitive advantage through product differentiation, low cost, rapid response, flexibility or a combination of these.



Think Point 5.1

Explain how a company can effectively create awareness and generate interest for a new product in a competitive market.

5.2 Objectives of Product and Service Design

Stevenson (2018) asserts that the objectives are:

- To bring new or revised products or services to the market as quickly as possible
- To design products/services that have customer appeal
- To increase the level of customer satisfaction
- To improve quality
- To reduce costs

5.3 Goods and Services Selection

Generation of new product Opportunities

Product selection, definition and design takes place on a continuing basis because so many new product opportunities exist. Heizer.et.al (2020) concludes that the following factors can influence market opportunities:

- Understanding the customer is the premier issue in new product development. Many commercially important products are initially thought of and even prototyped by users rather than

producers

- Economic change, which brings increasing levels of affluence in the long run, but economic cycles and price changes in the short run
- Sociological and demographic change which may occur as a result of decreasing family size, for example AIDS
- Technological change, which makes possible everything from home computers to cellular phones to artificial hearts
- Political/legal change, which brings about new trade agreements, tariffs and government contract requirements
- Other changes, which may be brought about through market practice, professional standards, suppliers and distributors



Think Point 5.2

Illustrate ideas to generate new products.

5.4. The Design Process

The design process begins with a motivation for a design. Ultimately, the customer is the driving force for product and service design. Failure to satisfy customers can result in customer complaints, returns, warranty/guarantee claims, amongst other things. Ideas for new or improved designs come from a variety of sources including:

- **Customer** – The marketing department can tap this source of ideas through focus groups, surveys and an analysis of buying patterns
- **Research and development** – Which is an organised effort to increase scientific knowledge or product innovation
- **Competitors** – By studying our competitor's product or service, we can learn a great deal in improving our own products
- **Reverse engineering** – Dismantling and inspecting a competitor's product

Ideas for new or improved design cannot work in isolation. We must ensure that we have the capability of production which include, equipment, skills, types of material, technology and special abilities.

Manufacturability is a key concern for manufactured goods. Ease of assembly is important for costs,

productivity and quality. In general, design, marketing and production must work closely together, keeping each other informed and taking into account the needs of the customer.



Activity 5.1

Explain how the production process has enabled competitive advantage.



Activity 5.2

Illustrate your organisation's products in the form of a catalogue and outline into which stage of the product life cycle it falls.

5.5 Product Life Cycles

Many new products exhibit a product life cycle. Heizer et al (2020) contend that there are four stages in product/service life cycles, and these are:

- **Introduction**

When an item is first introduced it may be treated as a curiosity. Demand is generally high because potential buyers may not be familiar with the product/service.

- **Growth**

As products or services survive the rigours of their introduction to the market, they will begin to be more widely accepted. Increasing numbers of customers accept the value of the product or service and volume starts to grow.

- **Maturity**

After a period of rapid growth, customers may become bored with products or services. Demand starts to level off as many customers have already been supplied.

- **Decline**

Sales start to decline, and the product life cycle is at the end. No new capital investments are made in the product.

Video Activity 5.1

Watch the video using the link below and answer the question that follows.



Product Life Cycle Explained | Apple iPhone & Coca Cola Examples

Available at: <https://www.youtube.com/watch?v=ob5KWs3l3aY>

Questions:

1. Analyse how the product life cycle applies to the Apple iPhone.
2. Outline some limitations of the product life cycle model as discussed in the video.

The strategies and issues within the product life cycle is depicted in the figure below:

	Introduction	Growth	Maturity	Decline
OM Strategy/Issues	Product design and development critical Frequent product and process design changes Short production runs High production costs Limited models Attention to quality	Forecasting critical Product and process reliability Competitive product improvements and options Increase capacity Shift toward product focus Enhance distribution	Standardization Less rapid product changes – more minor changes Optimum capacity Increasing stability of process Long production runs Product improvement and cost cutting	Little product differentiation Cost minimization Overcapacity in the industry Prune line to eliminate items not returning good margin Reduce capacity

Source: Heizer.et.al (2020)



Think Point 5.3

Discuss why it is critical to track product performance? How do organisations ensure that they produce relevant products into the market?

Product Development Stages

Product concepts are developed from a variety of sources, both external and internal to the firm. Concepts that survive the product idea stage progress through various stages, with nearly constant review, feedback and evaluation in a highly participative environment to minimise failure.



Activity 5.3

1. Discuss how effective operations management supports the growth phase of the product life cycle.
2. Explain the role of market research during the introduction phase of the product life cycle.
3. Provide reasons for why it is important for companies to focus on cost control during the maturity phase of the product life cycle.

5.6. Key Terms

Product Development teams are teams responsible for moving from market requirements for a product to achieving product success

- Concurrent Engineering uses participating teams in the design and engineering activities
- Computer-Aided Design (CAD) is the use of a computer to interactively develop, design and document products
- Modular Design are parts or components of a product which are subdivided into modules that are easily interchanged or replaced
- Robust Design is a design that can be produced to requirements even with unfavourable conditions in the production process
- Value Analysis is a review of successful products that takes place during the production process
- Standardisation is the extent to which there is absence of variety in the product, service or process

Quality Function Deployment (QFD) is a structured approach for integrating the customers' needs into the product development process. It is used to connect customer attributes to engineering characteristics. This is typically done by a technique called the House of Quality.



Practical Application 5.1

Use the link below and learn how 3M Science works.

3M. Science. Applied to Life. (2023) 3M Science. [online]. Available from: https://www.3m.co.za/3M/en_ZA/company-meal/

Explain the new product that 3M has introduced.

Case Study 5.1

Read the following article and answer the question below.

Product Development at Apple Inc.

Apple Inc. is renowned for its innovative products and successful product development strategies. This case study explores Apple's approach to product development, focusing on the launch of the iPhone, a product that revolutionised the mobile phone industry and established Apple as a leader in technology.

Market Research: Apple identified a gap in the market for a user-friendly, multifunctional mobile device. The company conducted extensive market research to understand consumer needs and preferences.

Vision: Steve Jobs envisioned a device that combined a phone, an iPod, and an internet communicator. This vision guided the product development process, aiming to create a seamless user experience.

Design Thinking: Apple employed design thinking to create a product that was not only functional but also aesthetically pleasing. The company focused on simplicity, elegance, and intuitive user interfaces.

Prototyping and Iteration: Apple's design team developed numerous prototypes, iterating on feedback and testing each version rigorously. This iterative process ensured that the final product met high standards of quality and usability.

Hardware and Software Integration: A key aspect of Apple's product development was the seamless integration of hardware and software. The development teams for iOS (the operating system) and the iPhone hardware worked closely to ensure compatibility and optimal performance.

Advanced Engineering: Apple invested heavily in advanced engineering to ensure the iPhone's performance. This included developing custom processors, optimising battery life, and ensuring robust build quality.

Quality Assurance: Apple's QA team conducted extensive testing to identify and resolve any issues. This included stress tests, drop tests, and usability tests to ensure the product's durability and functionality.

User Testing: Apple conducted user testing sessions to gather feedback from potential customers. This feedback was used to make final adjustments and improvements to the product.



Marketing Strategy: Apple's marketing strategy for the iPhone was meticulously planned. The company created a sense of anticipation and excitement through teaser campaigns and high-profile launch events.

Distribution: Apple ensured that the iPhone was widely available at launch through partnerships with mobile carriers and retail stores. The company also leveraged its own Apple Stores for distribution.

Customer Support: Apple emphasised customer support, providing extensive resources and training for its support staff to assist new iPhone users. This helped in building customer loyalty and trust.

Feedback Loop: After the launch, Apple continued to gather customer feedback and monitor product performance. This feedback informed updates and improvements in subsequent models.

Continuous Improvement: Apple adopted a strategy of continuous improvement, releasing new iPhone models regularly with enhanced features and capabilities. This ensured that the product remained competitive and met evolving customer needs.

Ecosystem Development: Apple focused on developing an ecosystem around the iPhone, including the App Store, iCloud, and integration with other Apple devices. This created a comprehensive user experience and increased customer retention.

1. User-Centred Design:

Focus on Experience: Apple's emphasis on user experience set the iPhone apart from competitors. The intuitive interface, elegant design, and seamless functionality attracted a wide range of consumers.

Attention to Detail: Apple's meticulous attention to detail in design and engineering resulted in a high-quality product that met and exceeded customer expectations.

2. Innovation:

Technological Advancements: The iPhone introduced several technological advancements, including multi-touch screens, a powerful mobile operating system, and an integrated ecosystem of apps and services.

Market Disruption: By combining multiple functionalities into one device, Apple disrupted the mobile phone market and set new industry standards.

3. Leadership and Vision:

Steve Jobs' Leadership: Steve Jobs' visionary leadership and unwavering commitment to excellence were crucial in driving the iPhone's development.

His ability to anticipate market trends and focus on innovation led to the iPhone's success.

Team Collaboration: The close collaboration between Apple's design, engineering, and marketing teams ensured that all aspects of the product were aligned and optimised for success.

Apple's product development process for the iPhone exemplifies the importance of innovation, user-centred design, and strong leadership in creating successful products. By focusing on quality, usability, and continuous improvement, Apple was able to revolutionise the mobile phone industry and establish a dominant market position. This case study highlights the critical factors that contribute to successful product development and the impact of strategic vision and execution on a company's success.

Available at: https://youtu.be/8oPXtfZlsto?si=o9m_AWrswlmEol_K

Available at: <https://youtu.be/C2tjnJOy7Hw?si=K7mqmFHeQooCRgpn>

Questions:

1. Determine the key factors that contributed to the successful product development of the iPhone at Apple.
2. Discuss how Apple ensures continuous improvement of the iPhone after its initial launch.

Knowledge Check Questions 5.1**Select the correct answer:**

1. Which phase of the product life cycle is characterised by intense competition and declining sales?
 - a. Introduction
 - b. Growth
 - c. Maturity
 - d. Decline
2. In the introduction phase, which of the following is a primary focus for operations management?
 - a. Reducing production costs
 - b. Maximising production efficiency
 - c. Ensuring product quality and managing initial production
 - d. Phasing out the product
3. What is a common strategy for companies to extend the life cycle of a product during the decline phase?
 - a. Increase prices
 - b. Invest heavily in marketing
 - c. Innovate and add new features to the product
 - d. Reduce production to save costs



Knowledge Check Questions 5.2

State if the following is True or False by selecting the correct answer.

1. A product typically reaches peak profitability during the decline phase.

True

False



2. Innovation is less critical during the growth phase compared to the introduction phase.

True

False

3. In the maturity phase, companies often introduce variations or improvements to the product to maintain interest.

True

False

Knowledge Check Questions 5.3

Select the correct answer:



1. In which phase of the product life cycle is a product likely to experience the highest sales growth.
 - a. Introduction
 - b. Growth
 - c. Maturity
 - d. Decline

2. Which strategy is most appropriate during the maturity phase of the product life cycle?
 - a. Heavy marketing to create product awareness
 - b. Cost-cutting and efficiency improvements
 - c. Rapid product development and innovation
 - d. Phasing out the product

3. During which phase of the product life cycle is it most important to focus on differentiating the product from competitors?
 - a. Introduction
 - b. Growth
 - c. Maturity
 - d. Decline

5.7 Summary

The design of goods and services is crucial in operations management because it directly impacts the efficiency, quality, and cost-effectiveness of production processes. Effective design takes into account factors such as functionality, manufacturability, and serviceability, while also aligning with market demand and sustainability goals. When an organisation focuses on design it can increase production and enhance customer satisfaction. The design phase sets the foundation for operational success and long-term profitability.

Revision Questions

1. You want to compete in the super premium ice cream market. The task is to determine the wants of the super-premium market and the attributes/how to be met by their firm. Use the house of quality concept. Market research has revealed that customers feel four factors are significant in making a buying decision. A “rich” taste is most important followed by smooth texture, distinct flavour, and a sweet taste. From a production standpoint, important factors are the sugar content, the amount of butterfat, low air content, and natural flavours.

2. Illustrate a bill-of-material for a ham and cheese sandwich

3. Illustrate an assembly chart for a ham and cheese sandwich.

Based on the BOM in Revision Question 2, an assembly chart might look like

4. Michael's Engineering, Inc. manufactures components for the ever-changing notebook computer business. He is considering moving from a small custom design facility to an operation capable of much more rapid design of components. This means that Michael must consider upgrading his CAD equipment.

Option 1 is to purchase two new desktop Computer Assisted Design systems at R100, 000 each.

Option 2 is to purchase an integrated system and the related server at R500, 000.

Michael's sales manager has estimated that if the market for notebook computers continues to expand, sales over the life of either system will be R1, 000,000. He places the odds of this happening at 40%.

He thinks the likelihood of the market having already peaked to be 60% and future sales to be only R700, 000.

Outline your suggestion to Michael and explain the EMV of this decision



Answers

Activity 5.1

Explain how the production process has enabled competitive advantage.

Answer: Efficient production reduces costs and speeds up time-to-market.

Activity 5.2

Illustrate your organisation's products in the form of a catalogue and outline into which stage of the product life cycle it falls.

Answer: To be discussed during the webinar(s) with lecturer.

Activity 5.3

1. Discuss how effective operations management supports the growth phase of the product life cycle.

Answer: Effective operations management supports the growth phase by ensuring efficient production processes, maintaining high product quality, scaling operations to meet increasing demand, and optimising supply chain management to reduce costs and improve delivery times

2. Explain the role of market research during the introduction phase of the product life cycle.

Answer: Market research plays a crucial role during the introduction phase by identifying target customer segments, understanding customer needs and preferences, informing product development, and guiding marketing strategies to create awareness and drive early adoption

3. Provide reasons for why it is important for companies to focus on cost control during the maturity phase of the product life cycle.

Answer: During the maturity phase, market saturation and increased competition often lead to price pressures. Focusing on cost control helps companies maintain profitability by optimising production processes, reducing waste, and improving operational efficiency, allowing them to offer competitive prices while preserving margins.

Case Study 5.1

1. Determine the key factors that contributed to the successful product development of the iPhone at Apple.

Answer: The key factors included user-centred design, innovation, and strong leadership. Apple focused on creating an intuitive user experience, introduced technological advancements like multi-

touch screens, and benefited from Steve Jobs' visionary leadership. Close collaboration between design, engineering, and marketing teams ensured a cohesive and high-quality product.

2. Discuss how Apple ensures continuous improvement of the iPhone after its initial launch.

Answer: Apple ensured continuous improvement through regular updates and the release of new iPhone models with enhanced features. The company gathered customer feedback to inform these updates and focused on building an ecosystem around the iPhone, including the App Store and iCloud, to create a comprehensive and evolving user experience.

Knowledge Check Questions 5.1

1. Which phase of the product life cycle is characterised by intense competition and declining sales?

Answer: d. Decline

2. In the introduction phase, which of the following is a primary focus for operations management?

Answer: c. Ensuring product quality and managing initial production

3. What is a common strategy for companies to extend the life cycle of a product during the decline phase?

Answer: c. Innovate and add new features to the product

Knowledge Check Questions 5.2

1. A product typically reaches peak profitability during the decline phase.

Answer: False

2. Innovation is less critical during the growth phase compared to the introduction phase.

Answer: False

3. In the maturity phase, companies often introduce variations or improvements to the product to maintain interest.

Answer: True

Knowledge Check Questions 5.3

1. In which phase of the product life cycle is a product likely to experience the highest sales growth.

Answer: b. Growth

2. Which strategy is most appropriate during the maturity phase of the product life cycle?

Answer: b. Cost-cutting and efficiency improvements

3. During which phase of the product life cycle is it most important to focus on differentiating the product from competitors?

Answer: b. Introduction

Practical Application 5.1

Explain the new product that 3M has introduced.

Answer: To be discussed during the webinar(s) with lecturer.

Think Point 5.1

Explain how a company can effectively create awareness and generate interest for a new product in a competitive market.

Answer: By the use of digital marketing, social media, and product demos.

Think Point 5.2

Illustrate ideas to generate new products.

Answer: It can be generated through customer feedback, competitor analysis, and brainstorming sessions.

Think Point 5.3

Discuss why it is critical to track product performance? How do organisations ensure that they produce relevant products into the market?

Answer: Tracking product performance helps organisations respond to customer needs and market trends

Video Activity 5.1**1. Analyse how the product life cycle applies to the Apple iPhone.**

Answer: The Apple iPhone demonstrates the product life cycle stages by introducing new models (introduction phase), gaining widespread popularity (growth phase), reaching a peak in sales (maturity phase), and then facing competition and market saturation, which leads to slower sales (decline phase). Apple mitigates the decline by launching newer versions.

2. Outline some limitations of the product life cycle model as discussed in the video.

Answer: The product life cycle model oversimplifies market realities and does not account for unpredictable factors like technological advancements or changing consumer preferences, making it difficult to predict the exact timing of each phase for every product.

Unit 6:

Process Strategy and Capacity Planning

Unit Learning Outcomes

CONTENT LIST	LEARNING OUTCOMES FOR THIS UNIT
6.1 Introduction	Introduces the topic areas covered for the unit
6.2 Process Strategy and Capacity Planning	Explore the concepts of process strategies and capacity planning within organisations
6.3 Process Analysis and Design	Analyse the applicability of the choice of process types in different operational environments
6.4 Lean Production	Analyse lean production in the development of process strategies and capacity planning
6.5 Capacity Management	Examine the productivity of your organisation
6.6 Summary	Summarises the topic areas covered for the unit

Prescribed and Recommended Reading(s)/Textbook(s)

Prescribed Reading(s)/Textbook(s)

- Heizer, J., Render, B. and Munson, C. (2020). *Operations Management: Sustainability and Supply Chain Management*. Thirteenth Edition. United Kingdom: Pearson.

Recommended Reading(s)/Textbook(s)

- Venkataraman, R.R. and Pinto, J.K. (2020). *Operations Management: Managing Global Supply Chains*. Second Edition. Sage Publications.
- Parksoy, T. and Deveci, M. (2023). *Smart and Sustainable Operations and Supply Chain Management in Industry 4.0*. First Edition. Boca Raton: CRC Press.
- Slack, N., Jones, B. and Burgess, N. (2022). *Operations Management*. Tenth Edition. United Kingdom: Pearson.
- Kareska, PhD Katerina. 'Operational Management Strategies for Sustainable Productivity Growth' (August 5, 2023). Available at SSRN: <https://ssrn.com/abstract=4532740> or <http://dx.doi.org/10.2139/ssrn.4532740>
- 'The Impact Of Ergonomics On Quality Of Life In The Workplace'. (2023). https://www.researchgate.net/publication/377267055_The_Impact_of_Ergonomics_on_Quality_of_Life_in_the_Workplace



6.1 Introduction

Effective process strategy and capacity planning are fundamental to achieving operational efficiency and sustaining competitive advantage. Process strategy defines how a company structures and organises its production or service processes to align with strategic goals, while capacity planning ensures that resources are effectively utilised to meet demand fluctuations. Together, these elements form the foundation of operational excellence in both manufacturing and service industries.

This unit explores key aspects of process strategy and capacity planning, including process analysis and design, lean production, and capacity management. Process strategies focus on selecting the most suitable production approach—whether project-based, batch, repetitive, or continuous flow—to optimise efficiency and flexibility. Process analysis and design help organisations identify inefficiencies, eliminate waste, and streamline workflows. Lean production principles emphasise value creation through waste reduction, just-in-time (JIT) inventory management, and continuous improvement. Finally, capacity management ensures that an organisation can meet demand without incurring excessive costs or bottlenecks.

By the end of this unit, students will be able to evaluate different process strategies, analyse and improve workflows, apply lean principles, and develop effective capacity planning techniques to enhance productivity and operational efficiency.



Activity 6.1

Explain the processing technology that enhanced productivity within organisations

6.2 Process Strategy and Capacity Planning

A process strategy is an organisation's approach to transform resources into goods and services. The objective of process strategy is to find a way to produce goods and services that meet and exceed customer requirements and product specification within cost and other managerial constraints. The process selected will have a long-term effect on operating efficiency, quality and production.

Heizer et al (2020) argues that there are basically four process strategies that an organisation can use, and they are:

- Process Focus
- Repetitive Focus
- Product Focus

- Mass Customisation



Think Point 6.1

Explain the process strategy that your organisation uses.

Process Focus

According to Heizer et al (2020), almost 75% of all global production is devoted to making low-volume, high-variety products in a job shop environment. Such facilities are organised around performing processes. In a factory, these processes might be departments such as welding, grinding, assembly and painting. In an office environment, the processes include accounts payable, sales and salaries department.

In a restaurant, it may be the bar, kitchen, grill and bakery. Such facilities are process-focused in terms of equipment, layout and supervision. They provide high degrees of product flexibility. Each process is designed to perform a wide variety of activities and to handle frequent changes. Therefore, they are sometimes called intermittent processes. These facilities have high variable costs with extremely low utilisation of facilities.

Repetitive Focus

Repetitive processes use modules, which are parts or components that have been previously prepared. The repetitive process line is usually an assembly-line producing goods such as cars and household appliances amongst other things. It is more structured and therefore has less flexibility than a process-focused facility.

Fast food outlets are examples of repetitive processing. There is a certain amount of pre-processing done, e.g. Meat, cheese, sauce, onions are prepared in advance.

Product Focus

High-volume, low variety processes are referred to as product focus. The facilities are organised around products, and it is also known as continuous processing. Products including glass, paper, tin sheets, light bulbs, beer and canned foods are made via a continuous process. It is through standardisation and effective quality control that companies have established product-focused facilities. The specialised nature of the facility requires high fixed cost, but low variable cost.

Mass Customisation

Rapid low-cost production that caters to constantly changing unique customer desires. Mass Customisation brings us the variety of products traditionally provided by low-volume manufacture (a process focus) at the cost of standardised high-volume (product focused) production.



Video Activity 6.1

Watch the video using the link below for further insights into Process Strategies.

Process Strategies and Design overview in Operations Management

Available at: <https://youtu.be/izrYFgsJPV4?si=BL2z2tlp9NuC046V>

Table 6.1: Comparison of Process Choices

PROCESS FOCUS (e.g. Dorbyl Ship Builders)	REPETITIVE FOCUS (e.g. Toyota Manufacturing)	PRODUCT FOCUS (e.g. South African Breweries)
Small quantity and large variety of products are produced	Long runs usually a standardised product	Large quantity and small variety of products are produced
Equipment used is general purpose	Special equipment used in the assembly line	Special purpose equipment used
Operators are broadly skilled	Employees are properly trained	Operators are less broadly skilled
There are many job instructions because each job changes	Repetitive operations reduce changes in job instruction	Work orders are few because they are standardised
Raw material inventories are high relative to the value of the product	Just-in-Time techniques are used	Raw material inventories are low
Work-in-progress is high compared to output	Just-in-Time reduces the need for carrying high stock	Inventory of work-in-progress is low
Units move slowly through the plant	Movement is measured in hours and in days	Swift movement of units through the plant
Finished goods are made to order and not stored	Finished goods are made to frequent forecasts	Finished goods are for stock

Scheduling of orders is complex	Scheduling is based on building various models	Scheduling is relatively simple
Fixed costs are low and variable costs are high	Fixed costs are dependent on the flexibility of the plant	Fixed costs are high and variable cost low
Costing is estimated prior to doing the job	Costs are usually known prior to starting the job	Costs are dependent on utilisation of capacity

Heizer.et.al (2020)

Activity 6.2



1. Determine how a company's process strategy impacts its competitive advantage.
2. Determine the factors that should be considered when planning for capacity in a seasonal business.
3. Discuss why customer involvement is important in process strategy, particularly in service industries.

6.3 Process Analysis and Design

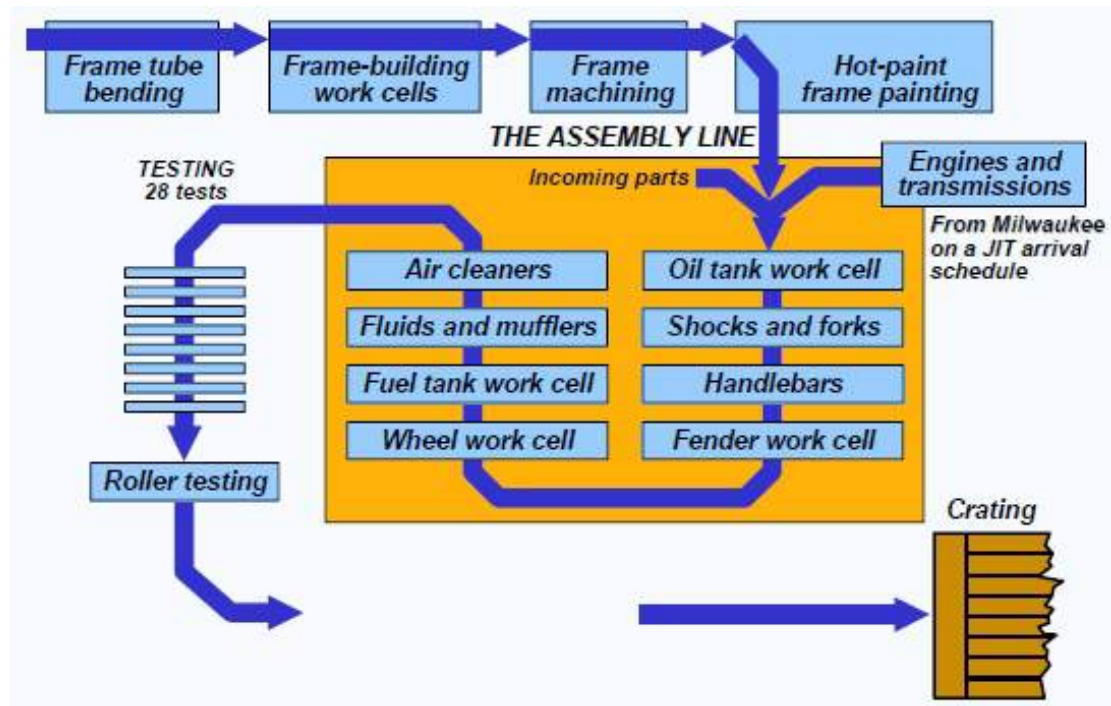
Heizer.et.al (2020) advise that when analysing and designing processes to transform resources into goods, the following questions should be asked:

- Is the process that is designed to achieve competitive advantage in terms of differentiation, response or low cost?
- Does the process eliminate steps that do not add value?
- Does the process Maximise customer value as perceived by the customer?
- Will the process win orders?

Heizer.et.al (2020) highlight the following tools that can be used to understand the complexities of process analysis and design:

Flow Diagrams

This is a schematic drawing of the movement of people, materials or product. These diagrams are useful in understanding the analysis and communication of a process. An example of a flow diagram is depicted below.

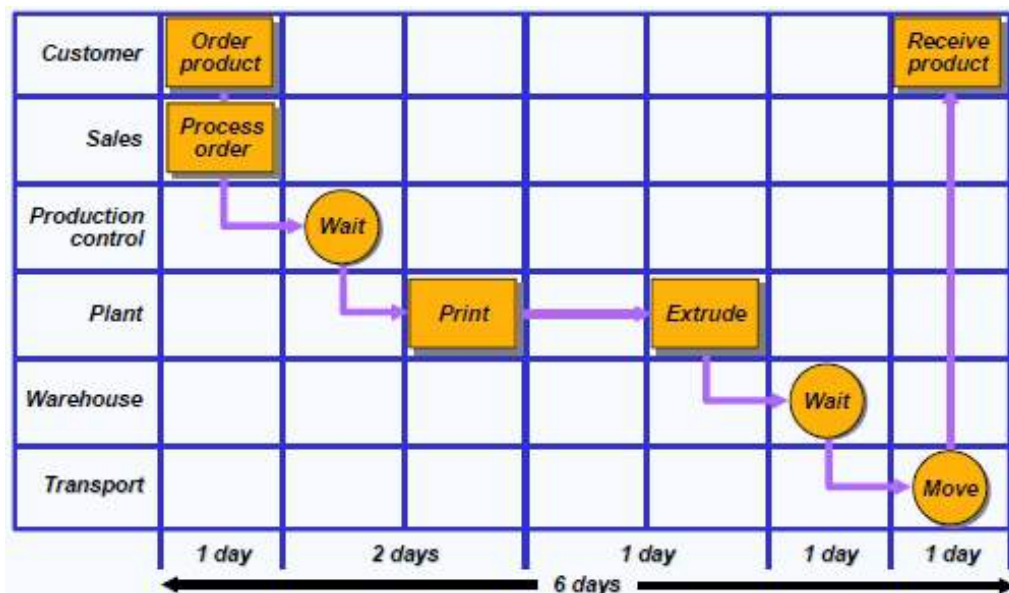


Source: Heizer.et.al (2020)

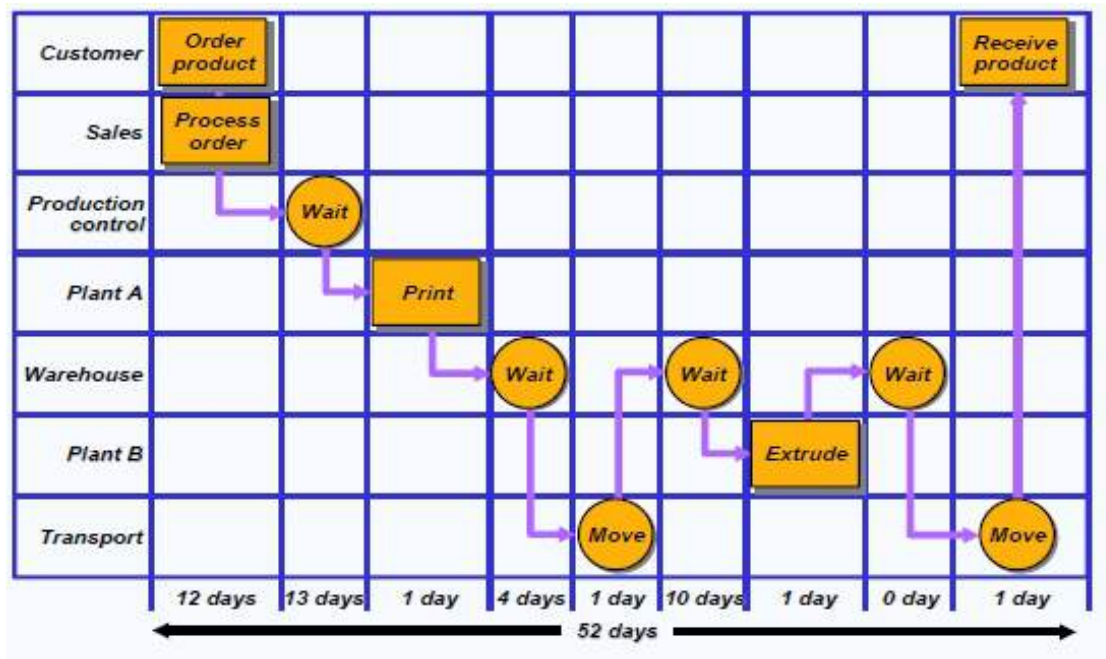
Time-Function Mapping

This is similar to a flow process chart, but with time added to the horizontal axis. With time-function mapping, notes indicate the activities and arrows indicate the flow direction, with time on the horizontal axis.

(a) "Baseline" Time-Function Map



Source: Heizer.et.al (2020)

(b) “Target” Time-Function Map

Source: Heizer.et.al (2020)

Value-Stream Mapping

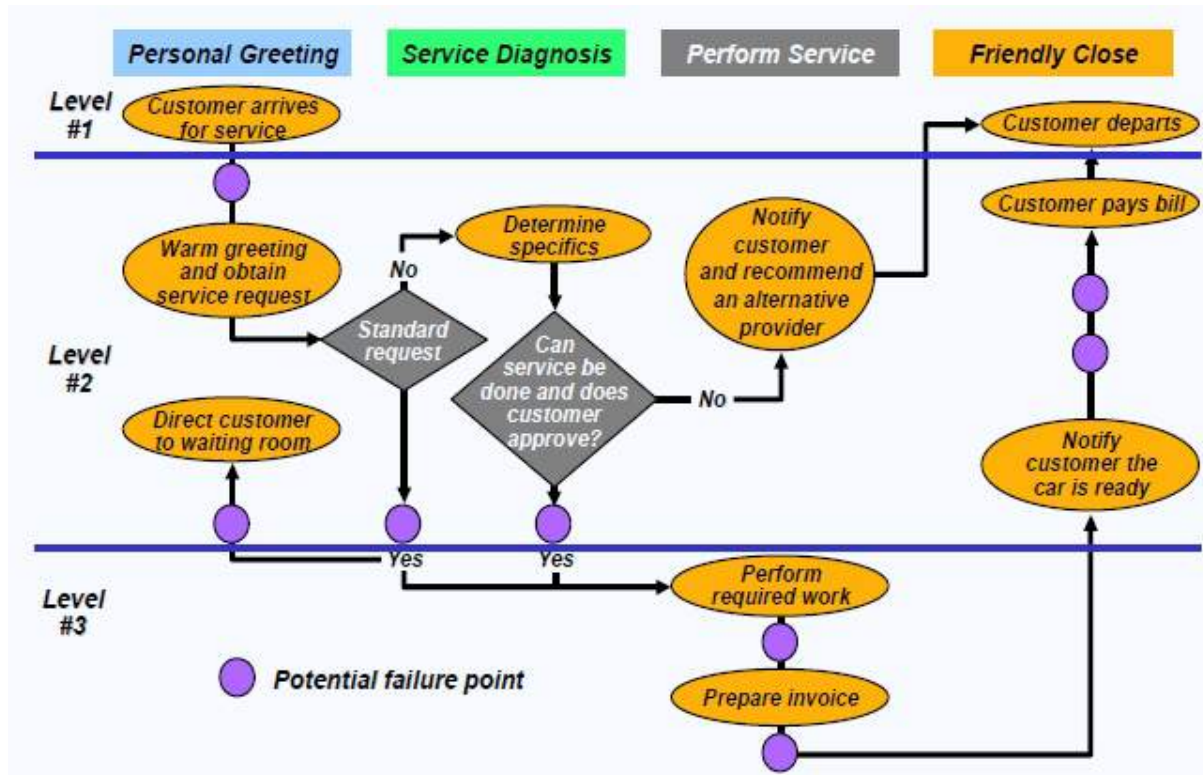
Helps managers understand adding value in the flow of material and information through the production process. Value-Stream Mapping (VSM) takes into account not only the process but also the management decisions and information systems that support the process.

Process Charts

Process charts use symbols and sometimes time and distance to provide an objective and structured way to analyse and record activities that make up a process. They allow you to focus on value-added activities.

Service Blueprinting

A process analysis technique that lends itself to a focus on the customer and the provider's interaction with the customer. An example of a Service Blueprint is shown below.



Source: Heizer.et.al (2020)



Think Point 6.2

Outline the ways in which you can improve the service in your company.

Case Study 6.1

Read the following article and answer the question below.

Capacity Planning and Process Strategy at Intel

Intel Corporation, a global leader in computer processing technology, attributes much of its success to its strategic approach to capacity planning. Capacity planning at Intel is crucial due to the volatile and highly competitive nature of the technology industry, where rapid advancements and fluctuating demand are the norms. Intel divides its capacity planning into three distinct phases: long-range planning, medium-term planning, and short-term planning. This approach allows Intel to manage the risk of overcapacity or under capacity while ensuring that production can adapt to changing demands. Intel's long-range planning phase focuses on identifying future trends and requirements, taking into account macroeconomic factors,



technology evolution, and competitor actions. In the medium-term planning phase, Intel addresses resource allocation and facility adjustments, focusing on aligning production capabilities with forecasted demand. Lastly, short-term planning involves fine-tuning operations to meet immediate production needs, including inventory adjustments, labour scheduling, and equipment maintenance. To keep pace with rapid industry changes, Intel revisits its long-range plans every quarter. This process allows the company to make quick adjustments in response to emerging market trends, ensuring that production remains aligned with demand. Intel has invested heavily in its capacity planning, considering it a core element of its competitive edge. The substantial resources allocated to this process reflect Intel's belief that well-structured capacity planning is fundamental to maintaining its market leadership.

Questions:

1. Determine why capacity planning is important for Intel, particularly in the volatile computer processing industry?
2. Analyse how Intel's approach to capacity planning differs from a standard, single-phase planning process.
3. Explain how Intel's quarterly review of its long-range planning process contributes to its overall process strategy.
4. Examine the potential risks if Intel were to skip one of the capacity planning phases.
5. Discuss how Intel's heavy investment in capacity planning resources aligns with its broader corporate goals.

6.4 Lean Production

The objective of lean production is to achieve perfection through continuous learning, creativity and teamwork. Lean producers share the following attributes:

- They focus on inventory reduction to remove waste using Just-in-Time techniques
- They build systems that help employees produce a perfect part every time
- They reduce space requirements
- They develop close relationships with their suppliers
- They educate suppliers to accept responsibility
- They strive to eliminate non-value-adding activities

- They develop the workforce by constantly improving job design, training, employee participation and work teams
- They make jobs more challenging, empowering people at the lowest levels

Opportunities to improve Service Processes

Layout design is an integral part of any service process be it retailing, banking or restaurants. In retailing, for example, layouts can provide not only product exposure but also customer education and product enhancement.

STRATEGY	TECHNIQUE	EXAMPLE
Separation	• Restrict offerings • Customising delivery • Structuring service so customers must go where the service is needed • Self-service • Modular section of service • Product-focus production	• Limit menu in restaurant • Customising vans at delivery rather than production • Banks where customers go to the manager to open a new account, to loan officers for loans and tellers for deposits. • Supermarkets and department stores. • Investment and insurance selection. • Pre-packaged foods in restaurants
Automation	• Separating services that may lend themselves to some type of automation	• Automatic teller machines (ATMs)
Scheduling	• Precise personnel scheduling	• Scheduling ticket counter personnel at 15 minute intervals at airlines

Training	• Clarifying the service options • Explaining problems	• Investment counsellor, funeral directors • After-sales maintenance people
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Heizer.et.al (2020)

Practical Application 6.1**Read the following scenario and answer the questions below.****Process Strategy and Capacity Planning****Scenario:**

A mid-sized electronics manufacturer, TechCo, is experiencing a surge in demand for its latest gadget. To maintain competitiveness, they need to align their process strategy with capacity planning to meet growing customer orders. They currently rely on a batch production process, which may soon be insufficient to handle future demands.

**Questions:**

1. Discuss the process strategy that TechCo should consider meeting increasing demand efficiently.
2. Illustrate how capacity planning helps TechCo manage this surge in demand.
3. Discuss the challenges that TechCo may face when implementing a new process strategy and capacity planning.

6.5 Capacity Management

Capacity refers to the maximum output of a system in a given period. It is normally expressed as a rate, such as the number of products that can be produced in a day, week or month. Capacity can also be measured in terms of beds (in a hospital) or active members in a church, amongst others. Most organisations operate their facilities at rates of less than their maximum output. This assists them in operating more efficiently if their resources are not stretched to the limit – this is called effective capacity. It can be shown as follows:

$$\text{Effective capacity} = \frac{\text{Expected capacity}}{\text{Capacity}}$$

Effective capacity is the capacity that a firm can expect to achieve given its product mix, methods of scheduling, maintenance and standards of quality. Another consideration is efficiency. Typically,

efficiency is expressed as a percentage of effective capacity. Efficiency is therefore a measure of actual output over effective capacity:

$$\text{Efficiency} = \frac{\text{Actual output}}{\text{Effective capacity}}$$



Think Point 6.3

Determine why organisations conduct capacity planning

Forecasting Capacity Requirements

Determining future capacity requirements can be a somewhat complicated process – one based in large part on future demand. If demand for goods or services can be forecast with a reasonable degree of precision, then determining capacity requirements can be straightforward.

The capacity of an operating unit is important for planning purposes. It enables managers to quantify production capability in terms of either inputs or outputs. The basic questions in capacity planning are:

- What kind of capacity is required?
- How much is needed?
- When is it needed?

The link between marketing and operations is crucial in determining realistic capacity requirements. Through customer contacts, demographic analyses and forecasts, marketing can supply vital information to operations for determining capacity needs in both the short and long-term.

Developing Capacity Alternatives

There are some specific considerations that are relevant to developing capacity alternatives:

- Design flexibility into the system
- Take a 'big picture' approach to capacity changes
- Prepare to deal with capacity in 'chunks'
- Attempt to smooth out capacity requirements
- Identify the optimal operating level

Activity 6.3

Answer the following question using the information provided in the table below:

Product	Annual Demand	Standard Processing Time per Unit (Hr)	Processing Time Needed (Hr)
1	300	5.00	1.500
2	400	8.00	3.200
3	700	2.00	1.400
Total	6.100		



Use the information in the table to answer this question. Note: department is working one 8-hour shift 250 days a year. How many machines would be needed to handle the required volume? (Round your answer to the whole number)

- a) 3 machines
- b) 1 machine
- c) 5 machines
- d) 2 machines
- e) 4 machines

Table 6.2: Factors that determine effective capacity

FACTOR	DETERMINANTS
Facilities	Design Location Layout Environment
Product/Service	Design Product/service mix
Process	Quantity capabilities Quality capabilities

• Human factors	Job content Job design Training and experience Motivation Compensation Learning rates Absenteeism/ Labour turnover
• Operational	Scheduling Materials management Quality assurance Maintenance policies
• External factors	Product standards Safety regulations Unions Pollution control standards

Knowledge Check Questions 6.1**Select the correct answer:**

1. Which capacity planning strategy involves maintaining a level production rate while absorbing demand fluctuations with inventory?
 - a. Chase demand strategy
 - b. Level capacity strategy
 - c. Mixed strategy
 - d. Demand smoothing strategy
2. In process strategy, a focus on low-cost production typically involves:
 - a. High customisation and flexibility
 - b. High volume and standardised products
 - c. Low automation and manual processes
 - d. Decentralised decision-making
3. Capacity cushion is defined as:
 - a. The amount of excess capacity a company maintains to handle sudden increases in demand
 - b. The gap between actual and potential production capacity
 - c. The planned downtime for maintenance
 - d. The reduction in capacity due to inefficiencies



Knowledge Check Questions 6.2

State if the following is True or False by selecting the correct answer.

1. Process strategy should always prioritise automation to improve efficiency.

True

False



2. Capacity planning in a global market must consider factors such as international regulations and cultural differences.

True

False

3. Lean manufacturing principles are incompatible with high-variety production processes.

True

False

Knowledge Check Questions 6.3

Select the correct answer:

1. Which of the following is NOT a primary consideration in process strategy?
 - a. Process structure
 - b. Customer involvement
 - c. Flexibility of workforce
 - d. Market segmentation
2. Capacity planning primarily involves determining:
 - a. The layout of the production floor
 - b. The maximum output that can be achieved
 - c. The marketing strategy for a product
 - d. The design of the product
3. Which of the following strategies is used to manage capacity in service organisations?
 - a. Inventory buffering
 - b. Back ordering
 - c. Adjusting service rates
 - d. Product differentiation



Readings

- Jacobs, F. R., & Chase, R. B. (2020). *Operations and Supply Chain Management*. McGraw-Hill Education.
- Chopra, S., & Meindl, P. (2020). *Supply Chain Management: Strategy, Planning, and Operation*. Pearson.



6.6 Summary

This unit examined how organisations develop and manage their process strategies and capacity planning efforts to improve efficiency and responsiveness. The key topics covered included process strategies for understanding different types of process strategies (project, batch, repetitive, and continuous) and how organisations select the most appropriate approach based on operational objectives. The process analysis and design was explored for applying analytical tools to map processes, identify inefficiencies, and optimise workflows to enhance productivity and reduce costs. The lean production explained the principles such as JIT, waste elimination, continuous improvement (Kaizen), and value stream mapping to maximise efficiency and minimise resource waste. Through case studies and practical applications, students gained insights into how process strategy and capacity planning influence operational success. By integrating lean production methods and strategic capacity decisions, organisations can improve agility, cost-effectiveness, and service quality.

Revision Questions

1. Jackson Custom Machine Shop has a contract for 130, 000 units of a new product. Sam Jumper, the owner, has calculated the cost for three process alternatives. Fixed costs will be for general-purpose equipment (GPE), R150, 000; flexible manufacturing (FMS), R350, 000; and dedicated automation (DA), R950, 000. Variable costs will be: GPE, R10; FMS, R8; and DA, R6. Select the one he should choose.

2. Illustrate Problem 1 graphically.

3. Using either your analytical solution found in Problem 1, or the graphical solution found in Problem 2, outline the volume ranges where each process should be used.

4. If Jackson Custom Machine is able to convince the customer to renew the contract for another one or two years, explain what implications this may have on his decision.

5. The design capacity for engine repair in our company is 80 trucks/day. The effective capacity is 40 engines/day and the actual output is 36 engines/day.

Examine the utilisation and efficiency of the operation. If the efficiency for next month is expected to be 82%, explain what the expected output is.

7. Find the break-even point in R and in units.

8. Develop the break-even chart for Self-Assessment Activity 6.

9. Jack's Grocery is manufacturing a "store brand" item that has a variable cost of \$0.75 per unit and a selling price of \$1.25 per unit. Fixed costs are \$12,000. Current volume is 50,000 units. The Grocery can substantially improve the product quality by adding a new piece of equipment at an additional fixed cost of \$5,000. Variable cost would increase to \$1.00, but their volume should increase to 70,000 units due to the higher quality product. Should the company buy the new equipment?

10. What are the break-even points (\$ and units) for the two processes considered in Self-Assessment Activity 8?

11. Good News! You are going to receive \$6,000 in each of the next 5 years for sale of used machinery. A bank is willing to lend you the present value of the money in the meantime at discount of 10% per year. Illustrate how much cash you would receive now.



Answers

Activity 6.1

Explain the processing technology that enhanced productivity within organisations

Answer: Processing technology: Automation and AI are examples that improve productivity by reducing labour costs and increasing precision.

Activity 6.2

1. Determine how a company's process strategy impacts its competitive advantage.

Answer: A company's process strategy impacts its competitive advantage by determining the efficiency and flexibility of its operations. A well-aligned process strategy can enhance product quality, reduce costs, and improve responsiveness to market changes, thereby strengthening the company's market position.

2. Determine the factors that should be considered when planning for capacity in a seasonal business.

Answer: In a seasonal business, factors such as demand variability, peak season capacity requirements, off-season flexibility, inventory management, and workforce scalability should be considered. Effective capacity planning ensures that the business can meet peak demand without excessive idle capacity during off-peak times.

3. Discuss why customer involvement is important in process strategy, particularly in service industries.

Answer: Customer involvement is crucial in-service industries because it directly influences service delivery and customer satisfaction. Involving customers in the process allows for personalised services, immediate feedback, and adjustments, which can enhance the overall service experience and build customer loyalty.

Activity 6.3

Use the information in the table to answer this question. Note: department is working one 8-hour shift 250 days a year. How many machines would be needed to handle the required volume? (Round your answer to the whole number)

Answer: 3 machines

Case Study 6.1

1. Determine why capacity planning is important for Intel, particularly in the volatile computer processing industry?

Answer: Capacity planning is crucial for Intel because it allows the company to balance supply with unpredictable demand in a rapidly changing industry. Proper capacity planning helps Intel avoid the risks associated with overcapacity, such as excess costs, and under capacity, which can lead to lost sales and reduced market share. By adjusting capacity in response to market conditions, Intel can maintain operational efficiency, minimise waste, and keep pace with industry advances.

2. Analyse how Intel's approach to capacity planning differs from a standard, single-phase planning process.

Answer: Intel uses a three-phase capacity planning process that includes long-range, medium-term, and short-term planning. This approach differs from single-phase planning as it allows Intel to address different timeframes and planning complexities at each stage. By breaking down capacity planning, Intel can strategically forecast long-term industry trends, allocate resources effectively in the medium term, and make operational adjustments in the short term to meet immediate production needs.

3. Explain how Intel's quarterly review of its long-range planning process contributes to its overall process strategy.

Answer: The quarterly review enables Intel to stay agile in a rapidly evolving industry. By revisiting long-term plans every three months, Intel can incorporate the latest market data, technology advancements, and competitor developments, allowing for timely adjustments. This frequent reassessment strengthens Intel's ability to anticipate demand shifts and fine-tune its processes, ensuring that capacity planning remains accurate and relevant, which supports continuous improvement in their process strategy.

4. Examine the potential risks if Intel were to skip one of the capacity planning phases.

Answer: Skipping any of the planning phases could lead to several risks. Omitting long-range planning may cause Intel to miss out on emerging trends and long-term opportunities. Ignoring medium-term planning could lead to misaligned resources and production inefficiencies, and skipping short-term planning might result in operational disruptions, labour shortages, or excess inventory. Each phase addresses a specific aspect of production alignment, so missing a phase would reduce Intel's adaptability and risk management capabilities.

5. Discuss how Intel's heavy investment in capacity planning resources aligns with its broader corporate goals.

Answer: Intel's investment in capacity planning aligns with its corporate goals of innovation, market leadership, and operational excellence. By dedicating resources to precise and adaptable planning, Intel ensures it can quickly respond to market demands and technological shifts, which is essential in the competitive computer processing industry. This investment supports Intel's goal to maintain a robust, scalable production capacity that aligns with growth and profitability objectives while driving continuous innovation.

Knowledge Check Questions 6.1

1. Which capacity planning strategy involves maintaining a level production rate while absorbing demand fluctuations with inventory?

Answer: b. Level capacity strategy

2. In process strategy, a focus on low-cost production typically involves:

Answer: b. High volume and standardised products

3. Capacity cushion is defined as:

Answer: a. The amount of excess capacity a company maintains to handle sudden increases in demand

Knowledge Check Questions 6.2

1. Process strategy should always prioritise automation to improve efficiency.

Answer: False

2. Capacity planning in a global market must consider factors such as international regulations and cultural differences.

Answer: True

3. Lean manufacturing principles are incompatible with high-variety production processes.

Answer: False

Knowledge Check Questions 6.3

1. Which of the following is NOT a primary consideration in process strategy?

Answer: d. Market segmentation

2. Capacity planning primarily involves determining:

Answer: b. The maximum output that can be achieved

3. Which of the following strategies is used to manage capacity in service organisations?

Answer: c. Adjusting service rates

Practical Application 6.1**1. Discuss the process strategy that TechCo should consider meeting increasing demand efficiently.**

Answer: TechCo should consider shifting from batch production to an assembly line process strategy. This strategy will streamline operations, improve throughput, and reduce costs as demand continues to rise.

2. Illustrate how capacity planning helps TechCo manage this surge in demand.

Answer: Capacity planning will allow TechCo to forecast future demand, determine if additional equipment or labour is needed, and ensure they can meet customer orders without delays or excessive costs by scaling production efficiently.

3. Discuss the challenges that TechCo may face when implementing a new process strategy and capacity planning.

Answer: TechCo could face challenges like the cost of equipment upgrades, training employees for new processes, and maintaining flexibility to respond to fluctuating demand while ensuring the new strategy supports long-term growth.

Think Point 6.1

Explain the process strategy that your organisation uses.

Answer : To be discussed during the webinar(s) with lecturer.

Think Point 6.2

Outline the ways in which you can improve the service in your company.

Answer : To be discussed during the webinar(s) with lecturer.

Think Point 6.3

Determine why organisations conduct capacity planning

Answer: It ensures optimal resource use and meets demand fluctuations

**Unit
7:**

Location Strategies

Unit Learning Outcomes

CONTENT LIST	LEARNING OUTCOMES FOR THIS UNIT
7.1 Introduction	Introduces the topic areas covered for the unit
7.2 General Procedure for Making Location Decisions	Outline the relevant procedure for assessing the suitability of a business location
7.3 Factors that affect Location Decisions	Analyse the critical success factors when choosing a location
7.4 Trends in Location Strategies	Outline both local and global trends in location strategies that improve decision making
7.5 Methods of Evaluating Location Alternatives	Analyse appropriate methods of evaluating location alternatives to improve business profits
7.6 Service Location Strategy	Examine location strategies peculiar to the service industry
7.7 Summary	Summarises the topic areas covered for the unit

Prescribed and Recommended Reading(s)/Textbook(s)

Prescribed Reading(s)/Textbook(s)

- Heizer, J., Render, B. and Munson, C. (2020). *Operations Management: Sustainability and Supply Chain Management*. Thirteenth Edition. United Kingdom: Pearson.

Recommended Reading(s)/Textbook(s)

- Venkataraman, R.R. and Pinto, J.K. (2020). *Operations Management: Managing Global Supply Chains*. Second Edition. Sage Publications.
- Parksoy, T. and Deveci, M. (2023). *Smart and Sustainable Operations and Supply Chain Management in Industry 4.0*. First Edition. Boca Raton: CRC Press.
- Slack, N., Jones, B. and Burgess, N. (2022). *Operations Management*. Tenth Edition. United Kingdom: Pearson.
- Kareska, PhD Katerina. 'Operational Management Strategies for Sustainable Productivity Growth' (August 5, 2023). Available at SSRN: <https://ssrn.com/abstract=4532740> or <http://dx.doi.org/10.2139/ssrn.4532740>
- 'The Impact Of Ergonomics On Quality Of Life In The Workplace'. (2023). https://www.researchgate.net/publication/377267055_The_Impact_of_Ergonomics_on_Quality_of_Life_in_the_Workplace



7.1 Introduction

Selecting the right location is a critical strategic decision that significantly impacts an organisation's operational efficiency, cost structure, and long-term success. The location of a business influences accessibility to markets, availability of resources, transportation costs, labour supply, and overall profitability. Whether for manufacturing, distribution, or service operations, organisations must systematically evaluate location alternatives to align with strategic and operational objectives.

Location strategies are a critical component of operations management because the physical placement of facilities, such as factories, warehouses, and retail outlets, has a significant impact on cost, efficiency, and service delivery. The right location can reduce transportation costs, improve access to resources and labour, and enhance customer satisfaction by reducing delivery times. The location decisions are influenced by factors like proximity to suppliers, demand centres, government regulations and the availability of infrastructure. An effective location strategy helps businesses optimise their supply chains, reduce operational costs, and increase their competitive advantage, making it a key element for success in operations management.

This unit explores the general procedure for making location decisions, focusing on key factors that influence site selection, emerging trends in location strategy, and quantitative and qualitative methods used to assess location alternatives. Location strategies help businesses determine optimal locations based on cost minimisation, revenue maximisation, or competitive advantage. Various economic, demographic, and regulatory factors affect location decisions, requiring organisations to conduct a thorough analysis. Additionally, trends such as globalisation, nearshoring, and sustainability considerations have reshaped location strategies in recent years. Businesses use different decision-making methods, including factor-rating systems, centre-of-gravity techniques, and cost-profit analysis, to evaluate potential locations. Lastly, service location strategy emphasises accessibility, customer proximity, and market potential to enhance service delivery. By the end of this unit, students will be equipped with the knowledge and analytical skills to assess location strategies, evaluate critical location factors, apply decision-making techniques, and develop location plans that align with business goals.

7.2 General Procedure for Making Location Decisions

Location Strategies

Organisations become involved in location decisions for many reasons. Many firms including banks, retailers, fast-food chains and supermarkets view location as part of a marketing strategy and they look for locations that will assist them to expand their respective markets. A similar situation occurs when an organisation experiences a growth in demand for its products which cannot be satisfied with the expansion of current facilities. Some firms are forced to relocate because of the depletion of their markets, e.g. fishing, mining or logging operations. For other firms, a shift in their markets would

force them to relocate.

Location options include:

- Expanding a current facility instead of moving
- Maintaining current sites whilst adding another facility elsewhere
- Closing the current facility and moving elsewhere

The way in which an organisation approaches location decisions often depends on its size and the nature of its operations. New and small organisations tend to adopt rather informal approaches to location decisions. Stevenson (2018) infers that the general procedure for making location decisions consist of the following steps:

- Decide on the criteria that will be used to evaluate location alternatives, such as increased revenues or community service
- Identify factors that are important, such as location of markets or raw materials
- Develop location alternatives
- Evaluate alternatives and make a selection

7.3 Factors that affect Location Decisions

Selecting a facility location is becoming much more complex due to the rapid globalisation of many countries. Many factors can influence location decisions, and these include:

Regional Factors

The primary regional factors involve raw materials, markets and labour considerations.

Location of Raw Materials

There are three primary reasons why firms locate near the source of raw materials: necessity, perishability and transport costs.

Location of Markets

Firms tend to locate in proximity to the markets they intend to serve as part of their competitive strategies.

Labour Factors

This relates to availability and cost of labour (wage/salary rates) in an area, or if there is a serious potential problem with the unions. Skills of potential employees may also be a factor.

Other factors one has to consider, include:

- Climate and Taxes
- Business and Personal Income Tax
- Cost of Energy and Services
- Tax Incentives
- Tariff Protection
- Effectiveness of Governments
- Housing
- Educational Systems
- Local Customs
- Language
- Exchange Rates
- Municipal by-laws
- Proximity to Suppliers
- Environmental Regulations
- Site/ Land Costs
- Availability of Transport

Practical Application 7.1

Read the following excerpt and answer the following questions:

A start-up company, EcoFresh Organics, is looking to set up its first production facility for organic health snacks. The company's primary target market is young, health-conscious consumers in urban areas across the country. They need to make a strategic decision about where to locate their facility and are considering three potential sites:

City A – Near a large metropolitan area with a high population of young professionals. The costs are relatively high, but the transportation infrastructure is excellent, with major highways and rail connections.

City B – A smaller city with moderate costs, fewer young professionals, but substantial government incentives for new businesses. The transportation infrastructure is moderate, with good road networks but limited rail access.

City C – A rural area with very low operational costs and ample space for future expansion. However, it is far from major markets and has limited access to transportation networks.

EcoFresh Organics needs to consider several factors, including costs, access to the target market, transportation infrastructure, and growth potential.

Questions:

1. Examine which factors EcoFresh Organics should prioritise when choosing the location for its production facility?
2. Analyse the benefits and drawbacks of each location (City A, B and C) for EcoFresh Organics.
3. If EcoFresh Organics chooses City C for its low costs and future expansion potential, what strategies could it employ to manage the drawbacks of limited market proximity and transportation?
4. Based on the scenario, which location (City A, B and C) would be the most strategic choice for EcoFresh Organics if the company's primary objective is to quickly reach urban consumers with minimal logistics costs? Justify your answer.
5. Suppose EcoFresh Organics is willing to balance moderate costs with accessibility to its target market. How would City B's government incentives impact the location decision, and what additional factors might EcoFresh consider?



7.4 Trends in Location Strategies

Recent trends in locating manufacturing facilities reflect a combination of competitive and technological factors. One such trend is for foreign producers, especially Japanese automotive firms, to locate factories in the United States. One reason for this is the fact that the United States represents a tremendous market for Japanese cars, trucks and recreational vehicles. By locating in the United States, these firms can reduce delivery times and costs. Furthermore, they can avoid any tariffs or quotas that may be applied to imports.

Another trend is Just-in-Time manufacturing techniques, which encourages suppliers and customers to locate in proximity to each other in order to reduce lead times. We can expect to see the establishment of micro-factories with narrow product focus that will be located near major markets in order to reduce response times. Advances in information technology have enhanced the ability of companies to gather, track and distribute information. With the introduction of the Internet, many companies are now global players.



Video Activity 7.1

Watch the video using the link below

Location Decisions

Available from: <https://youtu.be/5KFT2Vm1bw8>



Think Point 7.1

Explain in terms of strategic objectives, how do goods producing, and service location decisions differ.

7.5 Methods of evaluating Location Alternatives

Heizer et al (2020) suggest four methods that can be used to solve location problems:

- **The Factor-Rating Method**

This is a location method that instils objectivity into the process of identifying hard-to-evaluate costs. It has six steps:

1. Develop a list of relevant factors.
2. Assign a weight to each factor that reflects its relative importance in the company's objectives.
3. Develop a scale for each factor (for example 1 to 10 or 1 to 100).

4. Have management score each location for each factor, using the scale in step (iii).
5. Multiply the score by the weights for each factor and the total score for each location.
6. Make a recommendation based on the maximum point score.



Activity 7.1

Scenario: You are interested in opening an up-market restaurant in the Johannesburg area. Outline the factors that should be considered, with a choice of suburb in the city that would best in ensuring sustainable profits.

• The Locational Break-Even Analysis

This is a cost-volume analysis to make an economic comparison of location alternatives. There are three steps:

- Determine the fixed and variable costs for each location
- Plot the costs for each location, with costs on the vertical axis of the graph and the annual volume on the horizontal axis
- Select the location that has the lowest total cost for the expected production volume

Centre-of-Gravity Method

This is a mathematical technique used for finding the best location for a single distribution point that services several stores in the area.

Transportation Model

The objective of the transportation model is to determine the best pattern of shipments from several points of supply (sources) to several points of demand (destinations).



Think Point 7.2

Determine the importance of a distribution centre why must it be strategically placed?

Case Study 7.1

Read the following article and answer the question below.

Shoprite's Supply Chain Resilience and Efficiency

Shoprite's extensive supply chain provides a competitive advantage through resilience, agility, responsibility, and strategic partnerships. These factors help Shoprite serve customers better and ensure that products remain affordable and accessible. At the same time, the company is committed to reducing its environmental impact through operational and supply chain efficiency.

Technological Investments

Shoprite continuously invests in technologically enabled systems and infrastructure to maintain and enhance efficiencies throughout its supply chain. This includes advanced planning and forecasting tools, automated procurement processes, sophisticated warehousing solutions, and efficient distribution networks.

Supply Chain Design

Shoprite's supply chain is designed to meet the demands of its expanding network, improving efficiencies, enabling business growth, minimising waste, mitigating risks, and advancing centralisation levels. By expanding, adapting, and securing its supply chain, Shoprite enhances its resilience and agility to meet the growing needs of its customers.

Shoprite manages efficiencies by concentrating on three critical areas:

1. Building a Resilient Supply Chain: Ensuring the supply chain can withstand disruptions and continue to operate effectively.
2. Improving and Managing Systems and Processes: Continuously refining systems and processes to improve performance.
3. Transparency and Accountability: Maintaining clear and accountable operations to foster trust and reliability.



Diligent supply chain management at Shoprite includes:

- Planning: Strategic planning to anticipate demand and align supply chain activities
- Forecasting: Accurate demand forecasting to ensure appropriate inventory levels
- Procurement: Efficient procurement processes to source materials and

services

- Warehousing: Optimised warehousing to manage inventory and distribution
- Distribution: Reliable distribution networks to deliver products to customers

By focusing on resilience, agility, responsibility, and partnerships, and by investing in technology and efficient systems, Shoprite continues to enhance its supply chain's ability to meet the needs of its expanding network. This not only improves operational efficiency but also supports business growth and environmental sustainability.

Available from: <https://www.shopriteholdings.co.za/suppliers/supply-chain-management.html>

Questions:

1. Determine how Shoprite's focus on building a resilient supply chain can contribute to its overall business strategy and competitive advantage?
2. In what ways do technological investments enhance efficiencies throughout Shoprite's supply chain, and what are some specific examples of these technologies?
3. Explain how Shoprite's commitment to transparency and accountability in its supply chain operations impacts its relationships with stakeholders.

Activity 7.2



1. Discuss the factors that a company should consider when making a location decision for a new manufacturing plant?
2. Describe the Centre of Gravity method and how it can be used for location decisions.
3. Analyse the importance of considering both qualitative and quantitative factors in location decisions?
4. Scenario: A company is planning to open a new distribution centre. The coordinates and shipment volumes of three major customers are as follows:
 Customer A: Coordinates (2, 3), Shipment Volume: 50 units
 Customer B: Coordinates (5, 8), Shipment Volume: 30 units
 Customer C: Coordinates (9, 6), Shipment Volume: 20 units
 - a. Using the centre of Gravity method, calculate the optimal location for the distribution centre.
 - b. Explain why the centre of Gravity method is appropriate for determining the location of a distribution centre.
 - c. Discuss the limitations that should be considered when using the centre of Gravity method?

7.6 Service Location Strategy

Whilst the focus in the manufacturing sector location analysis is to minimise costs, the focus in the service sector is to Maximise revenue. This is because manufacturing firm's costs tend to vary substantially between locations, whilst service firms find location has more impact on revenue than cost. This means that the location for service firms should be based on determining the volume of business and revenue. Heizer.et.al (2020) submits that there are eight major components of volume and revenue for service firms, and these are:

- Purchasing power of the customer drawing area
- Service and image compatibility with demographics of the customer drawing area
- Competition in the are
- Quality of competition
- Uniqueness of the firm's and competitors' locations
- Physical qualities of facilities and neighbouring businesses
- Operating policies of the firm

- Quality of management



Think Point 7.3

When developing a service location strategy, businesses must carefully consider how the chosen location will affect both customer experience and operational efficiency. Why?

Knowledge Check Questions 7.1

Select the correct answer:

1. Which of the following is a quantitative factor in location decisions?
 - a. Community attitudes
 - b. Quality of life
 - c. Transportation costs
 - d. Political stability
2. The Centre of Gravity method helps to determine:
 - a. The most cost-effective transportation route
 - b. The optimal facility location based on transportation costs
 - c. The best inventory management strategy
 - d. The ideal product mix for a facility
3. Which of the following is a qualitative factor that should be considered in a location decision?
 - a. Labour costs
 - b. Proximity to suppliers
 - c. Local regulations
 - d. Community attitudes



Knowledge Check Questions 7.2

State if the following is True or False by selecting the correct answer.



1. The Centre of Gravity method is primarily used to optimise labour costs.
 True
 False
2. A company's location strategy can significantly impact its competitive advantage.
 True
 False
3. Proximity to customers is less important for service businesses compared to manufacturing businesses.
 True
 False

7.7 Summary

This unit provided an in-depth exploration of the key principles and methodologies involved in making location decisions. The major topics covered included understanding different strategic approaches to location selection, including cost-driven, market-driven, and supply chain-driven strategies, examining key determinants such as labour availability, infrastructure, economic policies, political stability, and proximity to suppliers and customers and analysing recent developments such as globalisation, restoring, sustainability-focused site selection, and the impact of digital transformation on location decisions.

The methods of evaluating location alternatives was explored for applying various decision-making techniques, including factor-rating analysis, centre-of-gravity models, break-even analysis, and location cost-profit analysis. The service location strategy was explained for exploring how service-oriented businesses select locations based on customer accessibility, demographic trends, and service quality considerations.

Through case studies and practical applications, students gained insights into how organisations make informed location decisions that enhance competitiveness and operational efficiency. The ability to assess location alternatives and align them with organisational strategy is essential for long-term business success.

Revision Questions

1. A major drug store chain wishes to build a new warehouse to serve the whole Midwest. Currently, it is looking at three possible locations. The factors, weights, and ratings being considered are given below:

Explain which city they should choose.

		Ratings		
Factor	Weights	Peoria	Des Moines	Chicago
Nearness to markets	20	4	7	5
Labour cost	5	8	8	4
Taxes	15	8	9	7
Nearness to suppliers	10	10	6	10

2. Balfour's is considering building a plant in one of three possible locations.

They have estimated the following parameters for each location:

Illustrate for what unit sales volume they should choose each location.

Location	Fixed Cost	Variable Cost
Waco, Texas	R300,000	R5.75
Tijuana, Mexico	R800,000	R2.75
Fayetteville, Arkansas	R100,000	R8.00

3. Our main distribution centre in Phoenix, AZ is due to be replaced with a much larger, more modern facility that can handle the tremendous needs that have developed with the city's growth. Fresh produce travels to the seven store locations several times a day making site selection critical for efficient distribution.

Using the data in the following table, illustrate the map coordinates for the proposed new distribution centre.

Store Locations	Map Coordinates (x,y)	Truck Round Trips per Day
Mesa	(10,5)	3
Glendale	(3,8)	3



Camelback	(4,7)	2
Scottsdale	(15,10)	6
Apache Junction	(13,3)	5
Sun City	(1,12)	3
Pima	(5,5)	10

4. A company is planning on expanding and building a new plant in one of three countries in Middle or Eastern Europe. The General Manager, Patricia Donegal, has decided to base her decision on six critical success factors including: technology availability and support, availability and quality of public education, legal and regulatory aspects, social and cultural aspects, economic factors, and political stability. Using a rating system of 1 (least desirable) to 5 (most desirable) she has arrived at the following ratings (you may, of course, have different opinions).

Select the country where the plant should be built.

Critical Success Factor	Turkey	Serbia	Slovakia
Technology availability and support	4	3	4
Availability and quality of public education	4	4	3
Legal and regulatory aspects	2	4	5
Social and cultural aspects	5	3	4
Economic factors	4	3	3
Political stability	4	2	3

5. Assume that Patricia decides to use the following weights for the critical success factors:

- Technology availability and support 0.3
- Availability and quality of public education 0.2
- Legal and regulatory aspects 0.1
- Social and cultural aspects 0.1
- Economic factors 0.1

- Political stability 0.2

Explain if this would change her decision.

6. Patricia's advisors have suggested that Turkey and Slovakia might be better differentiated by either:

Explain how you would advise Ms. Donegal.

- a. Doubling the number of critical success factors, or
 - b. Breaking down each of the existing critical success factors into smaller, more narrowly defined items, e.g., Availability and quality of public education might be broken into primary, secondary, and post-secondary education.
7. Examine the methods of evaluating location alternatives.

Answers

Activity 7.1

Scenario: You are interested in opening an up-market restaurant in the Johannesburg area.

Outline the factors that should be considered, with a choice of suburb in the city that would best in ensuring sustainable profits.

Answer: To be discussed during the webinar(s) with lecturer

Activity 7.2

1. Discuss the factors that a company should consider when making a location decision for a new manufacturing plant?

Answer: A company should consider factors such as proximity to suppliers and customers, transportation and logistics costs, availability of skilled labour, local business regulations and taxes, infrastructure quality, community attitudes, and economic incentives offered by local governments.

2. Describe the Centre of Gravity method and how it can be used for location decisions.

Answer: The Centre of Gravity method is a quantitative technique used to find the optimal location for a facility by minimising transportation costs. It involves plotting the locations of suppliers and customers on a coordinate grid, assigning weights based on shipment volumes, and calculating the weighted average of the coordinates to determine the best location.

3. Analyse the importance of considering both qualitative and quantitative factors in location decisions?

Answer: It is important to consider both qualitative and quantitative factors because quantitative factors like costs and distances can be measured and analysed numerically, while qualitative factors like community attitudes, quality of life, and political stability can impact employee satisfaction and long-term operational success. A comprehensive evaluation ensures a balanced and strategic location decision.

4. Examine the methods of evaluating location alternatives.

Answer: The prescribed textbook suggested four methods outlined below but some other texts have five methods separating the CVP method (cost volume profit analysis) from Break even method. For purposes of this module, only four methods are suggested and are as follows:

The Factor-Rating Method

This is a location method that instils objectivity into the process of identifying hard-to-evaluate costs. It has six steps:

1. Develop a list of relevant factors.
2. Assign a weight to each factor that reflects its relative importance in the company's objectives.
3. Develop a scale for each factor (for example 1 to 10 or 1 to 100).
4. Have management score each location for each factor, using the scale in step (iii).
5. Multiply the score by the weights for each factor and the total score for each location.
6. Make a recommendation based on the maximum point score.

The Locational Break-Even Analysis

This is a cost-volume analysis to make an economic comparison of location alternatives. There are three steps:

- Determine the fixed and variable costs for each location
- Plot the costs for each location, with costs on the vertical axis of the graph and the annual volume on the horizontal axis
- Select the location that has the lowest total cost for the expected production volume

Centre-of-Gravity Method

This is a mathematical technique used for finding the best location for a single distribution point those services several stores in the area.

Transportation Model

The objective of the transportation model is to determine the best pattern of shipments from several points of supply (sources) to several points of demand (destinations).

5. a. Using the centre of Gravity method, calculate the optimal location for the distribution centre.

Answer:

$$X - \text{coordinate} = (2 \times 50) + (5 \times 30) + (9 \times 20) / 50 + 30 + 20 = 100 + 150 + 180 / 100 = 430 / 100 = 4.3$$

$$Y - \text{coordinate} = (3 \times 50) + (8 \times 30) + (6 \times 20) / 50 + 30 + 20 = 150 + 240 + 120 / 100 = 510 / 100 = 5.1$$

b. Explain why the centre of Gravity method is appropriate for determining the location of a distribution centre.

Answer: The centre of Gravity method is appropriate for determining the location of a distribution

centre because it minimises transportation costs by considering the coordinates and shipment volumes of multiple locations. This method ensures that the distribution centre is placed in a position that reduces the total distance travelled for deliveries, optimising logistics efficiency.

c. Discuss the limitations that should be considered when using the centre of Gravity method?

Answer: Limitations of the centre of Gravity method include the assumption that transportation costs are directly proportional to distance, ignoring other factors like road conditions and traffic. It also doesn't account for qualitative factors such as local regulations, community attitudes, and the availability of infrastructure. Additionally, it assumes a flat plane, which may not represent the actual geography of the area.

Case Study 7.1

1. Determine how Shoprite's focus on building a resilient supply chain can contribute to its overall business strategy and competitive advantage?

Answer: Shoprite's focus on building a resilient supply chain ensures that it can withstand disruptions, such as natural disasters, political instability, or supply shortages, which might otherwise halt operations. This reliability is crucial for maintaining continuous product availability, thereby enhancing customer satisfaction and trust. Additionally, a resilient supply chain supports business continuity and minimises losses during unforeseen events, giving Shoprite a competitive edge in the market. This strategy aligns with the broader business goals of sustainable growth and customer retention.

2. In what ways do technological investments enhance efficiencies throughout Shoprite's supply chain, and what are some specific examples of these technologies?

Answer: Technological investments enhance efficiencies in Shoprite's supply chain by improving accuracy, reducing costs, and increasing responsiveness. Examples of these technologies include:

- **Advanced Planning and Forecasting Tools:** These help predict customer demand more accurately, ensuring optimal inventory levels and reducing the risk of stockouts or overstock
- **Automated Procurement Processes:** Streamline the purchasing process, reduce errors, and ensure timely procurement of goods
- **Sophisticated Warehousing Solutions:** Implementing automation in warehousing operations, such as robotic picking systems and warehouse management software, improves inventory management and order fulfilment speed
- **Efficient Distribution Networks:** Utilising route optimisation software and real-time tracking systems to ensure timely and cost-effective delivery of products to stores

These technologies collectively contribute to a more efficient, cost-effective, and responsive supply chain, which supports Shoprite's goal of maintaining affordability and accessibility for its customers

3. Explain how Shoprite's commitment to transparency and accountability in its supply chain operations impacts its relationships with stakeholders.

Answer: Shoprite's commitment to transparency and accountability in its supply chain operations fosters trust and reliability among its stakeholders, including customers, suppliers, and partners. Clear and accountable operations ensure that stakeholders have a precise understanding of Shoprite's processes, which builds confidence in the company's ability to deliver on its promises. For customers, this means trust in the quality and availability of products. For suppliers and partners, it means reliable and predictable business dealings, which can lead to stronger, more collaborative relationships. Additionally, transparency and accountability help Shoprite adhere to regulatory requirements and ethical standards, further enhancing its reputation and stakeholder trust. This commitment ultimately supports long-term business sustainability and success.

Knowledge Check Questions 7.1

1. Which of the following is a quantitative factor in location decisions?

Answer: c. Transportation costs

2. The Centre of Gravity method helps to determine:

Answer: b. The optimal facility location based on transportation costs

3. Which of the following is a qualitative factor that should be considered in a location decision?

Answer: d. Community attitudes

Knowledge Check Questions 7.2

1. The Centre of Gravity method is primarily used to optimise labour costs.

Answer: False

2. A company's location strategy can significantly impact its competitive advantage.

Answer: True

3. Proximity to customers is less important for service businesses compared to manufacturing businesses.

Answer: False

Practical Application 7.1

1. Examine which factors EcoFresh Organics should prioritise when choosing the location for its production facility?

Answer: EcoFresh Organics should prioritise costs, proximity to the target market, transportation infrastructure, and growth potential. Proximity to the target market (young urban professionals) is important for efficient distribution and faster delivery times. Transportation infrastructure should also be a priority to ensure cost-effective logistics. Additionally, the company should consider operational costs and potential for future expansion.

2. Analyse the benefits and drawbacks of each location (City A, B and C) for EcoFresh Organics.

Answer:

City A:

- Benefits: Close to a large urban market, strong transportation infrastructure, and easy access to suppliers and distributors
- Drawbacks: High operational costs, including rent and wages

City B:

- Benefits: Moderate costs, government incentives, and good road networks
- Drawbacks: Smaller population of the target market and limited rail access

City C:

- Benefits: Low operational costs, abundant space for expansion
- Drawbacks: Far from the primary market, which may increase logistics costs and delivery times, and limited transportation options

3. If EcoFresh Organics chooses City C for its low costs and future expansion potential, what strategies could it employ to manage the drawbacks of limited market proximity and transportation?

Answer:

- Set up a central distribution hub closer to urban markets to reduce delivery times
- Partner with a third-party logistics provider to handle distribution in key urban areas
- Invest in predictive demand planning to ensure efficient inventory management and reduce transportation frequency
- Consider co-location agreements with logistics providers to improve supply chain efficiency

4. Based on the scenario, which location (City A, B and C) would be the most strategic choice for EcoFresh Organics if the company's primary objective is to quickly reach urban consumers with minimal logistics costs? Justify your answer.

Answer: City A would be the most strategic choice for EcoFresh Organics if the primary objective is to reach urban consumers quickly and minimise logistics costs. City A's proximity to a large metropolitan area means that EcoFresh Organics would be close to its target market, reducing transportation costs and ensuring timely deliveries. Although operational costs are higher, the savings on logistics and the increased likelihood of customer retention in the urban area could offset these expenses.

5. Suppose EcoFresh Organics is willing to balance moderate costs with accessibility to its target market. How would City B's government incentives impact the location decision, and what additional factors might EcoFresh consider?

Answer: Government incentives in City B could reduce operational costs, making it an attractive middle ground between high-cost City A and distant City C. This could allow EcoFresh Organics to allocate more resources toward marketing, product development, or logistics within a manageable budget. Additionally, EcoFresh should evaluate factors like availability of skilled labour for its production facility, local regulations impacting production, and potential competition in City B to make a well-rounded decision.

Think Point 7.1

Explain in terms of strategic objectives, how do goods producing, and service location decisions differ.

Answer: Goods-producing locations prioritise cost efficiency, while service locations emphasise customer accessibility and experience. In terms of strategic objectives, goods-producing and service location decisions differ primarily in their focus:

Goods-Producing Location Decisions:

- Objective: Minimise costs
- Focus: Proximity to raw materials, transportation networks, and labour markets to reduce production and distribution costs
- Example: A factory might be located near suppliers or major transportation hubs to optimise supply chain efficiency. Think Amazon

Service Location Decisions:

- Objective: Maximise customer accessibility and satisfaction
- Focus: Proximity to customers, visibility, and convenience to attract and serve clients effectively
- Example: A retail store or hospital is typically located in high-traffic areas or close to the target market to ensure easy access and high customer traffic

Think Point 7.2

Determine the importance of a distribution centre why must it be strategically placed?

Answer: Ensures timely delivery, reduces logistics costs, and improves inventory management

Think Point 7.3

When developing a service location strategy, businesses must carefully consider how the chosen location will affect both customer experience and operational efficiency. Why?

Answer: A poor location can deter customers and increase operational costs

Unit 8:

Human Resources, Job Design and Organisational Effectiveness

Unit Learning Outcomes

CONTENT LIST	LEARNING OUTCOMES FOR THIS UNIT
8.1 Introduction	Introduces the topic areas covered for the unit
8.2 Human Resource for Competitive Advantage	Analyse the relationship between human resources and the impact on a firm's competitiveness
8.3 Labour Planning	Analyse the impact that labour planning has on the workforce and company productivity
8.4 Job Design and Expansion	Relate the concepts of job design and job expansion in improving staff morale and productivity
8.5 Self-directed Teams	Analyse the dynamics and mechanisms of self-directed teams are explored to enable alternate labour management
8.6 Motivation and Incentive Methods	Analyse different methods of staff motivation methods
8.7 Ergonomics and Work Methods	Outline and relate suitable ergonomic and work methods to improve productivity and staff morale
8.8 Organisational Effectiveness	Examine the effectiveness of the use of resources and work measurement techniques
8.9 Summary	Summarises the topic areas covered for the unit

Prescribed and Recommended Reading(s)/Textbook(s)

Prescribed Reading(s)/Textbook(s)

- Heizer, J., Render, B. and Munson, C. (2020). *Operations Management: Sustainability and Supply Chain Management*. Thirteenth Edition. United Kingdom: Pearson.

Recommended Reading(s)/Textbook(s)

- Venkataraman, R.R. and Pinto, J.K. (2020). *Operations Management: Managing Global Supply Chains*. Second Edition. Sage Publications.
- Parksoy, T. and Deveci, M. (2023). *Smart and Sustainable Operations and Supply Chain Management in Industry 4.0*. First Edition. Boca Raton: CRC Press.
- Slack, N., Jones, B. and Burgess, N. (2022). *Operations Management*. Tenth Edition. United Kingdom: Pearson.
- Kareska, PhD Katerina. 'Operational Management Strategies for Sustainable Productivity Growth' (August 5, 2023). Available at SSRN: <https://ssrn.com/abstract=4532740> or <http://dx.doi.org/10.2139/ssrn.4532740>
- 'The Impact Of Ergonomics On Quality Of Life In The Workplace'. (2023). https://www.researchgate.net/publication/377267055_The_Impact_of_Ergonomics_on_Quality_of_Life_in_the_Workplace



8.1 Introduction

In the field of Operations Management, human resources play a crucial role in driving productivity, efficiency, and overall organisational success. A well-designed human resource strategy aligns workforce planning, job design, motivation, and work methods with the company's operational goals, ultimately providing a competitive advantage. Managing human resources effectively ensures that employees are engaged, well-trained, and equipped with the necessary skills to contribute to process efficiency and continuous improvement.

This unit explores key aspects of human resource management in an operational context, including constraints on human resource strategy, which may arise from labour laws, economic conditions, or organisational structure. Labour planning focuses on ensuring the right number of employees with the required skills are available to meet operational demands. Job design and expansion determine how tasks are structured to maximise efficiency and employee satisfaction, while self-directed teams enhance workplace autonomy and collaboration. Additionally, motivation and incentive methods are critical for improving employee performance and retention. The unit also examines ergonomics and work methods, which optimise job design to improve safety, efficiency, and worker well-being. Finally, the impact of human resource decisions on organisational effectiveness is explored, demonstrating how workforce management contributes to achieving business objectives.

By the end of this unit, students will understand how human resource strategies influence operational efficiency, workforce productivity, and overall business competitiveness.

8.2 Human Resource for Competitive Advantage

The objective of a human resource strategy is to manage labour and design jobs, so people are effectively and efficiently utilised. As we focus on a human resource strategy, we want to ensure that people:

- Are efficiently utilised within the constraints of other operations management decisions
- Have a reasonable quality of work life in an atmosphere of mutual commitment and trust

By reasonable quality of work life, we mean a job that is not only reasonably safe and for which the pay is equitable, but which also achieves an appropriate level of both physical and psychological requirements. Mutual commitment means that both management and employee strive to meet common objectives. Mutual trust is reflected in reasonable, documented employment policies that are honestly and equitably implemented to the satisfaction of both management and employee. When management has genuine respect for its employees and their contributions to the firm, establishing a reasonable quality of work life and mutual trust are not particularly difficult.

Constraints on Human Resource Strategy

Many decisions made about people are constrained by other decisions. Firstly, the product mix may determine seasonality and stability of employment. Secondly, technology, equipment, and processes may have implications for safety and job content. Thirdly, the location decision may have an impact on the ambient environment in which the employees work. Finally, layout decisions, such as assembly line versus work cell, influence job content. Technology decisions impose substantial constraints. For instance, some of the jobs in steel mills are dirty, noisy, and dangerous; slaughterhouse jobs may be stressful and subject workers to stomach crunching stench; assembly-line jobs are often boring and mind-numbing; and high capital expenditures such as Lucent's fabricating plants may require 24 – hour, 7-day-a-week operation in restrictive clothing. We are not going to change these jobs without making changes in our other strategic decisions. So, the trade-offs necessary in order to reach a tolerable quality of work life are difficult to ascertain. Effective managers consider such decisions simultaneously. The result: an effective, efficient system in which both individual and team performance are enhanced through optimum job design.

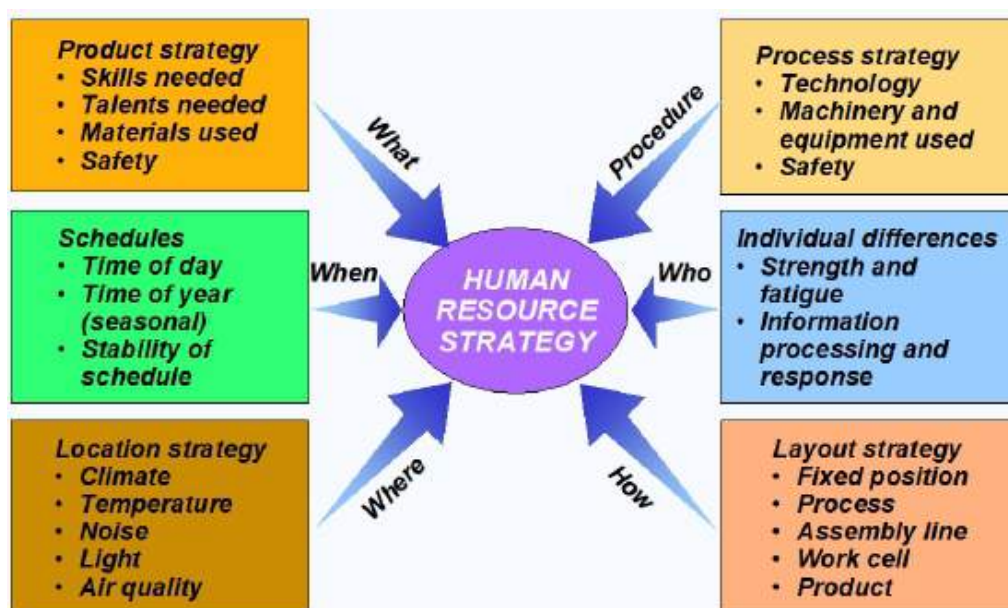


Figure: Constraints on Human Resource Strategy

Source: Heizer and Render (2020)

The effective operations manager understands how decisions blend together to constrain the human resource strategy.

Acknowledging the constraints imposed on human resource strategy, we now look at three distinct decision areas of human resource strategy: labour planning, job design, and labour standards. The supplement to this chapter expounds upon the discussion of labour standards and introduces work

measurement.



Think Point 8.1

Analyse the factors that contribute to job satisfaction

8.3 Labour Planning

Labour planning is determining staffing policies that deal with employment stability and work schedules:

Employment Stability Policies

Employment stability deals with the number of employees maintained by an organisation at any given time. There are two very basic policies for dealing with stability:

- **Follow demand exactly.** Following demand exactly keeps direct labour costs tied to production but incurs other costs. These other costs include (a) hiring and termination costs; (b) unemployment insurance; and (c) premium wages in order to entice personnel to accept unstable employment. This policy tends to treat labour as a variable cost
- **Hold employment constant.** Holding employment levels constant maintains a trained workforce and keeps hiring, termination and unemployment costs to a minimum. However, with employment held constant, employees may not be utilised fully when demand is low, and the firm may not have the human resources it needs when demand is high. This policy tends to treat labour as a fixed cost

Maintaining a stable workforce may allow a firm to pay lower wages than a firm that follows demand. This saving may provide a competitive advantage. However, a fluctuating workforce may best serve firms with highly seasonal work and little control over demand. For example, a salmon canner on the Columbia River only processes salmon when the salmon are running.

However, the firm may find complementary labour demands in other products or operations, such as making cans and labels or repairing and maintaining facilities.

Firms must determine policies about employment stability. The above policies are only two of many that can be efficient and provide a reasonable quality of work life.



Activity 8.1

Outline your understanding of a 'good quality of work life'

Work Schedules

Although the standard work schedule in the United States is still five eight-hour days, variations do exist. A currently popular variation is a work schedule called flexitime. Flexitime allows employees, within limits to determine their own schedules. A flexitime policy might allow an employee (with proper notification) to be at work at 8 am plus or minus 2 hours. This policy allows more autonomy and independence on the part of the employee. Some firms have found flexitime to be a low-cost fringe benefit that enhances job satisfaction.

The problem from the OM perspective is that much production work requires full staffing for efficient operations. A machine that requires 3 people cannot run at all if only 2 operators show up. Having a waiter show up to serve lunch at 1:30pm rather than 11:30 am is not much help either. Similarly, some industries find that their process strategies severely constrain their human resource scheduling options.

For instance, paper manufacturing, petroleum refining, and power stations must be staffed around the clock except for maintenance and repair shutdown.

Another option is the flexible work week. This plan often calls for fewer but longer days, such as four 10-hour days or, for example, in the case of Lucent Technologies, 12-hour shifts are sometimes called compressed work weeks. These schedules are viable for many operations functions – as long as suppliers and customers can be accommodated. Firms that have high process start-up times (say, to get a boiler up to operating temperature) find longer workday options particularly appealing.

Compressed work weeks have long been common in fire and utility departments, where physical exertion is modest but 24-hour coverage desirable. A recent Gallup survey showed that two thirds of working adults would prefer toiling four 10-hour days to the standard five-day eight-hour schedule. Duke Power Co., Los Angeles County, AT&T and General Motors are just a few organisations to offer the 4-day week.

Another option is shorter days rather than longer days. This plan often moves employees to part-time status. Such an option is particularly attractive in-service industries, where staffing for peak loads is necessary. Banks and restaurants often hire part-time workers. In addition, many firms reduce labour costs by reducing fringe benefits for part-time employees.

Job Classifications and Work Rules

Many organisations have strict job classifications and work rules that specify who can do what, when they can do it, and under what conditions they can do it, often as a result of union pressure. These job classifications and work rules restrict employee flexibility on the job, which in turn reduces the flexibility of the operations function. However, part of an operations manager's task is to manage the unexpected.

Therefore, the more flexibility a firm has when staffing and establishing work schedules, the more efficient and responsive it can be. This is particularly true in-service organisations where extra

capacity often resides in extra or flexible staff. Building morale and meeting staffing requirements that result in efficient, responsive operations are easier if managers have fewer job classifications and work-rule constraints.

If the strategy is to achieve a competitive advantage by responding rapidly to the customer, a flexible workforce may be a prerequisite.

8.4 Job Design and Expansion

Job design specifies the tasks that constitute a job for an individual or a group. We examine five components of job design as outlined by Heizer et al (2020):

- Job specialisation
- Job expansion
- Psychological components
- Self-directed teams
- Motivation and incentive systems

Labour Specialisation

The importance of job design as a management variable is credited to the eighteenth-century economist Adam Smith who suggested that a division of labour, also known as labour specialisation (or job specialisation), would assist in reducing labour costs of multi-skilled artisans. This is accomplished in several ways:

- Development of dexterity and faster learning by the employee because of repetition
- Less loss of time because the employee would not be changing jobs or tools
- Development of specialisation tools and the reduction of investment because each employee has only a few tools needed for a particular task

The nineteenth-century British mathematician Charles Babbage determined that a fourth consideration was also important for labour efficiency. Because pay tends to follow skill with a rather high correlation, Babbage suggested paying exactly the wage needed for the skill required. If the entire job consists of only one skill, then we would pay for only that skill. Otherwise, we would tend to pay for the highest skill contributed by the employee. These four advantages of labour specialisation are still valid today.

A classic example of labour specialisation is the assembly line. Such a system is often very efficient, although it may require employees to do repetitive, mind-numbing jobs. The wage rate for many of these jobs, however, is very good. Given the relatively high wage rate for the modest skills required in many of these jobs, there is often a large pool of employees from which to choose.

This is not an incidental consideration for the manager with responsibility for staffing the operations function. It is estimated that 2% to 3% of the workforce in industrialised nations perform highly specialised, repetitive assembly line jobs. The traditional way of developing and maintaining worker commitment under labour specialisation has been good selection (matching people to the job), good wages, and incentive systems.

From the manager's point of view, a major limitation of specialised jobs is their failure to bring the 'whole person' to the job. Job specialisation tends to bring only the employee's manual skills to work. In an increasingly sophisticated knowledge-based society, managers may want employees to bring their mind to work as well.

Job Expansion

In recent years, there has been an effort to improve the quality of work life by moving from labour specialisation towards more varied job design. Driving this effort is the theory that variety makes the job more enjoyable, and that the employee therefore enjoys a higher quality of work life. It is assumed that this flexibility thus benefits the employee and the organisation.

We modify jobs in a variety of ways. The first approach is job enlargement, which occurs when we add tasks requiring similar skill to an existing job. Job rotation is a version of job enlargement that occurs when the employee is allowed to move from one specialised job to another. Variety has been added to the employee's perspective of the job. Another approach is job enrichment, which adds planning and control to the job. An example is to have department store salespeople responsible for ordering, as well as selling, their goods. Job enrichment can be thought of as *vertical expansion*, as opposed to job enlargement, which is horizontal.

A popular extension of job enrichment, employee empowerment is the practice of enriching jobs so that employees accept responsibility for a variety of decisions normally associated with staff specialists. Empowering employees helps them take "ownership" of their jobs so that they have personal interests in improving performances.



Activity 8.2

Differentiate between job enlargement, job rotation, job specialisation and employee empowerment.

Hackman and Oldham have incorporated much of that work into five desirable characteristics of job design. Their summary suggests that jobs should include the following characteristics:

- Skill variety, requiring the worker to use a variety of skills and talents
- Job identity, allowing the worker to perceive the job as a whole and recognise a start and a finish
- Job significance, providing a sense that the job has impact on the organisation and society

- Autonomy, offering freedom, independence, and discretion
- Feedback, providing clear, timely information about performance

Including these five ingredients in job design is consistent with job enlargement, job enrichment, and employee empowerment. We now look at some of the ways in which teams can be used to expand jobs and achieve these five job characteristics.

8.5 Self-Directed Teams

Many world-class organisations have adopted teams to foster mutual trust and commitment and provide the core job characteristics. One team concept of particular note is the self-directed team: a group of empowered individuals working together to reach a common goal. These teams may be organised for long-term or short-term objectives.

Teams are effective primarily because they can easily provide employee empowerment, ensure core job characteristics, and satisfy many of the psychological needs of individual team members.

Of course, many good job designs can provide these psychological needs. Therefore, to maximise team effectiveness, managers do more than just form 'teams'. For instance, they:

- Ensure that those who have legitimate contributions appear on the team
- Provide management support
- Ensure the necessary training
- Endorse clear objectives and goals

Successful teams should also receive financial and non-financial rewards. Finally, managers must recognise that teams have life cycles and that achieving objectives may suggest disbanding the teams. However, teams may be renewed with changes in members or new assignments.

Teams and other approaches to job expansion should not only improve the quality of work life and job satisfaction but also motivate employees to achieve strategic objectives.

Both managers and employees need to be committed to achieving strategic objectives. However, employee contribution is fostered in a variety of ways, including organisational climate, supervisory action, and job design.

Expanded job designs allow employees to accept more responsibility. For employees who accept this responsibility, one may well expect some enhancement in productivity and product quality. Amongst the other positive aspects of job expansion, are reduced turnover, tardiness and absenteeism.

Managers who expand jobs and build communication systems that elicit suggestions from employees have added potentials for efficiency and flexibility to meet market demands. However, these job designs have a number of limitations, including:

Higher Capital Cost

- Individual differences
- Higher wage rates
- Smaller labour pool
- Increased accident rates
- Current technology may not lend itself to job expansion

Source: Heizer.et.al (2020)



Activity 8.3

Explain the importance of teamwork in improving efficiency

8.6 Motivation and Incentive Methods

Our discussion of the psychological components of job design provides insight into the factors that contribute to job satisfaction and motivation. In addition to these psychological factors, there are monetary factors. Money often serves as a psychological as well as financial motivator. Monetary rewards take the form of bonuses, profit and gain-sharing and incentive systems. Bonuses, typically in cash or stock options, are often used at executive levels to reward management. Profit-sharing systems provide some part the profit for distribution to employees. A variation of profit sharing is gain-sharing, which rewards employees for improvements made in an organisation's performance. The most popular of these is the Scanlon plan, where any reduction in the cost of labour is shared between management and labour.

The gain-sharing approach used by Panhandle Eastern Corp. of Houston; Texas allows for employees to receive a bonus of 2% of their salary at year's end if the company earns at least R2.00 per share. When Panhandle earns R2.10 per share, the bonus climbs to 3%. Employees have become much more sensitive about costs since the plan began.

Incentive systems based on individual, or group productivity are used in nearly half of the manufacturing firms in America. These systems often require employees or crews to achieve production above a predetermined standard. The standard can be based on a standard time per task or number of pieces made. Standard time systems are sometimes called *measured daywork*, where employees, payments are based on the amount of standard time accomplished.

A piece-rate system assigns a standard time for the production of each piece, and the employee is paid based on the number of pieces made. Both measured daywork and piece-rate systems typically guarantee the employee at least a base rate.

With the increasing use of teams, various forms of team-based pay are also being developed. Many are based on traditional pay systems supplemented with some form of bonus or incentive system. However, because many team environments require cross-training of enlarged jobs, knowledge-based pay systems have also been developed. With respect to knowledge-based (or skill-based) pay systems, a portion of the employee's pay depends on demonstrated knowledge or skills possessed. Knowledge-based pay systems are designed to reward employees for the enlarged scope of their jobs. Some of these pay systems have three dimensions: horizontal skills, which reflect the variety of tasks the employee can perform; vertical skills, which reflect the planning and control aspects of the job; and depth of the skills, which reflects quality and productivity. At Wisconsin's Johnsonville Sausage Co., employees receive pay raises only by mastering new skills such as scheduling, budgeting and quality control.



Think Point 8.2

Explain if money is an incentive and if there are any drawbacks to purely financial incentives.

Practical Application 8.1

Read the scenario below and answer the questions that follow:

Company XYZ (Pty) Ltd is experiencing a decline in employee productivity and morale. The HR department is considering implementing a new incentive program to address these issues



Questions:

1. Explain the factors that HR should consider when designing an effective incentive program to improve productivity and morale at Company XYZ (Pty) Ltd?
2. Determine how Company XYZ (Pty) Ltd can ensure that the new incentive program is fair and transparent to all employees?
3. Discuss the potential challenges might Company XYZ (Pty) Ltd face when implementing the new incentive program, and how can they be mitigated?

8.7 Ergonomics and Work Methods

According to Heizer.et.al (2020), Frederick W Taylor began the era of scientific management in the late 1800s. He and his contemporaries began to examine personnel selection, work methods, labour standards and motivation.

With the foundation provided by Taylor, a body of knowledge has developed with respect to people's capabilities and limitations. This knowledge is necessary because humans are hand/eye animals possessing exceptional capabilities and some limitations Heizer.et.al (2020). Because managers must design jobs that can be done, we now introduce a few of the issues related to people's capabilities and limitations.

Ergonomics The operations manager is interested in building a good interface between human and machine. Studies of this interface are known as ergonomics. Ergonomics means the "study of work" or "work study". In the United States, the term human factors are often substituted for the word ergonomics. Understanding ergonomics issues helps to improve human performance.

Male and female adults come in limited configurations. Therefore, design of tools and the workplace

depends on the study of people to determine what they can and cannot do. Substantial data have been collected that provide basic strength and measurement data needed to design tools and the workplace. The design of the workplace can make the job easier or more difficult. Additionally, we now have the ability, through the use of computer modelling, to analyse human motions and efforts.

Let's look briefly at one instance of human measurements: determining the proper height for a writing desk. The desk has an optimum height depending on the size of the individual and the task to be performed. The common height for a writing desk is 29 inches. For typing of data entry at a computer, the surface should be lowered. The preferred chair and desk height should result in a very slight angle between the body and arm when the individual is viewed from the front and when the back is straight.

This is the critical measurement; it can be achieved via adjustment in either table or chair height.

Video Activity 8.1

Watch the video using the link below and answer the question that follows.

This video illustrates ergonomics tips and safety in the workplace

Available from: https://www.youtube.com/watch?v=2Eeqliwt0_Y (Accessed: 04 May 2025)

Questions:

1. Define ergonomics and explain why it is important in the context of Operations Management.
2. Identify three ergonomic tips discussed in the video that can help reduce strain for employees working at desks. Describe how each of these tips contributes to workplace efficiency and employee health.
3. Create an ergonomic checklist for setting up a computer workstation. Include at least five items based on the tips provided in the video.
4. Discuss the potential costs associated with poor ergonomics in a workplace from an OM perspective. What might be the long-term implications if these issues aren't addressed?
5. Propose a small improvement project an operations manager could implement to improve ergonomics in a workplace where employees work on computers all day. Outline the steps and potential benefits of your project.



Operator Input to Machines - Operator response to machines, be they hand tools, pedals, levers or buttons, needs to be evaluated. Operations managers need to be sure that operators have the

strength, reflexes, perceptions and mental capacity to provide necessary control. Such problems as carpal tunnel syndrome result when a tool as simple as a keyboard is poorly designed.

Feedback to Operators - Feedback to operators is provided by sight, sound, and feel; it should not be left to chance. The mishap at the Three Mile Island nuclear facility, America's worst nuclear experience, was in large part the result of poor feedback to the operators about reactor performance. Non-functional groups of large, unclear instruments and inaccessible controls, combined with hundreds of confusing warning lights, contributed to that nuclear failure. Such relatively simple issues make a difference in operator response and, therefore, performance.

The Work Environment - The physical environment in which employees work affects their performance, safety, and quality of work life. Illumination, noise and vibration, temperature, humidity, and air quality are work-environment factors under the control of the organisation and the operations manager. The manager must approach them as controllable.

Illumination is necessary, but the proper level depends upon the work being performed. However, other lighting factors are important. These include reflective ability, contrast of the work surface with surroundings, glare, and shadows.

Noise of some form is usually present in the work area, but most employees seem to adjust well. However, high levels of sound will damage hearing. Extended periods of exposure to decibel levels above 85 dB are permanently damaging. The Occupational Safety and Health Administration (OSHA) requires ear protection above this level if exposure equals or exceeds 8 hours. Even at low levels, noise and vibration can be distracting.

Therefore, most managers make substantial effort to reduce noise and vibration through good machine design, enclosures, or segregation of sources of noise and vibration.

Temperature and humidity parameters have been well established. Managers with activities operating outside the established comfort zone should expect adverse effect on performance.

Methods Analysis focusses on how a task is accomplished. Whether controlling a machine or making or assembling components, how a task is done makes a difference in performance, safety and quality. Using knowledge from ergonomics and methods analysis, methods engineers are charged with ensuring that quality and quantity standards are achieved efficiently and safely. Methods analysis and related techniques are useful in office environments as well as the factory. Methods techniques are used to analyse:

- Movement of individuals or material. The analysis is performed using flow diagrams and process charts with varying amounts of detail.
- Activity of human and machine and crew activity. This analysis is performed using activity charts (also known as man-machine charts or crew charts).
- Body movement (primarily arms and hands). This analysis is performed using micro-motion charts.

Flow diagrams are schematics (drawings) used to investigate movement of people or material.

Process charts use symbols, to help us understand the movement of people or material. In this way, movement and delays can be reduced, and operations made more efficient.

Activity charts are used to study and improve the utilisation of an operator and a machine or some combination of operators (a “crew”) and machines. The typical approach is for the analyst to record the present method through direct observation and then propose the improvement on a second chart.

An Operations Chart analyses body movement. It is designed to show economy of motion by pointing out wasted motion and idle time (delay). The operations chart (also known as right-hand-left-hand chart).

Case Study 8.1

Read the following article and answer the questions below.

Also, check out the link to the video of this case study at the end of the article.

Unilever is a British-Dutch transnational consumer goods company co-headquartered in London, England, and Rotterdam, Netherlands. With over 400 brands and around 170,000 employees globally, they recruit 30,000 people annually and receive approximately 1.8 million job applications. Managing each application, which involves coordination, filtering, and interviews, is a massive task even for a large company like Unilever.

To make recruitment easier, more cost-effective, time-saving, and efficient, Unilever has turned to AI and automation. They partnered with two firms specializing in AI technology: Pymetrics and HireVue.

By implementing AI and automation in their hiring process, Unilever saved over 50,000 hours in candidate interview time, achieved more than £1 million in annual savings, and improved candidate diversity through machine analysis of video-based interviews.

Available from: https://youtu.be/O5ID6OPbVLk?si=bGbnKT6iYOJS_Ggo

Questions:

1. Discuss how Unilever utilised AI and automation in their recruitment process?
2. Outline the benefits that Unilever has experienced by using AI/automation in their recruitment process?
3. Explain why it was necessary for Unilever to implement AI and automation in their recruitment process?



The Visual Workplace

The Visual Workplace uses low-cost visual devices to share information quickly and accurately. Well-designed displays and graphs root out confusion and replace difficult-to-understand printouts and paperwork. Because workplace data change quickly and often, operations managers need to share accurate and up-to-date information. Workplace dynamics, with changing customer requirements, specifications, schedules, and other details on which an enterprise depends, must be rapidly communicated.

Visual systems can include statistical process control (SPC) charts, details of quality, accidents, service levels, delivery performance, costs, cycle time, and such traditional variables as attendance

and tardiness. All visual systems should focus on improvement because progress almost always has motivational benefits.

An assortment of visual signals and charts is an excellent tool for communication not only amongst people doing the work, but also amongst support people, management, visitors, and suppliers. All these stakeholders deserve feedback on the organisation. Managers need to think in terms of visual management.

The visual workplace can take many forms. Kanban are a type of visual signal indicating the need for more production. The three-minute clocks found in Burger Kings are a type of visual standard indicating the acceptable wait for service. Painted symbols indicating the place for tools are another visual standard to aid housekeeping. Some organisations have found it helpful to have performance standards indicated by hourly quota numbers for all to see.

Andon lights are another visual signal. An Andon is a signal that there is a problem. Employees can manually initiate Andon's when they notice a problem or defect. They can also be triggered automatically when machine performance drops below a certain pace or when the numbers of cycles indicate that it is time for maintenance.

Visual systems also communicate the larger picture, helping employees to understand the link between their day-to-day activities and the organisation's overall performance. At Baldor Electric Co. in Fort Smith, Arkansas, the prior day's closing price Baldor's stock is posted for all to see. The stock price is to remind employees that a portion of their pay is based on profit sharing and stock options, and to encourage them to keep looking for ways to increase productivity.

Similarly, Missouri's Springfield Re Manufacturing Corp. has developed a concept called "open book management", where every employee is trained to understand the importance of financial measures (such as return on equity) and is provided with these measures regularly. When a huge copper mining company from Zambia asked its managers to benchmark the Springfield Re visual workplace, the mine company managers returned to spread this philosophy to their 55 000 employees.

Now as workers go in the front entrance of the mine, they cannot help but spot a 50-foot-high scoreboard that lists monthly and year-to-date financials.

The purpose of the visual workplace is to eliminate non-value-added activities and other forms of waste by making problems, abnormalities, and standards visual. This concept enhances communication and feedback by providing immediate information. The visual workplace needs less supervision because employees understand the standard, see the results, and know what to do.

Activity 8.4

1. Explain how HR departments can leverage ergonomics to improve organisational effectiveness?
2. Determine why it is important to consider ergonomic principles in job design, and what are the potential outcomes?
3. Describe the relationship between ergonomics, employee well-being, and organisational effectiveness
4. Explain how financial incentives impacts employee motivation and organisational performance?
5. Discuss the role of non-financial incentives in employee motivation, and how can they be effectively implemented?
6. Examine how HR can design incentive programs to align with organisational goals and enhance employee motivation?

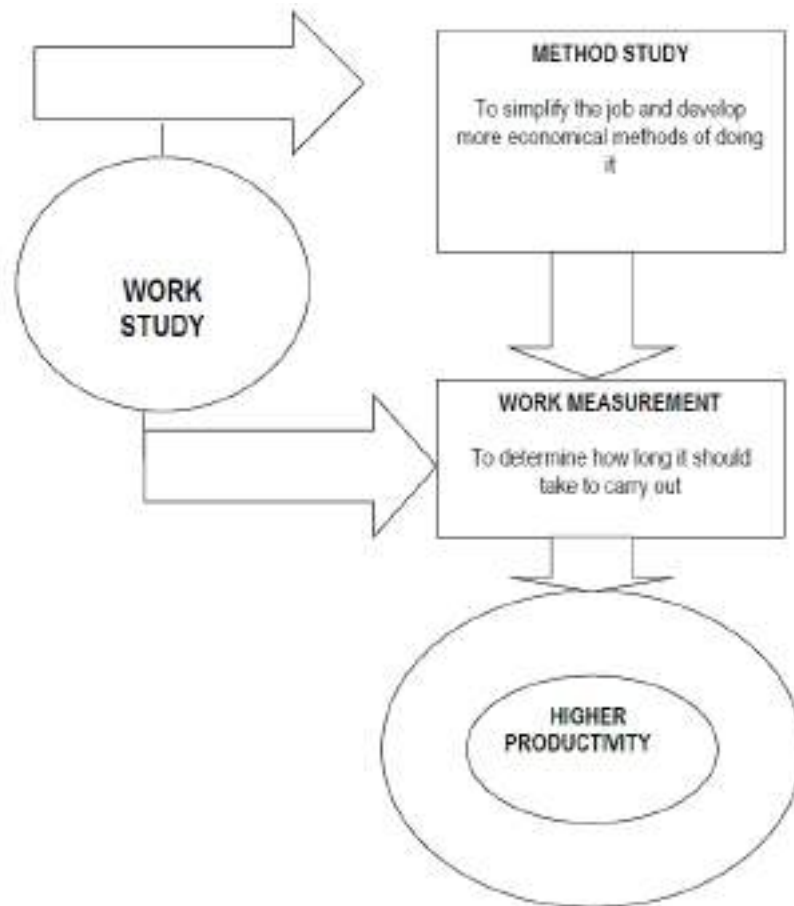
8.8 Organisational Effectiveness

Organisational effectiveness also known as Work Study is the systematic examination of the methods of carrying out activities to improve the effective use of resources and to set up standards of performance for the activities being carried out. Work Study aims at examining the way an activity is being carried out, simplifying or modifying the method of operation to reduce unnecessary or excess work, or the wasteful use of resources and setting time studies for performing that activity.

Work Study comprises two categories, Work Measurement and Method Study. The Figure on the following page depicts the typical work study process.

**Readings**

Here is a link to an article by McKinsey and Company regarding Organisational Effectiveness: Available from: [Organizational effectiveness for sustained performance and health \(mckinsey.com\)](https://www.mckinsey.com/organizational-effectiveness-for-sustained-performance-and-health)



Source: Kanawaty (2017)

Basic Work Content or A Product or Operation

The basic work content is the time taken to manufacture a product or to perform the operation if the design or specification was perfect, or if the process of the operation was perfectly carried out. Therefore, the basic work content is the irreducible minimum time required producing one unit of output. This is obviously a perfect condition, which seldom, if ever, occurs in practice.

- Work content means the amount of work contained in a given product or a process, measured in human work or machine hours
- A work hour is the labour for one person for one hour
- A machine hour is the running of a machine for one hour

Excess Work Content

The following issues can result increased work content:

- Poor design and frequent design changes

The product may be designed in such a way that it may require numerous non-standard parts, thereby causing lengthy times to assemble. An excessive variety of products and lack of standardisation means that work has to be produced in small batches, with time lost as the operator adjusts and shifts from one batch to the next.

- Waste of Materials

The components of the products may be so designed that an excessive amount of material has to be removed to bring them to their final shape.

The Uses of Work Measurement

We use Work Measurement to:

- Compare the efficiency of different methods
- Balance the work of team members
- Determine the ideal number of machines a worker can operate
- Provide the basis for planning and control
- Choose an improved layout for process planning and for establishing a Just-in-Time system
- Provide information for tender estimates, selling prices and delivery dates
- Set standards for machine utilisation and labour performance that can be used for incentive schemes
- Provide information for labour cost control and to enable standard costs to be fixed and maintained



Think Point 8.3

Explain how we establish a “fair day’s” work

Basic Procedure for Work Study

Kanawathy (2017) suggests the following procedure:

- **Select** the job or process to be studied
- **Record** and collect all the relevant data about the job or process, using the most suitable data collection techniques, so that the data will be in the most convenient form to be analysed
- **Examine** the recorded facts critically and challenge everything that is done including the purpose of the activity, the place where it’s performed, the sequence in which it is done, the person who is doing it and the means by which it is done
- **Develop** the most economical method, taking into account all the circumstances
- **Evaluate** the results attained by the improved method compared with the quantity of work

involved and calculate a standard time for it

- **Define** the new method and present it to all concerned, either verbally or in writing, using demonstrations
- **Install** the new method, training those involved, as an agreed practice with allocated time of operation
- **Maintain** the new standard practice by monitoring the results and comparing them with the original targets

Work Measurement Techniques

These include:

- **Activity Sampling (Work Sampling)**

Activity sampling is a method of finding a percentage occurrence of a certain activity by statistically sampling and random observations. The basis of activity sampling is to observe numerous machines or workers at random intervals, noting which are working and which are idle. The more frequent the observations, the more accurate the results.

- **Time Studies**

A time study procedure involves timing a sample of a worker's performance and using it to set a standard.

- **Predetermined Time Studies**

This is an approach that divides manual work into small basic elements that have been established with widely accepted times.

- **Synthesis**

A work measurement technique used to build up the time for a task at a defined rate of work using previously established elemental times.



Activity 8.5

List the various Work Measurement techniques

Method Study

Method Study is the systematic recording and critical examination of ways of doing things in order to make improvements. The basic approach to method study consists of the following steps:

- Select the work to be studied and define its boundaries

- Record the relevant facts about the job by direct observation and collect such additional data as may be needed from appropriate sources
- Examine the way the job is being performed and challenge its purpose, place, sequence and method of performance
- Develop the most practical, effective and economical method, seeking help from those concerned
- Evaluate the various alternatives in order to develop a new improved method. Cost these alternatives to see which yields the best results
- Define the new method in a clear manner and communicate it to all concerned – workers, supervisors, management
- Install the new method as standard practice and train persons involved in applying it
- Maintain the new method and install control procedures to ensure that the previous method has been changed

Knowledge Check Questions 8.1

State if the following is True or False by selecting the correct answer.

1. Effective ergonomic practices can reduce workplace injuries and improve overall organisational effectiveness

True

False

2. Implementing ergonomic solutions has little impact on employee satisfaction and retention.

True

False

3. Job design that incorporates ergonomic principles can lead to a decrease in absenteeism and an increase in productivity

True

False



Knowledge Check Questions 8.2**Select the correct answer:**

1. Which of the following is a primary goal of ergonomics in the workplace from an HR perspective?
 - a. Reducing production costs
 - b. Enhancing employee well-being and productivity
 - c. Increasing product variety
 - d. Minimising employee training time

2. In job design, which approach focuses on redesigning tasks to make them more engaging and less physically demanding for employees?
 - a. Job enlargement
 - b. Job enrichment
 - c. Job simplification
 - d. Job rotation

8.9 Summary

This unit provided an in-depth exploration of how human resource management contributes to an organisation's competitive advantage within Operations Management. The key topics covered included examining external and internal factors such as labour laws, union regulations, market conditions, and technological changes that impact HR decision-making, understanding workforce demand forecasting, scheduling, and capacity planning to align labour supply with operational requirements and exploring techniques such as job enrichment, job rotation, and job enlargement to improve employee engagement and productivity.

The key explanations also included so analysing the role of autonomous workgroups in improving collaboration, problem-solving, and process innovation, evaluating financial and non-financial incentives that enhance employee performance, including performance-based pay, recognition programs, job satisfaction strategies and applying ergonomic principles and efficient work designs to improve workplace safety, reduce fatigue, and enhance productivity.

Through real-world case studies and practical applications, students gained insights into how workforce management decisions impact operational efficiency and long-term competitiveness. By integrating human resource planning with Operations Management principles, organisations can optimise performance, reduce costs, and enhance employee satisfaction.

Revision Questions

1. Kanawaty, G. (2020) in his book “Introduction to Work Study” laid much of the foundation for recognising the work force and critical to the sustainability of the business.

Critically analyse the basic procedure he proposes for a work study.

2. Read the following scenario and answer the questions that follow:

Company B has a diverse workforce with varying motivational drivers. The management wants to implement a motivation strategy that caters to different employee needs.

a. Determine the strategies that Company B can use to cater to the diverse motivational needs of its workforce?

b. Analyse how Company B can measure the effectiveness of its new motivation strategy?

c. Discuss the role that leadership plays in the success of the motivation strategy at Company B, and how can leaders be equipped to support it?

3. Read the following scenario and answer the questions that follow:

Company CEB LTD is launching a new product and needs to ensure that its production team is highly motivated to meet tight deadlines and high-quality standards.

a. Determine the specific incentives that Company CEB LTD can offer to motivate the production team during the new product launch?

b. Discuss how Company CEB LTD can maintain high motivation levels among the production team throughout the entire product launch process?

c. Examine the steps that Company CEB LTD can take to evaluate the success of its motivational efforts during the product launch?

4. Determine what influences to job satisfaction

5. Explain if money is an incentive and if there are any drawbacks to purely financial incentives



Answers

Activity 8.1

Outline your understanding of a 'good quality of work life'

Answer: A good quality of work life refers to the overall well-being of employees in their work environment, encompassing both their professional and personal satisfaction. A good quality of work life enhances employee motivation, job satisfaction, productivity, and overall well-being, contributing to a positive and sustainable work environment.

The following list is not exhaustive some students may mention factors not outlined below:

1. Healthy Work Environment:

- Safe, comfortable, and supportive physical and psychological conditions at work

2. Work-Life Balance:

- The ability to balance work responsibilities with personal life, including flexible work hours and reasonable workload

3. Job Security:

- Stability and assurance of continued employment, reducing stress and uncertainty

4. Fair Compensation:

- Competitive pay and benefits that reflect the value of the work performed

5. Opportunities for Growth:

- Access to career development, training, and clear paths for advancement

6. Job Satisfaction:

- Engagement in meaningful, challenging work that aligns with personal skills and interests

7. Supportive Leadership:

- Management that is approachable, communicative, and supportive of employees' needs and aspirations

8. Positive Work Relationships:

- Strong, collaborative relationships with colleagues and a sense of community at work

9. Recognition and Appreciation:

- Regular acknowledgement of achievements and contributions, fostering a sense of value

10. Organisational Culture:

- A culture that aligns with personal values and promotes well-being, inclusion, and a shared sense of purpose

Activity 8.2

Differentiate between job enlargement, job rotation, job specialisation and employee empowerment.

Answer: In recent years, there has been an effort to improve the quality of work life by moving from labour specialisation towards more varied job design. Driving this effort is the theory that variety makes the job more enjoyable, and that the employee therefore enjoys a higher quality of work life. It is assumed that this flexibility thus benefits the employee and the organisation.

We modify jobs in a variety of ways. The first approach is job enlargement, which occurs when we add tasks requiring similar skill to an existing job. Job rotation is a version of job enlargement that occurs when the employee is allowed to move from one specialised job to another. Variety has been added to the employee's perspective of the job. Another approach is job enrichment, which adds planning and control to the job. An example is to have department store salespeople responsible for ordering, as well as selling, their goods. Job enrichment can be thought of as vertical expansion, as opposed to job enlargement, which is horizontal.

A popular extension of job enrichment, employee empowerment is the practice of enriching jobs so that employees accept responsibility for a variety of decisions normally associated with staff specialists. Empowering employees helps them take "ownership" of their jobs so that they have personal interests in improving performances. In short, this is how they could be distinguished:

1. Job enlargement expands the number of tasks.
2. Job rotation varies tasks by rotating roles.
3. Job specialisation narrows tasks to focus on specific skills.
4. Employee empowerment increases autonomy and decision-making authority

Activity 8.3**Explain the importance of teamwork in improving efficiency**

Answer: It enhances collaboration, increases efficiency, and improves problem-solving.

Activity 8.4**1. Explain how HR departments can leverage ergonomics to improve organisational effectiveness?**

Answer: HR departments can leverage ergonomics by ensuring workstations and tools are designed to reduce strain and fatigue, promoting employee health and safety. This leads to fewer injuries, higher employee satisfaction, and improved productivity, all of which enhance organisational effectiveness.

2. Determine why it is important to consider ergonomic principles in job design, and what are the potential outcomes?

Answer: Considering ergonomic principles in job design is important because it helps create tasks that are physically comfortable and safe for employees, reducing the risk of injury. Potential outcomes include increased job satisfaction, higher productivity, lower absenteeism, and reduced turnover rates.

3. Describe the relationship between ergonomics, employee well-being, and organisational effectiveness

Answer: Ergonomics directly impacts employee well-being by reducing physical strain and preventing injuries. When employees feel comfortable and safe, their job satisfaction and productivity increase. This, in turn, leads to improved organisational effectiveness, as healthy and satisfied employees are more engaged and efficient.

4. Explain how financial incentives impacts employee motivation and organisational performance?

Answer: Financial incentives, such as bonuses, commissions, and profit-sharing, can significantly boost employee motivation by directly linking performance to rewards. This can lead to increased productivity, higher levels of engagement, and improved organisational performance as employees are more likely to go above and beyond to achieve their targets.

5. Discuss the role of non-financial incentives in employee motivation, and how can they be effectively implemented?

Answer: Non-financial incentives, such as recognition programs, professional development opportunities, flexible work arrangements, and career advancement prospects, play a crucial role in motivating employees by addressing intrinsic needs for appreciation, growth, and work-life balance. They can be effectively implemented by creating a culture of recognition, providing ongoing learning opportunities, and offering clear pathways for career progression.

6. Examine how HR can design incentive programs to align with organisational goals and enhance employee motivation?

Answer: HR can design incentive programs to align with organisational goals by ensuring that the incentives are directly tied to the achievement of specific, measurable objectives that support the company's strategic priorities. This can be done by setting clear performance metrics, regularly reviewing and adjusting the incentive schemes, and ensuring that employees understand how their contributions impact the organisation's success. Additionally, combining both financial and non-financial incentives can address a broader range of motivational drivers.

Activity 8.5

List the various Work Measurement techniques

Answer: The work measurement techniques include:

- Time studies
- Predetermined motion time systems
- Work sampling

Case Study 8.1

1. Discuss how Unilever utilised AI and automation in their recruitment process?

Answer: Unilever has utilised AI and automation by partnering with Pymetrics and HireVue. These technologies help streamline the hiring process by coordinating, filtering, and interviewing candidates through machine analysis of video-based interviews.

2. Outline the benefits that Unilever has experienced by using AI/automation in their recruitment process?

Answer: By using AI and automation, Unilever saved over 50,000 hours in candidate interview time, achieved more than £1 million in annual savings, and improved candidate diversity

3. Explain why it was necessary for Unilever to implement AI and automation in their recruitment process?

Answer: With 30,000 new recruits annually and around 1.8 million job applications to process, managing each application was a massive task. Implementing AI and automation was necessary to make the recruitment process easier, more cost-effective, time-saving, and efficient.

Knowledge Check Questions 8.1

1. Effective ergonomic practices can reduce workplace injuries and improve overall organisational effectiveness

Answer: True

2. Implementing ergonomic solutions has little impact on employee satisfaction and retention.

Answer: False

3. Job design that incorporates ergonomic principles can lead to a decrease in absenteeism and an increase in productivity

Answer: True

Knowledge Check Questions 8.2

1. Which of the following is a primary goal of ergonomics in the workplace from an HR perspective?

Answer: b. Enhancing employee well-being and productivity

2. In job design, which approach focuses on redesigning tasks to make them more engaging and less physically demanding for employees?

Answer: b. Job enrichment

Practical Example 8.1

1. Explain the factors that HR should consider when designing an effective incentive program to improve productivity and morale at Company XYZ (Pty) Ltd?

Answer: HR should consider factors such as employee preferences, the company's strategic goals, the balance between financial and non-financial incentives, and the need for fairness and transparency. They should also gather employee feedback to ensure the program addresses their motivational needs and aligns with their values.

2. Determine how Company XYZ (Pty) Ltd can ensure that the new incentive program is fair and transparent to all employees?

Answer: Company XYZ (Pty) Ltd can ensure fairness and transparency by clearly communicating the criteria for earning incentives, involving employees in the design process, setting achievable and measurable goals, and regularly reviewing and adjusting the program based on feedback and performance data.

3. Discuss the potential challenges might Company XYZ (Pty) Ltd face when implementing the new incentive program, and how can they be mitigated?

Answer: Potential challenges include resistance to change, perceived unfairness, and administrative complexity. These can be mitigated by engaging employees early in the process, providing thorough training and communication, ensuring the program is easy to understand and administer, and continuously monitoring and adjusting the program based on feedback and outcomes.

Think Point 8.1

Analyse the factors that contribute to job satisfaction

Answer: Fair compensation, growth opportunities, work-life balance, and supportive management.

Think Point 8.2

Explain if money is an incentive and if there are any drawbacks to purely financial incentives.

Answer: Money as an effective incentive but may not address intrinsic motivators like job satisfaction. The drawbacks to financial incentives include creating competition and conflict among employees, encouraging short-term focus and unethical behavior, and reducing intrinsic motivation and creativity.

Think Point 8.3

Explain how we establish a “fair day’s” work

Answer: Establishing a "fair day's work" involves determining a reasonable amount of work that an average employee should be able to complete in a standard workday, under normal working conditions. This process ensures that expectations are realistic, achievable, and aligned with both employee well-being and organisational goals. The steps typically include:

- Compare the efficiency of different methods
- Balance the work of team members
- Determine the ideal number of machines a worker can operate
- Provide the basis for planning and control
- Choose an improved layout for process planning and for establishing a Just-in-Time system
- Provide information for tender estimates, selling prices and delivery dates

- Set standards for machine utilisation and labour performance that can be used for incentive schemes
- Provide information for labour cost control and to enable standard costs to be fixed and maintained

Video Activity 8.1

1. Define ergonomics and explain why it is important in the context of Operations Management.

Answer: Ergonomics is the science of designing workspaces and tools to fit the needs and capabilities of people, aiming to minimise discomfort and reduce the risk of injury. In Operations Management, ergonomics is crucial because it enhances worker safety, reduces fatigue, increases productivity, and minimises downtime related to injury or discomfort.

2. Identify three ergonomic tips discussed in the video that can help reduce strain for employees working at desks. Describe how each of these tips contributes to workplace efficiency and employee health.

Answer: Examples may include:

- Chair adjustments: Adjusting the chair height so that feet are flat on the floor and knees are at a right angle. This promotes proper posture, reducing back strain and discomfort, which can lead to fewer sick days and higher productivity
- Screen position: Positioning the screen at eye level helps avoid neck strain, which keeps employees more comfortable and focused, reducing the likelihood of repetitive strain injuries
- Breaks and stretching: Taking regular breaks to stretch can prevent stiffness and improve circulation, keeping employees alert and less prone to fatigue throughout the day

3. Create an ergonomic checklist for setting up a computer workstation. Include at least five items based on the tips provided in the video.

Answer:

- Ensure the chair height allows feet to be flat on the floor
- Set the monitor at eye level to avoid neck strain
- Position the keyboard and mouse so elbows are at a 90-degree angle
- Keep frequently used items within arm's reach to prevent overextension
- Encourage taking breaks every hour to stretch and move around

4. Discuss the potential costs associated with poor ergonomics in a workplace from an OM perspective. What might be the long-term implications if these issues aren't addressed?

Answer: Poor ergonomics can lead to increased absenteeism due to injury, reduced productivity from discomfort, and potential workers' compensation claims. In the long term, it may also result in higher employee turnover and recruitment costs as employees seek more comfortable work environments, ultimately affecting the company's operational efficiency and profitability.

5. Propose a small improvement project an operations manager could implement to improve ergonomics in a workplace where employees work on computers all day. Outline the steps and potential benefits of your project.

Answer: An example project could be the "Ergonomic Workspace Enhancement Project":

Step 1: Conduct a survey to identify common ergonomic discomforts among employees.

Step 2: Arrange ergonomic assessments of each workspace and recommend adjustments.

Step 3: Provide ergonomic equipment, like adjustable chairs and monitor stands.

Step 4: Implement a training session on ergonomic best practices.

Benefits: Increased productivity, reduced injury-related absenteeism, improved employee morale, and lower long-term healthcare costs

**Unit
9:**

**Supply Chain
Management**

Unit Learning Outcomes

CONTENT LIST	LEARNING OUTCOMES FOR THIS UNIT
9.1 Introduction	Introduces the topic areas covered for the unit
9.2 Supply-Chain Management	Explore the concepts of supply chain management
9.3 Global Supply-Chain Issues	Examine supply chain issues as a barrier to globalisation
9.4 Purchasing	Analyse make or buy decisions
9.5 Supply-Chain Strategies	Examine the different supply chain strategies available
9.6 Vendor Selection	Relate the vendor selection criterion to ensure a reliable vendor database
9.7 Materials Management	Analyse the different requirements that enable profitable materials management
9.8 Summary	Summarises the topic areas covered for the unit

Prescribed and Recommended Reading(s)/Textbook(s)

Prescribed Reading(s)/Textbook(s)

- Heizer, J., Render, B. and Munson, C. (2020). *Operations Management: Sustainability and Supply Chain Management*. Thirteenth Edition. United Kingdom: Pearson.

Recommended Reading(s)/Textbook(s)

- Venkataraman, R.R. and Pinto, J.K. (2020). *Operations Management: Managing Global Supply Chains*. Second Edition. Sage Publications.
- Parksoy, T. and Deveci, M. (2023). *Smart and Sustainable Operations and Supply Chain Management in Industry 4.0*. First Edition. Boca Raton: CRC Press.
- Slack, N., Jones, B. and Burgess, N. (2022). *Operations Management*. Tenth Edition. United Kingdom: Pearson.
- Kareska, PhD Katerina. 'Operational Management Strategies for Sustainable Productivity Growth' (August 5, 2023). Available at SSRN: <https://ssrn.com/abstract=4532740> or <http://dx.doi.org/10.2139/ssrn.4532740>
- 'The Impact Of Ergonomics On Quality Of Life In The Workplace'. (2023). https://www.researchgate.net/publication/377267055_The_Impact_of_Ergonomics_on_Quality_of_Life_in_the_Workplace

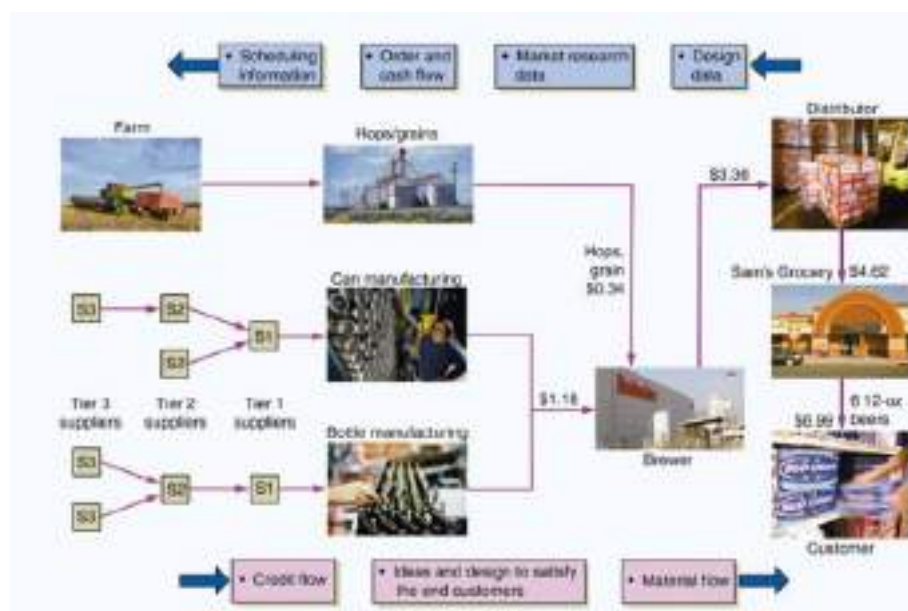


9.1 Introduction

Supply-chain management is the integration of the activities that procure materials, transform them into products and deliver them to the customer via a distribution system. Supply-chain management includes:

- Transportation
- Credit and cash transfers
- Suppliers
- Distributors and banks
- Accounts payable and receivable
- Warehousing and inventory levels
- Order fulfilment
- Forecasting
- Production

As companies strive to improve their competitiveness, quality, cost reductions and speed to market, they place added emphasis on supply-chain management. The key to effective supply-chain management is to make suppliers partners.



Source: Heizer.et.al (2020)

Let's examine how supply-chain can support an organisation's overall strategy:

	LOW-COST STRATEGY	RESPONSE STRATEGY	DIFFERENTIATION STRATEGY
Supplier's goal	Supply demand at lowest possible cost	Respond quickly to changing requirements	Jointly develop products and options
Primary selection criteria	Select primarily for cost	Select for capacity, speed and flexibility	Select for product development skills
Process characteristics	Maintain high utilisation	Invest in excess capacity and flexible processes	Develop modular processes
Inventory characteristics	Minimise inventory throughout the chain	Develop responsive systems with buffer stock	Minimise inventory to avoid obsolescence
Lead time characteristics	Shorten lead times	Reduce production lead-time	Reduce development lead-time
Product design characteristics	Maximise performance and Minimise costs	Reduce set-up times	Use modular design

Source: Heizer.et.al (2020)

Practical Application 9.1

Watch the following video and answer the following Questions:

Available from: <https://youtu.be/Lpp9bHtPAN0?si=hiyMgrVh3bWvB-k4>

Questions:

1. Examine the primary causes of supply chain disruptions for Apple's iPhone 13 Pro, as discussed in the video? How do these causes illustrate the complexities of global supply chains?
2. Describe one strategy Apple might use to mitigate the impact of supply chain delays and production cuts for future product releases. How would this strategy improve resilience in their supply chain?
3. Imagine you are a Supply Chain Manager at Apple facing similar challenges. Determine the steps you would take to optimise the flow of components and reduce delays, and how would these steps impact customer satisfaction?



9.2 Global Supply-Chain Issues

When companies enter the global market, expanding their supply chains becomes a strategic challenge. Distribution systems in certain countries may be inferior or unreliable. Additionally, companies may be faced with tariff quotas. Furthermore, market instabilities, such as the devaluation of the rand, are common in newly emerging industrial economies. Heizer et al (2020) suggests that supply-chains in a global environment must be:

- Flexible enough to react to sudden changes in parts availability, distribution or shipping channels, import duties and currency rates
- Able to use latest computer and transmission technologies to manage the shipment of parts
- Staffed with local specialists to handle duties, trade, freight, customs and political issues

Case Study 9.1

Read the following article and answer the questions below.

Amazon's Competitive Edge in Supply Chain Management

Amazon has established itself as a global leader in e-commerce by leveraging an advanced and highly efficient supply chain. The video discusses various strategies and innovations that Amazon employs to stay ahead in the competitive retail landscape. These include the use of cutting-edge technology, strategic warehouse placements, and a focus on customer satisfaction.

Technological Integration: Amazon utilises sophisticated algorithms, artificial intelligence (AI), and machine learning to optimise its supply chain operations. These technologies help predict customer demand, manage inventory levels, and streamline delivery routes.

Strategic Warehousing: The company has a vast network of fulfilment centres strategically located near key markets. This proximity allows Amazon to offer rapid delivery services, including same-day and next-day delivery options.

Automation and Robotics: Amazon's warehouses are equipped with advanced robotics that enhance efficiency by automating various tasks such as picking, packing, and sorting items. This reduces labour costs and improves order accuracy.

Prime Membership Program: The Amazon Prime program plays a crucial role in maintaining customer loyalty. Members receive benefits like free and fast shipping, which is made possible by Amazon's robust supply chain.



Sustainability Initiatives: Amazon is also focusing on sustainability by investing in electric delivery vehicles and renewable energy sources for its operations, aiming to reduce its carbon footprint.

Chopra, S., & Meindl, P. (2020). *Supply Chain Management: Strategy, Planning, and Operation*. Pearson.

You may watch the video for more insight:

Available from: <https://youtu.be/HxXJ8Q2GCs4>

Questions:

1. Determine how Amazon's use of advanced technology contributes to its supply chain efficiency and customer satisfaction?
2. Analyse the impact of Amazon's strategic warehousing on its delivery capabilities and overall competitive advantage.
3. Discuss the role of automation and robotics in Amazon's supply chain and its impact on operational efficiency.



Think Point 9.1

Explain what supply-chain strategy you would use, if your company were to enter the global market.

9.3. Purchasing

Purchasing can be defined as the acquisition of goods and services. For both goods and services, the cost of purchases as a percent of sales is often substantial. Since such a huge portion of revenue is devoted to purchasing, an effective purchasing strategy is vital. Purchasing provides a major opportunity to reduce costs and increase profit margins. The objective of purchasing activity is:

- To help identify the products and services that can be obtained externally
- To develop, evaluate and determine the best supplier, price and delivery for those products and services

Make-or-Buy Decisions

This entails choosing between producing a component or service or purchasing it from an outside source. The purchasing department's role is to evaluate alternative suppliers and provide current, accurate and complete data relevant to the buy alternative. The Table below highlights considerations for the make-or buy decision.

Reasons for making	Reasons for buying
1. Maintain core competence	1. Frees management to deal with its primary business
2. Lower production cost	2. Lower acquisition costs
3. Unsuitable suppliers	3. Preserve supplier commitment
4. Assure adequate supply (quantity or delivery)	4. Obtain technical or management ability
5. Utilise surplus labour or facilities	5. Inadequate capacity
6. Obtain desired quality	6. Reduce inventory costs
7. Remove supplier collusion	7. Ensure alternative sources
8. Obtain unique item that would entail a prohibitive commitment for a supplier	8. Inadequate managerial or technical resources
9. Protect personnel from a layoff	9. Reciprocity
10. Protect proprietary design or quality	10. Item is protected by a patent or trade secret
11. Increase or maintain the size of the company	

Source: Heizer.et.al (2020)



Activity 9.1

Explain if it would be easier for you to make or buy in terms of your operation.

9.4 Supply-Chain Strategies

Heizer.et.al (2020) maintains the following strategies are being used:

Many Suppliers

With the many-supplier strategy, the supplier responds to demands and specification of a 'request for quotation', with the order usually going to the lowest bidder. This strategy plays one supplier against another. Suppliers tend to compete aggressively with each other. This approach holds the supplier responsible for maintaining the necessary technology, expertise, quality, cost and delivery schedules.

Few Suppliers

A strategy of few suppliers implies that rather than looking for short-term attributes, such as low cost, a buyer is better off forming a long-term relationship with a few dedicated suppliers. Using few suppliers can create value by allowing suppliers to have economies of scale. Few suppliers, each with large commitment to the buyer, may also be more willing to participate in JIT systems, as well as provide innovations and technological expertise.



Video Activity 9.1

Watch the video using the link

Toyota changed the supply chain landscape

Available from: <https://youtu.be/0wVAPXswE6E>

Vertical Integration

This simply means developing the ability to produce goods or services previously purchased or actually buying out (acquiring) a supplier or distributor. This strategy has the advantage of improving research and development, quality and product flexibility.

Keiretsu Networks

Manufacturers sometimes offer financial support to their suppliers through loans. The supplier then becomes part of the company coalition known as Keiretsu. Members of Keiretsu are assured long-term relationships and are therefore to function as partners.

Virtual Companies

Virtual companies rely on a variety of supplier relationships to provide services on demand. These companies have fluid, moving organisational boundaries that allow them to create unique enterprises in order to meet changing market demands. These relationships may provide a variety of vendor services such as doing the payroll, recruiting personnel, designing products, providing consulting services, manufacturing components, conducting tests or distributing products.



Think Point 9.2

Determine how virtual companies leverage their operational flexibility and scalability to optimise supply chain management?



Think Point 9.3

Describe the strategies that virtual companies can employ to manage risks and enhance resilience in their supply chain operations?

Practical Application 9.2

Read and answer the following questions:

You are the Materials Manager for a manufacturing company, ABC Industries, which produces high-quality electronic components for the automotive sector. ABC Industries spends over 60% of its capital on raw materials, such as metals and microchips, many of which have long lead times and are costly to store. Recently, ABC has been facing frequent production delays due to shortages of key materials and overstock of less critical ones, leading to increased costs and reduced efficiency. The company's leadership has tasked you with improving materials management processes to reduce costs and avoid these delays.



Questions:

1. Given the challenges ABC Industries is facing with material shortages and overstock, what materials management strategies could you implement to balance inventory levels and reduce lead times? Explain how each strategy would address these issues.
2. ABC Industries has been asked to cut down on its storage costs without risking production interruptions. Propose a solution that would help ABC reduce inventory holding costs while ensuring materials are available for production.
3. Imagine that a key supplier of microchips is facing global shortages, increasing lead times and costs for ABC Industries. How would you manage this risk to prevent disruption in ABC's production schedule?

9.5. Vendor Selection

A firm that decides to buy components rather than make them, must select suitable vendors. Vendor selection considers numerous factors, such as:

- Inventory and transportation costs
- Availability of supply
- Delivery performance
- Quality and reputation of suppliers
- Financial strength of the supplier
- Manufacturing range

- Technical assistance
- After-sales service
- Labour/trade relations
- Packaging
- Warranties and guarantees

Activity 9.2

1. Explain how a supply chain can improve competitive advantage.

Answer the following scenario-based questions:

2. A manufacturing company is looking to expand its supplier base to reduce risk. They are evaluating potential vendors based on quality, delivery performance, and cost.

Discuss some of the additional criteria the companies should consider when selecting new vendors to ensure a robust and reliable supply chain?

3. An electronics firm faces frequent supply chain disruptions due to the volatility of demand in the market. They are considering adopting an agile supply chain strategy.

Discuss how an agile supply chain strategy helps the electronics firm better manage demand volatility and improve responsiveness?

4. A retailer aims to implement a lean supply chain strategy to reduce costs and improve efficiency. They are currently dealing with high inventory levels and slow turnover rates.

4. Discuss the steps that the retailer should take to transition to a lean supply chain strategy and achieve their goals?



9.6 Materials Management

The purpose of materials management is to obtain efficiency of operations through the integration of all material purchases, movement and storage activities. If transportation and inventory costs are high, the emphasis should be placed on materials management. Due to the high costs of moving materials, firms constantly evaluate their means of distribution.

Companies usually move materials through the following channels:

Road

Trucks move a large amount of materials. Companies have put pressure on truckers to pick up and deliver on time, with no damage to goods. Trucking firms are using computer technology to monitor

driving, weather patterns, loading and unloading methods, reducing fuel consumption and finding the most effective routes.

Rail

Containerisation has made inter-modal shipping of truck trailers on railroad flat cars a popular means of distribution. With the growth of JIT, rail transportation has suffered a setback because small-batch manufacturers require frequent, smaller shipments.

Airfreight

Seldom used because of the high cost. It is used mainly for national or international movement of lightweight items such as medical and emergency supplies, flowers, documents, etc. Airfreight offers speed and reliability.

Shipping

Shipping remains one of the oldest means of freight transportation. The usual cargo on ships are bulky products such as iron-ore, grains, cement, coal, chemicals and petroleum products.

Pipelines

Pipelines are an important form of transporting crude oil, natural gas and other petroleum and chemical products.



Activity 9.3

Analyse the method of transportation you are currently using and explain, in your opinion, if this is the best mode. Suggest practical alternatives, if any

Knowledge Check Questions 9.1**Select the correct answer:**

1. Which of the following is a key criterion for selecting vendors in a supply chain?
 - a. Marketing skills
 - b. Financial stability
 - c. Social media presence
 - d. Office location
2. In supply chain strategies, which approach focuses on forming close relationships with a few key suppliers?
 - a. Competitive bidding
 - b. Single sourcing
 - c. Multiple sourcing
 - d. Dual sourcing
3. Which supply chain strategy emphasises the importance of reducing inventory levels and improving efficiency?
 - a. Lean supply chain
 - b. Agile supply chain
 - c. Resilient supply chain
 - d. Green supply chain



Knowledge Check Questions 9.2

State if the following is True or False by selecting the correct answer.



1. Vendor selection should prioritise price over all other factors to ensure cost savings.

True

False

2. An agile supply chain strategy is best suited for industries with stable and predictable demand.

True

False

3. Developing long-term relationships with suppliers can lead to improved quality and reliability in the supply chain.

True

False

9.7 Summary

The supply chain is a critical element of operations management that directly impacts an organisation's efficiency, cost-effectiveness and ability to meet customer demands. Effective supply chain management involves strategic decisions in areas like purchasing, vendor selection, and the implementation of supply-chain strategies to ensure the smooth flow of goods and services. However, global supply-chain issues have highlighted the importance of flexibility and risk mitigation. In stating so, managing the complexities of the supply chain is essential for long-term success and sustainability in operations management.

Revision Questions



1. Determine the sales necessary to equal a dollar of savings on purchases for a company that has a net profit of 6% and spends 70% of its revenues on purchases.
2. Determine the sales necessary to equal a dollar of savings on purchases for a company that has a net profit of 8% and spends 40% of its revenues on purchases.
3. Phil Carter, President of Carter Computer Components, Corp. has the option of shipping computer transformers from its Singapore plant via container ship or airfreight. The typical shipment has a value of R75,000. A container ship takes 24 days and costs R5,000; airfreight takes 1 day and costs R8,000. Holding cost is estimated to be 40% in either case. Explain how shipments should be made.

Answers

Activity 9.1

Explain if it would be easier for you to make or buy in terms of your operation.

Answer: To be discussed during the webinar(s) with lecturer.

Activity 9.2

1. Explain how a supply chain can improve competitive advantage.

Answer: Efficient supply chains reduce costs and improve product delivery times.

2. Discuss some of the additional criteria the companies should consider when selecting new vendors to ensure a robust and reliable supply chain?

Answer: The company should consider criteria such as the vendor's financial stability, technological capabilities, ability to scale production, compliance with regulatory standards, and overall reputation in the industry. Evaluating these factors will help ensure that the selected vendors can consistently meet the company's requirements and contribute to a resilient supply chain.

3. Discuss how an agile supply chain strategy helps the electronics firm better manage demand volatility and improve responsiveness?

Answer: An agile supply chain strategy can help the electronics firm by enhancing its ability to quickly adapt to changes in demand. This includes maintaining flexible production processes, utilising real-time data for decision-making, and collaborating closely with suppliers to ensure rapid response to market shifts. By adopting agility, the firm can reduce lead times, minimise stock-outs, and better align supply with fluctuating demand.

4. Discuss the steps that the retailer should take to transition to a lean supply chain strategy and achieve their goals?

Answer: The retailer should start by conducting a thorough analysis of their current inventory levels and identifying areas of waste. Implementing just-in-time (JIT) inventory practices, improving demand forecasting accuracy, and enhancing communication with suppliers are key steps. Additionally, they should streamline their internal processes, reduce lead times, and focus on continuous improvement to eliminate inefficiencies and achieve a lean supply chain.

Activity 9.3

Analyse the method of transportation you are currently using and explain, in your opinion, if this is the best mode. Suggest practical alternatives, if any

Answer: To be discussed during the webinar(s) with lecturer.

Case Study 9.1

1. Determine how Amazon's use of advanced technology contributes to its supply chain efficiency and customer satisfaction?

Answer: Amazon's integration of advanced technology such as AI and machine learning significantly enhances supply chain efficiency by accurately predicting customer demand and optimising inventory management. These technologies enable Amazon to forecast sales trends, reduce stock-outs, and ensure that popular items are always available. Additionally, AI-driven algorithms optimise delivery routes, minimising delivery times and costs. This technological prowess translates into high levels of customer satisfaction as customers receive their orders quickly and reliably, reinforcing Amazon's reputation for excellent service.

2. Analyse the impact of Amazon's strategic warehousing on its delivery capabilities and overall competitive advantage.

Answer: Amazon's strategic placement of fulfilment centres near major markets is pivotal to its delivery capabilities. By positioning warehouses close to urban centres and high-demand areas, Amazon reduces the distance products need to travel, enabling faster delivery times. This network supports services like same-day and next-day delivery, which are critical for maintaining customer satisfaction and loyalty. The proximity also lowers transportation costs and enhances operational efficiency. This strategic warehousing approach gives Amazon a significant competitive edge over rivals who may not have as extensive or optimally located fulfilment networks, allowing Amazon to offer superior delivery services at competitive prices.

3. Discuss the role of automation and robotics in Amazon's supply chain and its impact on operational efficiency.

Answer: Automation and robotics are integral to Amazon's supply chain, driving significant improvements in operational efficiency. The use of robots in fulfilment centres automates repetitive tasks such as picking, packing, and sorting, which speeds up the order fulfilment process and reduces human error. This automation allows Amazon to handle large volumes of orders efficiently, especially during peak shopping periods like holidays. Furthermore, automation reduces labour costs and enhances scalability, enabling Amazon to quickly adapt to changing demand without

compromising service quality. The efficiency gains from automation and robotics support Amazon's ability to maintain low prices and fast delivery times, solidifying its market leadership.

Knowledge Check Questions 9.1

1. Which of the following is a key criterion for selecting vendors in a supply chain?

Answer: b. Financial stability

2. In supply chain strategies, which approach focuses on forming close relationships with a few key suppliers?

Answer: b. Single sourcing

3. Which supply chain strategy emphasises the importance of reducing inventory levels and improving efficiency?

Answer: a. Lean supply chain

Knowledge Check Questions 9.2

1. Vendor selection should prioritise price over all other factors to ensure cost savings.

Answer: False

2. An agile supply chain strategy is best suited for industries with stable and predictable demand.

Answer: False

3. Developing long-term relationships with suppliers can lead to improved quality and reliability in the supply chain.

Answer: True

Practical Application 9.1

1. Examine the primary causes of supply chain disruptions for Apple's iPhone 13 Pro, as discussed in the video? How do these causes illustrate the complexities of global supply chains?

Answer: The primary causes of disruptions were supply delays and production cuts due to factors like global semiconductor shortages, COVID-19 restrictions, and logistic bottlenecks. These issues highlight global supply chain complexities as they show how a single disruption, such as a delay in chip manufacturing, can affect multiple stages of production and lead to delays in delivering products to the end consumer. It also emphasises the dependence on global suppliers and the challenges of

managing components sourced from different countries.

2. Describe one strategy Apple might use to mitigate the impact of supply chain delays and production cuts for future product releases. How would this strategy improve resilience in their supply chain?

Answer: Apple could use a dual-sourcing strategy for critical components, such as chips, by partnering with multiple suppliers to reduce dependency on a single source. This approach would increase resilience by providing backup options if one supplier faces delays. Additionally, Apple could maintain strategic inventory reserves of essential components, ensuring they have enough stock to buffer against short-term disruptions, which would allow production to continue with minimal delays.

3. Imagine you are a Supply Chain Manager at Apple facing similar challenges. Determine the steps you would take to optimise the flow of components and reduce delays, and how would these steps impact customer satisfaction?

Answer:

- Optimise transportation logistics by collaborating with logistics partners to prioritise Apple shipments and improve efficiency
- Increase transparency by implementing real-time tracking systems for component shipments, allowing better monitoring and proactive response to delays
- Negotiate contracts with suppliers that include flexibility for adjustments based on demand fluctuations

Practical Application 9.2

1. Given the challenges ABC Industries is facing with material shortages and overstock, what materials management strategies could you implement to balance inventory levels and reduce lead times? Explain how each strategy would address these issues.

Answer: Two effective strategies could be:

- Demand Forecasting and Inventory Planning: By analysing historical data and customer orders, ABC can predict material needs more accurately. This would allow the company to purchase materials based on actual demand trends, reducing the chances of both overstocking and shortages
- Supplier Relationship Management (SRM): Building stronger relationships with suppliers can help in negotiating faster delivery times or setting up consignment inventory options, where

suppliers hold inventory until it's needed. SRM would help reduce lead times and ensure that critical materials are available as needed, minimising delays

2. ABC Industries has been asked to cut down on its storage costs without risking production interruptions. Propose a solution that would help ABC reduce inventory holding costs while ensuring materials are available for production.

Answer: Implementing a Just-In-Time (JIT) Inventory System would be effective. Under JIT, materials are ordered and received only shortly before they are needed in production, thus reducing the amount of inventory stored at any time. For ABC Industries, this approach would lower storage costs, reduce capital tied up in inventory, and free up warehouse space. To ensure production is not interrupted, close communication with suppliers and reliable transportation are crucial, allowing ABC to receive materials promptly as needed.

3. Imagine that a key supplier of microchips is facing global shortages, increasing lead times and costs for ABC Industries. How would you manage this risk to prevent disruption in ABC's production schedule?

Answer: To manage this risk, I would employ Dual Sourcing and Strategic Safety Stock strategies:

- **Dual Sourcing:** By identifying and onboarding an additional supplier for microchips, ABC can reduce its dependency on one source. This strategy adds flexibility and redundancy, which is crucial during supply shortages
- **Strategic Safety Stock:** ABC could maintain a small buffer stock of microchips to cover production needs during lead time delays. Although safety stock requires additional storage, its strategic use in limited quantities could prevent costly production stoppages, especially for critical components like microchips

Think Point 9.1

Explain what supply-chain strategy you would use, if your company were to enter the global market.

Answer: To be discussed during the webinar(s) with lecturer.

Think Point 9.2

Determine how virtual companies leverage their operational flexibility and scalability to optimise supply chain management?

Answer: Virtual companies can leverage supply chain strategies by flexibility, scalability, and using digital platforms for coordination.

Think Point 9.3

Describe the strategies that virtual companies can employ to manage risks and enhance resilience in their supply chain operations?

Answer: Diversify suppliers, use automation, and maintain agile systems to manage risks.

Unit 10:

Inventory Management and JIT Systems

Unit Learning Outcomes

CONTENT LIST	LEARNING OUTCOMES FOR THIS UNIT
10.1 Introduction	Introduces the topic areas covered for the unit
10.2 Inventory Management	- Identify the functions and requirements of inventory systems - Explore the concept of inventory accounting systems
10.3 Cost Information	Analyse the different costs that make up the total inventory cost
10.4 Cost Classification System	Select ABC costing to analyse and effectively manage different stock items
10.5 Economic Order Quantities	Examine the EOQ and related formulas
10.6 Just-in-Time Systems	Analyse the importance of JIT systems on business profits
10.7 Summary	Summarises the topic areas covered for the unit

Prescribed and Recommended Reading(s)/Textbook(s)

Prescribed Reading(s)/Textbook(s)

- Heizer, J., Render, B. and Munson, C. (2020). *Operations Management: Sustainability and Supply Chain Management*. Thirteenth Edition. United Kingdom: Pearson.

Recommended Reading(s)/Textbook(s)

- Venkataraman, R.R. and Pinto, J.K. (2020). *Operations Management: Managing Global Supply Chains*. Second Edition. Sage Publications.
- Parksoy, T. and Deveci, M. (2023). *Smart and Sustainable Operations and Supply Chain Management in Industry 4.0*. First Edition. Boca Raton: CRC Press.
- Slack, N., Jones, B. and Burgess, N. (2022). *Operations Management*. Tenth Edition. United Kingdom: Pearson.
- Kareska, PhD Katerina. 'Operational Management Strategies for Sustainable Productivity Growth' (August 5, 2023). Available at SSRN: <https://ssrn.com/abstract=4532740> or <http://dx.doi.org/10.2139/ssrn.4532740>
- 'The Impact Of Ergonomics On Quality Of Life In The Workplace'. (2023). https://www.researchgate.net/publication/377267055_The_Impact_of_Ergonomics_on_Quality_of_Life_in_the_Workplace



10.1 Introduction

Effective inventory management is essential for maintaining operational efficiency, reducing costs, and ensuring the smooth flow of goods and services. Inventory represents a significant investment for organisations, and managing it strategically can enhance profitability, customer satisfaction, and competitive advantage. This unit explores the principles and techniques of Inventory Management and Just-In-Time (JIT) Systems, focusing on how businesses balance inventory levels to meet demand while minimising holding costs.

Key topics include cost information and cost classification systems, which help businesses categorise and track inventory-related expenses for better financial control. The unit also covers Economic Order Quantities (EOQ)—a fundamental inventory model that determines the optimal order size to minimise total inventory costs. Finally, the JIT system is examined as a lean inventory approach that reduces waste, improves efficiency, and aligns production with actual demand.

By the end of this unit, students will gain a comprehensive understanding of inventory management techniques, cost analysis, EOQ models, and JIT principles. This knowledge will enable them to apply best practices in inventory control, optimise stock levels, and improve supply chain efficiency.

10.2 Inventory Management

An inventory is a stock or store of goods. Inventory can take various forms, and these include:

- Raw Material Inventory – which has been purchased, but not processed
- Work-in-Process inventory (WIP), which is incomplete products or components that are no longer considered raw material but have yet to become finished products
- Maintenance, repair and operating materials which are supplies necessary to keep machines and processes productive
- Finished goods inventory which comprise end items, ready to be sold

Functions of Inventory

There are six basic functions of inventory:

1. To provide a stock of goods to meet expected customer demand
2. To de-couple components from the production-distribution system
3. To take advantage of quantity discounts as purchases in larger quantities may reduce the costs of goods or delivery
4. To strike a balance against inflation and upward price changes
5. To protect against delivery variation due to weather, supplier shortages, quality problems or improper deliveries

6. To permit operations to continue easily

Requirements for Effective Inventory Management

Management has two basic functions with respect to inventory. One is to establish a system of accounting for items in inventory, and the other is to make decisions on how much to order and when to order. To be effective, companies should have the following:

- A system to keep track of inventory on hand and on order
- A reliable forecast of demand that includes the forecast error
- Knowledge of lead times and lead-time variability
- Reasonable estimates of inventory holding costs, shortage costs and order costs
- A classification system for inventory items

Inventory Accounting Systems

Inventory accounting systems can be periodic or perpetual. Under a periodic system, a physical count of inventory items is made at periodic intervals (weekly, monthly, quarterly, etc.) in order to decide how much to order of each item. A perpetual inventory system keeps track of removals from inventory on a continuous basis, so that the system can provide information on the current level of inventory for each item.

When the amount of inventory reaches a certain level, an order is triggered. The Two-Bin system is a very simplified system. It involves the use of two containers for inventory. Items are withdrawn from the first bin, when its contents are finished, it's time to order. Cycle counting is a continuous reconciliation of inventory with inventory records.



Think Point 10.1

Explain if inventory is an advantage or disadvantage.

10.3 Cost Information

There are three basic costs associated with inventory:

Holding or Carrying Costs

These relate to physically holding items in storage. They include insurance, interest, depreciation, obsolescence, deterioration, spoilage, pilferage, and warehousing costs (electricity, services, security and rent, rates, staff... right down to cleaning materials, coffee/tea/milk/water usage).

Ordering Costs

These are costs associated with the ordering and receiving of inventory. It includes determining how much is needed, typing up invoices, inspecting goods upon arrival for quantity and quality and moving goods to temporary storage.

Shortage Costs

This often results when demand exceeds the supply of inventory on hand. The costs can include the lost opportunity of not making a sale, loss of customer goodwill, lateness charges, etc. Shortage costs are usually difficult to measure.

Case Study 10.1

Read the following article and answer the question below.

ABC Electronics is a manufacturer of consumer electronic devices, such as smartphones and laptops. Due to the highly competitive and fast-changing nature of the electronics industry, ABC Electronics decided to adopt a Just-In-Time (JIT) inventory system to minimise costs and improve flexibility. The company's primary goals were to reduce inventory holding costs, shorten production lead times, and respond quickly to market changes.

With JIT, ABC only orders parts like microchips, screens, and batteries shortly before they are needed on the production line. This approach reduces storage costs and waste, as there are fewer unused materials in stock. ABC has also fostered close relationships with suppliers to ensure rapid, on-time deliveries.

However, challenges emerged as supply chain disruptions and transportation delays led to shortages of key components. In one instance, a delayed shipment of microchips caused a halt in production for a week, resulting in lost revenue and backorders. This experience forced ABC Electronics to consider adjustments to its JIT strategy to manage risks effectively.

Questions:

1. Evaluate how the JIT inventory system benefits ABC Electronics, and what risks does it introduce?
2. Considering the risks faced by ABC Electronics, determine what adjustments could be made to their JIT system to improve resilience against supply chain disruptions?



10.4 Cost Classification System

Operations managers need to establish systems for managing inventory. We will briefly investigate two systems:

- How inventory items can be classified and
- How to accurately maintain inventory records

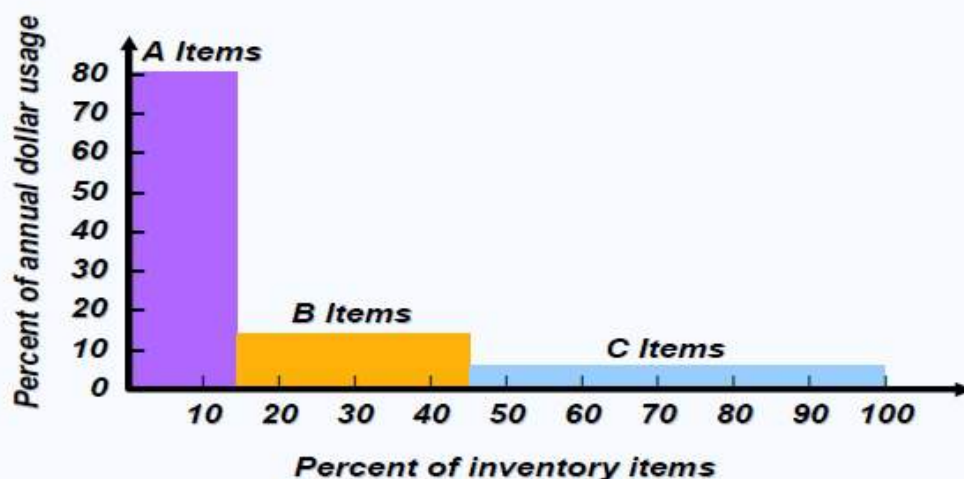
ABC Analysis

This method classifies inventory into three categories according to some measure of importance with the intention of allocating control efforts accordingly. The idea is to establish inventory policies that focus resources on a few critical inventory parts and not the many trivial ones. It is not realistic to monitor inexpensive stock with the same intensity as very expensive items.

We classify items into three categories:

- Class A items (very important)
- Class B items (moderately important) and
- Class C items (least important) based on annual rand value

The following is a graphic representation of ABC Analysis



Source: Heizer.et.al (2020)



Activity 10.1

Classify the inventory items in your Company using the ABC Approach.

Record Accuracy

Good inventory policies are meaningless if management does not know what inventory items are on hand. Accuracy of records is a critical ingredient in production and inventory systems. Record accuracy allows organisations to focus on those items of inventory that are needed. Only if organisations can determine accurately what it has on hand can it make precise decisions about ordering, scheduling and distribution. To ensure accuracy, proper documentation must be kept of incoming and outgoing stock.

It is important to have a well-organised warehouse that has limited access and good security systems. Computerisation, bar coding, scanning equipment, tracking technology and automatic storage and retrieval systems have greatly enhanced inventory management.

10.5 Economic Order Quantities

The Economic Order Quantity (EOQ) model is one of the oldest, most commonly known inventory-control technique. EOQ models identify the optimal order quantity in terms of minimising the sum of certain annual costs that vary with order size. This technique is relatively easy to use and is based on several assumptions:

- There is only one product involved
- Annual usage (demand) requirements are known
- Usage is spread evenly throughout the year so that the usage rate is reasonably constant
- Lead time does not vary
- Each order is received in a single delivery
- There are no quantity discounts

The Inventory Cycle



Source: Heizer.et.al (2020)

We use the formula below to calculate this technique:

$$Q_0 = \sqrt{\frac{2DS}{H}}$$

Where:

D	=	Demand usually expressed as units per year
Q	=	Order quantity in units
S	=	Order costs in rands
H	=	Holding costs usually in rands per unit per year.

The formula for the number of orders per year is shown as:

$$\frac{D}{Q_0}$$

The formula for the length of the order cycle is shown as:

Practical Application 10.1



A local distributor for a national tyre company expects to sell approximately 9600 steel radial tyres of a certain size and tread design next year. Annual carrying costs are R16.00 per tyre and the order cost is R75.00. The distributor works 288 days per year.

1. Determine the EOQ.
2. Illustrate how many times per year the store re-orders.
3. Determine the length of the order cycle.

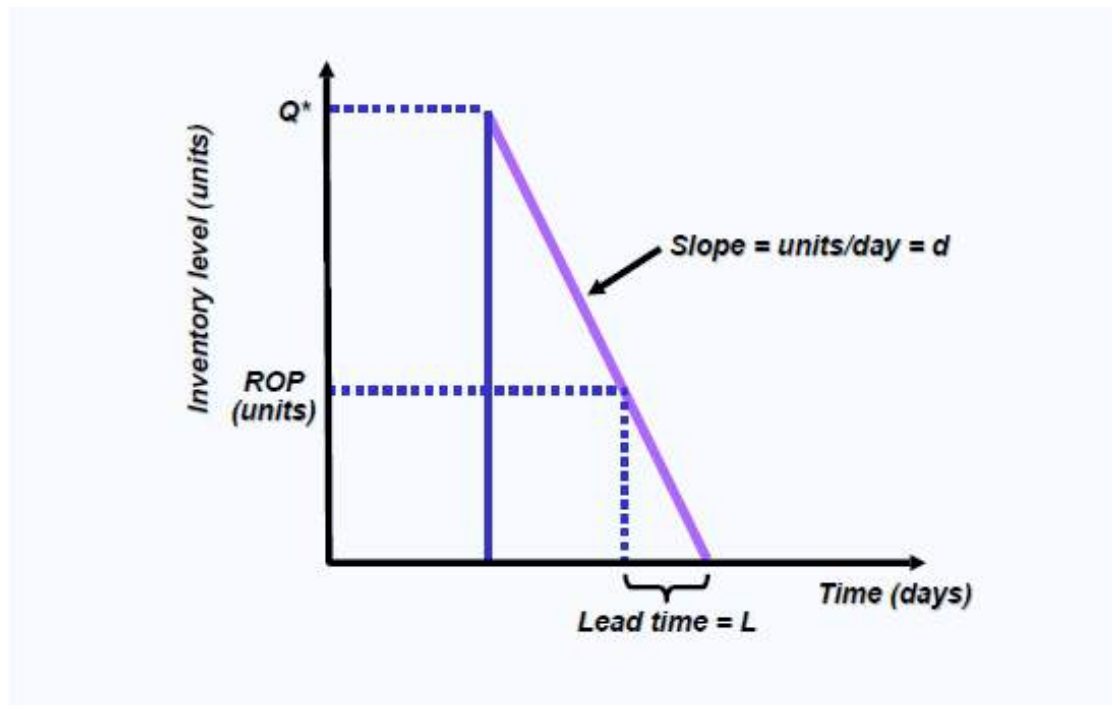
When to Re-order

EOQ models generally answer the question of how much to order, but not when to order. To answer the question of when to re-order, we must first determine the re-order point (ROP). The re-order point occurs when the quantity on hand reaches a pre-determined amount. This amount includes expected demand during lead time and safety stock. There are four determinants of the reorder point quantity:

- The rate of demand
- The length of lead time
- The extent of demand and lead time variability
- The degree of stock out risk acceptable to management

Lead time is defined as the time between placing an order and actually receiving it.

Safety stock is extra stock to allow for uneven demand (or buffer stock). The illustration on the following page depicts the re-order point curve.



Source: Heizer.et.al (2020)

10.6 Just-In-Time Systems

Just-in-Time (JIT) is a philosophy of continuous problem solving. With JIT, supplies and components are 'pulled' through a system to arrive where they are needed, when they are needed and in the right quantity. JIT is an excellent tool to assist operations managers add value by eliminating waste and unwanted variability.

Because there is no surplus inventory, costs associated with unwanted inventory are eliminated and throughput improved. Mr. Taiichi Ohno and several of his colleagues developed the JIT approach at the Toyota Motor Company of Japan.



Video Activity 10.1

Watch the video using the link below and answer the question that follows.

The JIT system in action

Available from: <https://youtu.be/HIEufX83EEY>

Key Elements of a JIT System

JIT systems are designed to achieve a smooth flow of production using minimal inventories. The systems are fairly flexible, with a high degree of worker participation. The most important aspect of a JIT system is that quality is built into the system. Stevenson (2018) maintains that the key elements to the system are:

- High quality levels
- Production smoothing
- Low inventories
- Small lot/batch sizes
- Quick, low-cost setups
- Layout
- Preventative maintenance and repair
- Multi-skilled workers
- A cooperative spirit
- Few, reliable suppliers
- Problem solving approach
- Continuous improvement

Benefits of a JIT System

The main benefits are:

- Reduced levels of in-process stock, purchased and finished goods
- Less space required
- Increased product quality and reduced scrap
- Reduced manufacturing lead times
- Greater flexibility in changing the production mix
- Smoother production flow
- Increased productivity and utilisation of plant and equipment
- Worker participation in problem solving
- Good relationships built with suppliers
- Reduced need for indirect labour such as materials handlers

JIT contributes to competitive advantage. The table below outline the contributions of JIT

JIT Requires:

Suppliers:	Reduced number of vendors; Supportive supplier relationships; Quality deliveries on time
Layout:	Work-cell layouts with testing at each step of the process; Group technology; Movable, changeable, flexible machinery; High level of workplace organization and neatness; Reduced space for inventory; Delivery directly to work areas
Inventory:	Small lot sizes; Low setup time; Specialized bins for holding set number of parts
Scheduling:	Zero deviation from schedules; Level schedules; Suppliers informed of schedules; Kanban techniques
Preventive maintenance:	Scheduled; Daily routine; Operator involvement
Quality production:	Statistical process control; Quality suppliers; Quality within the firm
Employee empowerment:	Empowered and cross-trained employees; Training support; Few job classifications to ensure flexibility of employees
Commitment:	Support of management, employees, and suppliers

**Which Results In:**

Queue and delay reduction speeds throughput, frees assets, and wins orders

Quality improvement reduces waste and wins orders

Cost reduction increases margin or reduces selling price

Variability reduction in the workplace reduces wastes and wins orders

Rework reduction reduces wastes and wins orders

**Which Yields:**

Faster response to the customer at lower cost and higher quality—

A Competitive Advantage

Source: Heizer, J. et al 2020

Activity 10.2

1. A manufacturing company experiences frequent stockouts and high inventory holding costs. The company decides to implement a Just-in-Time (JIT) system.

Discuss how the implementation of a JIT system helps the company address these issues?

2. A retail business uses the ABC analysis method for inventory management. The business classifies its products into three categories: A, B, and C, based on their annual consumption value.

Outline how the business can manage each category of inventory differently to optimise inventory levels and costs?

3. Illustrate the steps that the company can take to ensure supplier reliability when implementing a JIT system?

4. Calculate the reorder point for the company.

Converting to a JIT System

To successfully convert to a JIT system companies should adopt the following approach:

- Ensure that top management is committed to the conversion, and it knows what will be required in terms of cost and time frames and what results can be expected
- Study the operations carefully; decide in advance which parts will need the most effort to convert
- Enlist the support and cooperation of the workers by preparing training in appropriate programmes. Make certain workers understand the concepts of JIT and why it is desirable. Assure them that their jobs are secure.
- Start by trying to reduce setup times whilst maintaining the current system
- Gradually convert operations, beginning at the end of the process and working backwards. At each stage, make sure that the conversion has been successful before moving on
- Convert your suppliers to a JIT system as one of the last steps – work closely with them. Identify those suppliers who are willing to embrace JIT philosophy and give them preference
- Be prepared to encounter 'obstacles' to the conversion

JIT in Services

JIT in service can be a major competitive advantage for companies that can achieve it. The key element is to provide the service when it is needed. This requires flexibility on the part of the provider, which generally means short setup times and clear communication on the part of the

customer. If a customer can determine when it will need a particular service, a JIT server can then schedule service to correspond to those needs.



Activity 10.3

Analyse the impact JIT has had on operations

Knowledge Check Questions 10.1

Select the correct answer:

1. Which of the following is a primary objective of inventory management?
 - a. To maximise inventory levels
 - b. To minimise holding costs and avoid stockouts
 - c. To increase production lead times
 - d. To reduce the number of suppliers
2. Which inventory management technique uses a fixed order quantity but varies the time between orders?
 - a. Economic Order Quantity (EOQ)
 - b. Just-in-Time (JIT)
 - c. Periodic Review System
 - d. ABC Analysis
3. What is the primary goal of a Just-in-Time (JIT) inventory system?
 - a. To increase inventory levels
 - b. To reduce lead times and increase flexibility
 - c. To decrease the frequency of orders
 - d. To improve supplier relationships



10.7 Summary

This unit provided an in-depth understanding of Inventory Management and JIT Systems, highlighting their role in optimising operations and reducing costs. The key topics covered included exploring different types of inventory (raw materials, work-in-progress, finished goods) and strategies to maintain optimal stock levels, understanding the financial impact of inventory decisions, including holding costs, ordering costs, and stockout costs and applying mathematical models to determine the most cost-effective order size that balances holding and ordering costs. A key concept also included the lean inventory practices that focus on reducing waste, improving efficiency, and synchronising

production with demand.

Through practical applications and case studies, students gained insights into how effective inventory management contributes to cost savings, operational efficiency, and supply chain optimisation. Understanding EOQ and JIT principles enables businesses to enhance productivity, reduce excess stock, and improve customer responsiveness.

Revision Questions

1. Illustrate the appropriate ABC groups of inventory items.

ABC Analysis		
Stock Number	Annual R Volume	Percent of Annual R Volume
J24	12 500	46.2
R26	9 000	33.3
L02	3 200	11.8
M12	1 550	5.8
P33	620	2.3
T72	65	0.2
S67	53	0.2
Q47	32	0.1
V20	30	0.1
		=100.0



2. A firm has 1,000 “A” items (which it counts every week, i.e., 5 days), 4,000 “B” items (counted every 40 days), and 8,000 “C” items (counted every 100 days). Explain how many items should be counted per day.

3. Assume you have a product with the following parameters:

Demand = 360 Holding cost per year = \$1.00 per unit Order cost = \$100 per order

Explain what the EOQ is.

4. Given the data from Revision Question 3, and assuming a 300-day work year; explain how many orders should be processed per year. Explain what the expected time between orders is.

5. Determine the advantages and disadvantages of inventory.

Answer**Activity 10.1**

Classify the inventory items in your Company using the ABC Approach.

Answer: To be discussed during the webinar(s) with lecturer.

Activity 10.2

1. Discuss how the implementation of a JIT system helps the company address these issues?

Answer: Implementing a JIT system can help the company reduce inventory levels, thereby lowering holding costs. By receiving materials and components only when needed, the company can minimise stock-outs and ensure a smoother production flow. Additionally, JIT can improve responsiveness to demand changes and increase overall efficiency.

2. Outline how the business can manage each category of inventory differently to optimise inventory levels and costs?

Answer: For category A items, which have the highest annual consumption value, the business should maintain strict control, frequent reviews, and tighter inventory levels to avoid stock-outs. For category B items, moderate control and regular reviews are sufficient. Category C items, with the lowest consumption value, can be managed with less stringent controls and larger safety stock to reduce administrative costs.

3. Illustrate the steps that the company can take to ensure supplier reliability when implementing a JIT system?

Answer: To ensure supplier reliability in a JIT system, the company can develop strong relationships with key suppliers, establish clear communication channels, and collaborate closely on production schedules. It can also implement supplier performance metrics, conduct regular audits, and develop contingency plans to address any potential disruptions in the supply chain.

4. Calculate the reorder point for the company.

Answer:

Reorder Point=Demand during lead time

Weekly demand = 500 units

Lead time = 2 weeks

Reorder Point=500 units/week*2 weeks=1,000 units

The reorder point is 1,000 units.

Activity 10.3

Analyse the impact JIT has had on operations

Answer: JIT reduces waste and inventory costs but requires reliable suppliers

Case Study 10.1

1. Evaluate how the JIT inventory system benefits ABC Electronics, and what risks does it introduce?

Answer: JIT benefits ABC Electronics by significantly reducing inventory holding costs, as materials are only ordered when needed. This leads to leaner inventory levels, lower storage costs, and less waste from obsolete components, especially valuable in a fast-paced industry like electronics. Additionally, JIT allows the company to be more responsive to market changes by adjusting production quickly without excess inventory. However, JIT also introduces risks, particularly supply chain vulnerability. Since ABC maintains minimal inventory, any delay in material delivery, such as the microchip shortage experienced, can halt production, leading to lost revenue and reduced customer satisfaction. JIT's reliance on timely supplier deliveries makes it sensitive to supply chain disruptions, suggesting that JIT works best when supported by a reliable supply chain and robust contingency plans.

2. Considering the risks faced by ABC Electronics, determine what adjustments could be made to their JIT system to improve resilience against supply chain disruptions?

Answer: To enhance resilience, ABC Electronics could consider a hybrid approach that combines JIT with strategic safety stock for critical, high-risk components, like microchips, that have experienced past supply issues. Maintaining a small reserve of these key items would reduce the risk of production delays without significantly increasing holding costs. Another adjustment could be diversifying suppliers to reduce reliance on a single source. By establishing relationships with multiple suppliers for critical parts, ABC can switch to an alternative supplier if one faces delays. Finally, real-time inventory tracking and forecasting would help ABC better predict demand and manage supplier lead times, allowing for pre-emptive action before delays impact production.

Knowledge Check Questions 10.1

1. Which of the following is a primary objective of inventory management?

Answer: b. To minimise holding costs and avoid stock-outs

2. Which inventory management technique uses a fixed order quantity but varies the time between orders?

Answer: a. Economic Order Quantity (EOQ)

3. What is the primary goal of a Just-in-Time (JIT) inventory system?

Answer: b. To reduce lead times and increase flexibility

Practical Application 10.1

1. Determine the EOQ.

Answer:

$$= \sqrt{\frac{2DS}{H}}$$

$$= \sqrt{\frac{2 \times 9600 \times 75}{16}}$$

$$= 300 \text{ tyres}$$

2. Illustrate how many times per year the store re-orders.

Answer:

$$\text{Number of orders per year} = \frac{D}{Q_0}$$

$$= \frac{9600}{300}$$

$$= 32 \text{ orders}$$

3. Determine the length of the order cycle.

Answer:

$$\begin{aligned}\text{Length of order cycle} &= \frac{Q_0}{D} \\ &= \frac{300}{9600} \\ &= 3.125\% \times 288 \text{ days per annum} \\ &= 9 \text{ days}\end{aligned}$$

Think Point 10.1

Explain if inventory is an advantage or disadvantage.

Answer: Inventory allows for smooth production but ties up capital and risks obsolescence.

**Unit
11:**

Aggregate Scheduling

Unit Learning Outcomes

CONTENT LIST	LEARNING OUTCOMES FOR THIS UNIT
11.1 Introduction	Introduces the topic areas covered for the unit
11.2 Aggregate Scheduling	Outline the purpose and scope of aggregate scheduling
11.3 Aggregate Scheduling Strategies	Analyse the different aggregate scheduling strategies
11.4 Techniques for Aggregate Scheduling	Examine the applicability of aggregate scheduling techniques in different business environments
11.5 Aggregate Scheduling in Services	Examine the importance of aggregate scheduling in the service sector
11.6 Summary	Summarises the topic areas covered for the unit

Prescribed and Recommended Reading(s)/Textbook(s)

Prescribed Reading(s)/Textbook(s)

- Heizer, J., Render, B. and Munson, C. (2020). *Operations Management: Sustainability and Supply Chain Management*. Thirteenth Edition. United Kingdom: Pearson.

Recommended Reading(s)/Textbook(s)

- Venkataraman, R.R. and Pinto, J.K. (2020). *Operations Management: Managing Global Supply Chains*. Second Edition. Sage Publications.
- Parksoy, T. and Deveci, M. (2023). *Smart and Sustainable Operations and Supply Chain Management in Industry 4.0*. First Edition. Boca Raton: CRC Press.
- Slack, N., Jones, B. and Burgess, N. (2022). *Operations Management*. Tenth Edition. United Kingdom: Pearson.
- Kareska, PhD Katerina. 'Operational Management Strategies for Sustainable Productivity Growth' (August 5, 2023). Available at SSRN: <https://ssrn.com/abstract=4532740> or <http://dx.doi.org/10.2139/ssrn.4532740>
- 'The Impact Of Ergonomics On Quality Of Life In The Workplace'. (2023). https://www.researchgate.net/publication/377267055_The_Impact_of_Ergonomics_on_Quality_of_Life_in_the_Workplace



11.1 Introduction

Efficient scheduling of production and resources is a key aspect of Operations Management, ensuring that businesses meet demand while optimising costs and capacity. Aggregate scheduling is a medium-term planning process that aligns production output, workforce levels, and inventory to balance supply and demand over a specific period. This strategic approach helps organisations manage fluctuations in demand while maintaining efficiency and cost-effectiveness.

This unit explores the fundamentals of aggregate scheduling, including its importance in coordinating operations, improving resource utilisation, and maintaining service levels. Different aggregate scheduling strategies—such as level, chase, and hybrid strategies—are examined to understand their advantages and applications in various industries. The unit also introduces techniques for aggregate scheduling, including mathematical models, graphical methods, and heuristic approaches that help managers develop optimal schedules. Lastly, the application of aggregate scheduling in services is discussed, focusing on how service organisations manage workforce capacity, appointment scheduling, and demand variability.

By the end of this unit, students will be equipped with the knowledge to analyse scheduling challenges, evaluate different scheduling strategies, and apply appropriate techniques to enhance operational efficiency.

11.2 Aggregate Scheduling

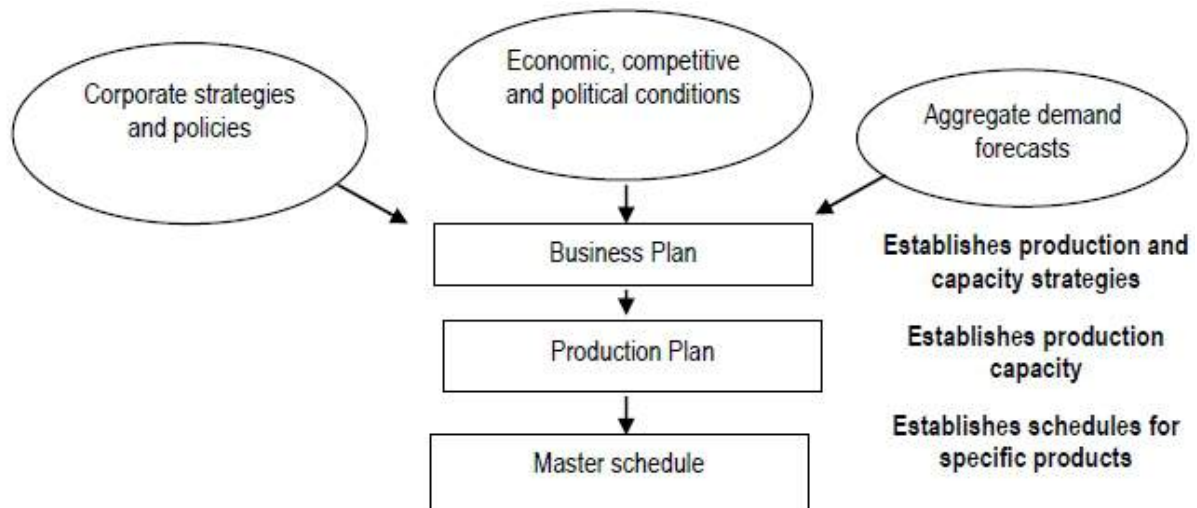
Aggregate Scheduling

Aggregate scheduling, also known as aggregate planning, is essentially a 'big picture' approach to planning. The main focus is to determine the quantity and timing of production for the intermediate future – between three and 18 months. Operations managers try to determine the best way to meet forecasted demand by adjusting production rates, labour levels, inventory levels, overtime work, subcontracting rates and other controllable variables.

The objective of aggregate planning is to Minimise costs over the planning period. According to Heizer.et.al (2020), four requirements are needed for aggregate planning:

- A logical overall unit for measuring sales and output
- A demand forecast for the intermediate planning period
- A method of determining the various costs
- A model that combines forecasts and costs so that scheduling decisions can be made for the planning period

Often aggregation is done by product groupings, where products (or services) with similar requirements are lumped together for planning purposes. As an example, let's consider how aggregate planning might work in a large departmental store. The manager may decide to allocate 20% of the available space to men's clothing, 50% to female clothing 20% to children's clothing and 10% to cosmetics. As you can conclude from this example no mention is made as to what type, size, colour, etc. of clothing is to be stocked.



Source: Stevenson (2018)



Activity 11.1

Examine the three levels of planning that operation's managers are involved in and list what kind of decisions are made at each level.

The Purpose and Scope of Aggregate Scheduling

Aggregate planning seeks to address the balance between supply and demand. If supply and demand are not in balance, the organisation will incur cost penalties. There will be added costs of adjusting the system as well as opportunity costs.



Video Activity 11.1

Watch the video using the link below and answer the question that follows.

Aggregate planning

Available from: <https://youtu.be/qy--zXxOwQ4>

Demand Options

The basic demand options are the following:

- Pricing

Adjusting of prices is commonly used to shift demand from peak periods to off-peak periods. Hotels, for example, offer lower rates for off-peak periods. Some restaurants offer specials for children's meals.

- Promotion

Advertising and other forms of promotion, such as displays and direct marketing, can be extremely effective in shifting demand so that it matches existing capacity.

- Back-orders

Using this system, orders are taken in one period and deliveries promised for a later period. The success of this approach depends on how willing customers are to wait for deliveries.

- New Demand

Organisations try to influence demand by creating a new demand for its product or service. For example, demand for bus transportation is more intense during morning and afternoons. In this case, bus companies need to create new demand during the off-peak period by offering their buses for school trips, senior citizen groups and other facilities.

Case Study 11.1

Read the following article and answer the question below.

Aggregate Scheduling at Acme Manufacturing

Acme Manufacturing, a mid-sized company specialising in consumer electronics, has experienced fluctuating demand for its products. To address these challenges, the company has implemented an aggregate scheduling strategy to balance demand and capacity, ensuring efficient production and customer satisfaction.

Acme Manufacturing produces a variety of electronic gadgets, including smartphones, tablets, and smartwatches. The company faces seasonal demand variations, with peaks during holiday seasons and back-to-school periods. Managing these fluctuations while maintaining operational efficiency and minimising costs is critical for Acme's success.

Aggregate Scheduling Strategy

To manage demand and capacity effectively, Acme Manufacturing employs several key concepts and strategies:

1. **Demand Forecasting:** Acme uses historical sales data, market trends, and advanced forecasting techniques to predict future demand. This allows the company to anticipate peaks and troughs in demand and plan accordingly.
2. **Capacity Planning:** The company assesses its production capacity, including workforce availability, machinery, and raw materials. By understanding its capacity constraints, Acme can develop strategies to match production levels with forecasted demand.
3. **Demand Options:**
 - **Promotions and Discounts:** To smooth demand, Acme runs promotions and offers discounts during off-peak periods to encourage purchases
 - **Backorders:** In cases where demand exceeds capacity, Acme offers customers the option to place backorders, ensuring that orders are fulfilled as soon as additional capacity is available
 - **New Product Introductions:** Acme times the launch of new products to align with periods of lower demand for existing products, balancing overall demand across the product portfolio.
4. **Capacity Options:**
 - **Overtime and Extra Shifts:** During peak demand periods, Acme schedules overtime and adds extra shifts to increase production capacity
 - **Temporary Workforce:** Acme hires temporary workers during high-demand seasons to supplement the regular workforce without committing to long-term employment costs
 - **Subcontracting:** For certain components, Acme outsources production to subcontractors to increase capacity without investing in additional equipment or facilities
 - **Inventory Management:** Acme builds up inventory during low-demand periods to ensure sufficient stock is available during peak times, reducing the need for drastic capacity adjustments



By implementing these aggregate scheduling strategies, Acme Manufacturing has achieved several key benefits:

- **Improved Customer Satisfaction:** By balancing demand and capacity, Acme can fulfil customer orders more reliably, reducing lead times and backorders
- **Cost Efficiency:** The company minimises costs associated with capacity adjustments by strategically using overtime, temporary workers, and subcontracting
- **Operational Stability:** Aggregate scheduling allows Acme to maintain a stable production environment, avoiding the disruptions caused by drastic changes in production levels.

Acme Manufacturing's aggregate scheduling approach effectively balances demand and capacity, ensuring efficient production and high customer satisfaction. By leveraging demand forecasting, capacity planning, and various demand and capacity options, the company navigates the challenges of fluctuating demand while maintaining operational and cost efficiency.

Questions:

1. Analyse how Acme Manufacturing can use demand forecasting to manage production?
2. Determine the capacity options that Acme Manufacturing can employ to handle peak demand periods?
3. Discuss how Acme Manufacturing balances demand during off-peak periods?

Capacity Options

Varying Workforce size by Hiring and Laying off staff

- One way to alter capacity is to hire or lay off workers to match production rates. We must remember that as new employees need to be trained, productivity tends to decrease, as they are absorbed into the operations

Working Overtime / Short-Time

- Using overtime or short time is a less severe way of changing capacity than hiring and firing workers. The use of overtime is fairly attractive when dealing with seasonal demand peaks. Overtime permits the company to maintain a skilled workforce base and gives employees an opportunity to increase their earnings. It is advisable to use short time when demand is less than

capacity. Some organisations use this time for training workers, assisting in general maintenance of plant and equipment, etc

Part-time or Casual Workers

- In some cases, the use of part-time workers is a viable option. Seasonal work that requires low or moderate skills lends itself to part-time workers, who generally cost less than regular workers do. Department stores, restaurants and supermarkets make extensive use of part-time worker

Inventories

- By using finished goods inventories firms can produce goods in one period and sell or deliver them later. This, however, carries a cost penalty, as you would incur storage or holding costs. This strategy lends itself to manufacturing, as it is relatively easy to store tangible products. One cannot use this approach in a service industry, as services can't be stored

Outsourcing

- Outsourcing enables planners to acquire temporary capacity. The question of hiring someone else to do your work generally depends on factors such as available capacity, relative experience, quality considerations, costs and the stability of demand



Think Point 11.1

How can companies effectively balance fluctuating demand with their production capacity without incurring excessive costs or compromising on customer satisfaction?

11.3 Aggregate Scheduling Strategies

Arising out of the above, we can conclude that operations managers have a wide range of decision options that they can consider in achieving a balance between demand and capacity. Some of the strategies used according Heizer.et.al (2020) are:

- Maintaining a level workforce
- Maintaining a steady output rate
- Matching demand period-by-period
- Using a combination of decision variables

Under a level capacity strategy, variations in demand are met by using some combination of inventories, overtime, part-time workers, backorders and outsourcing. Matching capacity to demand implies a chase demand strategy, which is the planned output for a period, is set at the expected demand for that period.

11.4 Techniques for Aggregate Scheduling

To help decision-makers with the tasks of aggregate scheduling, several techniques have been developed. These generally fall into one of two categories: informal trial-and-error techniques and mathematical techniques. In practice, informal techniques are more commonly used. A general procedure for aggregate planning consists of the following steps:

- Determine the demand for each period
- Determine the capacities for each period – this includes regular time, overtime, sub-contracting, etc
- Identify company or departmental policies that apply (e.g. the amount of safety stock that the organisation has to carry, employment/ retrenchment policies and other issues)
- Calculate the unit cost for regular time, overtime, sub-contracting, carrying stock, retrenchments and other related issues
- Develop alternative plans and calculate the cost for each plan
- Select the plan that best satisfies the objectives

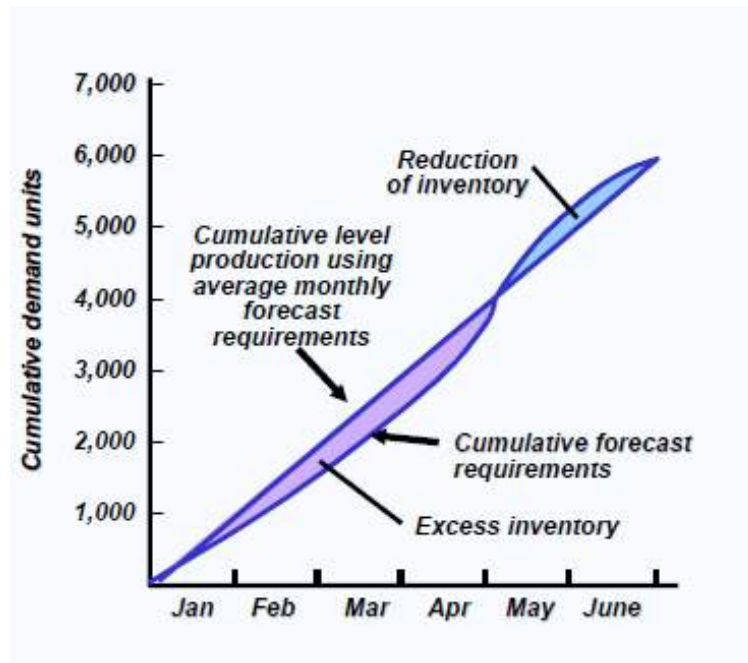


Think Point 11.2

Explain if there is a need for aggregate scheduling in your organisation and substantiate your response.

Informal Techniques

Informal techniques consist of developing simple tables or graphs that enable operations managers to visually compare projected demand requirements with existing capacity. Frequently, graphs can be used to guide the development of alternatives.



Source: Heizer.et.al (2020)

Activity 11.2



A firm forecasts a demand of 20,000 units for the next year. The firm has two options: (a) Produce at a steady rate of 1,800 units per month and carry inventory, or (b) vary production to meet monthly demand. The carrying cost for inventory is R2 per unit per month.

Calculate the annual carrying cost if the firm chooses to produce at a steady rate.

Mathematical Techniques

Several mathematical techniques have been developed to deal with aggregate scheduling. These include:

- **Linear Programming**

These are mathematical models used to obtain optimal solutions to problems involving the allocation of scarce resources in terms of cost minimisation or profit maximisation.

- **Linear Decision Rule**

This is an optimisation technique that seeks to Minimise combined costs, using a set of cost approximating functions to obtain a single quadratic equation.

- **Simulation Models**

Computerised models that can be tested under different scenarios to identify acceptable solutions to

problems.

- **Management Coefficient Models**

This is a formal planning model built around a manager's experience and performance



Activity 11.3

Assess the impact that proper scheduling has on improving sustainability.

11.5 Aggregate Scheduling in Services

Services occur in industries such as banking, transport, hospitals, education, fast foods, etc. There are several issues relating to the differences between manufacturing and services:

- Services occur when they are rendered. Services cannot be stored but they can be delayed. This removes the option of building up inventories during slack periods in anticipation of future demand
- Demand for service can be difficult to predict. The volume of demand for services is often quite variable. In some instances, customers may require prompt service e.g. ambulances, police, fire services and other related emergency disciplines, whilst in others, they simply want prompt service, e.g. pizza deliveries, and may go elsewhere if their demands are not met
- Capacity availability can be difficult to predict. Processing requirements for services can also be extremely variable. Bank tellers, for example, are often called upon to handle a wide variety of transactions and requests for information thus making it difficult to establish a suitable measure of their capacities
- The flexibility of labour can be an advantage in services. Service providers are capable of handling a wide variety of service requirements and therefore planning is easier



Think Point 11.3

Examine the key considerations for selecting and implementing capacity options such as overtime, temporary workers, and subcontracting to manage peak demand periods?

Knowledge Check Questions 11.1**Select the correct answer:**

1. Which of the following is NOT a common strategy for managing demand in aggregate planning?
 - a. Pricing strategies
 - b. Promotion and advertising
 - c. Subcontracting
 - d. Backordering
2. In aggregate planning, the primary objective is to:
 - a. Minimise production costs
 - b. Maximise inventory levels
 - c. Balance demand and capacity
 - d. Reduce workforce size
3. Which aggregate planning strategy adjusts capacity by varying the size of the workforce?
 - a. Chase strategy
 - b. Level strategy
 - c. Hybrid strategy
 - d. Demand management strategy



Knowledge Check Questions 11.2

State if the following is True or False by selecting the correct answer.



1. The level strategy in aggregate planning maintains a constant production rate regardless of demand fluctuations.
True
False
2. Back ordering is a capacity option used to manage demand by allowing orders to be fulfilled at a later date.
True
False
3. Aggregate planning is primarily concerned with long-term decisions about production capacity
True
False

11.6 Summary

This unit provided a comprehensive understanding of aggregate scheduling and its role in balancing supply and demand within operations. The key topics covered included understanding the importance of medium-term planning in managing workforce levels, production rates, and inventory, examining different strategies such as the level strategy, the chase strategy and the hybrid strategy and exploring various methods, including graphical approaches, mathematical models, and decision rules, to optimise scheduling decisions. The aggregate scheduling in services was also explored for analysing how service-based businesses manage workforce capacity, seasonal demand, and appointment-based scheduling to improve service efficiency.

Through practical case studies and real-world applications, students gained insights into how organisations develop aggregate schedules to improve operational performance, cost efficiency, and customer satisfaction. Mastery of these concepts enables managers to make informed scheduling decisions that align with business goals and market demands.

Revision Questions

1. Illustrate the following problem in transportation format and solve for the minimum cost plan.

	Period		
	Feb	Mar	Apr
Demand	55	70	75
Capacity			
Regular	50	50	50
Overtime	5	5	5
Subcontract	12	12	10
Beginning Inventory	10		
Costs			
Regular Time	R60 per unit		
Overtime	R80 per unit		
Subcontract	R90 per unit		
Inventory carrying cost	R1 per unit per month		
Back order cost	R3 per unit per month		



2. A company faces seasonal demand for its products, with high demand during the winter holiday season and lower demand during the summer months. The company has decided to use a chase strategy for its aggregate planning. Determine how the company can adjust its workforce and production levels to align with the chase strategy during these seasonal fluctuations?

3. A manufacturing firm has decided to build up inventory during low-demand periods to meet high demand during peak seasons without changing its workforce size. Discuss the aggregate planning strategy that the firm is using, and what are the potential benefits and drawbacks of this approach?

4. An electronics company plans to introduce a new product next year and anticipates a significant increase in demand. The company currently operates

at near full capacity. Analyse the capacity options that the company could consider meeting the expected increase in demand?

5. A company needs to produce 10,000 units over the next 5 months. The current workforce can produce 1,500 units per month. The company can use overtime to produce an additional 500 units per month at an extra cost of R10 per unit.

Calculate the total overtime cost if the company uses overtime to meet the production target.

Answers

Activity 11.1

Examine the three levels of planning that operation's managers are involved in and list what kind of decisions are made at each level.

Answer: Three levels of planning:

- Strategic planning: Long-term goals (e.g., market expansion)
- Tactical planning: Medium-term resource allocation
- Operational planning: Day-to-day operations and scheduling

Activity 11.2

Calculate the annual carrying cost if the firm chooses to produce at a steady rate.

Answer:

Monthly production rate = 1,800 units

Monthly demand = 20,000 units / 12 months = 1,667 units

Average inventory per month = (1,800 units - 1,667 units) / 2 = 66.5 units

Total inventory per month = 66.5 units * 12 months = 798 units

Annual carrying cost = 798 units * R2/unit = R1,596

Activity 11.3

Assess the impact that proper scheduling has on improving sustainability.

Answer: Proper scheduling reduces waste and energy consumption.

Case Study 11.1

1. Analyse how Acme Manufacturing can use demand forecasting to manage production?

Answer: Acme Manufacturing uses historical sales data, market trends, and advanced forecasting techniques to predict future demand. This helps the company anticipate peaks and troughs in demand, allowing for better planning and resource allocation.

2. Determine the capacity options that Acme Manufacturing can employ to handle peak demand periods?

Answer: During peak demand periods, Acme Manufacturing employs capacity options such as scheduling overtime, adding extra shifts, hiring temporary workers, subcontracting certain components, and managing inventory to ensure sufficient stock is available.

3. Discuss how Acme Manufacturing balances demand during off-peak periods?

Answer: Acme Manufacturing balances demand during off-peak periods by running promotions and offering discounts to encourage purchases, timing new product launches to coincide with lower demand for existing products and allowing backorders to manage excess demand.

Knowledge Check Questions 11.1

1. Which of the following is NOT a common strategy for managing demand in aggregate planning?

Answer: c. Subcontracting

2. In aggregate planning, the primary objective is to:

Answer: c. Balance demand and capacity

3. Which aggregate planning strategy adjusts capacity by varying the size of the workforce?

Answer: a. Chase strategy

Knowledge Check Questions 11.2

1. The level strategy in aggregate planning maintains a constant production rate regardless of demand fluctuations.

Answer: True

2. Back ordering is a capacity option used to manage demand by allowing orders to be fulfilled at a later date.

Answer: True

3. Aggregate planning is primarily concerned with long-term decisions about production capacity

Answer: False

Think Point 11.1

How can companies effectively balance fluctuating demand with their production capacity without incurring excessive costs or compromising on customer satisfaction?

Answer: By using forecasting methods, flexible scheduling, and maintain safety stock to avoid overproduction and shortages.

Think Point 11.2

Explain if there is a need for aggregate scheduling in your organisation and substantiate your response.

Answer:

- Allocate resources for specific activities
- Price service contracts for maximum margin right with customer value and market acceptance
- Acquire, train, and terminate staff effectively
- Accurately forecast demand to produce or deliver services

Temporary workers, overtime, and subcontracting help meet demand without long-term costs.

Think Point 11.3

Examine the key considerations for selecting and implementing capacity options such as overtime, temporary workers, and subcontracting to manage peak demand periods?

Answer: To be discussed during the webinar(s) with lecturer

A graphic for Unit 12. It features a black triangle on the left with the text 'Unit 12:' in white. To the right of the triangle is a grey rectangle with the text 'Materials Resources Planning (MRPII)' in white. The triangle and rectangle are separated by a diagonal line.

**Unit
12:**

**Materials Resources Planning
(MRPII)**

Unit Learning Outcomes

CONTENT LIST	LEARNING OUTCOMES FOR THIS UNIT
12.1 Introduction	Introduces the topic areas covered for the unit
12.2 Materials Requirement Planning	Outline the concept of materials requirements planning
12.3 Production Plan	Analyse the benefits of the production plan
12.4 Master Production Schedule (MPS)	Outline the MOPS in terms of the activities in MRP
12.5 Kanban	Analyse the methodology that Kanban proposes in the current business environment
12.6 Inventory Rules	Examine the process involved in acquiring inventory in terms of authorisation of orders
12.7 Bill of Material (BOM)	Compare the two basic elements that the BOM must fulfil in order to ensure uninterrupted production
12.8 Summary	Summarises the topic areas covered for the unit

Prescribed and Recommended Reading(s)/Textbook(s)

Prescribed Reading(s)/Textbook(s)

- Heizer, J., Render, B. and Munson, C. (2020). *Operations Management: Sustainability and Supply Chain Management*. Thirteenth Edition. United Kingdom: Pearson.

Recommended Reading(s)/Textbook(s)

- Venkataraman, R.R. and Pinto, J.K. (2020). *Operations Management: Managing Global Supply Chains*. Second Edition. Sage Publications.
- Parksoy, T. and Deveci, M. (2023). *Smart and Sustainable Operations and Supply Chain Management in Industry 4.0*. First Edition. Boca Raton: CRC Press.
- Slack, N., Jones, B. and Burgess, N. (2022). *Operations Management*. Tenth Edition. United Kingdom: Pearson.
- Kareska, PhD Katerina. 'Operational Management Strategies for Sustainable Productivity Growth' (August 5, 2023). Available at SSRN: <https://ssrn.com/abstract=4532740> or <http://dx.doi.org/10.2139/ssrn.4532740>
- 'The Impact Of Ergonomics On Quality Of Life In The Workplace'. (2023). https://www.researchgate.net/publication/377267055_The_Impact_of_Ergonomics_on_Quality_of_Life_in_the_Workplace



12.1 Introduction

Effective planning and control of manufacturing resources are essential for optimising production efficiency, reducing costs, and ensuring timely delivery of goods. Materials Resource Planning (MRP II) is an advanced system that extends beyond traditional Material Requirements Planning (MRP) by integrating broader business functions such as production scheduling, inventory management, and capacity planning. By incorporating financial, operational, and supply chain aspects, MRP II serves as a comprehensive tool for coordinating materials, labour, and production schedules.

This unit explores core components of MRP II, including the Production Plan, which aligns production capacity with market demand, and the Master Production Schedule (MPS), which provides a detailed breakdown of what needs to be produced and when. Additionally, the unit examines Kanban, a lean inventory management approach that supports just-in-time (JIT) production, and Inventory Rules, which define optimal stock levels to avoid shortages and excesses. The Bill of Materials (BOM) is also discussed as a crucial component that outlines the raw materials and components required for production.

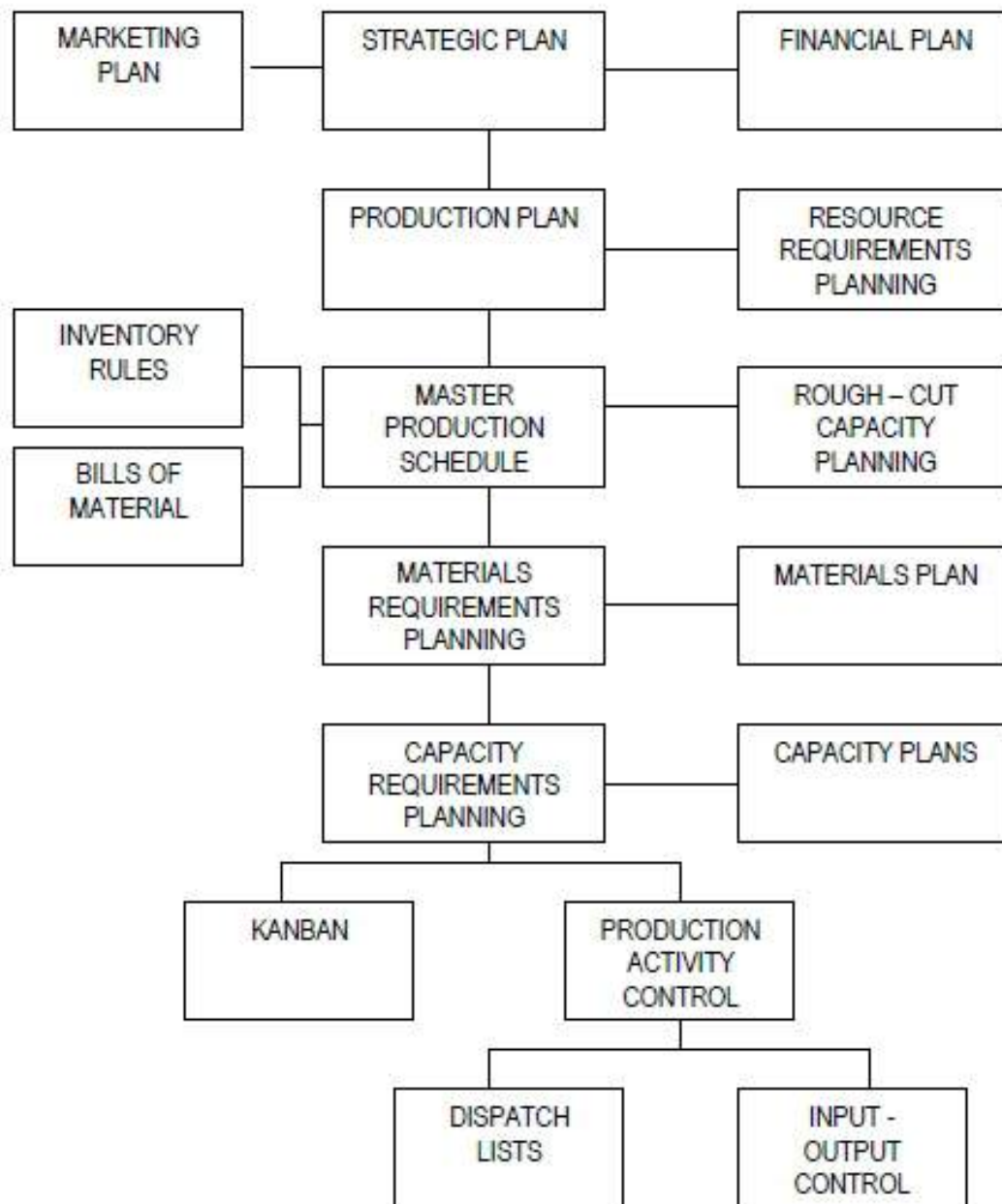
By the end of this unit, students will gain an in-depth understanding of how MRP II integrates production planning, scheduling, and inventory management to enhance efficiency and responsiveness in modern manufacturing environments.

12.2 Materials Requirements Planning

Materials Requirements Planning

Material planning is the very heart of an MRP, II system for a manufacturing company. It is important to first understand where it fits into the overall system and the part the Bills and Routings play in material planning. The diagram below is a simplified flow chart of the manufacturing portion of an MRP II system.

MRP FLOW CHART



Video Activity 12.1

Watch the video using the link below and answer the question that follows.

Complete Material Requirements Planning (MRP) Problem

Available from: <https://www.youtube.com/watch?v=PXyK6yVbxA8>

Scenario:

You are tasked with managing the production of a custom-built bicycle. To successfully manufacture the bicycle, you need to plan for multiple components like the frame, wheels, and handlebars. After watching a video on MRP, you will use MRP grids to ensure that all components are available for the assembly process.

Questions:

1. Explain the main purpose of using MRP in manufacturing?
2. Determine how you could use MRP grids to plan the production of the bicycle's components?
3. Determine what could happen if one key component's MRP grid is not properly aligned with the production schedule?



Think Point 12.1

How can Materials Requirements Planning (MRP) optimise production schedules to improve operational efficiency and reduce costs?



Video Activity 12.2

Watch the video using the link below.

What is the Material Requirement planning (MRP)? | MRP Process

Available from: <https://www.youtube.com/watch?v=u3P6YMI5Ah0>

12.3 Production Plan

A production plan is a formal plan of where the company is going, created by the company executive. The main inputs comprise the forecast or marketing plan, the long-term or strategic plan and the budget or financial plan. The production plan is defined in family groups and in coarse time periods (months or quarters). It is the executive's plan for the master scheduler to turn into a manufacturing plan. Therefore, it must be signed off to authorise the commitment for purchasing material to meet the plan.

The production plan is the agreed-upon plan that it created by the company management. It uses family groups and covers the horizon of the Master Production Schedule (MPS) in monthly periods and the time required to change the resources in quarters. It is used by the Master scheduler to create the Master Production Schedule. The activity to create this plan is now being referred to as 'production planning' or 'sales and operations planning'.

Resource Requirements Planning (RRP)

As the executive sitting in the boardroom generates the production plan, how practical is it?

Resource Requirements Planning Objectives

Resource requirements planning is the capacity or resource check of the production plan using load profiles for each family group. It allows you to:

- Anticipate the long-term resource requirements
- Picture the load at critical work centre
- Balance the requirements across the available capacity
- Check out the plan to verify if it will give you the required performance measurements

The Calculation

Resources requirements planning calculates the resources required by taking the quantities per time product family and multiplying these by the contents of the load profile. Each load profile identifies the key resources and quantities needed for one of that family. This is calculated for each family group. The total requirements are then added for each resource per time period. The total is then compared to that which is stated to be available in the work centre or resource file. The result is a comparison report that can be produced and that indicates the amount of a resource that is required against that which is available.



Activity 12.1

Explain how MRP can contribute to improving productivity as a theoretical query, or as in your own organisation in a practical manner.

12.4 Master Production Schedule (MPS)

This is performed by the master scheduler and is the act of converting the production plan into a detailed manufacturing plan.

The MPS is defined normally at the finished goods part number level in much finer time periods, either actual dates or weeks. The master production schedule is the plan on which all subsequent plans are built.

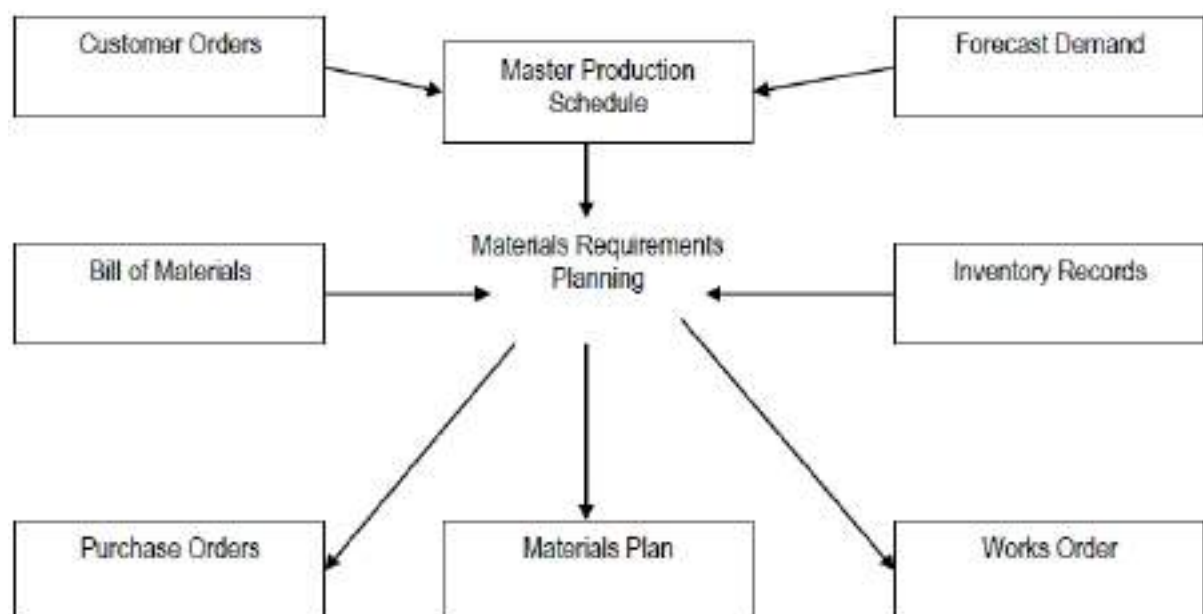
The MPS is often termed 'the anticipated build schedule'. This first short definition sums up the whole concept of the MPS. It is what we believe we will need to build in order to meet the company's objectives.



Think Point 12.2

Explore the role of MRP in synchronising production activities to meet customer demand.

The Master Production Schedule in the MRP 1 Schematic



Source: Pycraft et al (2010)

Practical Application 12.1



Read the following scenario and answer the questions that follow:

Delta Furniture Co. is planning its next three months of production based on customer orders and projected demand. The Master Production Schedule (MPS) is critical in this process, as it helps Delta determine when to produce each product and aligns these plans with material availability. The MPS for tables is set to produce 500 units in the first month, but a large, urgent order requires an additional 200 tables in the second month. The MPS needs adjustment to meet this new demand, but the team is concerned about the impact on other scheduled production activities and materials availability.

Question:

Establish how should Delta Furniture can adjust its Master Production Schedule (MPS) to meet the new demand for tables without disrupting other production schedules, and what considerations are essential in making these adjustment?

The purposes of MPS

The MPS is the key to an MRP II success because everything else within the system is working to meet this plan. The MPS is where marketing and sales meet manufacturing and the link man is the master production scheduler. The concept is that order entry and customer promises are made from information on the MPS, and the task of the manufacturing department is to meet the schedule, which will then result in satisfied customers. Whilst the master production schedule will differ from company-to-company dependent upon the software used, the manufacturing process, and the requirements of the particular industry, the overall reasoning is similar.

Rough Cut Capacity Planning (RCCP)

One of the problems the master scheduler has when converting the production plan into the MPS is to have a practical balance schedule. Rough cut capacity planning is the capacity check module for the MPS. It analyses the resources required by the MPS and uses load profiles for the critical resources. Rough cut capacity planning is an interactive what-if approach.

Material Requirements Planning (MRP)

This part of the system is straightforward; in that it is based purely on arithmetic. It is the add-up, take-away part, which calculates both the purchased material requirements and the lower-level manufacturing requirements. MRP creates planned orders in line with the inventory rules and goes out into the future to the full horizon of the MRP.



Video Activity 12.3

Watch the video using the link below.

Watch the below video explaining MRP

Available from: <https://youtu.be/u3P6YMI5Ah0>



Activity 12.2

Illustrate the types of resources that can be scheduled using MRP II.

Capacity Requirements Planning (CRP)

CRP is a comparison of the standard hours required with the hours stated to be available. This is summarised in time periods but calculated on a date resolution by manufacturing order operation, per work centre. The logic of CRP is that it receives all manufacturing orders from MRP and then breaks each order down into individual operations.

For each operation, the batch quantity is multiplied by the standard time required. In this manner, a number of standard hours required per work centre per operation can be calculated. This can be achieved for each manufacturing order operation. The sub-totals can then be accumulated per work centre, per time period. The final figure can then be compared to that time which is stated to be available.

Capacity requirements planning is a comparison of the standard hours required to those stated to be available. Capacity requirements planning is a comparison of the standard hours required to those stated to be available in a summarised form, either as a histogram or as a percentage load report.

Production Activity Control (PAC)

This section was previously called 'shop floor control', but nowadays since it is fully realised that much of the control is not situated on the shop floor but in the planning office, it is called Production Activity Control.

This means that our manufacturing managers need information so they can do their respective jobs better. They must have the information to make logical decisions timeously.

Input/output control is used to control the volume of work on the shop floor, and operation sequencing makes detailed scheduling and priority control feasible.

Case Study 12.1

Read the following article and answer the question below.

Delta Furniture Co. manufactures custom furniture, including tables, chairs, and cabinets. To meet customer demands without overstocking inventory, Delta adopted a Material Requirements Planning (MRP) system. By using MRP, Delta can manage its raw materials and production schedules more efficiently, ensuring each piece of furniture is completed on time with minimal excess inventory.

The MRP system at Delta Furniture relies on three main data sources: the Master Production Schedule (MPS) to plan production, the Bill of Materials (BOM) to determine required materials, and Inventory Records to track stock levels. Delta's MRP process starts with the MPS, which outlines upcoming production for the next three months based on current orders and sales forecasts. Then, it uses the BOM to calculate the exact quantity of wood, screws, hinges, and varnish required. Finally, Inventory Records help Delta decide when to order more materials.

However, Delta recently experienced a setback: a shortage of a specific type of hardwood delayed the production of tables. The hardwood supplier faced unexpected disruptions, and Delta's MRP system didn't account for this shortage in advance, leading to delayed orders and customer dissatisfaction. The incident highlighted the need for Delta to adapt its MRP system to handle unpredictable supply chain issues.

Questions:

1. Outline the main advantages of MRP for Delta Furniture Co., and how did the recent supply disruption highlight limitations in their MRP system?
2. Evaluate how Delta Furniture Co. can improve its MRP system to reduce the risk of production delays due to material shortages?



12.5 Kanban

The Japanese word 'Kanban' has become familiar to many of us in recent years. The Kanban, or pull system, serves two purposes:

- To get material from the 'next point of use'
- To authorise the producer to make more material

The essence of this methodology is the standard container with a card attached. This creates a 'pull system' in which work centres signal with a card that they wish to withdraw parts from feeding operations or vendors.



Think Point 12.3

Evaluate the impact of employee training and engagement in leveraging MRP tools and methodologies for continuous improvement.

'Pull' signifies the production of items only as demanded for use.

In practice, the Kanban can be something other than a card, such as a returnable container, a circle/square on the ground or a tag. When a Kanban is empty or free, it authorises material to be replenished. For example, when material is collected by the shop feeder of the assembly line, he returns the empty container that held the parts they have already used. The receipt of the empty container is the instruction to the producing work centre to make another container full.

12.6 Inventory Rules

For any planning calculation to be made, the company's batching, ordering and inventory rules must be clearly stated so the system can use them to make the ordering plan both practical and in agreement with the company's policy. The rules normally have to be formally approved by the financial director as they have serious consequences on the inventory holding and cash flow.

The inventory rules provide management with the means to dictate to the system how the company's policies should be applied. Various order methods are available to the user, such as order point, MRP, time phased order point, and no system control (user planning and control). This means that you have a choice in the method used by the system for the ordering of material.

You are given a choice in the method by which the order quantities are calculated, for example, least total cost, part period balancing or discrete. The most commonly used of all methods is period order quantity that is a means of ordering a number of days' supply instead of certain quantities. All of these methods apply different parameters for the grouping of daily requirements for parts so that the purchase orders, or manufacturing order quantities are economical or desirable's quantities to be ordered. To make the resulting quantities practical, batching rules are used.



Activity 12.3

Explain the use of the informal and formal system for MRP planning and scheduling.



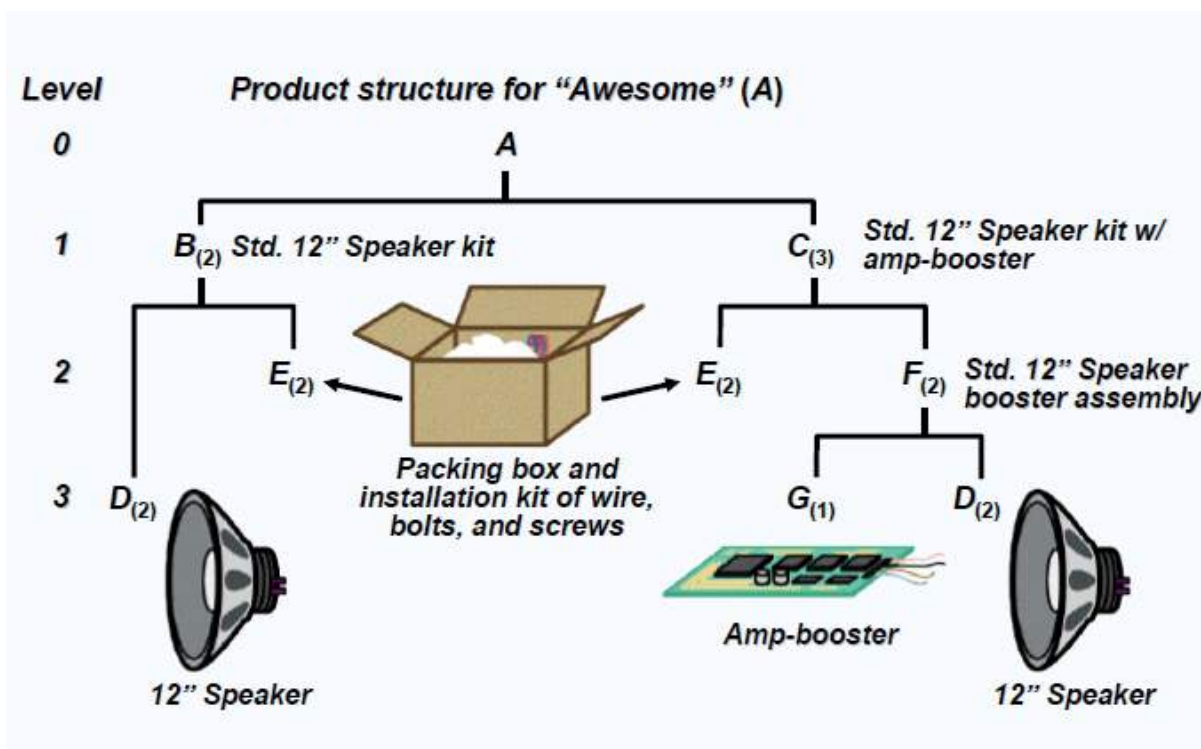
Practical Application 12.2

Explain the use of the system for Material Requirements planning and scheduling.

12.7 Bill of Material (BOM)

For any calculation to be made to establish what material is required, the BOM must be available, because it is in the BOM that the material content of each product is specified. For a works order to be scheduled the operation lead times are needed, and these are found on the routing file.

The BOM consists of two elements: an item master file where the details of each part number used in the system are identified, and the bill of material itself (or product structure record as it is sometimes known), where the content of an assembly is defined. The routing file is where the operations that the product must pass through during the manufacturing process are identified. The example on the following page represents how to develop the product structure and “explode” it to reveal the requirements for each component.



Source: Heizer.et.al (2020)

The item master file is where the information about a part is held. It is not uncommon to find over 200 fields in this file. For example, it will hold engineering, planning, inventory, costing and stores related data for each part number.

Each manufactured part or sub-assembly has a bill of material and a routing. A bill of material consists of a record for each component or raw material used on that parent. The parent is the item to be made and the raw materials or piece parts used are known as its components. A whole

complex assembly is built up of sub-assemblies and piece parts, but each sub-assembly is entered into the BOM and controlled as an entity in itself. They are known as single level bills of material. For each BOM there is routing which holds the detail of each operation, such as set-up time and unit run time. It is from these times that the operation time is established for the order quantity and is then used for scheduling the work through the plant.

Other MRP II Modules

A full MRP II system will either include or provide interface points to other modules which are needed to run a manufacturing business. Such modules are:

- Customer order entry
- Accounts payable (creditors)
- Accounts receivable (debtors)
- General ledger
- costing
- Purchasing
- Forecasting
- Sales analysis
- CAD/CAM interfaces
- Distribution (DRP)

It depends on the particular software whether these are further modules of the MRP II package or if a financial package is made to interface. A typical approach is to provide an interface to a well-known specialised package as in the case of payroll.

Practical Example 12.3

Example of MRP Use in Practice



For instance, a furniture manufacturer using MRP would first input their production schedule for tables, chairs, etc. MRP would then refer to the BOM to determine the quantity of wood, nails, and varnish required. Based on existing inventory and lead times, the system would generate a schedule for ordering materials in time for production, optimising inventory levels and reducing costs.

MRP II Summary

It should be realised by now that MRP II is not just a production control system but a system which provides the tools for people throughout the whole company to manage their business better. It alerts you to potential problems to solve, before it is too late.

Why Material Planning?

As we have seen, material planning is at the heart of the MRP II system for manufacturing companies, but perhaps the question is, why do we need MRP II anyway?

To obtain and implement an MRP II system is not a small job, so why do companies embark on this long and expensive journey?

It has been established internationally over the last few years that on average, the following benefits can be gained if the implementation is taken seriously.

Tangible Benefits

- An inventory reduction of 20% or an increase in stock turnaround of 50% is normal. This is obviously dependent upon how well you have managed your inventory prior to the start of the implementation
- A saving of 5% in purchased material cost due to working to a plan and knowing what is wanted when
- A company can expect productivity improvement from 10% to 30% depending upon complexity of material requirements. In the electronics assembly business, with hundreds of components going onto a printed circuit board, the problem has always been how to assemble at the right time. Working to a plan and using MRP II to calculate the component requirements reduces the number of material shortages considerably
- The overtime used to overcome late shipments is typically reduced by 50%. By working to a plan, the system can identify future problems and we can solve them before they affect production
- Due to the fact that we will be able to produce to the master plan, we will be delivering what is needed to the finished goods store on time. This will result in us having the correct items in stock when needed. With the correct stock available, sales should increase by 5% for a company that sells ex-stock
- If we now apply each of these savings to the company, it will result in an overall profitability increase of 30% or more

These figures are not sales talk but well established over a number of years from companies that have made a reasonable job of implementing MRP II.

Intangible Benefits

Having looked at the hard numbers with the tangible benefits we now need to examine the intangible benefits:

- Because we will now know what we have and *what* will be available *when*, it will mean that our sales and customers service staff can talk more confidently to our customers and give more realistic promises
- Typically, management can become 'highly paid progress chasers', as they are called upon to solve supply problems with our supplier's integrated system. Much of this need will fall away and they can then do what they are paid to do, which is to manage the business
- The 'what if' facility will allow the company to manage its future growth effectively by computing what resources will be required to meet the proposed plan
- The MRP logic will allow us to manage the changes in business volumes and product mix before they seriously affect the output programme and inventory levels
- If it is a requirement to keep track of which lot of material has been used to produce a particular product, the 'Lot Tractability Module' will support this
- Many hours are often spent in 'Output Meetings' where the main topic is not output but shortages. Often, much time is wasted arguing whose figures are correct, when we are all getting our information from the same common source. With the introduction of MRP, this problem will disappear

Overall, we can say that with everything moving in the same direction in a coordinated way, the problems will reduce and subsequently the quality of life for everyone in the company will improve.



Activity 12.4

Determine the effect of MRP in improving company profits.

The Scheduling Problem

Having now stated what the benefits can be after implementing an MRP II system, we need to understand what the main cause of our problem is. Generally, in most manufacturing companies it can be put down to the inability of scheduling to manage a continual stream of changes. Let's examine what can go wrong with scheduling:

- Schedules start with a forecast, and by definition they are only estimates so these will not necessarily be accurate. It means that our basic material plan starts with unreliable numbers
- We then find that the dates on which products are required keep changing and this plays havoc with the material plan

- Our customers change their minds not only on what they *want*, but *when* they want it. This causes multiple changes to the manufacturing plan and to the material requirements
- The engineering design department keeps updating products and modifying them continually. To inform every one of the alterations they produce what is known as an Engineering Change Notes (ECNs). Effectively, they are telling the planner that the material he ordered is no longer required but now they need something else urgently. Again, the plan is being changed
- Our manufacturing team does its part to upset the plan by scrapping the odd batch or so. This results in more material being needed and an extra capacity requirement which has to be fitted into an already busy schedule
- Our cycle counters find errors with the stock balances so put through stock adjustment transactions which alter the basic numbers on which our initial plans were built
- The management team goes missing by attending conferences or courses or participants may even be sick, but while they are away more alterations are made to the plan
- Last, but not least is our group of suppliers which has been known to deliver late on the odd occasion. When this occurs it requires the planner to re-schedule work that cannot be built but also to bring in work that can be built

With all these changes taking place on a daily basis, is it any wonder that scheduling is a problem, MRP II is not going to stop this from happening, but the occurrences should be reduced, and it will give us the means to manage them better.

What else do we need to do to improve the situation? As the changes are occurring by the hour in manufacturing, at least our system must endeavour to keep up with them so at least we are working to the latest information. This means that we need to run MRP II system nightly so that it can identify what needs to be done each day, to keep on top of the ever-changing situation.

The Formal or Informal System

In many companies, although each person or department is trying his or her best, things still don't work out too well. No matter who you ask, they will quickly tell you why they have not met their objectives and which department is causing their problem. This is commonly known as 'chronic finger pointing'. It is a common 'ailment' that companies suffer when their operations are not well coordinated.

We also find that the numbers in the system are not reliable. For example, the stock balances on the system do not reflect what is really in the stock bin. The purchase order file tells us that the following quantities are due on certain dates, but these orders were completed some weeks ago.

Overall, we call this the informal system and with it goes the problem of accountability. Who is really responsible for what, and do they have the means to be responsible and accountable for these

activities? When we take on MRP II we need to accept that if we want it to work then we must operate in the *formal* system. This does not mean that the system takes over and stops us from doing what the business needs, it means that we do things properly at the right time, in the agreed manner and we do it 'right' the first time.

Knowledge Check Questions 12.1

Select the correct answer:

1. Which of the following is a primary objective of Materials Requirements Planning (MRP)?
 - a. Minimise production costs
 - b. Maximise inventory levels
 - c. Schedule production activities
 - d. Reduce supplier lead times
2. What is the primary input required for Materials Requirements Planning (MRP)?
 - a. Sales forecasts
 - b. Production capacity
 - c. Bill of materials (BOM)
 - d. Inventory turnover rates
3. In MRP, what does the lead time offset represent?
 - a. The time required to deliver materials after placing an order
 - b. The duration between production scheduling and actual production
 - c. The time needed for quality control inspections
 - d. The delay in payment to suppliers



Knowledge Check Questions 12.2

State if the following is True or False by selecting the correct answer.



1. MRP is primarily concerned with determining the optimal order quantity for materials.

True

False

2. A Master Production Schedule (MPS) is a key output of the MRP process.

True

False

3. MRP systems are mainly used for managing finished goods inventory.

True

False

12.8 Summary

This unit provided a comprehensive exploration of Materials Resource Planning (MRP II) and its role in managing manufacturing resources efficiently. The key topics covered included understanding how organisations align production schedules with customer demand and available resources, examining the detailed scheduling of production activities to ensure timely fulfilment of orders and exploring the application of this lean inventory technique in just-in-time (JIT) systems to minimise waste and improve workflow.

The inventory rules was explored for discussing strategies for maintaining optimal inventory levels, balancing supply and demand while reducing holding costs and the bill of materials (BOM) was explained for analysing how BOM structures help organisations track and manage raw materials and component requirements for production.

Through practical case studies and industry applications, students gained valuable insights into how MRP II supports decision-making, streamlines production planning, and enhances supply chain coordination. Understanding these concepts equips operations managers with the tools to develop efficient, cost-effective manufacturing strategies in a dynamic business environment

Revision Questions

1. The Hunicut and Hallock Corporation makes two versions of the same basic file cabinet, the TOL (Top-of-the-line) five drawer file cabinet and the HQ (High-quality) five drawer filing cabinet. The TOL and HQ use the same cabinet frame and locking mechanism. The drawer assemblies are different although both use the same drawer frame assembly. The drawer assemblies for the TOL cabinet use a sliding assembly that requires four bearings per side whereas the HQ sliding assembly requires only two bearings per side. (These bearings are identical for both cabinet types.) 100 TOL and 300 HQ file cabinets need to be assembled in week #10. No current stock exists.

Illustrate a material structure tree for the TOL and the HQ file cabinets.

2. Illustrate net material requirements plan for the TOL and HQ file cabinets in the previous problems assuming a current on-hand finished goods inventory of 100 TOL cabinets. The lead times are given below.

Painting and final assembly of both HQ and TOL requires 2 weeks.

Both cabinet frames and lock assembly require 1 week for manufacturing.

Both drawer assemblies require 2 weeks for assembly.

Both sliding assemblies require 2 weeks for manufacturing.

Bearings require 2 weeks to arrive from the supplier.

3. A manufacturing company is implementing Materials Requirements Planning (MRP) to manage its production processes. The company faces challenges with supplier reliability and lead times. How can the company address these challenges using MRP principles?

4. A company produces a product that requires 3 units of component A and 2 units of component B. The company plans to produce 500 units of the final product. Calculate the total quantity of component A needed using Materials Requirements Planning (MRP).

5. A company uses MRP to plan its production schedule for the next month. It has a lead time of 2 weeks for receiving materials after placing an order. Calculate the reorder point for a component that has a weekly demand of 200 units.

6. A company uses MRP to plan its production activities for a complex product. During implementation, the company encounters discrepancies between planned and actual production output. What steps should the company take to improve the accuracy of its MRP system?



Answers

Activity 12.1

Explain how MRP can contribute to improving productivity as a theoretical query, or as in your own organisation in a practical manner.

Answer: To be discussed during the webinar(s) with lecturer

Activity 12.2

Illustrate the types of resources that can be scheduled using MRP II.

Answer: To be discussed during the webinar(s) with lecturer

Activity 12.3

Explain the use of the informal and formal system for MRP planning and scheduling.

Answer: Formal systems are automated, while informal rely on manual processes

Activity 12.4

Determine the effect of MRP in improving company profits.

Answer: Material Requirements Planning (MRP) Improves profits by optimising production and reducing waste.

Case Study 12.1

1. Outline the main advantages of MRP for Delta Furniture Co., and how did the recent supply disruption highlight limitations in their MRP system?

Answer: MRP offers Delta Furniture significant advantages, such as precise control over inventory and materials, which reduces waste and holding costs. By coordinating materials with production needs, MRP allows Delta to plan efficiently, ensuring timely availability of materials based on the production schedule. This reduces excess inventory and streamlines ordering, which is particularly beneficial in custom furniture production with specific material requirements. However, the recent hardwood shortage exposed limitations in Delta's MRP system, primarily its vulnerability to supply chain disruptions. Since MRP relies on fixed lead times and regular supplier performance, any unexpected delays or shortages can disrupt production. Delta's MRP system lacked a mechanism for adjusting to supply fluctuations, highlighting the need for incorporating contingency plans, such as safety stock or multiple supplier sources, for critical materials.

2. Evaluate how Delta Furniture Co. can improve its MRP system to reduce the risk of production delays due to material shortages?

Answer: Delta could implement several enhancements to improve resilience in its MRP system:

- **Strategic Safety Stock:** By maintaining a small safety stock of high-risk materials, such as hardwood, Delta could prevent production delays during temporary shortages. This approach would provide a buffer for essential materials without significantly raising holding costs
- **Supplier Diversification:** Establishing relationships with multiple suppliers for critical materials would allow Delta to source from an alternative supplier if one faces delays
- **Dynamic Lead Time Adjustment:** Delta could update its MRP system to reflect real-time supplier lead times, allowing the MRP to account for potential delays. Incorporating forecasted variability in lead times can help Delta plan proactively for supply disruptions

Knowledge Check Questions 12.1**1. Which of the following is a primary objective of Materials Requirements Planning (MRP)?**

Answer: c. Schedule production activities

2. What is the primary input required for Materials Requirements Planning (MRP)?

Answer: c. Bill of materials (BOM)

3. In MRP, what does the lead time offset represent?

Answer: a. The time required to deliver materials after placing an order

Knowledge Check Questions 12.2**1. MRP is primarily concerned with determining the optimal order quantity for materials.**

Answer: False

2. A Master Production Schedule (MPS) is a key output of the MRP process.

Answer: True

3. MRP systems are mainly used for managing finished goods inventory.

Answer: False

Practical Application 12.1

Establish how should Delta Furniture can adjust its Master Production Schedule (MPS) to meet the new demand for tables without disrupting other production schedules, and what considerations are essential in making these adjustment?

Answer: Delta should adjust its MPS by balancing the increased demand for tables with the availability of materials and production capacity. To meet the new demand, Delta could consider pulling production resources from products with lower demand or scheduling additional shifts for table production, provided labour resources allow.

- **Material Availability:** Delta's MRP system should be consulted to confirm there are enough raw materials (e.g., hardwood) to meet the updated production schedule. If not, Delta may need to expedite material orders or reallocate materials from other products with lower demand
- **Production Capacity:** The MPS should consider whether Delta has enough capacity to produce the extra tables without significantly delaying other items. If capacity is limited, alternative solutions might include prioritising the most urgent orders or temporarily adjusting production on lower-priority items
- **Lead Time Adjustments:** If suppliers need extra lead time to fulfil the increased demand for raw materials, Delta should update the MPS to reflect these new timelines. Additionally, setting a contingency within the MPS can help Delta adapt to potential changes in supplier delivery times

Practical Application 12.2

Explain the use of the system for Material Requirements planning and scheduling.

Answer: Material Requirements Planning (MRP) is a system used in manufacturing to plan and schedule the necessary materials and production steps to meet demand. It ensures that materials and components are available for production when needed, optimises inventory levels, and organises production schedules effectively. Here's how MRP is used for planning and scheduling:

- **Demand Forecasting and Input Data:**
 - **Demand Forecasting:** MRP begins with demand forecasting or actual customer orders to determine what products are needed and when
 - **Input Data:** The MRP system uses three primary types of data:
 - **Master Production Schedule (MPS):** Outlines what to produce, how much, and when
 - **Bill of Materials (BOM):** A detailed list of all raw materials, components, and sub-assemblies required to make a product
 - **Inventory Records:** Keeps track of stock levels, lead times, and item availability to avoid shortages and overstock.

- Planning for Materials Requirements:
 - The MRP system calculates the specific quantities of each material needed at each production stage, ensuring components are available when production requires them. It determines the timing and quantity for ordering materials by “exploding” the BOM. For example, if a product requires three parts, MRP will check the availability of each part, then calculate the order quantities needed to meet production demand
 - Scheduling Production Activities:
 - Lead Time Management: MRP uses lead time data to ensure materials are ordered early enough to arrive before production starts, considering procurement or manufacturing lead times
 - Order Release Schedules: Based on calculated needs, the system creates a schedule of when each component or material must be ordered or produced
 - Capacity Planning: Some advanced MRP systems incorporate capacity planning, taking into account the production floor's resources to ensure that there's enough capacity to fulfil the production schedule
- Inventory Control and Cost Optimisation
 - MRP minimises holding costs by ordering materials just in time for production, reducing excess inventory
 - Lot Sizing: It can recommend optimal order quantities (lot sizes) to balance holding costs against ordering costs, helping companies maintain lean inventory levels without risking stock-outs
- Adjusting for Changes
 - Dynamic Rescheduling: MRP systems are often updated regularly to reflect changes in demand, lead times, or material availability, allowing production schedules to adjust dynamically to real-time conditions
 - What-If Analysis: MRP systems often include what-if scenarios, allowing planners to assess how changes in demand, lead times, or supplier issues might affect production

Think Point 12.1

How can Materials Requirements Planning (MRP) optimise production schedules to improve operational efficiency and reduce costs?

Answer: MRP and production schedules helps synchronise production with demand, improving efficiency.

Think Point 12.2

Explore the role of MRP in synchronising production activities to meet customer demand.

Answer: It aligns resources with production needs, ensuring timely delivery.

Think Point 12.3

Evaluate the impact of employee training and engagement in leveraging MRP tools and methodologies for continuous improvement.

Answer: Employee training in MRP improves efficiency and supports continuous improvement in operations.

Video Activity 12.1**1. Explain the main purpose of using MRP in manufacturing?**

Answer: MRP ensures that the right quantities of materials are available at the right time to meet production schedules, reducing waste and improving efficiency.

2. Determine how you could use MRP grids to plan the production of the bicycle's components?

Answer: MRP grids help map out the material requirements for each component (e.g., frame, wheels) over time, ensuring materials are ordered and delivered just in time for assembly.

3. Determine what could happen if one key component's MRP grid is not properly aligned with the production schedule?

Answer: If a component's MRP grid is not aligned, it could lead to delays in the production process due to missing parts, potentially causing missed deadlines and increased costs.

Answers to Revision Questions

Unit 1

1. Mance Fraily, the Production Manager at Ralts Mills, can currently expect his operation to produce 1000 square yards of fabric for each ton of raw cotton. Each ton of raw cotton requires 5 labour hours to process. He believes that he can buy a better quality raw cotton, which will enable him to produce 1200 square yards per ton of raw cotton with the same labour hours. Outline the impact on productivity (measured in square yards per labour-hour) if he purchases the higher quality raw cotton.

Answer:

$$\text{Current labor productivity} = \frac{1000 \text{ sq yds}}{1 \text{ ton} \times 5 \text{ hours}} = 200 \text{ sq yds per hour}$$

$$\text{New labor productivity} = \frac{1200 \text{ sq yds}}{1 \text{ ton} \times 5 \text{ hours}} = 240 \text{ sq yds per hour}$$

2. C. A. Ratchet, the local auto mechanic, finds that it usually takes him 2 hours to diagnose and fix a typical problem. Illustrate his daily productivity (assume an 8-hour day).

Answer: Mr. Ratchet believes he can purchase a small computer trouble-shooting device, which will allow him to find and fix a problem in the incredible (at least to his customers!) time of 1 hour. He will, however, have to spend an extra hour each morning adjusting the computerised diagnostic device.

3. Mr. Ratchet believes he can purchase a small computer trouble-shooting device, which will allow him to find and fix a problem in the incredible (at least to his customers!) time of 1 hour. He will, however, have to spend an extra hour each morning adjusting the computerised diagnostic device. Explain the impact on his productivity if he purchases the device.

Answer:

$$\text{Current productivity} = \frac{8 \text{ hours per day}}{2 \text{ hours per problem}} = 4 \text{ problems per day}$$

$$\text{Productivity with computer} = \frac{7 \text{ hours per day}}{1 \text{ hour per problem}} = 7 \text{ problems per day}$$

$$\text{Productivity improves } 75\% \left(\frac{7-4}{4} = \frac{3}{4} = .75 \right)$$

4. Joanna French is currently working a total of 12 hours per day to produce 240 dolls. She

thinks that by changing the paint used for the facial features and fingernails that she can increase her rate to 360 dolls per day. Total material cost for each doll is approximately R3.50; she has to invest R20 in the necessary supplies (expendables) per day; energy costs are assumed to be only R4.00 per day; and she thinks she should be making R10 per hour for her time. Viewing this from a total (multi-factor) productivity perspective, outline her productivity at present and with the new paint.

Answer:

	Currently	Using the new paint
Labour	$12 \text{ hrs} * \text{R}10 = \text{R}120$	$12 \text{ hrs} * \text{R}10 = \text{R} 120$
Material	$240 * \text{R}3.50 = \text{R}840$	$360 * \text{R}3.50 = \text{R}1260$
Supplies	$= \text{R} 20$	$= \text{R} 20$
Energy	$= \text{R} 4$	$= \text{R} 4$
Total Inputs	$= \text{R}984$	$= \text{R}1404$
Productivity	$240/984 = 0.24$	$360/1404 = .26$

5. Illustrate how total (multi-factor) productivity would change if using the new paint raised Ms. French's material costs by R0.50 per doll.

Answer:

Using the new paint

Labour	$12 \text{ hrs} * \text{R}10 = \text{R} 120$
Material	$360 * \text{R}4.00 = \text{R}1440$
Supplies	$= \text{R} 20$
Energy	$= \text{R} 4$
Total Inputs	$= \text{R}1584$
Productivity	$360/1584 = 0.23$

6. If she uses the new paint, outline by what amount Ms. French's material costs could increase without reducing total (multi-factor) productivity.

Answer:

	Currently	Using the new paint
Labour	12 hrs * R10 = R120	12 hrs * R10 = R 120
Material	240 * R3.50 = R840	360 * R3.50 = R1260
Supplies	= R 20	= R 20
Energy	= R 4	= R 4
Total Inputs	= R884	= R1404
Productivity	$240/884 = 0.24$	$360/1404 = 0.26$

It follows then that the new paint could raise Materials cost by no more than approximately R0.27 (the difference between R3.77 and R3.50) before Ms. French would experience a decrease in multifactor productivity.

Unit 2

1. Explain how changes in the external environment may affect the OM strategy for a company. For example, examine the impact that the following factors are likely to have on OM strategy.

a.The occurrence of a major storm or hurricane.

Answer: A major storm or hurricane may have considerable impact on a company's facilities and scheduling. Flooding and wind damage can make a facility unusable or significantly reduce its capacity. Stocks of raw materials, especially agricultural products, might be damaged or in short supply. The long-term availability of some materials might be significantly reduced. There may be a shortage of important services during the recovery. For example, the demand for roofers and builders is high after a major storm and they would like to be able to rapidly increase their capacity to handle the higher demand.

b.Terrorist attacks of 9/11/01.

Answer: Terrorist activity has forced organisations to rethink, and in many cases expand, their security systems. Firms have also had to reevaluate their supply networks and consider increasing their inventory safety stock. They may also reassess the risks of foreign locations and expansion.

c.The much-discussed decrease in the quality of American primary and secondary school systems.

Answer: A decrease in the skill levels of Americans entering the labour market requires that organisations place more emphasis on training, turn to automation to obviate the need for human labour, and hire from outside the United States.

d.Trade Legislation such as WTO and NAFTA and changes in tariffs and quotas.

Answer: World Trade Organisation (WTO) and North American Free Trade Agreement (NAFTA) changed the rules for trading, opened new markets, and in some instances, changed the role of labour versus capital (where labour is especially low cost, emphasis often shifts from the use of capital to the use of labour).

e. The rapid rate at which the cost of health insurance is increasing.

Answer: The increasing cost of health insurance adds significantly to the cost of labour. Some large US organisations are passing on this increased cost to the employees or reducing other parts of the benefit package in response to these pressures.

f. The Internet.

Answer: The Internet has promoted Globalisation of markets, and eliminated barriers of geography and time.

2. Explain how the changes in the internal environment affect the OM strategy for a company.

For example, examine the impact that the following factors are likely to have on OM strategy.

a. The increased use of Local and Wide Area Networks (LANs and WANs).

Answer: The increased use of Local Area Networks (LANs) and Wide Area Networks (WANs) has, among other things, enabled new organisational structures, the movement of the locus of responsibility further down the organisational hierarchy (elimination of middle management), and the increasing practicality of JIT operations, mass customisation, etc.

b. An increased emphasis on service.

Answer: The increased emphasis on service has, among other things, fostered an increased information or information technology content of many products. Firms are also increasing training because so much of the service economy is dependent upon individual competence.

c.The increased role of women in the workplace.

Answer: The increased role of women in the workplace is requiring an increased emphasis on the creation and communication of appropriate human resource policies. It may also be fostering the creation of flexible work schedules and, to a lesser degree, telecommuting.

d. The seemingly increasing rate at which both internal and external environments change.

Answer : Some companies seem to be adopting the perspective that their main problem is now the “management of change” as opposed to the management of a specific process or product. If nothing else, the management of change is becoming a formal part of the manager’s responsibility.

3. Operations managers are called upon to support the organisation's strategy. OM does this with some combination of one of three strategies. Outline these three strategies.

Answer: OM managers support the firm's strategy by achieving a competitive advantage through some combination of:

- Differentiation, (different product/service offerings)
- Low-cost leadership, and, (providing a product/service at a price lower than other competitors)
- Response. (quickly responding products/services to the changing needs and wants of customers)

4. Determine how globalisation impacts the standardisation of products in OM?

Answer: Globalisation allows companies to source materials from different countries, reducing costs and expanding supplier options. However, it also increases complexity, with longer lead times and risks like political instability or trade restrictions.

5. Discuss the challenges of managing global operations.

Answer: Globalisation introduces challenges such as managing cross-border logistics, dealing with different regulatory environments, currency fluctuations, and ensuring consistent quality across various locations.

6. Analyse how trade policies and tariffs influence operations management under globalisation.

Answer: Globalisation pushes companies to adopt more efficient inventory strategies, like Just-in-Time (JIT), to reduce holding costs and respond to international demand fluctuations, though it also makes them vulnerable to supply chain disruptions.

7. Explain how globalisation impacts the standardisation of products in OM.

Answer: Globalisation encourages companies to standardise products to cater to international markets, achieving economies of scale. However, it may also require localisation strategies to meet specific customer preferences or regulatory requirements in different regions.

8. Analyse some of the challenges of managing global operations.

Answer: Managing global operations presents challenges such as cultural differences, varying legal and regulatory frameworks, and managing a dispersed workforce. Companies must adapt to different business environments while maintaining operational efficiency

9. How do trade policies and tariffs influence operations management under globalisation?

Answer: Trade policies and tariffs affect the cost of importing and exporting goods, which can disrupt global supply chains and impact profitability. Companies need to develop strategies like sourcing locally or adjusting prices to mitigate these effects.

10. Explain the advantages and disadvantages of globalisation.

Answer: Globalisation can be described as the increasing interconnectedness and interdependence of the world's economies, cultures, and populations. Here are some key advantages and disadvantages:

Advantages of Globalisation	Disadvantages of Globalisation
• Economic Growth: Boosts economic activity and GDP • Access to Markets: Expands customer base • Increased Efficiency: Drives innovation and improvement • Job Creation: Generates employment opportunities • Cultural Exchange: Enhances cultural understanding and diversity • Technological Advancements: Spreads technology and innovation	• Economic Inequality: Can increase income disparity • Job Displacement: Local jobs may be lost to international competition • Cultural Erosion: May overshadow local cultures and traditions • Environmental Impact: Can lead to pollution and resource depletion • Economic Dependence: Vulnerable to global market fluctuations • Loss of Sovereignty: Limits national policy flexibility

Unit 3

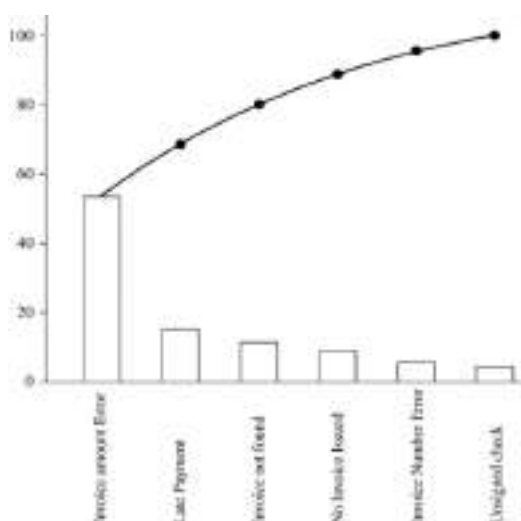
1. Explain the techniques that you would use, and the conclusions drawn about defects in the accounts receivable department.

Answer:

Category	Frequency	Percent
Invoice amount does not agree with the check amount	108	54
Invoice not on record (not found)	24	12
No formal invoice issued	18	9
Check (payment) not received on time	30	15
Check not signed	8	4
Invoice number and invoice referenced do not agree	12	6
? =	200	100

2. Use a Pareto chart to organise the defects and conclude that the obvious problem (about half the defects) is the failure of the check to agree with the company's records as to the correct amount. Other problems are late payments and an apparent invoice-filing problem in the office. Notice that 27% of these common errors appear to be the result of procedural problems within accounts receivable (invoice not on record, no invoice issued, and invoice numbering problems). This value could be considerably higher depending on how much of the problem of disagreement between invoice and check amounts is the result of accounts receivable process problems.

Answer:



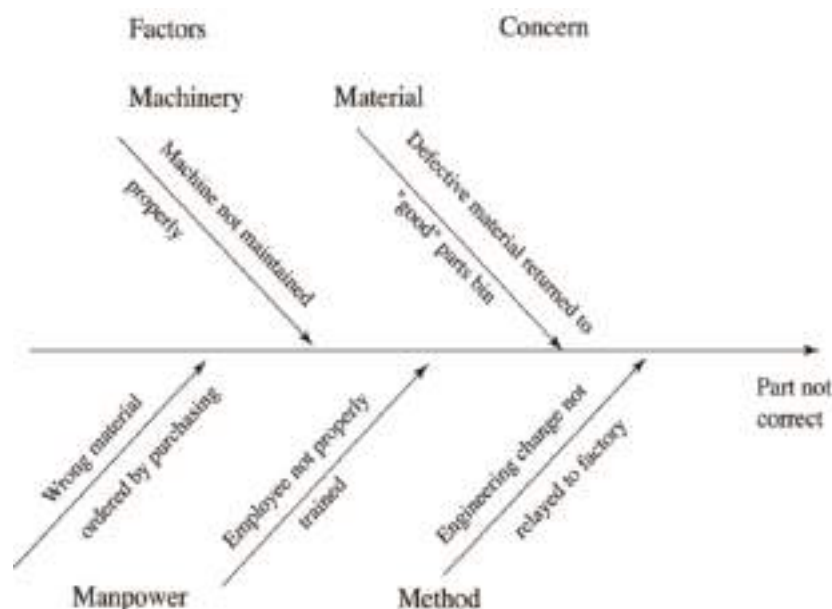
3. Outline a flow chart for purchasing a Big Mac at the drive-through window at McDonald's.

Answer:

Distance	Symbol	Activity
-	↓	Pull up to speaker
-	○	Press button
-	D	Wait for response
-	○	Verbalize order
-	□	Get confirmation of order and cost
20	↓	Move car up in line
-	D	Wait
20	↓	Move car up in line
-	D	Wait
-	□	Verify order and cost
-	○	Pay and receive order
-	↓	Leave
-	●	Realize they forgot the extra cheese!

4. Illustrate a fishbone chart detailing reasons why a part might not be correctly machined.

Answer:



5. Examine how Mercedes-Benz benefits from benchmarking L.L. Bean's warehouse practices, and outline the changes that were implemented.

Answer: Mercedes-Benz observed that L.L. Bean's warehouse employees used flowcharts to identify and eliminate wasted motions. By benchmarking this practice, Mercedes-Benz adopted the use of flowcharts and other visual tools to enhance efficiency in their own operations. This led to a

greater reliance on problem-solving at the worker level, empowering employees to identify inefficiencies and improve workflows autonomously. The practice of benchmarking enabled Mercedes-Benz to integrate effective process management techniques from another industry, highlighting the value of cross-industry benchmarking in fostering continuous improvement.

6. Discuss the primary goal of Just-in-Time (JIT) production.

Answer: To reduce inventory levels and minimise waste by producing only what is needed, when it is needed.

7. Determine how Benchmarking help a organisations improve their performance.

Answer: By comparing their performance metrics to those of industry leaders or best practices, identifying gaps, and implementing changes to reach or exceed those standards.

8. Discuss the reasons why the Taguchi method is important in quality management.

Answer: It focuses on robust design and reducing variation, leading to higher quality products and lower costs.

9. Examine the implications of high failure costs for an organisation practising TQM.

Answer: High failure costs, particularly external failure costs, can have several negative implications for an organisation practising TQM:

- **Financial Impact:** High failure costs can result in significant financial losses due to increased expenses for warranty claims, product recalls, repairs, and customer refunds. This can erode profit margins and strain the company's resources
- **Reputation Damage:** Frequent quality issues can damage an organisation's reputation, leading to a loss of customer trust and loyalty. Negative publicity and dissatisfied customers can hurt the brand image and make it challenging to attract and retain customers
- **Operational Disruptions:** Internal failure costs such as rework and scrap can disrupt production schedules, reduce efficiency, and increase operational complexity. This can lead to higher production costs and delays in delivering products or services to customers
- **Regulatory and Legal Risks:** Persistent quality problems can attract regulatory scrutiny and legal challenges, especially in industries with strict compliance requirements. This can result in fines, penalties, and increased oversight, further adding to the costs and complexities of doing business
- By effectively managing and reducing failure costs, organisations can enhance their overall quality, financial performance, and market competitiveness

Unit 4**1. List the qualitative methods of forecasting.**

Answer: Qualitative methods: Include Delphi method, market research, and expert judgment.

2. Auto Sales at Carmen's Chevrolet are shown below. Develop a 3-week moving average.

Week	Auto Sales
1	8
2	10
3	9
4	11
5	10
6	13
7	-

Answer:

$$\text{Moving average} = \frac{\sum \text{demand in previous } n \text{ periods}}{n}$$

Week	Auto Sales	Three-Week Moving Average
1	8	
2	10	
3	9	
4	11	$(8 + 9 + 10) / 3 = 9$
5	10	$(10 + 9 + 11) / 3 = 10$
6	13	$(9 + 11 + 10) / 3 = 10$
7	-	$(11 + 10 + 13) / 3 = 11 \frac{1}{3}$

3. Carmen decides to forecast auto sales by weighting the three weeks as follows:

Weights Applied	Period
3	Last week
2	Two weeks ago
1	Three weeks ago
6	Total

Answer:

$$\text{Weighted moving average} = \frac{\sum (\text{weight for period } n)(\text{demand in period } n)}{\sum \text{weights}}$$

Week	Auto Sales	Three-Week Moving Average
1	8	
2	10	
3	9	
4	11	$[(3 \times 9) + (2 \times 10) + (1 \times 8)] / 6 = 9.16$
5	10	$[(3 \times 11) + (2 \times 9) + (1 \times 10)] / 6 = 10.16$
6	13	$[(3 \times 10) + (2 \times 11) + (1 \times 9)] / 6 = 10.16$
7	-	$[(3 \times 13) + (2 \times 10) + (1 \times 11)] / 6 = 11.23$

4. A firm uses simple exponential smoothing with $\alpha = 0.1$ to forecast demand. The forecast for the week of January 1 was 500 units whereas the actual demand turned out to be 450 units. Calculate the demand forecast for the week of January 8.

Answer:

$$F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1}) = 500 + 0.1(450 - 500) = 495 \text{ units}$$

5. Exponential smoothing is used to forecast auto mobile battery sales. Two values of $\alpha = 0.8$ and $\alpha = 0.5$.

Evaluate the accuracy of each smoothing constant. Which is preferable? (Assume the forecast for January was 22 batteries.) Actual sales are given below:

Month	Actual Battery Sales	Forecast
January	20	22
February	21	
March	15	
April	14	
May	13	
June	16	

Answer:

Month	Actual Battery Sales	Rounded Forecast with $\alpha = 0.8$	Absolute Deviation with $\alpha = 0.8$	Absolute Error ² with (0.8 Alpha)	Rounded Forecast with $\alpha = 0.5$	Absolute Deviation with $\alpha = 0.5$	Absolute Error ² (with 0.5 Alpha)
Jan	20	22	2	$2^2 = 4$	22	2	
Feb	21	20.4	0.6	$0.6^2 = 0.36$	21	0	
Mar	15	20.88	5.88	$5.88^2 = 34.5744$	21	6	
Apr	14	16.176	2.176	$2.176^2 = 4.734976$	18	4	
May	13	14.4352	1.4352	$1.4352^2 = 2.05979904$	16	3	
Jun	16	13.28704	2.71296	$2.71296^2 = 7.36015196$	14.5	1.5	

				Sum of Absolute Error ² = 53.089327			
			S = 14.80416			S = 16.5	
MAD = ? Absolute deviations / n			2.46736			2.75	
MSE = ?(Forecast errors) ² / n				8.84822116 7			11.2083333 3

On the basis of this analysis, a smoothing constant of $a = 0.8$ is preferred to that of $a = 0.5$ because it has a smaller MAD and a smaller MSE.

6. To Minimise computations, transform the value of x (time) to simpler numbers. In this case, designate year 1996 as year 1, 1997 as year 2, etc.

Answer:

Year	Time Period (X)	Sales (Units) (Y)	X ²	XY
1996	1	100	1	100
1997	2	110	4	220
1998	3	122	9	366
1999	4	130	16	520
2000	5	139	25	695
2001	6	152	36	912
2002	7	164	49	1148
	S X = 28	S Y = 917	S X ² = 140	S XY = 3961

$$\bar{x} = \frac{\sum x}{n} = \frac{28}{7} = 4$$

$$\bar{y} = \frac{\sum y}{n} = \frac{917}{7} = 131$$

$$b = \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n\bar{x}^2} = \frac{3961 - (7)(4)(131)}{140 - (7)(4^2)} = \frac{293}{28} = 10.46$$

$$a = \bar{y} - b\bar{x} = 131 - (10.46 \times 4) = 89.16$$

Therefore, the least squares trend equation is:

$$\hat{y} = a + bx = 89.16 + 10.46x$$

To project demand in 2003, we denote the year 2003 as $x = 8$, and:

$$\text{Sales in 2003} = 89.16 + 10.46 \times 8 = 172.84$$

Unit 5

1. You want to compete in the super premium ice cream market. The task is to determine the wants of the super-premium market and the attributes/how to be met by their firm. Use the house of quality concept. Market research has revealed that customers feel four factors are significant in making a buying decision. A “rich” taste is most important followed by smooth texture, distinct flavour, and a sweet taste. From a production standpoint, important factors are the sugar content, the amount of butterfat, low air content, and natural flavours.

Answer:

					
					
		Sugar content	High butterfat	Low air content	Natural flavor
"Rich" taste	1	○	⊗		
Smooth texture	2		●	●	
Distinct flavor	4			○	
Sweet taste	3	○			
		12	5	2	14

2. Illustrate a bill-of-material for a ham and cheese sandwich

Answer:

One possible BOM would be:

Bill of Material

Bread 2 slices

Ham 1 slice

Swiss Cheese 1 slice

Lettuce 1/26 head of lettuce

Mustard 2 teaspoon

Pickle relish 1 teaspoon

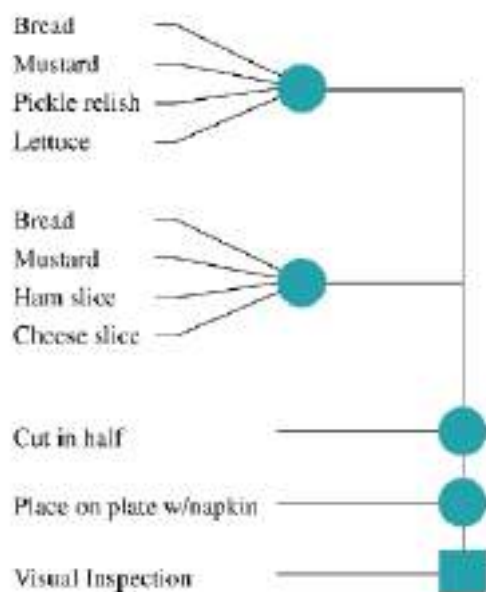
Paper plate 1

Paper napkin 1

3. Illustrate an assembly chart for a ham and cheese sandwich.

Based on the BOM in Revision Question 2, an assembly chart might look like

Answer:



3. Michael's Engineering, Inc. manufactures components for the ever-changing notebook computer business. He is considering moving from a small custom design facility to an operation capable of much more rapid design of components. This means that Michael must consider upgrading his CAD equipment.

Option 1 is to purchase two new desktop Computer Assisted Design systems at R100, 000 each.

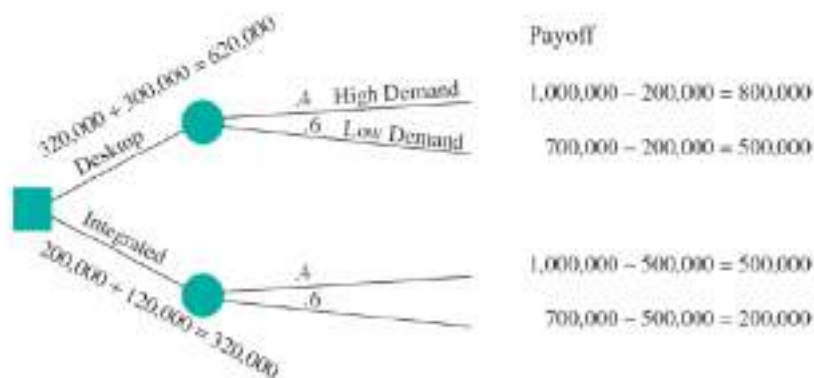
Option 2 is to purchase an integrated system and the related server at R500, 000.

Michael's sales manager has estimated that if the market for notebook computers continues to expand, sales over the life of either system will be R1, 000,000. He places the odds of this happening at 40%.

He thinks the likelihood of the market having already peaked to be 60% and future sales to be only R700, 000.

Outline your suggestion to Michael and explain the EMV of this decision

Answer:



The EMV for the desktop systems is R620, 000 vs. R320,000 for the integrated system. Therefore, Michael should purchase the desktop systems.

Unit 6

1. Jackson Custom Machine Shop has a contract for 130, 000 units of a new product. Sam Jumper, the owner, has calculated the cost for three process alternatives. Fixed costs will be for general-purpose equipment (GPE), R150, 000; flexible manufacturing (FMS), R350, 000; and dedicated automation (DA), R950, 000. Variable costs will be: GPE, R10; FMS, R8; and DA, R6.

Select the one he should choose.

Answer:

Solve for the crossover between GPE and FMS:

$$10X + 150000 - 8X + 350000$$

OR

$$2X = 200000$$

$$x = 100,000 \text{ units}$$

Solve for the crossover between FMS and DA:

$$8X + 350000 = 6X + 950000$$

OR

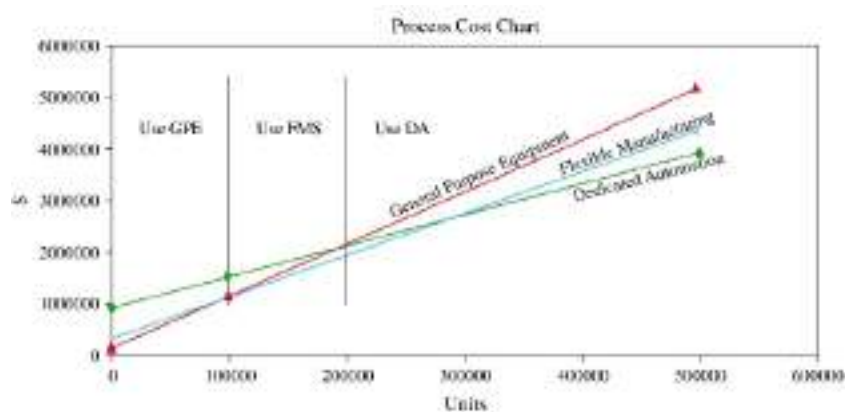
$$2X = 600000$$

$$X = 300000$$

Therefore, at a volume of 300,000 units, FMS is the appropriate strategy

2. Illustrate Problem 1 graphically.

Answer:



3. Using either your analytical solution found in Problem 1, or the graphical solution found in Problem 2, outline the volume ranges where each process should be used.

Answer: Below 100,000 units use GPE, between 100,000 and 300,000 use FMS, above 300,000 use DA

4. If Jackson Custom Machine is able to convince the customer to renew the contract for another one or two years, explain what implications this may have on his decision.

Answer: If Jackson Custom Machine is able to get the customer to extend the contract for another two years, the owner would certainly wish to take advantage of the savings using Dedicated Automation

5. The design capacity for engine repair in our company is 80 trucks/day.

The effective capacity is 40 engines/day and the actual output is 36 engines/day.

Examine the utilisation and efficiency of the operation. If the efficiency for next month is expected to be 82%, explain what the expected output is.

Answer:

$$\text{Utilization} = \frac{\text{Actual output}}{\text{Design capacity}} = \frac{36}{80} = 45\%$$

$$\text{Efficiency} = \frac{\text{Actual output}}{\text{Effective capacity}} = \frac{36}{40} = 90\%$$

$$\begin{aligned} \text{Expected output} &= (\text{Effective capacity}) (\text{Efficiency}) \\ &= (40)(0.90) = 36 \text{ engines/day} \end{aligned}$$

7. Find the break-even point in R and in units.

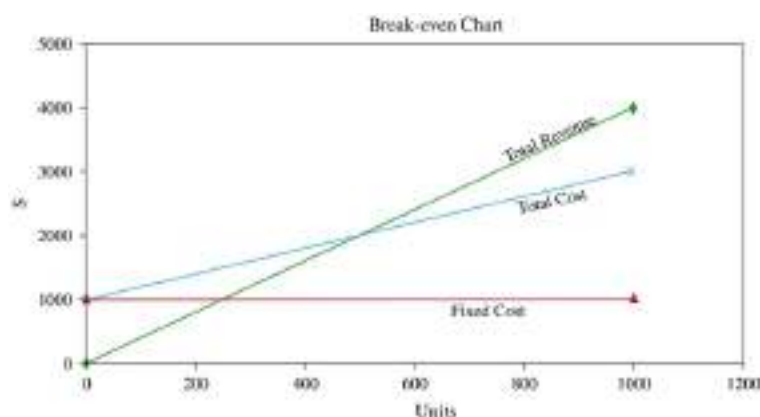
Answer:

$$\text{Break-even point}(\$) = BEP(\$) = \frac{F}{1 - \frac{P}{P}} = \frac{1000}{1 - \frac{2}{4}} = \frac{1000}{0.5} = \$2,000$$

$$\text{Break-even point}(x) = BEP(x) = \frac{F}{P - V} = \frac{1000}{4 - 2} = 500$$

8. Develop the break-even chart for Self-Assessment Activity 6.

Answer:



9. Jack's Grocery is manufacturing a "store brand" item that has a variable cost of \$0.75 per unit and a selling price of \$1.25 per unit. Fixed costs are \$12,000. Current volume is 50,000 units. The Grocery can substantially improve the product quality by adding a new piece of equipment at an additional fixed cost of \$5,000. Variable cost would increase to \$1.00, but their volume should increase to 70,000 units due to the higher quality product. Should the company buy the new equipment?

Answer:

$$\text{Profit} = IR - TC$$

Option A: Stay as is:

$$\text{Profit} = 50,000 * (1.25 - .75) - 12,000 = \$13,000.$$

Option B: Add equipment:

$$\text{Profit} = 70,000 * (1.25 - 1.00) - 17,000 = \$500.$$

Therefore, the company should continue as is with the present equipment as this returns a higher profit.

10. What are the break-even points (\$ and units) for the two processes considered in Self-Assessment Activity 8?

Answer:

Using current equipment:

$$BEP(\$) = \frac{F}{1 - \frac{V}{P}} = \frac{12,000}{1 - \frac{0.75}{1.25}} = \frac{12,000}{1 - 0.60} = \frac{12,000}{0.40} = \$30,000$$

$$BEP(x) = \frac{F}{P - V} = \frac{12,000}{1.25 - 0.75} = 24,000$$

Using the new equipment

$$BEP(\$) = \frac{F}{1 - \frac{V}{P}} = \frac{17,000}{1 - \frac{1.00}{1.25}} = \frac{17,000}{1 - .80} = \frac{17,000}{0.2} = \$85,000.$$

$$BEP(x) = \frac{F}{P - V} = \frac{17,000}{1.25 - 1.00} = \frac{17,000}{0.25} = 68,000.$$

11. Good News! You are going to receive \$6,000 in each of the next 5 years for sale of used machinery. A bank is willing to lend you the present value of the money in the meantime at discount of 10% per year. Illustrate how much cash you would receive now.

Answer:

The net present value factor for 10% and 5 years is 3.79

$$(3.79 = 0.909 + 0.826 + 0.751 + 0.683 + 0.621)$$

Therefore, the present value is: $3.79 \times R\ 6000 = R\ 22\ 740$

The Bad News is you do have to pay back the loans!

Unit 7

1. A major drug store chain wishes to build a new warehouse to serve the whole Midwest. Currently, it is looking at three possible locations. The factors, weights, and ratings being considered are given below:

Explain which city they should choose.

		Ratings		
Factor	Weights	Peoria	Des Moines	Chicago
Nearness to markets	20	4	7	5
Labour cost	5	8	8	4
Taxes	15	8	9	7
Nearness to suppliers	10	10	6	10

Answer:

		Ratings			Weighted Ratings		
Factor	Weights	Peoria	Des Moines	Chicago	Peoria	Des Moines	Chicago
Nearness to markets	20	4	7	5	80	140	100
Labour cost	5	8	8	4	40	40	20
Taxes	15	8	9	7	120	135	105
Nearness to suppliers	10	10	6	10	100	60	100
		Sum of Weighted ratings:			340	375	325

Therefore, it appears that based upon the weights and rating, Des Moines should be chosen.

2. Balfour's is considering building a plant in one of three possible locations. They have estimated the following parameters for each location:

Illustrate for what unit sales volume they should choose each location.

Location	Fixed Cost	Variable Cost
Waco, Texas	R300,000	R5.75
Tijuana, Mexico	R800,000	R2.75
Fayetteville, Arkansas	R100,000	R8.00

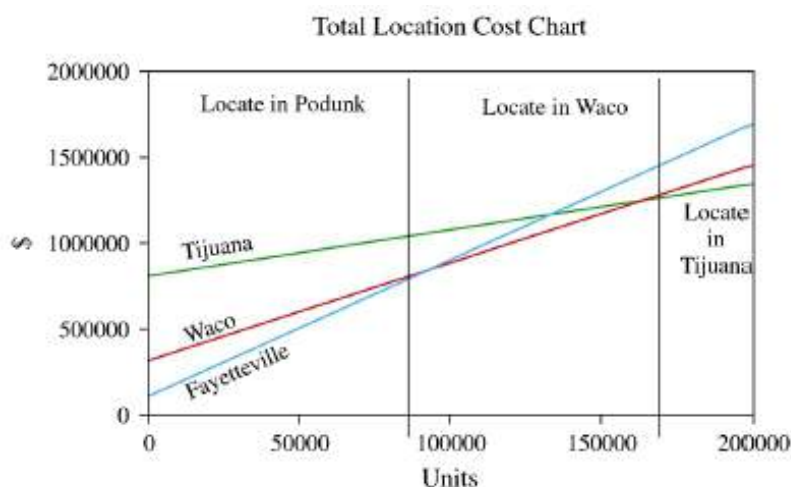
Answer:

Transition between Waco and Tijuana:

$$\begin{aligned}
 300,000 + (5.75x) &= 800,000 + (2.75x) \\
 3x &= 500,000 \\
 x &= 166,000
 \end{aligned}$$

Transition between Waco and Fayetteville:

$$\begin{aligned}
 300,000 + (5.75x) &= 100,000 + (8.00x) \\
 200,000 &= 2.25x \\
 88,888 &= x
 \end{aligned}$$



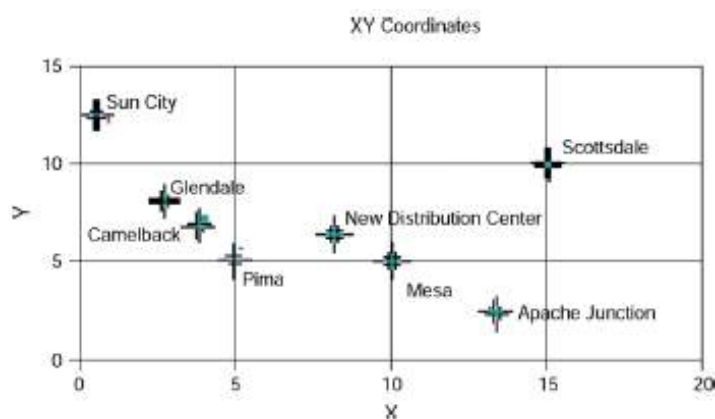
3. Our main distribution centre in Phoenix, AZ is due to be replaced with a much larger, more modern facility that can handle the tremendous needs that have developed with the city's growth. Fresh produce travels to the seven store locations several times a day making site selection critical for efficient distribution. Using the data in the following table, Illustrate the map coordinates for the proposed new distribution centre

Store Locations	Map Coordinates (x,y)	Truck Round Trips per Day
Mesa	(10,5)	3
Glendale	(3,8)	3
Camelback	(4,7)	2
Scottsdale	(15,10)	6
Apache Junction	(13,3)	5
Sun City	(1,12)	3
Pima	(5,5)	10

Answer:

$$C_x = \frac{(10*3)+(3*3)+(4*2)+(15*6)+(13*5)+(1*3)+(5*10)}{3+3+2+6+5+3+10} = \frac{255}{32} = 7.97$$

$$C_y = \frac{(5*3)+(8*3)+(7*2)+(10*6)+(3*5)+(12*3)+(5*10)}{3+3+2+6+5+3+10} = \frac{214}{32} = 6.69$$



4. A company is planning on expanding and building a new plant in one of three countries in Middle or Eastern Europe. The General Manager, Patricia Donegal, has decided to base her decision on six critical success factors including: technology availability and support, availability and quality of public education, legal and regulatory aspects, social and cultural aspects, economic factors, and political stability. Using a rating system of 1 (least desirable) to 5 (most desirable) she has arrived at the following ratings (you may, of course, have different opinions).

Select the country where the plant should be built.

Critical Success Factor	Turkey	Serbia	Slovakia
Technology availability and support	4	3	4
Availability and quality of public education	4	4	3
Legal and regulatory aspects	2	4	5
Social and cultural aspects	5	3	4
Economic factors	4	3	3
Political stability	4	2	3

Answer:

Critical Success Factor	Turkey	Serbia	Slovakia
Technology availability and support	4	3	4
Availability and quality of public education	4	4	3
Legal and regulatory aspects	2	4	5
Social and cultural aspects	5	3	4
Economic factors	4	3	3
Political stability	4	2	3
=	23	19	22

Based upon her ratings of the critical success factors, Patricia should choose Turkey. From a practical perspective, given the small difference between the scores for Turkey and Slovakia, and the subjectivity of the ratings themselves, Patricia would be better advised to develop additional critical success factors, more carefully weigh the individual factors; or, in general, to acquire more information before making her decisions.

5. Assume that Patricia decides to use the following weights for the critical success factors:

- Technology availability and support 0.3
- Availability and quality of public education 0.2
- Legal and regulatory aspects 0.1
- Social and cultural aspects 0.1
- Economic factors 0.1

- **Political stability 0.2**

Explain if this would change her decision.

Answer:

Critical Success Factor	Wgt	Turkey		Serbia		Slovakia	
Technology availability and support	0.3	4	1.2	3	0.9	4	1.2
Availability and quality of public education	0.2	4	0.8	4	0.8	3	0.6
Legal and regulatory aspects	0.1	2	0.2	4	0.4	5	.5
Social and cultural aspects	0.1	5	0.5	3	0.3	4	0.4
Economic factors	0.1	4	0.4	3	0.3	3	0.3
Political stability	0.2	4	0.8	2	0.4	3	0.6
=			3.9		3.1		3.6

No, in this case, use of the weighting factors does not change the recommendation. One might again suggest that additional information be considered in making the decision.

6. Explain how you would advise Ms. Donegal.

a. Doubling the number of critical success factors, or

Answer :

Doubling the number of critical success factors. There are two issues here. First, from a practical perspective there are a limited number of truly “critical” success factors – and these should be the ones presently being considered. Any additional factors should be of secondary or tertiary importance. Second, given the subjective nature of the rating process, adding additional factors would also increase the overall margin of error of the final ratings to a degree that may eliminate any gain in differentiation arising from the use of the additional factors. The use of a maximum of seven to nine critical success factors is usually appropriate.

b. Breaking down each of the existing critical success factors into smaller, more narrowly defined items, e.g., Availability and quality of public education might be broken into primary, secondary, and post-secondary education.

Answer :

Given that one’s ability to estimate or rate an aggregate is usually better than one’s ability to estimate or rate the individual components of the aggregate, this approach is unlikely to provide much help.

7. Examine the methods of evaluating location alternatives.

Answer: The prescribed textbook suggested four methods outlined below but some other texts have five methods separating the CVP method (cost volume profit analysis) from Break even method. For purposes of this module, only four methods are suggested and are as follows:

The Factor-Rating Method

This is a location method that instils objectivity into the process of identifying hard-to-evaluate costs. It has six steps:

1. Develop a list of relevant factors.
2. Assign a weight to each factor that reflects its relative importance in the company's objectives.
3. Develop a scale for each factor (for example 1 to 10 or 1 to 100).
4. Have management score each location for each factor, using the scale in step (iii).
5. Multiply the score by the weights for each factor and the total score for each location.
6. Make a recommendation based on the maximum point score.

The Locational Break-Even Analysis

This is a cost-volume analysis to make an economic comparison of location alternatives. There are three steps:

1. Determine the fixed and variable costs for each location
2. Plot the costs for each location, with costs on the vertical axis of the graph and the annual volume on the horizontal axis
3. Select the location that has the lowest total cost for the expected production volume

Centre-of-Gravity Method

This is a mathematical technique used for finding the best location for a single distribution point those services several stores in the area.

Transportation Model

The objective of the transportation model is to determine the best pattern of shipments from several points of supply (sources) to several points of demand (destinations).

Unit 8

1. Kanawaty, G. (2020) in his book "Introduction to Work Study" laid much of the foundation for recognising the work force and critical to the sustainability of the business.

Critically analyse the basic procedure he proposes for a work study.

Answer:

- Select the job or process to be studied
- Record and collect all the relevant data about the job or process, using the most suitable data collection techniques, so that the data will be in the most convenient form to be analysed
- Examine the recorded facts critically and challenge everything that is done including the purpose of the activity, the place where it's performed, the sequence in which it is done, the person who is doing it and the means by which it is done
- Develop the most economical method, taking into account all the circumstances
- Evaluate the results attained by the improved method compared with the quantity of work involved and calculate a standard time for it
- Define the new method and present it to all concerned, either verbally or in writing, using demonstrations
- Install the new method, training those involved, as an agreed practice with allocated time of operation
- Maintain the new standard practice by monitoring the results and comparing them with the original targets

2. Read the following scenario and answer the questions that follow:

Company B has a diverse workforce with varying motivational drivers. The management wants to implement a motivation strategy that caters to different employee needs.

a. Determine the strategies that Company B can use to cater to the diverse motivational needs of its workforce?

Answer: Company B can use a combination of financial incentives (e.g., bonuses, profit-sharing), non-financial incentives (e.g., recognition programs, career development opportunities), and personalised approaches (e.g., flexible working hours, tailored career paths). They should also consider conducting surveys to understand specific motivational drivers and preferences of different employee groups.

b. Analyse how Company B can measure the effectiveness of its new motivation strategy?

Answer: Company B can measure effectiveness by tracking key performance indicators (KPIs) such as employee productivity, engagement scores, turnover rates, and satisfaction levels. They can also gather direct feedback through surveys and focus groups to assess employee perceptions and experiences with the new strategy.

c. Discuss the role that leadership plays in the success of the motivation strategy at Company B, and how can leaders be equipped to support it?

Answer: Leadership plays a critical role by modelling the desired behaviours, communicating the importance of the strategy, and actively participating in its implementation. Leaders can be equipped through training on motivational techniques, ongoing support from HR, and clear guidelines on how to recognise and reward employees effectively.

3. Read the following scenario and answer the questions that follow: Company CEB LTD is launching a new product and needs to ensure that its production team is highly motivated to meet tight deadlines and high-quality standards.

a. Determine the specific incentives that Company CEB LTD can offer to motivate the production team during the new product launch?

Answer: Company CEB LTD could offer incentives such as performance-based bonuses, overtime pay, additional time off, public recognition for exceptional work, team-building activities, and opportunities for professional development. These incentives should be clearly linked to meeting deadlines and maintaining quality standards.

b. Discuss how Company CEB LTD can maintain high motivation levels among the production team throughout the entire product launch process?

Answer: Company CEB LTD can maintain high motivation levels by providing continuous feedback and support, celebrating small milestones, ensuring a positive and collaborative work environment, and addressing any issues or concerns promptly. Regular communication and involvement of the team in decision-making can also help sustain motivation.

c. Examine the steps that Company CEB LTD can take to evaluate the success of its motivational efforts during the product launch?

Answer: Company CEB LTD should track metrics such as production output, quality levels, employee attendance and punctuality, and overall morale. They should also conduct post-launch surveys or focus groups to gather feedback from the production team on the effectiveness of the motivational efforts and identify areas for improvement for future projects.

4. Determine what influences to job satisfaction

Answer:

Job satisfaction is influenced by several key factors, which can significantly impact an employee's motivation, performance, and overall well-being. These factors collectively contribute to how

employees perceive their job, influencing their overall satisfaction, productivity, and retention. Some of the factors are outlined below:

Work Environment:

Description: A positive, safe, and supportive work environment enhances job satisfaction. This includes comfortable working conditions, appropriate resources, and a positive organisational culture.

Impact: A good work environment reduces stress and fosters a sense of belonging.

Compensation and Benefits:

Description: Competitive salaries, bonuses, and benefits (like health insurance, retirement plans, and paid leave) are crucial for job satisfaction.

Impact: Fair compensation reflects the value of an employee's work and provides financial security.

Job Security:

Description: Stability and the assurance of continued employment contribute to job satisfaction.

Impact: Employees who feel secure in their jobs are more likely to be motivated and loyal.

Work-Life Balance:

Description: Flexibility in work hours, remote work options, and time off are important for maintaining a healthy work-life balance.

Impact: Good work-life balance reduces burnout and increases overall happiness.

Recognition and Appreciation:

Description: Regular recognition and appreciation for work well done boosts morale and job satisfaction.

Impact: Feeling valued and acknowledged reinforces positive behaviour and motivation.

Opportunities for Growth:

Description: Access to professional development, training, and career advancement opportunities are key motivators.

Impact: Employees who see a clear path for growth are more engaged and committed.

Job Design and Content:

Description: Jobs that are meaningful, challenging, and utilise an employee's skills contribute to higher satisfaction.

Impact: Engaging work that offers variety and autonomy increases fulfilment.

Management and Leadership:

Description: Effective, supportive, and communicative leadership is crucial for a positive work experience.

Impact: Good relationships with supervisors and clear communication foster trust and satisfaction.

Interpersonal Relationships:

Description: Positive relationships with colleagues create a supportive and collaborative work environment.

Impact: Good teamwork and social support at work enhance job satisfaction.

Organisational Culture:

Description: A culture that aligns with an employee's values and provides a sense of purpose contributes to satisfaction.

Impact: A strong, positive culture fosters loyalty and a shared sense of mission.

5. Explain if money is an incentive and if there are any drawbacks to purely financial incentives

Answer:

Incentive Role: Money is a powerful motivator and is often used as an incentive in the form of salaries, bonuses, commissions, and other financial rewards. It can drive behaviour, encourage performance, and align employee efforts with organizational goals. For many employees, financial incentives fulfil basic needs, provide a sense of security, and serve as recognition for their hard work.

Drawbacks of Purely Financial Incentives:

Short-Term Motivation:

Explanation: Financial incentives can boost motivation temporarily but may not sustain long-term engagement or commitment. Once the financial reward is received, the motivation may diminish, leading to fluctuating performance levels.

Undermining Intrinsic Motivation:

Explanation: Over-reliance on financial rewards can reduce intrinsic motivation, where employees might begin to focus solely on the reward rather than finding satisfaction in the work itself. This can lead to a decline in creativity, innovation, and job satisfaction.

Encouraging Unethical Behaviour:

Explanation: When financial incentives are tied to specific targets, there is a risk that employees might engage in unethical behaviour, such as cutting corners or manipulating results, to meet the criteria and secure the reward.

Inequality and Resentment:

Explanation: If financial incentives are perceived as unfair or are unevenly distributed, they can create resentment among employees, leading to a negative work environment, reduced collaboration, and lower morale.

Focus on Quantity over Quality:

Explanation: Financial incentives often emphasize meeting quantitative targets, which can lead employees to prioritise speed and volume over quality, potentially harming the overall product or service.

Inflation of Expectations:

Explanation: Over time, employees may come to expect financial rewards, leading to a situation where the same level of incentive no longer motivates them, requiring increasingly larger rewards to achieve the same effect.

Ignoring Non-Financial Needs:

Explanation: Employees have needs beyond financial compensation, such as recognition, work-life balance, job satisfaction, and opportunities for growth. Purely financial incentives may neglect these aspects, leading to dissatisfaction despite good pay.

Unit 9

1. Determine the sales necessary to equal a dollar of savings on purchases for a company that has a net profit of 6% and spends 70% of its revenues on purchases.

Answer: From Table 11.3, we see that this company would have to increase sales by approximately R5.56.

2. Determine the sales necessary to equal a dollar of savings on purchases for a company that has a net profit of 8% and spends 40% of its revenues on purchases.

Answer: From Table 11.3, we see that this company would have to increase sales by approximately R2.94.

3. Phil Carter, President of Carter Computer Components, Corp. has the option of shipping computer transformers from its Singapore plant via container ship or airfreight. The typical shipment has a value of R75,000. A container ship takes 24 days and costs R5,000; airfreight takes 1 day and costs R8,000. Holding cost is estimated to be 40% in either case.

Explain how shipments should be made.

Answer:

Cost via container ship:

$$\left[24 * \frac{(.40 * 75,000)}{365}\right] + 5,000 = (24 * 82.19) + 5,000 = 1,972.56 + 5,000 = \$6,972.56$$

Cost via air freight:

$$\left[1 * \frac{(.40 * 75,000)}{365}\right] + 8,000 = (1 * 82.19) + 8,000 = 82.19 + 8,000 = \$8,082.19$$

Unit 10

1. Illustrate the appropriate ABC groups of inventory items.

ABC Analysis		
Stock Number	Annual R Volume	Percent of Annual R Volume
J24	12 500	46.2
R26	9 000	33.3
L02	3 200	11.8
M12	1 550	5.8
P33	620	2.3
T72	65	0.2
S67	53	0.2
Q47	32	0.1
V20	30	0.1
		=100.0

Answer:

ABC Groups			
Class	Items	Annual Volume	Percent of R volume
A	J24, R26	21 500	79.5
B	L02, M12	4 750	17.6
C	P33, T72, S67, Q47, V20	800	2.9
			= 100.0

2. A firm has 1,000 “A” items (which it counts every week, i.e., 5 days), 4,000 “B” items (counted every 40 days), and 8,000 “C” items (counted every 100 days). Explain how many items should be counted per day.

Answer:

Item Class	Quantity	Policy	Number of items to Count Per Day
A	1 000	Every 5 days	$1000/5 = 200/\text{day}$
B	4 000	Every 40 days	$4000/40 = 100/\text{day}$
C	8 000	Every 100 days	$8000/100 = 80/\text{day}$
		Total items to count: 380/day	

3. Assume you have a product with the following parameters:

Demand = 360 Holding cost per year = \$1.00 per unit Order cost = \$100 per order

Explain what the EOQ is.

Answer:

$$\text{EOQ} = \sqrt{\frac{2 * \text{Demand} * \text{Order cost}}{\text{Holding cost}}} = \sqrt{\frac{2 * 360 * 100}{1}} = \sqrt{72000} = 268 \text{ items}$$

4. Given the data from Revision Question 3, and assuming a 300-day work year; explain how many orders should be processed per year.

Explain what the expected time between orders is.

Answer:

$$N = \frac{\text{Demand}}{Q} = \frac{360}{268} = 1.34 \text{ orders per year}$$

$$T = \frac{\text{Working days}}{\text{Expected number of orders}} = 300 / 1.34 = 224 \text{ days between orders}$$

5. Determine the advantages and disadvantages of inventory.

Answer:

Advantages of Inventory:	Disadvantages of Inventory:
• Meeting demand: Ensures products are available to meet customer demand, reducing the risk of stockouts and lost sales • Buffer against uncertainty: Acts as a buffer against fluctuations in demand, supply chain disruptions, and lead time variability • Economies of scale: Allows for bulk purchasing and production, which can reduce costs per unit • Smooth production: Supports continuous production processes by ensuring that raw materials and components are readily available	• Carrying costs: Involves significant costs, such as storage, insurance, and obsolescence, which can reduce profitability • Risk of obsolescence: Inventory may become outdated, particularly in industries with fast-changing products, leading to potential losses • Tied-up capital: Capital invested in inventory could be used elsewhere, leading to reduced financial flexibility • Space requirements: Requires physical storage space, which incurs additional costs and may lead to inefficiencies if space is limited

Unit 11

1. Illustrate the following problem in transportation format and solve for the minimum cost plan.

	Period		
	Feb	Mar	Apr
Demand	55	70	75
Capacity			
Regular	50	50	50
Overtime	5	5	5
Subcontract	12	12	10
Beginning Inventory	10		
Costs			
Regular Time	R60 per unit		
Overtime	R80 per unit		
Subcontract	R90 per unit		
Inventory carrying cost	R1 per unit per month		
Back order cost	R3 per unit per month		

Answer:

Demand for						
		Feb	Mar	Apr	Unused Capacity (dummy)	Total Capacity Available (Supply)
	Beginning Inventory	10	1	2	0	10
February	Regular time	60 45	61 5	62	0	50
	Overtime	80	81 5	82	0	5
	Subcontract	90	91 3	92	9 0	12
March	Regular time	63	60 50	61	0	50
	Overtime	83	80 5	81	0	5
	Subcontract	93	90 2	91 10	0	12
April	Regular time	66	63	60 50	0	50
	Overtime	86	83	80 5	0	5
	Subcontract	96	93	90 10	0	10
	Demand	55	70	75	9	209

2. A company faces seasonal demand for its products, with high demand during the winter holiday season and lower demand during the summer months. The company has decided to use a chase strategy for its aggregate planning. Determine how the company can adjust its workforce and production levels to align with the chase strategy during these seasonal fluctuations?

Answer: To align with the chase strategy, the company can increase its workforce and production levels during the winter holiday season to meet high demand and reduce the workforce and production during the summer months when demand is lower. This approach helps the company minimise inventory costs and avoid overproduction.

3. A manufacturing firm has decided to build up inventory during low-demand periods to meet high demand during peak seasons without changing its workforce size. Discuss the aggregate planning strategy that the firm is using, and what are the potential benefits and drawbacks of this approach?

Answer: The firm is using a level strategy. The potential benefits include stable employment levels and consistent production rates, which can improve worker morale and production efficiency. However, drawbacks may include higher inventory holding costs and the risk of obsolescence if demand forecasts are inaccurate.

4. An electronics company plans to introduce a new product next year and anticipates a significant increase in demand. The company currently operates at near full capacity. Analyse the capacity options that the company could consider meeting the expected increase in demand?

Answer: The company could consider the following capacity options

- Overtime – schedule overtime for current employees to increase production
- Temporary workers – hire temporary workers to handle the increased workload
- Subcontracting – outsource part of the production process to subcontractors
- Invest in new equipment – purchase additional equipment to expand production capacity

Each option has its own cost, quality and flexibility implications that the company needs to evaluate

5. A company needs to produce 10,000 units over the next 5 months. The current workforce can produce 1,500 units per month. The company can use overtime to produce an additional 500 units per month at an extra cost of R10 per unit.

Calculate the total overtime cost if the company uses overtime to meet the production target.

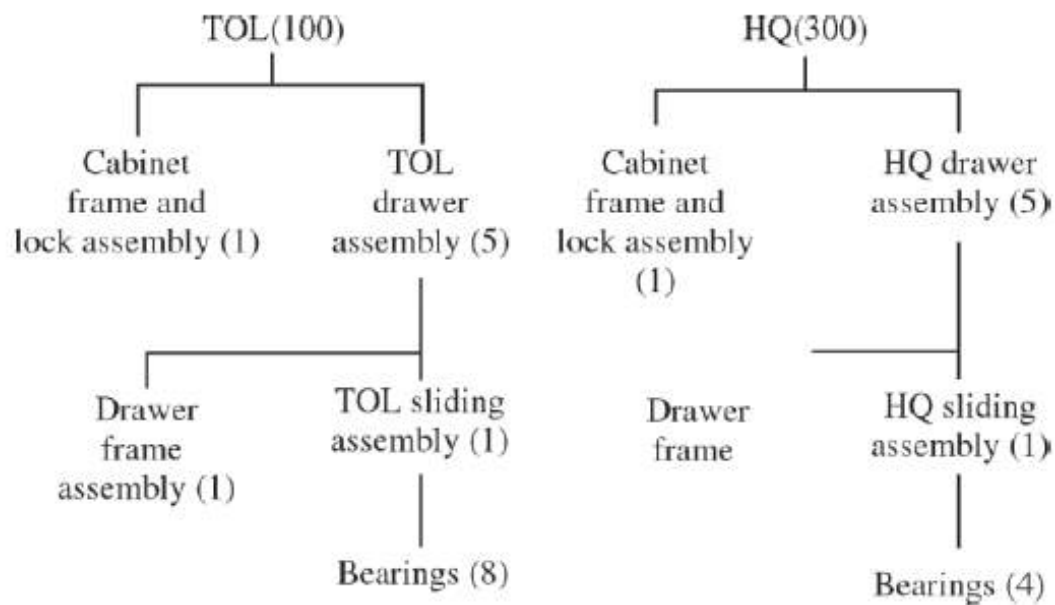
Answer:

- Monthly production requirement = $10,000 \text{ units} / 5 \text{ months} = 2,000 \text{ units per month}$
- Additional units needed per month = $2,000 \text{ units} - 1,500 \text{ units} = 500 \text{ units}$
- Total overtime cost = $500 \text{ units/month} * R10/\text{unit} * 5 \text{ months} = R25,000$

Unit 12

1. The Hunicut and Hallock Corporation makes two versions of the same basic file cabinet, the TOL (Top-of-the-line) five drawer file cabinet and the HQ (High-quality) five drawer filing cabinet. The TOL and HQ use the same cabinet frame and locking mechanism. The drawer assemblies are different although both use the same drawer frame assembly. The drawer assemblies for the TOL cabinet use a sliding assembly that requires four bearings per side whereas the HQ sliding assembly requires only two bearings per side. (These bearings are identical for both cabinet types.) 100 TOL and 300 HQ file cabinets need to be assembled in week #10. No current stock exists. Illustrate a material structure tree for the TOL and the HQ file cabinets.

Answer:



2. Illustrate net material requirements plan for the TOL and HQ file cabinets in the previous problems assuming a current on-hand finished goods inventory of 100 TOL cabinets. The lead times are given below.

Painting and final assembly of both HQ and TOL requires 2 weeks.

Both cabinet frames and lock assembly require 1 week for manufacturing.

Both drawer assemblies require 2 weeks for assembly.

Both sliding assemblies require 2 weeks for manufacturing.

Bearings require 2 weeks to arrive from the supplier.

Answer:

		Week										
		1	2	3	4	5	6	7	8	9	10	Lead Time
TOL	Required date										100	2 weeks
	Order release date											
HQ	Required date										300	2 weeks
	Order release date							300				
Cabinet frame and lock	Required date								300			1 week
	Order release date						300					
HQ drawer assembly	Required date								1500			2 weeks
	Order release date						1500					
Drawer frame assembly	Required date						1500					2 weeks
	Order release date				1500							
HQ sliding assembly	Required date						1500					2 weeks
	Order release date											
	Order release date				1500							
Bearings	Required date				6000							2 weeks
	Order release date		6000									

Receipts:

300 cabinet frames and locks in week 8

1500 HQ drawer assemblies in week 8

1500 drawer frame assemblies in week 6

1500 HQ sliding assemblies in week 6

6000 bearings in week 4

3. A manufacturing company is implementing Materials Requirements Planning (MRP) to manage its production processes. The company faces challenges with supplier reliability and lead times. How can the company address these challenges using MRP principles?

Answer:

The company can use MRP to develop a reliable production schedule based on accurate demand forecasts and inventory levels. By maintaining clear communication with suppliers and sharing production schedules, the company can reduce lead times and ensure timely delivery of materials. Additionally, MRP can help optimise inventory levels to minimise the impact of supplier delays.

4. A company produces a product that requires 3 units of component A and 2 units of component B. The company plans to produce 500 units of the final product. Calculate the total quantity of component A needed using Materials Requirements Planning (MRP).

Answer:

Bill of Materials (BOM):

Component A: 3 units per final product

Component B: 2 units per final product

Total production requirement = 500 units

Total quantity of component A needed = 500 units * 3 units = 1,500 units

5. A company uses MRP to plan its production schedule for the next month. It has a lead time of 2 weeks for receiving materials after placing an order. Calculate the reorder point for a component that has a weekly demand of 200 units.

Answer:

Weekly demand = 200 units

Lead time = 2 weeks

Reorder point = Weekly demand * Lead time = 200 units/week * 2 weeks = 400 units

6. A company uses MRP to plan its production activities for a complex product. During implementation, the company encounters discrepancies between planned and actual production output. What steps should the company take to improve the accuracy of its MRP system?

Answer:

To improve MRP accuracy, the company should regularly update input data such as sales forecasts, inventory levels, and lead times. Conducting regular reviews and audits of the Bill of Materials (BOM) and Master Production Schedule (MPS) can identify discrepancies and ensure alignment with actual production capabilities. Training staff on MRP system usage and ensuring robust communication between departments are also critical steps.

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